ABSTRACT

Title of Dissertation:	DIAGNOSING SMALL NOMINALS: THEORETICAL IMPLICATIONS FROM MOKSHA AND HILL MARI
	Polina Pleshak, Doctor of Philosophy, **2025**
Dissertation directed by:	Professor Maria Polinsky, Department of Linguistics

This dissertation investigates the morphosyntactic behavior of small nominals (SNs) — reduced nominal structures lacking certain morphosyntactic properties characteristic of full DPs — across languages. Based primarily on three detailed case studies (bare complements of Moksha postpositions, Moksha bare direct objects, and bare direct objects in Hill Mari non-finite clauses), I propose diagnostics that can reliably identify small nominals cross-linguistically and explore their implications for syntactic theory, particularly regarding the structure of the extended nominal projection.

The dissertation begins with a carefully delimited theoretical model of nominal phrase structure, emphasizing the role of functional projections in determining nominal size. Each nominal projection between NP and DP is examined in turn, with explicit predictions about its morphosyntactic reflexes.

The core of the dissertation is dedicated to Moksha. I first provide an overview of Moksha nominal morphology and map its features onto a hierarchical DP structure, demonstrating how Moksha nominal phrases align with theoretical expectations. In the first case study, I analyze the contrast between genitive-marked and bare complements of spatial postpositions in Moksha, arguing that the former are full DPs, while the latter are SNs displaying reduced morphosyntactic marking and agreement capabilities. In the second case study, I examine genitive-marked vs. bare direct objects in Moksha. Although bare DOs superficially resemble bare postpositional complements, I analyze them as full DPs with null case marking rather than SNs. In the third case study, I turn to related Hill Mari, analyzing the contrast between accusative-marked and bare DOs in non-finite clauses. I contend that bare DOs in Hill Mari are true SNs, with nominal size varying according to syntactic environment.

Drawing on these case studies, I propose new cross-linguistic diagnostics for identifying SNs, addressing weaknesses in existing criteria. The refined diagnostics focus on definiteness, number, and possession-marking mismatches, improving the identification of SNs across languages.

DIAGNOSING SMALL NOMINALS: THEORETICAL IMPLICATIONS FROM MOKSHA AND HILL MARI

by

Polina Pleshak

Dissertation submitted to the Faculty of the Graduate School of the
University of Maryland, College Park, in partial fulfillment
of the requirements for the degree of
Doctor of Philosophy
2025

Advisory Committee:
Professor Maria Polinsky, Chair
Professor Eric Potsdam
Professor Jeffrey Lidz
Professor Howard Lasnik
Professor Paolo Santorio

Acknowledgements

I am deeply grateful to my advisor and mentor, Masha Polinsky, for her unwavering support throughout every aspect of my PhD program—from our regular meetings filled with fruitful discussions of data and analysis to her care and understanding of my personal well-being. I am also grateful to Eric Potsdam for his insightful comments and invaluable guidance during the later stages of this dissertation. I owe special thanks to my former advisors Svetlana Toldova and Ekaterina Lyutikova, whose early mentorship shaped the foundations of this work. It was under their guidance that I first developed the ideas, collected the initial data, and formulated the hypotheses that ultimately grew into this dissertation. I am also grateful to Omer Preminger for his advice and support during the early stages of my PhD.

This research would not have been possible without my language consultants, who worked tirelessly with me during fieldwork and continued to answer my follow-up questions online in later years. I am especially grateful to Valentina Liskina and Larisa Izitsina, who bore the greatest share of my many questions with patience and generosity.

I would like to thank my committee members — Howard Lasnik, Jeffrey Lidz, and Paolo Santorio — for their careful reading of my dissertation and for our inspiring and productive discussion of my findings. I owe particular gratitude to Howard Lasnik for expanding my knowledge of the relevant scholarship and for his meticulous attention to the logical flow of my exposition and argumentation.

I would like to express my immense gratitude to my friends and colleagues Irina Burukina and Marcel den Dikken, who read early drafts of this work and offered invaluable feedback. I am also grateful to my friends and fieldwork collaborators Anastasiia Sirotina, Tatiana Davidyuk, Maria Kholodilova, and Egor Kashkin for their help with data collection, discussions, and constant support.

Finally, I would never have been able to complete my dissertation without my parents, Natalia and Sergey Pleshak, and my parents-in-law, Sharon and David Neumann, who took care of my daughter Natalie, as well as without my friend and Natalie's baby-sitter Eden Petri. I also wish to thank my friends Harrison Nix, Samantha Ferguson, Nick Roberts, Isabella Zutrau-Pell, Wes Michalsky, Min Zhang, and Sasha Lushtak for offering me support, shelter, and comfort during the hardest stages of this journey.

I acknowledge financial support from the following NSF grants to Maria Polinsky: BCS-2116344, BCS-1619857, and BCS-1941733.

Table of Contents

List of Abbreviations

1, 2, 3	= 1st, 2nd, 3rd person
ABIL	= ability
ABL	= ablative
ABS	= absolutive
ACC	= accusative
ACT	= active
ADD	= additive
ADESS	= adessive
ADJ	= adjective
ADV	= adverb
AGR	= agreement
ANT	= anteriority
AOR	= aorist
ATR	= atributivizer
ATT	= attenuative
ATTR	= atributivizer
CAR	= caritive
CAUS	= causative
CN	= connegative
COMPL	= complementizer
CONV	= converb
CVB	= converb
DAT	= dative
DEF	= definite
DENOM	= denominal
DIM	= diminutive

EL	= elative
EMPH	= emphatic particle
EQU	= equative
ERG	= ergative
ESS	= essive
EX	= existential
F	= feminine
FEM	= feminine
FREQ	= frequentative
FULL	= full form
GEN	= genitive
ILL	= illative
IMP	= imperative
IN	= inessive
INCH	= inchoative
INDEF	= indefinite
INF	= infinitive
INSTR	= instrumental
IPFV	= imperfective
ITER	= iterative
KIN	= kinship
LAT	= lative
LIM	= limitive
LOC	= locative
M	= masculine
MAN	= manner verb
MCMDI	= mismatch between case marking and definite interpretation
MDMI	= mismatch between definiteness marking and interpretation
MED	= middle voice

MIR	= mirative
MNMI	= mismatch between number marking and interpretation
MPMI	= mismatch between possession marking and interpretation
N	= neuter
NEG	= negation
NFIN	= non-finite
NMLZ	= nominalizer
NOM	= nominative
NPST	= non-past
NZR	= nominalizer
O	= object
OM	= object marker
P	= possession
PART	= partitive
PAST	= past
PASS	= passive
PL	= plural
PP	= postpositional
PRES	= present
PRET	= preterite
PROL	= prolative
PRON	= pronoun
PROP	= proprietive
PRS	= present
PRT	= preterite
PST	= past
PTCL	= particle
PTCP	= participle
PUN	= punctuative

REFL = reflexive

RES = resultative

RETR = retrospective shift

RUS = Russian

S = subject

SIM = similative

SG = singular

ST = stem

TMPR = temporalis

TRANS = translative

Chapter 1: Introduction

This dissertation focuses on **small nominals (SNs)**, reduced nominal structures that lack certain morphosyntactic properties characteristic for full DPs. While the concept of reduced structures has been extensively explored in the clausal domain, particularly in the context of clausal complements, the investigation of small nominals remains relatively underdeveloped. This study aims to fill this gap by examining the morphosyntactic behavior of SNs across languages, with a particular focus on the Uralic languages Moksha and Hill Mari.

It is a well-established theoretical assumption that all DPs must bear case. However, more and more studies reveal that often we find a bare nominal in a context where case would be expected — such as the absence of case on direct objects, which would otherwise bear an object case (1).

(1) TATAR (Lyutikova & Pereltsvaig 2015: 303)

a. *marat* ***mašina-nı*** *sat-ıp* *al-dı*
Marat car-ACC sell-CONV take-PAST
'Marat bought a (specific)/the car.'

b. *marat* ***mašina*** *sat-ıp* *al-dı*
Marat car sell-CONV take-PAST
'Marat bought a car/cars.'

One possible explanation for the absence of case on certain arguments is that such caseless arguments simply do not project DP or even more of the higher functional projections, such as NumP. This challenges the traditional view that all argumental nominal phrases are full DPs.

While the explanation of the cased vs. bare nominal (or triggering agreement vs. not triggering agreement nominal) pattern has been explored in several works (Déchaine and Wiltschko 2002; Danon 2006; Pereltsvaig 2006; Lyutikova and Pereltsvaig 2015; Levin 2015; Halm 2021; Lyskawa & Ranero 2022, a.o.), besides a short overview paper by Pereltsvaig (2021), research on the reduced nominal structure as a general phenomenon remains scarce. In this dissertation, I address some of the gaps in the understanding of SNs as a part of the grammatical inventory of the language. Specifically, I propose diagnostics that can reliably identify small nominals cross-linguistically and explore their implications for syntactic theory, particularly regarding the structure of the extended nominal projection. My diagnostics are based on the mismatches between overt marking and interpretation of the main nominal categories — definiteness, number and nominal agreement, as well as the ability of SNs to combine with certain modifiers. Additionally, I critique diagnostics that rely on the category restriction, non-referential interpretation and movement.

The three case studies presented in this dissertation — bare complements of Moksha postpositions, Moksha bare DOs and bare DOs in Hill Mari non-finite clauses — further expand the empirical domain of the small nominal research. To the best of my knowledge, there has been no in-depth discussion of bare complements of adpositions as SNs (Chapter 4); nor have previous studies demonstrated the presence of small DOs of different sizes within the same language depending on the specific construction (Chapter 6).

The main objectives of this research are as follows:

- To analyze the expected properties of functional heads within the nominal spine and distinguish those that indicate an SN structure from those that do not contradict it.
- To propose diagnostics for identifying SNs cross-linguistically.
- To determine the theoretical implications of the possibility of SNs in argument positions.

My theoretical analysis is primarily based on my own fieldwork data from two Uralic Finno-Ugric languages, Moksha and Hill Mari. The Moksha data were collected in the villages of Lesnoje Tsibajevo and Lesnoje Ardashevo (Republic of Mordovia, Russia) in 2014–2019 and via online elicitation from the same consultants in 2020–2024. Some supplementary data come from the grammatical description of the same Moksha dialect (Toldova et al. 2018), in which I am also one of the contributors, as well as from the text corpus associated with it. The Hill Mari data were collected in the village of Kuznetsovo (Mari El Republic, Russia) in 2015 – 2019, with further online elicitation from the same consultants in 2020–2024.

In this chapter, I provide an overview of studies of small structures in syntax, with a particular focus on research that featured nominals of reduced structure. I also introduce the fundamentals of late insertion models, specifically Distributed Morphology, which are crucial for the SN analysis in general and my illustrations in particular. Finally, I outline my specific theoretical assumptions regarding case assignment and agreement.

1.1. Precursors: Small structures in Syntax

In this section, I introduce the idea that not all syntactic configurations require a full structure, such as a full CP for a clause or a full DP for a nominal. The absence of higher projections can result from two kinds of processes: either deleting a previously built structure or not building the full structure in the first place. While acknowledging structure removal approaches (from Ross 1967 and Evers 1975 on pruning, to Müller 2015 on Remove and Pesetsky 2021 on exfoliation), I focus on approaches that do not involve deletion, since they do not violate Chomsky's (2004) No Tampering Condition, which prohibits any changes made to an already built syntactic structure.

First, I give a brief overview of the idea that not all sentential arguments must be full CPs. Then, I discuss proposals that analyze certain nominal arguments as being smaller than a full DP.

1.1.1. Restructuring

Since the works of Perlmutter (1968), Evers (1975), Rizzi (1976), and Haegeman and van Riemsdijk (1986), among others, it has been established that not all infinitival predicates exhibit properties expected of full clauses. The phenomenon in which infinitives lack clausal properties while combining with certain matrix predicates is known as **restructuring** (Rizzi 1976). Below, I outline some properties that suggest the absence of functional projections from vP to CP.

First, infinitival clauses analyzed as restructuring are incompatible with a complementizer. When combined with the ungrammaticality of wh-movement internal to the infinitival clause, this suggests that these clauses lack the CP layer. Second, restructuring infinitives do not encode tense: they cannot be modified by temporal adverbials, and their temporal interpretation is restricted. This suggests the absence of the corresponding functional projection T:

(2) GERMAN (adapted from Wurmbrand 2001: 80)

Der	*Kuchen*	*war*	*leicht*	[(**morgen*)	*zu*	*backen*]
DEF.M.NOM	cake.NOM	be.PST.3SG	easy	(*tomorrow)	to	bake.INF

‘The cake was easy to bake (*tomorrow).’

Third, in restructuring infinitives, DOs do not get accusative unless an alternative source, such as the matrix predicate, is available, and they participate in syntactic processes triggered in the matrix clause. This reveals the absence of the vP layer in such constructions, since most researchers agree that v head participates in the case assignment to the DO, as well as serves as a boundary for the processes between its complement and the structure above it.

Consider long passive, in which the matrix verb undergoes passivization, promoting the object of the embedded argument clause to the subject position of the matrix clause (3). If the v head were present, it would result in accusative case assignment, and the passivization of the matrix clause should not be able to affect the DO of the lower clause due to locality constraints.

(3) GERMAN (adapted from Wurmbrand 2001: 19)

a.	*dass*	***der***	***Traktor***	*zu*	*reparieren*	*versuch-t*	***wurde***
	that	DEF.M.NOM	tractor.NOM	to	repair.INF	try-PTCP	be.PASS.PST.3SG

‘that they tried to repair the tractor.’

b.	*dass*	***die***	***Traktor-en***	*zu*	*reparieren*	*versuch-t*	***wurde-n***
	that	DEF.PL.NOM	tractor-PL.NOM	to	repair.INF	try-PTCP	be.PASS.PST-3PL

‘that they tried to repair the tractors.’

Further evidence comes from nominative object constructions in Japanese, which show that DOs of restructuring infinitives receive case from the matrix clause. In (4), Japanese stative verbs assign nominative rather than accusative case to their DOs. Nevertheless, when embedded under a matrix predicate that can assign accusative, the DO appears in the accusative (4).

(4) JAPANESE (Wurmbrand 2001: 34 from Koizumi 1995: 66)

a.	**Emi-ga*	*nihongo-o*	*deki-ru*
	Emi-NOM	japanese-ACC	be.capable-PRES

‘Emi speaks Japanese.’

b.	*Emi-ga*	*nihongo-ga*	*deki-ru*
	Emi-NOM	japanese-NOM	be.capable-PRES

‘Emi speaks Japanese’

c.	*Emi-ga*	*nihongo-o*	*deki-hazime-ta*
	Emi-NOM	japanese-ACC	capable-begin-PAST

‘Emi began to be able to speak Japanese.’

The behavior in (3) and (4) is exactly what is predicted under the analysis that assumes the absence of vP, eliminating the need to posit exceptional long distance movement or case assignment rules. It is worth noting that the predicates selecting restructuring infinitives as their complements can also select non-restructuring infinitives (Wurmbrand 2001: 57).

Importantly, restructuring infinitives are only a subset of small structures in the infinitival constructions. Alongside restructuring infinitives (which are VPs), Wurmbrand (2001) distinguishes reduced non-restructuring infinitives, which can be TPs or vPs, as opposed to full CP clauses. For Wurmbrand, the difference between restructuring and reduced non-restructuring infinitives is in the ability of the latter to assign case to their complement.

The takeaway from this section is that syntactic complements, and clausal complements in particular, come in different sizes. Some are full structures, while others lack one or more higher functional layers. The smallest structures are the most easily identifiable, as they differ dramatically from full ones. For example, VPs, which lack all functional projections, have been identified as restructuring infinitives. By contrast, TPs lack only properties associated with C, and vPs lack properties associated with T and C. This classification of infinitival clauses parallels the analysis of nominal arguments proposed in Chapter 6.

1.1.2. Small nominals

As opposed to the reduced clausal structures serving as arguments, discussed in the previous section, reduced nominal structures have received less scholarly attention. However, analyses of certain nominal arguments as lacking higher projections have been sporadically proposed in the literature. In this section, I review some of the most prominent studies on this topic.

First, I provide an overview of objects in Niuean, which, according to Massam (2001) do not project a full DP structure. Then I move onto Hebrew objects that appear without the marker *et*, analyzed by Danon (2006) as being structurally smaller than DP. I then discuss Pereltsvaig's (2006) SNs in Russian. After that, I explore different caseless nominals in Tatar, all of which

Lyutikova and Pereltsvaig (2015) classify as SNs. Finally, I look at Meadow Mari bare direct objects (DOs) in non-finite clauses, which are analyzed as lacking DP by Serdobolskaya (2015).

1.1.2.1. Objects in Niuean

Discussing the distinction between marked (bearing an absolutive marker) and bare objects in Niuean, Massam (2001) notices that the latter exhibit a more restricted set of morphosyntactic properties than what would be expected from bona-fide DPs. Consider the contrast in (5). The DO in (5)a is marked as plural and absolutive, whereas the DO in (5)b is not marked for either. Given that the subject *e ia* 'he' in (5)a is ergative, the structure represents a prototypical transitive clause. In (5)b, by contrast, the subject *a ia* 'he' is absolutive, signaling the syntactic intransitivity of the clause, despite the presence of a DO.

The two objects also differ in the position they occupy within the sentence: in (5)a, the absolutive DO follows the subject and other clausal element, whereas in (5)b, the bare DO immediately follows the verb.

(5) NIUEAN (Massam 2001: 157)

a.	*takafaga*	*tūmau*	*nī*	*e*	*ia*	***e***	***tau***	***ika***
	hunt	always	EMPH	ERG	3SG	ABS	PL	fish
b.	*takafaga*	***ika***	*tūmau*	*nī*	*a*	*ia*		
	hunt	fish	always	EMPH	ABS	3SG		

'He is always fishing.'

The absence of overt expression of nominal categories, coupled with verb adjacency and non-specific interpretation, lead to the idea that Niuean bare objects are structurally smaller than DP.

Since Massam's work, the phenomenon in which DOs with reduced properties appear adjacent to the verb received the name pseudo-noun incorporation (PNI). This label underscores the similarities between bare DOs (such as those in Niuean) and incorporated objects, while also highlighting the key difference — the presence of structure on top of the head noun. I discuss the relationship between PNI and a small nominal analysis in Chapter 7. For now, I focus on the broader idea that certain argumental nominals can be analyzed as being structurally smaller than DP.

1.1.2.2. DOs in Hebrew

In Hebrew, DOs can either be preceded by a special marker *et* or appear bare. The choice between these two strategies depends on the presence of the definite article, a pronominal suffix or the DO being a proper name. Compare the contrast between the two definite DOs in (6). In (6)a, the DO carries a definite article and must be preceded by *et*. In (6)b, on the other hand, the DO lacks a definite article and *et* cannot appear.

(6) HEBREW (Danon 2001: 1002)

a.	*ha-memsala*	*daxata*	**(et)*	*ha-haca'a*	*ha-zo*
	the-government	rejected	OM	the-proposal	the-this
b.	*ha-memsala*	*daxata*	*(*et)*	*haca'a*	*zo*
	the-government	rejected	OM	proposal	this
	'The government rejected this proposal.'				

Regardless of whether the marker *et* is analyzed as the surface realization of accusative case (Shlonsky 1997: 17–20) or as a case assigner (Falk 1991; Danon 2001), what is important is that its presence on DOs depends on whether they are formally marked as definite.

To deal with a purely formal markedness of definiteness (DOs in both (6)a and (6)b are definite, yet only the first one is accompanied by *et*), Danon introduces a formal feature [+def]. The presence of this feature triggers the definite article within the noun phrase, and the presence of an article indicates the presence of the DP layer. According to Danon, in the absence of an article, the presence of the DP layer is not required.

The analysis of Hebrew bare DOs proposed by Danon is that the feature [+def] is situated in D, only DPs can bear definite articles, and *et* only can combine with DPs. Therefore, the absence of an article and the absence of the object marker *et* together indicate that the nominal lacks the DP projection.

1.1.2.3. Small nominals in Russian

While talking about nominals that are smaller than DP, I use the term "small nominals (SNs)", which was proposed by Pereltsvaig (2006). SN — "nominals that are not projected fully as DPs, but rather lack some or all functional projections" (Pereltsvaig 2006: 433). The term "small nominals" was inspired by the previously existing in the literature term "small clauses" (Williams 1974). Unlike Massam (2001) and Danon (2006), who propose reduced nominal structures to account for specific empirical problems, Pereltsvaig approaches the issue from a theoretical perspective, questioning whether argumental nominals can be structurally smaller than DP.

Pereltsvaig's theoretical stance aligns DP with TP, arguing that just as not all clauses must be CPs, not all nominals must be DPs. She supports this claim with Russian numerical constructions, which, according to her, can sometimes be smaller than DP.

Consider the contrast between (7)a and (7)b, which differ only in the presence of plural subject agreement on the verb. In (7)a, the plural subject *p'at' izvestnyx akterov* 'five famous actors' triggers plural agreement on the verb. However, in (7)b the same plural subject does not trigger agreement, and the verb instead appears in default singular neuter form.

(7) RUSSIAN (adapted from Pereltsvaig 2006: 438–439)

a.	*V*	*et-om*	*fil'm-e*	*igra-l-i*	[*p'at'*	*izvestn-yx*	*akter-ov*].
	in	this-SG.M.PP	film-SG.PP	play-PST-PL	five	famous-PL.GEN	actor-PL.GEN
b.	*V*	*et-om*	*fil'm-e*	*igra-l-o*	[*p'at'*	*izvestn-yx*	*akter-ov*].
	in	this-SG.M.PP	film-SG.PP	play-PST-SG.N	five	famous-PL.GEN	actor-PL.GEN

'Five famous actors played in this film.'

Pereltsvaig explains this difference by postulating two distinct structures for the subjects. In (7)a, the agreement triggering subject is a full DP, while in (7)b, the subject, which does not trigger agreement, is an SN.

To illustrate structural differences, Pereltsvaig proposes a simplified DP structure (9), where QP encodes number, numerals, and generalized quantifiers like *all*. This differs slightly from the structure I adopt in my research (8), though the differences are mostly terminological.

(8) My assumed structure: DP > NumP > FP > nP > NP

(9) Pereltsvaig's assumed structure: DP > QP > NP

Pereltsvaig's analysis of the contrast in (7) is that the subject in (7)a is a full DP, while the subject in (7)b is a QP. Based on her observations, Pereltsvaig identifies six key syntactic and semantic differences in the behavior of DP and QP arguments. First, DPs trigger agreement, and QPs do not. Second, DPs are referential, while QPs are not (10). Third, DPs can have non-isomorphic wide scope, while QPs cannot (11). Fourth, DPs can control PRO, and QPs cannot (12). Fifth, DPs can be antecedent to anaphors, while QPs cannot (13). Sixth, DPs and QPs require different pronominal forms (14). They are summarized in Table 1.

Table 1: *Syntactic behavior of DP and QP arguments*

	DP	QP
agreement triggering	OK	*
referentiality	referential	non-referential
non-isomorphic wide scope	OK	*
control of PRO	OK	*
pronominalization	personal, interrogative, indefinite	special quantitative (*this many, how many*)

In (10)a, the plural subject *konkretnye p'at' balerin* 'certain five ballerinas' can be modified by an adjective triggering a specific interpretation (*konkretnye* 'certain' in this case). The infelicity of this adjective in a sentence with a non-agreeing subject in (10)b suggests that small subjects must have non-specific interpretation.

(10) RUSSIAN (modified from Pereltsvaig 2006: 441–442)[1]

a. *V Mariinskom teatr-e tanceva-l-i* [*konkretnye p'at' balerin*].
In Mariinsky-SG.M.PP theater-SG.PP dance-PST-PL certain-PL.NOM five ballerina.PL.GEN

b. #*V Mariinskom teatr-e tanceva-l-o* [*konkretnyx p'at' balerin*].
In Mariinsky-SG.M.PP theater-SG.PP dance-PST-SG.N certain-PL.GEN five ballerina.PL.GEN
'A certain five ballerinas danced in the Mariinsky Theater.'

In (11)a, the plural subject can take the scope below the universal quantifier *każdyj den'* 'every day', as well as above it, which is supported by the possibility to modify the subject with the phrase 'the same'. In (11)b, on the other hand, only the narrow scope interpretation is available, as indicated by the unacceptability of 'the same'.

[1] As a native speaker of Russian, I have modified some of the examples originally presented in Pereltsvaig (2006), as I found certain reported judgments inconsistent with my own intuitions. These modifications reflect my own linguistic judgment and are not central to the core argument of this study.

(11) RUSSIAN

a. *Každyj den' na stojbišče ostava-l-i-s'* [(*odn-i i te*
each day on camp.SG.N.PP stay-PST-PL-MED one-PL.NOM and that.PL.NOM
že) *p'at' ženščin*].
PTCL five woman.PL.GEN

b. *Každyj den' na stojbišče ostava-l-o-s'* [*p'at'* (**odn-ix i*
each day on camp.SG.N.PP stay-PST-SG.N-MED five one-PL.GEN and
te-x že) *ženščin*].
that-PL.GEN PTCL woman.PL.GEN
'Every day, (the same) five women stayed at the camp.'

Unlike the DP subject which triggers agreement on the verb, the small subject cannot control PRO, which is shown by the ungrammaticality of (12)b containing a converbial clause in the case where there is no agreement on the main verb, compare to the grammatical (12)a where a converbial clause is felicitous. In a similar way, small subjects cannot bind an anaphor, which is manifested by the ungrammaticality of the reflexive in a construction with a non-agreeing verb (13), compare to the grammatical (13) with the bound reflexive anaphor *sebja*.

(12) RUSSIAN (adapted from Pereltsvaig 2006: 444)

a. [*P'at' banditov*]$_i$ *leža-l-i na zeml-e* [PRO$_i$ *ne otkryva-ja*
five thug-PL.GEN lay-PST-PL on ground-SG.PP NEG open-CVB
glaz].
eye.PL.GEN

b. *[*P'at' banditov*]$_i$ *leža-l-o na zeml-e* [PRO$_i$ *ne*
five thug-PL.GEN lay-PST-SG.N on ground-SG.PP NEG
otkryva-ja glaz].
open-CVB eye.PL.GEN
'Five thugs lay on the ground without opening their eyes.'

(13) RUSSIAN (adapted from Pereltsvaig 2006: 445)

a. [*P'at' bandit-ov*] *prikryva-l-i sebja ot pul'*
five thug-PL.GEN shield-PST-PL REFL from bullet.PL.GEN
Džejms-a Bond-a.
James-SG.GEN Bond-SG.GEN

b. *[*P'at' bandit-ov*] *prikryva-l-o* *sebja* *ot* *pul'*
five thug-PL.GEN shield-PST-SG.N REFL from bullet.PL.GEN
Džejms-a *Bond-a*.
James-SG.GEN Bond-SG.GEN
'Five thugs shielded themselves from James Bond's bullets.'

Finally, when the numerical phrase is replaced by a plural personal pronoun, the verb must agree with it in number, which means that SN subjects cannot be replaced by personal pronouns (14)a, and pronominalization with the pro-form *stol'ko* 'that much' is needed (14)b.

(14) RUSSIAN (adapted from Pereltsvaig 2006: 446–447)
a. [*Oni*] *tanceva-l-i* / **tanceva-l-o* *tango*.
3PL.NOM dance-PST-PL dance-PST-SG.N tango.SG.ACC
'They danced tango.'

b. *Emu* [*stol'ko*] *ne* *nužno* / **nužny*.
3SG.DAT that_much NEG needed.SG.N needed.PL
'He doesn't need that much.'

Considering the properties identified for Russian numerical constructions not triggering verbal agreement to be characteristic for SNs in general, Pereltsvaig looks at some other examples of SNs in languages other than Russian. In particular, she mentions bare singulars in Norwegian (15) and small PPs in Germanic and Romance (16). Both phenomena involve nominal properties similar to the ones of Russian numerical constructions. Bare nominals in (15) and (16) must have non-specific interpretation, cannot take a wide scope, control PRO, and have restrictions on pronominalization, see (Pereltsvaig 2006) for details.

(15) NORWEGIAN (Pereltsvaig 2006: 471 from Borthen 2003: 356)
Jeg bruker ikke ***nakent*** ***nomen***.
I use not naked nominal
'I don't use bare nominals.'

(16) ENGLISH (Pereltsvaig 2006: 475)
The child goes ***to school****, and her parents come* ***to *(the) school*** *to pick her up.*

To sum up, Pereltsvaig makes a proposal that there should be nominal arguments smaller than DP and comes up with ideas what theoretical expectations we should have regarding such nominals. She combines her theoretical expectations with empirical facts found in Russian and some other languages and comes up with a list of properties she considers to be representative of SNs. In Chapter 7, I address the question of whether the properties found by Pereltsvaig can be used as general diagnostics to detect SNs cross-linguistically, discuss the issues that I see in them, and suggest my modified version of the list of such diagnostics.

1.1.2.4. Small nominals in Tatar

Building on Pereltsvaig's (2006) idea that some nominal arguments are SNs, which manifests in certain properties of those nominals, Lyutikova and Pereltsvaig show that some Tatar nominals without overt case markers are SNs (see Lyutikova & Pereltsvaig 2015, Lyutikova 2019, and Lyutikova & Pereltsvaig 2023). In addition to arguing that these nominals lack the DP layer, they determine their exact structural size.

A comprehensive analysis of nominals of different sizes in Tatar is provided in (Lyutikova 2019). She demonstrates that case-marked maximal projections in Tatar, presumably DPs, can contain possessive markers, number marking, and various adnominal modifiers including genitive possessors (17). They can also be expressed by pronouns. Another kind of projections, which do not get marked with case, may contain most adnominal modifiers and number marking but lack genitive possessors and agreement (18). Such projections are identified as NumPs, and they occur as adnominal modifiers in Tatar. There are also nominals that can have some adnominal modifiers but cannot bear number marking and are number neutral (19). Such nominals are analyzed by Lyutikova as NPs. They occur as complements of the Tatar attributivizer -*le*, Finally, Tatar adnominal modifiers that neither trigger agreement on the head nominal nor permit any adnominal modification are analyzed as N heads (20).

(17) TATAR (Lyutikova 2019: 336)
[*ukučɤ-nɤŋ* *däftär-lär-e-neŋ*] *papka-sɤ*
student-GEN notebook-PL-3-GEN folder-3
'the folder of/for the student's notebooks'

(18) TATAR (Lyutikova 2019: 336)
[*ukučɤ däftär-lär*] *papka-sɤ* / *[*ukučɤ-nɤŋ* *däftär-lär-e*] *papka-se*
student notebook-PL folder-3 student-GEN notebook-PL-3 folder-3
'the folder for the student's notebooks'

(19) TATAR (Lyutikova 2019: 334)
kük *čäčäk-le* *čačka* / **kük* *čäčäk-lär-le* *čačka*
blue cup-ATR cup blue cup-PL-ATR cup
'a cup with a blue flower / with blue flowers'

(20) TATAR (adapted from Lyutikova 2019: 332)
čɤn [*altɤn* *jezek*] / *[*čɤn* *altɤn*] *jezek*
real gold ring real gold ring
'real golden ring / *ring of real gold'

If the analysis given by Lyutikova and Pereltsvaig of Tatar caseless nominals is on the right track, it tells us that not only may nominals with a DP and without the DP layer co-occur within a language, but there are also different sizes of the nominals that lack DP. In Chapter 6, I show that a similar situation with small nominals of different sizes based on the specific syntactic position is found in Hill Mari.

1.1.2.5. Small nominals in Meadow Mari

Looking at bare DOs in Meadow Mari non-finite clauses, Serdobolskaya (2015) concludes that they are best analyzed as pseudo-incorporation (PNI), which she defines as verbal arguments being small nominals (see Chapter 7 for my discussion of small nominals and PNI). Specifically, she argues that Meadow Mari bare DOs are NumPs.

She supports this claim by examining properties typically associated with PNI. In addition to lacking case marking, these DOs exhibit verb adjacency, category restrictions, and constraints on modification.

First, bare DOs in Meadow Mari must be adjacent to the verb. With the exception of certain enclitic particles, no intervening elements — such as manner adverbs — are permitted between the verb and the DO, as illustrated in (21).

(21) MEADOW MARI (Serdobolskaya 2015: 310)

a. *rvez-ən* *motor-ən* ***poč'elamut*** *lud-m-əž-lan* *tun-əkt-əšo* *kuan-en*
boy-GEN beautiful-ADV poem read-NZR-P.3SG-DAT learn-CAUS-PTCP.ACT rejoice-PRT.3SG

b. *rvez-ən* ***poč'elamut-əm*** / ****poč'elamut*** *motor-ən* *lud-m-əž-lan* ...
boy-GEN poem-ACC poem beautiful-ADV read-NZR-P.3SG-DAT
'The teacher is glad that the boy recites poems well.'

Second, personal pronouns cannot appear as bare DOs. Instead, they must be marked with accusative case, as shown in (22).

(22) MEADOW MARI (Serdobolskaya 2015: 314)
məjə ***təj-əm*** / ****təj*** *kəč'al-aš* *kaj-em*
I you-ACC you search-INF go-PRS.1SG
'I'm going to search for you.'

Third, bare DOs in Meadow Mari cannot be modified by restrictive relative clauses (23), demonstratives (24), and universal quantifiers (25).

(23) MEADOW MARI: RESTRICTIVE RELATIVE CLAUSE (Serdobolskaya 2015: 311)
teŋgeč'e *nal-me* ***ü-əm*** / ****ü*** *kond-aš* *küšt-əš-əm*
yesterday buy-NZR milk-ACC milk bring-INF order-PRS.3SG
'I asked (you) to bring the milk that we bought yesterday.'

(24) MEADOW MARI: DEMONSTRATIVE (Serdobolskaya 2015: 312)
joč'a-vlak ač'a-ž-ən tide pört-əm / **pört əšt-əm-əž-əm*
child-PL father-P.3SG-GEN this house-ACC house make-NZR-P.3SG-ACC
pal-at
know-PRS.3PL
'The children know that their father built this house.'

(25) MEADOW MARI: UNIVERSAL QUANTIFIER (Serdobolskaya 2015: 311)
keč'-mogaj ***kajək-əm*** / ****kajək*** *kuč'-aš lij-eš.*
every-what.kind.of bird-ACC bird catch-INF need-PRS.3SG
'Every bird can be caught (lit. it is possible to catch every bird).'

Despite these PNI-like properties, Serdobolskaya notes that Meadow Mari bare DOs can sometimes have a definite interpretation (26). This challenges the traditional view of PNI, where bare nouns are typically interpreted as indefinite or non-specific.

(26) MEADOW MARI: DEFINITE (UNIQUE) (Serdobolskaya 2015: 317)
keč'-əm / ***keč'e*** *onč'-aš jörat-əše jeŋ-vlak er kən'el-ət*
sun-ACC sun look-INF love-PTCP.ACT person-PL early get.up-PRS.3PL
'People who like to meet the sun get up early.'

Based on these observations, Serdobolskaya classifies Meadow Mari bare DOs as SNs or instances of PNI, aligning with prior research on asymmetric DOM in other languages. However, she rejects the idea that the absence of a DP layer necessarily prevents a definite interpretation, a stance consistent with Danon (2006) and the assumptions made in this dissertation.

1.1.2.6. Interim summary

In this section, I gave an overview of the previous research that had shown that a model of the language assuming that all nominal arguments must be DPs would be incomplete, and smaller structures occur in argumental positions. Nominal arguments that lack a DP layer are more common cross-linguistically than one might assume, appearing in a variety of languages. SNs can be found based on theoretical expectations regarding the properties obtained by nominals due to the presence of higher functional projections in them. These insights are crucial for my analysis of Moksha bare complements of postpositions and Hill Mari bare DOs.

1.2. Underlying theoretical assumptions

For the SN analysis I propose in my dissertation to be viable, it requires a non-lexicalist architecture of grammar, as well as a coherent theory of case and agreement. In this section, I outline my theoretical assumptions.

1.2.1. Non-lexicalist models: Distributed Morphology

Much contemporary theoretical syntactic research still assumes — either implicitly or explicitly — that syntactic computation operates on items drawn from the lexicon, where these items are "words", and where all morphological processes of word formation are internal to the lexicon and cannot be interleaved with syntactic operations. Under this lexicalist view, lexical items enter the syntax fully specified with their semantic and phonological features, which are then uniquely interpreted at the LF and PF interfaces.

An alternative to this lexicalist approach is the antilexicalist view, which holds that syntactic terminals do not inherently determine a unique interpretation (either semantic or phonological), as interpretations may be sensitive to syntactic context. In this sense, these terminals are not "lexical" but "abstract". This perspective underlies various theoretical frameworks, in which syntax operates with abstract morphemes, including Distributed Morphology (Halle & Marantz 1993), Nanosyntax (Caha 2009), Spanning (Svenonius 2012), and the Many-to-One Mapping approach (Preminger 2021).

The precise mechanics of how abstract syntactic nodes are lexicalized at the later stages of derivation are not crucial for my analysis. However, for demonstration purposes, I adopt the framework of Distributed Morphology (DM) throughout my dissertation. In this section, I provide an overview of its main theoretical components.

1.2.1.1. The main principles of DM

Distributed Morphology (DM) operates based on two fundamental principles: **Syntax-All-the-Way-Down** and **Late Insertion**. The first principle, **Syntax-All-the-Way-Down**, posits that both word-level and phrase-level structures are built through the same syntactic operations. Crucially, the nodes within this syntactic structure are abstract; they consist of bundles but lack phonological content. At the end of the syntactic derivation, in the morphological component, these abstract nodes are mapped onto their phonological forms through a process known as Vocabulary insertion. The principle that phonological realization occurs only after syntactic derivation is complete is referred to as **Late Insertion**.

A key aspect of DM is the mechanism by which abstract features are mapped to phonological exponents. The phonological forms corresponding to specific feature bundles are stored in the Vocabulary as Vocabulary Items (VIs), which have the general form shown in (27). This rule means that the node with the number feature [plural] is realized as *-t.*

(27) Vocabulary Items for Moksha plural (simplified):
[plural] <-> /t/

This mapping process follows the **Subset Principle**, formulated by Halle (1997) as given in (28). The final clause of this principle — requiring selection of the most highly specified item — corresponds to the **Elsewhere Condition** (Bobaljik 2015).

(28) Subset Principle (Halle 1997):

"The phonological exponent of a Vocabulary Item is inserted into a position if the item matches all or a subset of the features specified in that position. Insertion does not take place if the Vocabulary Item contains features not present in the morpheme. Where several Vocabulary Items meet the conditions for insertion, the item matching the greatest number of features specified in the terminal morpheme must be chosen."

Consider an example illustrating morpheme competition under the Subset Principle, as seen in Russian (29). In Russian, gender, number and case features are bundled. When a masculine singular nominative noun enters the morphological component, it is specified as [masculine; singular; nominative]. The vocabulary items in (29) all contain a subset of the relevant features: [singular; nominative] in (29)a-c, nominative in (29)d-e. However, (29)a, (29)b, (29)d and (29)e contain additional features that are not present on the abstract node that needs exponence; these are [feminine], [neuter], [neuter; plural] and [plural] respectively. Consequently, the VI that will be inserted there is (29)c, which is the least specified form.

(29) Vocabulary Items for a part of nominal paradigm in Russian (simplified for demonstration purposes):

a. [feminine; singular; nominative] → /a/
b. [neuter; singular; nominative] → /o/
c. [singular; nominative] → /Ø/
d. [neuter; plural; nominative] → /a/
e. [plural; nominative] → /i/

The Subset Principle operates efficiently due to the **Underspecification** of VIs — an exponent does not need to be fully specified for all features of a given morpheme; it only needs to match a subset. Proceeding with the Russian example, plural nouns receive the same marking regardless of whether they are masculine or feminine. The plural morpheme (29)e is underspecified for gender, allowing it to lexicalize nodes that are [feminine; plural; nominative], [neuter; plural; nominative] or [masculine; plural; nominative], as demonstrated in (30). The Subset Principle ensures that (29)e is not inserted for neuter nominals, as (29)d is a more specified morpheme, but feminine and masculine nouns receive the same exponent.

(30) RUSSIAN

a. *s't'ep'-i*	b. *kon'-i*	c. *pol'-a*
steppe[F]-PL	horse[M]-PL	field[N]-PL
'steppes'	'horses'	'fields'

As an alternative to the usage of underspecified morphemes to lexicalize nodes that have more features, like in the example above, DM offers application of the **Impoverishment rule** (Bonet 1991, 1995), which deletes specific features in the presence of certain other features. In the Russian example above, the Impoverishment rule would delete gender in the presence of plural:

(31) gender → Ø [_ plural] (Bobaljik 2015: 9)

In some cases, multiple exponents are specified for the same set of features but are restricted to specific environments. This phenomenon is known as **Contextual Allomorphy**. Returning to the Moksha plural example in (27). In fact, Moksha plural has two allomorphs: *-t'* occurs after palatalized consonants, while *-t* appears elsewhere. The formalization of this contextual allomorphy is given in (32).

(32) Vocabulary Items for Moksha plural:

a. [plural] <-> /t'/ /C^j _

b. [plural] <-> /t/ elsewhere

Contextual allomorphy can be conditioned by phonological features of the adjacent morphemes or by their syntactic features. For instance, in Russian, the diminutive suffix denoting offsprings of living creatures is sensitive to number features (33). Gouskova and Bobaljik (2020) propose the VIs in (34) to account for the alternation.

(33) RUSSIAN

a. *kot'-onok*	b. *kot'-at-a*
cat-DIM[SG.NOM]	cat-DIM-PL[NOM]
'kitten'	'kittens'

(34) RUSSIAN: VIS FOR BABY DIMINUTIVE

a. [dim] <-> /at/ /_ [plural]
b. [dim] <-> /onok/ elsewhere

In summary, DM assumes that morphology operates on the same abstract syntactic nodes as syntax. These nodes are assigned phonological exponents late in the derivation. The exponents, stored as VIs, compete for insertion according to the Subset Principle, which ensures that the most specific matching item is chosen. Additionally, Underspecification and Impoverishment enable a more flexible mapping of morphological exponents, while contextual allomorphy accounts for environment-sensitive variation in exponence.

1.2.1.2. Post-syntactic processes

Although DM departs from the lexicalist assumptions that syntax operates on items previously "precooked" by morphology and assumes that both morphology and syntax operate on the same abstract structure, it nonetheless recognizes certain morphological processes that occur after syntactic derivation. These include the **Insertion of dissociated** morphemes (also called "ornamental" morphemes) not present in syntax as separate projections, **Fission**, **Fusion**,

Lowering, and **Local Dislocation**. The first four happen before **Linearization** of structural nodes, and all five precede **Vocabulary Insertion**.

Some morphemes do not lexicalize independent syntactic heads but are associated with syntactic heads that have other exponents. These are known as **Dissociated morphemes** (Embick 1997) or **Ornamental morphemes** (Embick & Noyer 2007). Typical examples include case and agreement markers (Iatridou 1990, Marantz 1992, Chomsky 1995).

For instance, consider a syntactic tree where the finite T head contains an agreement probe (Figure 1a). In the morphological component, this representation is adjusted, inserting an independent Agr node to host agreement morphology (Figure 1b). Similarly, a syntactic tree where case features are associated with D (Figure 2a) is transformed into a morphological tree with an additional K node for case marking (Figure 2b).

Figure 1: *Syntactic trees with representation of morphological agreement*

(a) (b)

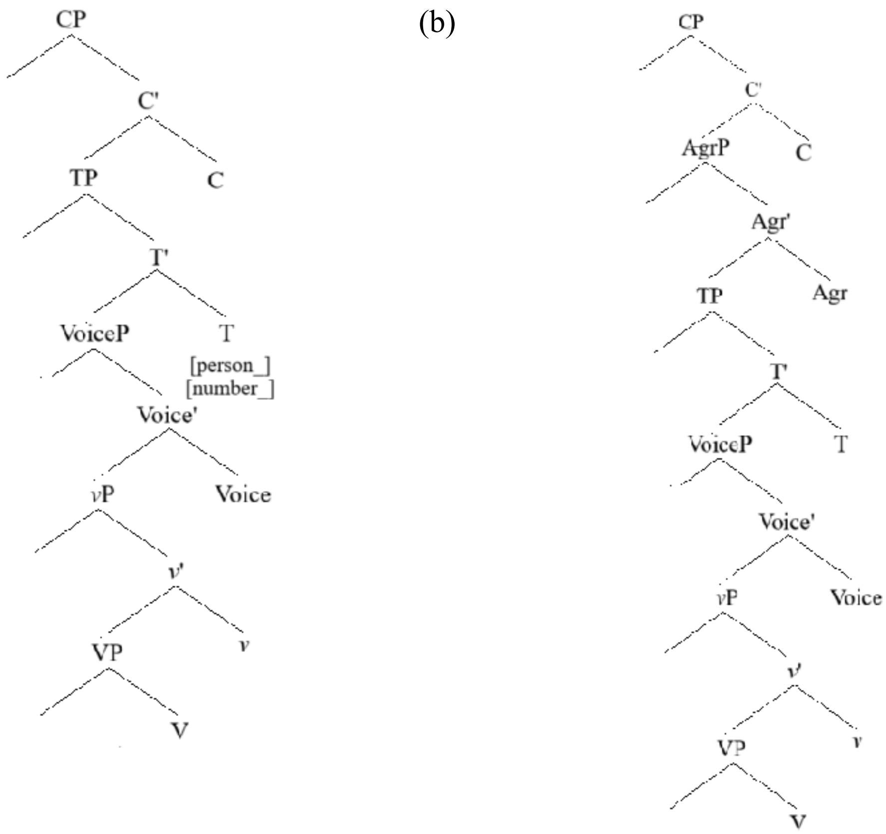
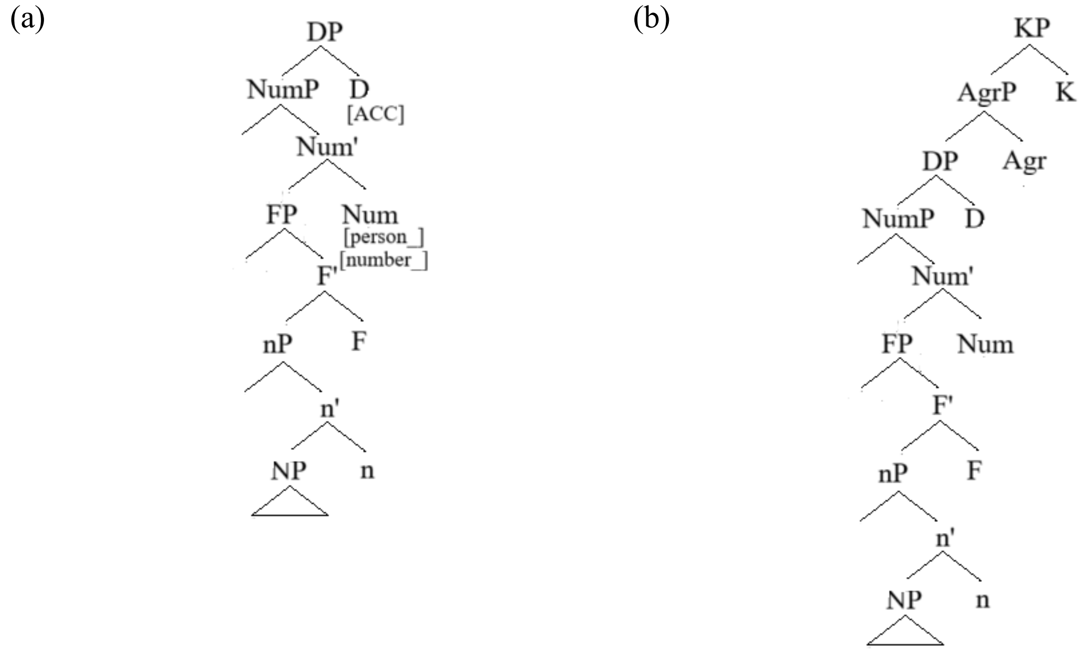


Figure 2: *Syntactic trees with representation of case*



Post-syntactic morphology can modify the structure by either splitting or merging nodes.

Fission (Embick 1997) is the process by which a single syntactic node is split into two during morphological processing. This operation accounts for cases where a single syntactic feature bundle is realized by multiple surface morphemes.

Fusion (Halle & Marantz 1993: 116) is the opposite operation, combining two adjacent nodes into one. The resulting fused node contains all the features of its constituent parts. For example, in many languages, number and case features originate on separate syntactic nodes (Num and D, respectively; see Chapter 2 for nominal structure). However, in languages like Russian, a single morpheme can encode both number and case, as illustrated in (29) above.

Other post-syntactic processes affect the linear arrangement of morphemes.

One of such operations is **Lowering**. **Lowering** is a post-syntactic movement operation in which a head moves downward and adjoins to the head immediately below it. Crucially, this movement requires **structural adjacency** but not necessarily **linear adjacency**. A well-known example is the realization of the English past tense suffix *-ed*, which surfaces on the verb due to the **Lowering** of T to V (Embick & Noyer 2001). This process remains unaffected by the presence of intervening elements such as adverbs:

(35) Mary [$_{TP}$ t_1 [$_{vP}$ loudly play-ed$_1$ the trumpet]] (Embick & Noyer 2001: 562)

Before Vocabulary Insertion takes place, the hierarchical structure gets linearized. As opposed to Lowering, **Local Dislocation** (Marantz 1988) is a post-syntactic process that operates after linearization. Unlike Lowering, which depends on syntactic hierarchy, Local Dislocation requires **linear adjacency** between elements and can reorder them accordingly.

To sum up, after syntactic derivation but before the abstract nodes get their exponents, several morphological operations can modify the structure. Dissociated morphemes may be inserted post-syntactically, and features may be split (*Fission*) or merged (*Fusion*), resulting in non-isomorphic mappings between syntax and morphology. Additionally, **Lowering** moves heads downward in the syntactic hierarchy, while **Local Dislocation** reorders linearly adjacent elements. This results in the absence of one-to-one correspondence between abstract nodes and surface forms. Some heads may also move down and adjoin to lower heads, which obscures the underlying syntactic structure.

1.2.2. Case assignment and agreement

My research focuses on the morphosyntactic properties of SNs as reduced structures. In particular, SNs lack the DP layer, which is widely assumed to host case features. Since other nominal features, including those responsible for phi-feature agreement, are also associated with the DP layer, its absence has significant consequences for the syntactic behavior of SNs, especially with respect to case assignment and agreement.

The lack of a DP layer in SNs is crucial for understanding their interaction with **case assignment** and **agreement**. Specifically:

- Case Assignment – If SNs lack a DP layer, they should not receive case under standard assumptions. If they do, this would necessitate a revision of case theory.
- Agreement – Without a DP layer, SNs should not participate in φ-feature agreement.

In this section, I outline theoretical foundations of case assignment and agreement, which will shape my analysis of SNs.

1.2.2.1. Case assignment

I assume that **only DPs receive case** (Lyutikova 2017: 227 and references therein). My framework does not adopt analyses where a KP layer is built above DP (Caha 2011). Given that SNs lack a DP layer, I do not expect them to undergo case assignment.

Two major theories of case currently dominate the field: **Case-by-Agree** (Chomsky 2000, 2001), and the **Dependent case theory** (DCT; Marantz 1991), see also (Baker 2015a) for a hybrid approach. I adopt DCT because it provides better empirical coverage for the languages in my study. Below, I outline its key principles and explain my choices.

In DCT, there are four basic types of cases, which constitute a disjunctive hierarchy (36) and get assigned according to this hierarchy, from left to right. The lexically governed case has the most requirements, and then the requirements of other types of cases decrease as we proceed to the right. With the exception of the lexically governed case, the cases are assigned due to a particular syntactic configuration of the nominal and its relations to other nominals in the syntactic context. Baker (2015) calls this syntactic context "domain", under which he means the Spell Out domain of a particular phase head.

(36) Case hierarchy
Lexically governed > 'Dependent' > Unmarked > Default

The **lexically governed** case is assigned first and is required by a specific head. Its assignment is closely tied to the theta-role. In general, this case corresponds to Chomsky's inherent case (1981, 1986). Such cases are also sometimes analyzed as Ps instead of features on D (e.g., Asbury 2008), and I am arguing for this analysis in Chapter 8.

After the lexically governed case, the **dependent case** is assigned if the structural conditions are met. The dependent case gets assigned to one of two still caseless DPs within a domain. The DP that gets assigned the dependent case should be either c-commanded by another caseless DP in the domain or c-command another caseless DP in the domain. The classic example of the former is the accusative case of direct objects, which are c-commanded by subjects within the clausal domain. The latter is usually exemplified by ergative subjects, which c-command objects, also in the clausal domain.

After the dependent case has been assigned, there is still a remaining DP in the domain. This DP gets assigned the **unmarked case** of that domain. Continuing with classic examples, unmarked cases of the clausal domain are nominative (in nominative-accusative languages) and absolutive (in ergative-absolutive languages). It is worth noting that this syntactic unmarkedness correlates with but does not equal morphological unmarkedness. Even though nominative and absolutive have indeed no morphological realization in many languages, this does not have to be true.

A good example of an unmarked case that usually does have morphological exponence is genitive. Genitive is the unmarked case of the nominal domain (Baker 2015b), and, in contrast to nominative, always has an overt marker.

Finally, if there are DPs that have not received the lexically governed, dependent or unmarked case, they receive the **default case**. Default case is not associated with any domain and is mostly assigned to DPs less integrated into syntactic structure, such as hanging topics, citation forms, etc. (Schütze 2001).

Crucially, DCT separates case from nominal licensing, unlike the Case Filter approach (Chomsky 1981), which requires all pronounced DPs to receive case. As one can see from the description above, in DCT, DPs always have at least some way to get case. This shift has broad implications. For instance, it excludes the possibility of case driven movement, or case-related explanation of null subjects in non-finite forms (see Levin 2015: 16–20 for the discussion of eliminating case from syntax).

Marantz (1991) originally proposed the disjunctive hierarchy as a post-syntactic phenomenon operating at PF, which cannot affect processes in narrow syntax. However, I follow (Preminger 2011, 2014) in integrating it into narrow syntax while preserving the dissociation of case and licensing. Preminger's argument goes as follows.

The purely morphological nature of case was argued for based on the claim that there is no syntactic process that is influenced by case. Preminger shows that this premise is erroneous. He provides evidence that movement to the canonical subject position in the languages with no quirky subjects is fed by phi-feature agreement, which is fed by morphological case (Bobaljik 2008). There is no doubt that movement to the canonical subject position is a syntactic process. Then, phi-feature agreement must be a syntactic process as well. By transitivity, case should also be in the narrow syntax. Based on this logic, the disjunctive case hierarchy (36) proposed by Marantz (1991) is a part of narrow syntax.

In the syntactic approach to DCT I adopt, case is represented as a syntactic feature on D realized phonologically after Spell Out. In Preminger's model, the unmarked case is the realization of an unvalued case feature. Elsewhere, he calls nominative "absence of case". Given that I distinguish between DPs and SNs, it becomes important not to confuse the absence of a case value and the absence of case features altogether. Subject DPs with an unmarked case are entities with unvalued case features, while SNs lack case features.

I finish this section with the argument for the superiority of the DCT, at least when it comes to Finno-Ugric languages I base my research on. In Case-by-Agree, the assignment of case is a byproduct of agreement — the operation of Agree. This means that every time a particular case is assigned, a successful operation of agreement has taken place. Unlike in English, agreement in Moksha (Mordvin < Finno-Ugric < Uralic) is overt, with a full paradigm of person and number markers. Consider possessive agreement in the Moksha noun phrase (37)–(38).

(37) MOKSHA

a. *mon'* *kud-əz'ə*
1SG.GEN house-1SG.AGR.SG
'my house'

b. *son'* *kud-əc*
3SG.GEN house-3SG.AGR.SG
'his house'

If genitive were assigned via possessive agreement, it should not appear in contexts without agreement. This theoretical prediction is not borne out. There are syntactic contexts, in which genitive case appears despite the absence of possessive agreement. Consider the locative phrase in (38). The complement shows up as genitive despite the absence of agreement. Notice that agreement in this syntactic context is not null[2], and a special possessive marker can be optionally used (38).

(38) MOKSHA

a. *pet'a-n'* *kucə*
Petja-GEN house.IN
'in my house'

b. *pet'a-n'* *kucə-nzə*
Petja-GEN house-AGR.3SG
'in my house'

Since genitive surfaces independently of agreement, Case-by-Agree fails to account for these facts. In contrast, DCT correctly predicts that the complement DP in a PP receives the unmarked case of the nominal domain — genitive in Moksha. In (Pleshak 2022a), I argue that the differential subject marking in Hill Mari participial clauses (nominative vs. genitive) is also better explained by DCT, as either marking is completely independent of agreement.

In Chapter 4, I further elaborate on case assignment in Moksha PPs, but for now, this evidence supports the superiority of DCT over Case-by-Agree.

[2] I assume that in languages with overt agreement morphology, there are no systematic null allomorphs corresponding to every overt marker. In other words, when agreement is absent on the surface, it is not due to a silent (null) allomorph but rather indicates that agreement has not taken place at all.

1.2.2.2. Agreement

Now I move to outlining my view of agreement and to the issue of non-agreement with SNs. In the modern linguistic literature, the term *agreement* encompasses various linguistic phenomena, ranging from only clausal phi-feature agreement (Preminger 2014) to broader notions like noun-modifier concord (e.g., Baker 2008, Carstens 2000, Mallen 1997), negative concord (Zeijlstra 2004, 2008a), modal concord (Zeijlstra 2008b), fake indexicals (e.g., Kratzer 2009), and even binding theory (Reuland 2011, Rooryck & Vanden Wyngaerd 2011).

I adopt a narrow definition of agreement, following Preminger (2014), but extend it beyond clausal heads to include agreement between nominal and adpositional heads and their arguments (cf. Abney 1987: 37–52; Chung 1998 on possessor-noun agreement). Additionally, I assume, following Baker (2008), that agreement is restricted to functional heads.

(39) Agreement (or ϕ-agreement) is morphophonologically overt covariance in ϕ-features between a verb-like element and one or more nominal arguments, where

a. verb-like element = a lexical verb, auxiliary verb, or tense/aspect/mood marker

b. ϕ-features = some nonempty subset of {person, number, gender/noun class}

(Preminger 2014: 6)

Despite rejecting the Case-by-Agree case assignment theory, I model agreement in a probe-goal architecture of Agree (Chomsky 2000; 2001), where the functional head with uninterpretable phi-features looks for a goal with interpretable phi-features in its c-commanding domain. Agree is restricted by locality; only the structurally closest nominal with needed features can serve as the goal.

I follow Preminger (2014) in treating Agree as an obligatory operation, rather than a free operation that can be performed or not. In Preminger's proposal, Agree always applies when structural conditions are met. All sentences, where conditions are met, must show agreement. If no appropriate goal is found, Agree fails silently, resulting in default agreement or absence of agreement, rather than ungrammaticality. This contrasts with Chomsky's initial approach, where ungrammatical sentences were filtered out later in the derivation due to agreement failure.

While neither case nor agreement licenses DPs, Agree does license specific features on certain arguments. For instance:

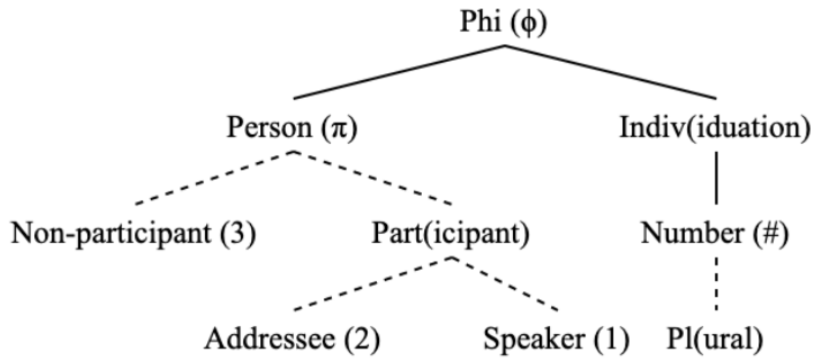
(40) Person Licensing Condition (Béjar and Rezac 2003 cited by Preminger 2014)
Interpretable 1st/2nd person features must be licensed by entering into an Agree relation with an appropriate functional category.

This principle explains why some person-marked elements are illicit in the absence of agreement.

When it comes to the distribution of features within the noun phrase, I assume that different phi-features are distributed among different functional heads within the nominal extended projection (Abney 1987; Valois 1991; Szabolcsi 1994; Bernstein 2001; Danon 2011, a.o), see Chapter 2 for a detailed discussion of the hierarchical structure of the noun phrase. Specifically, number features are located in Num, and person features are located in D. Although my dissertation does not address languages with grammatical gender, I tentatively place gender features in n.

Features are arranged hierarchically not only inside nominal goals, but also on the probes (Harley & Ritter 2002). Person features are highest in this hierarchy (Storment 2025 and references therein), see Figure 3.

Figure 3: *Possible ranges for feature geometry of English* (Storment 2025: 39)



Because of this feature geometry, combined with the hierarchical nature of the search mechanism (41), person probing universally takes place first (Bejar & Rezac 2009; Preminger 2014; Deal 2015; Kalin & van Urk 2015; Půskar 2019; Storment 2025).

(41) Hierarchical Search (Storment 2025: 43)

Phi-probes search first for the features located highest in their feature geometries, then work their way downward if the first search is unsuccessful in valuing all features.

Since person features are located in the highest nominal projection, and functional layers in the extended projection cannot be skipped, a phi-probe that encounters a goal specified for person will also encounter number (and gender, if present) on the same goal. In such cases, the probe is fully satisfied by that single goal. In absence of D and person features on it, SNs even as large as NumP cannot satisfy a phi-probe, unless the probe was previously satisfied by a phi-deficient goal specified only for person features, such as the English expletive *there* (Chomsky 2000: 128).

The takeaway from this section is that agreement operates with phi-features, which are distributed among functional heads in the extended nominal projection with person features located in the highest nominal head — D. Since person probing universally happens first, agreement probes do not see SNs as their goal, hence we never see agreement with SNs.

1.3. The structure of the dissertation

The main body of this dissertation is dedicated to the case-studies of bare nominals in Moksha and Hill Mari with the goal to determine whether they should be analyzed as SNs or full DPs. The findings from these case studies form the basis for broader theoretical generalizations presented in the final chapter.

Chapter 2 lays out the theoretical model of nominal phrase structure, emphasizing the role of functional projections in determining nominal size. I provide evidence for the hierarchical organization of nominal phrases using ellipsis, pronominal substitution, and agreement patterns as diagnostics. I also go through each nominal projection between NP and DP and show what is expected from a nominal that has that projection from the morphosyntactic point of view. I also critique the binary NP/DP classification of languages, showing that languages like Moksha and Estonian yield inconsistent results on standard DP tests, particularly those linked to article presence.

Chapters 3 – 5 are dedicated to my case study of Moksha bare nominals. In Chapter 3, I give an overview of the Moksha nominal morphology, which is crucial for understanding the problem and my analysis. I examine key categories expressed in the nominal domain — case, number, definiteness, and agreement — and highlight their interactions. I provide a detailed description of the behavior of adnominal modifiers, such as adjectives, numerals, and possessors. In the end, I map these features onto a hierarchical DP structure, demonstrating how Moksha nominal phrases align with theoretical models of syntax.

In Chapter 4, I analyze the contrast between genitive-marked and bare complements of spatial postpositions in Moksha. I argue that genitive-marked complements are full DPs, containing higher functional projections (DP, NumP) that enable them to express case, number, definiteness, and agreement. In contrast, bare complements are SNs, lacking these projections and displaying reduced morphosyntactic marking and agreement capabilities. I compare the SN analysis to alternative morphological and syntactic explanations, demonstrating that the SN analysis offers a more parsimonious account of the data.

In Chapter 5, I examine genitive-marked vs. bare direct objects (DOs) in Moksha. Although bare DOs superficially resemble bare postpositional complements, I argue that they are not SNs. Instead, I show that Moksha bare DOs are full DPs with null case marking. Through diagnostics such as causative constructions and case assignment patterns, I demonstrate that bare DOs exhibit DP-like behavior despite their lack of overt case marking. This challenges the assumption that all morphologically unmarked nominals are structurally reduced, underscoring the need for refined SN diagnostics.

In Chapter 6, I turn to Hill Mari, analyzing the contrast between accusative-marked and bare DOs in non-finite clauses. While all DOs in finite clauses are obligatorily accusative-marked, non-finite clauses allow both bare and accusative DOs. I show that bare DOs vary syntactically depending on the construction — for instance, bare DOs in nominalizations can bear number marking, while those in converbial clauses cannot. I argue that bare DOs in Hill Mari are SNs, with variation in nominal size depending on their syntactic environment. This highlights the internal diversity of SN structures even within a single language.

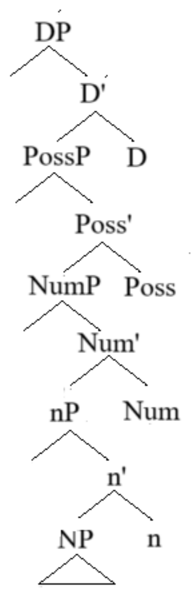
In Chapter 7, I propose new diagnostics for identifying SNs cross-linguistically, addressing weaknesses in existing criteria. I argue that many traditional SN properties fail to conclusively distinguish these structures. I introduce refined diagnostics, focusing on definiteness, number, and possession-marking mismatches, to improve the identification of SNs across languages. I also examine the connection between SNs and pseudo-noun incorporation (PNI), proposing that some PNI objects may be best analyzed as SNs.

In Chapter 8, I summarize the key findings of the dissertation and discuss their theoretical implications for the nominal architecture and the theory of case. I also outline directions for further research.

Chapter 2: Hierarchical structure of noun phrases

One of the core objectives of this dissertation is to demonstrate that a small nominal (SN) analysis of nominals in specific syntactic positions provides a parsimonious explanation of their behavior with minimal theoretical assumptions. However, a crucial assumption underlying my analysis is that nominal structure is hierarchical, consisting of multiple structural layers. Fortunately, this assumption is widely accepted in formal syntax. Since the seminal works of Abney (1987) and Szabolcsi (1983), the hierarchical organization of nominals has been extensively supported in the literature (Bernstein 2001; Alexiadou et al. 2007; Lyutikova 2017 and references therein). Namely, in addition to the lexical projection NP, it contains a number of functional projections responsible for phi-features of the nominal and its referential characteristics. These may include, for instance, nP introducing the possessor, #P or NumP introducing number, PossP, which is responsible for φ-feature agreement with the possessor, and DP relating the nominal to the external discourse, see Figure 4. Later in this section, I reconsider the necessity of PossP and argue for the absence of a specialized agreement projection in syntactic structure.

Figure 4: *Nominal structure*



The strongest position regarding the hierarchicity of the nominal structure is that the highest nominal projection is DP. This point of view is called the DP hypothesis (Abney 1987). Those researchers that adopt the DP hypothesis, assuming that D should be the actual head of a fully-projected nominal, hold different points of view. One point of view supports the universal DP hypothesis, meaning that all nominals in all languages must be DPs, regardless of the presence of overt D-elements, such as articles (Longobardi 1994). Another point of view is the parametrized DP hypothesis, claiming that only languages with articles project DP, while articleless languages only have NPs (Bošković 2008). I look at the latter hypothesis in more details in Section 2.4.

A weaker position would be that nominals contain some functional structure on top of the NP, but not necessarily DP. Then one can argue that all nominals in all languages must have some functional structure (in parallel to the strong universal DP hypothesis), or, the opposite, some languages do not project functional structure (in parallel to the strong parametrized DP hypothesis). A further refinement suggests that nominals of different sizes can co-occur within a single language, allowing for variation between DPs, NumPs, and other functional structures (Déchaine & Wiltschko 2002, Pereltsvaig 2006, Lyutikova & Pereltsvaig 2015, Erschler 2019; Pleshak 2023a).

Since my analysis assumes the hierarchical structure of nominals, this section provides an overview of key arguments supporting this view. While the general assumption of nominal hierarchicity is well-supported, many related claims about nominal structure remain controversial (Salzmann 2020). For instance, the assumption that each functional layer is headed by a distinct head is often justified by nominal-clausal parallels, but tests for headedness are inconclusive. It is also widely assumed that the highest projection is always a DP. However, the signs of the presence of D are usually just signs of the presence of some higher projection. A thorough examination of all flaws in the argumentation around the DP hypothesis is out of the scope of this dissertation, see (Salzmann 2020) and references therein for discussion. What is important is that, as argued by Salzmann, one of the strongest arguments in favor of the presence of functional heads in a nominal spine would be an argument involving selectional restrictions, i.e., selection for nominals of different sizes.

The two case studies presented in this dissertation involve a phenomenon that can be accounted for in terms of selectional restrictions, even though the restrictions on the nominal complement come from the number of higher functional heads rather than from some lexical property of the head immediately dominating the selected complement. I explore the ways of preserving locality of selection for the phenomena observed in Moksha and Hill Mari at the end of Chapter 6. In this section, I first review key arguments for the hierarchical structure of nominals before examining the specific functional projections that are central to my analysis.

2.1. Hierarchicity of the nominal structure

The strongest empirical argument for the hierarchical organization of nominal structure comes from ellipsis. Just as VP can be elided under T (Radford 1981: 67; Adger 2003: 65; Sportiche et al. 2014: 58–60, a.o), NP can be elided under D (Jackendoff 1971; Lobeck 1990, 2006; Saito & Murasugi 1990; Salzmann 2020, a.o.), as in (42). This suggests that the head noun forms a constituent with its closest modifiers, e.g., adjectives, to the exclusion of higher modifiers.

(42) ENGLISH

a. *Mary ate an apple, and I did* [*eat an apple*] *too.*

b. *Mary bought these green apples, and I bought those* [*green apples*].

A similar argument is pronominal substitution (Hornstein & Lightfoot 1981; Adger 2003: 63; Sportiche et al. 2014; Salzmann 2020, a.o.). Again, the proform *one* replaces a constituent that includes the noun and its closest modifiers but excludes higher ones:

(43) ENGLISH

Mary ate this [*green apple*], *and I ate that* [*one*].

Moving forward to other constituency tests, structures excluding demonstratives can be coordinated with the demonstrative scoping above both conjuncts (Adger 2003: 125; Salzmann 2020):

(44) ENGLISH (Eric Potsdam, p.c.)

this [*dog*] *and* [*cat*]

While these three diagnostics (ellipsis, pronominal substitution, and coordination) confirm the hierarchical nature of nominal structure, they do not specify whether the replaced or elided material corresponds to a maximal projection of a functional head (such as DP, NumP, etc.) or to an intermediate N-bar projection, as assumed in X-bar theory during the Government and Binding era.

Beyond demonstrating hierarchy, further arguments suggest that higher layers in the nominal structure correspond to functional projections rather than simple N-bar levels. One key argument comes from movement within the nominal phrase (Carstens 2017 and references therein). This was first noted by Ritter (1988) in Hebrew, where the head noun precedes both the internal and external argument (45). Since the external argument is base generated in the specifier position of the nominal head, it should precede the nominal. The fact that the external argument follows the head suggests that the head moved to the left. Therefore, there must be some functional head to which the noun head undergoes movement.

(45) HEBREW (Carstens 2017: 2)

reiyat	*ha-mora*	*et*	*ha-yalda*
view	the-teacher	ACC	the-girl

'the teacher's view of the girl'

Initially, Ritter (1988) proposed that the target of this head movement is D. However, in (Ritter 1991a), an intermediate numeral head was introduced to account for movement. Danon (2006) further refined this argument, showing that D is not the best candidate for movement due to Hebrew numeral constructions, where the numeral precedes the noun. If N-to-D movement were correct, the numeral would incorrectly follow the noun, as shown in (46). Instead, an intermediate numeral head provides a better explanation. Regardless of the exact labels assigned to these projections, the evidence strongly supports the presence of some functional head above the lexical noun, and I adopt this idea.

(46) HEBREW

a. *šloša sifrey balšanut*
three books linguistics
'three linguistics books' (Danon 2006: 999)

b. **sifrey šloša balšanut*
books three linguistics
'three linguistics books' (Gabi Danon, p.c.)

Another argument for functional projections in the nominal structure comes from agreement patterns, which reveal how nominal features are distributed along the nominal spine. In Finnish, predicate agreement differs depending on whether a demonstrative is present. Only numeral phrases with demonstratives trigger plural agreement on the predicate (47)a, while numeral phrases without demonstratives trigger singular agreement (47)b. These facts were taken to show that nominal number features do not originate in N or the numeral head itself, but rather in a higher functional projection (Danon 2011).

(47) FINNISH (Danon 2011: 301; from Brattico 2010)

a. *Ne kaksi pien-tä auto-a seiso-ivat tiellä.*
Those.PL two.SG small-PART.SG car-PART.SG stand-PAST.3PL road.ADESS
'Those two small cars stood at the road.'

b. *Kolme auto-a aja-a tiellä.*
three car-PART.SG drive-SG road
'Three cars drive on the road.'

Taken together, these diagnostics — ellipsis, pronominal substitution, coordination, movement, and agreement patterns — provide strong evidence that nominal structure is hierarchical and consists of multiple functional projections above the lexical noun. While the precise labels of these projections remain open to debate, the existence of functional structure above NP is well-supported.

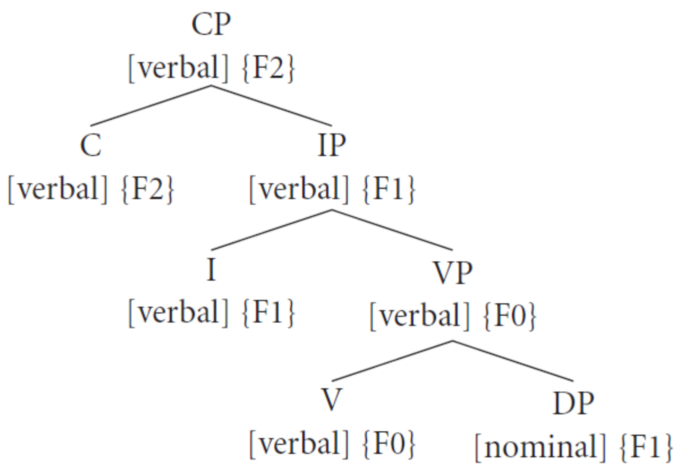
2.2. Extended projection

In talking about functional layers built on top of a lexical projection, I will be using the notion of **extended projection** (Grimshaw 1993). This framework is particularly useful for two reasons. First, it resolves issues related to selection — it is not a special D but a special N that gets selected (Baltin 1989). Second, it explains why non-full projections can serve as complements, but no functional layers can be cut out from the middle, a pattern that aligns with empirical observations.

An extended nominal projection consists of the projection of the lexical head, e.g., N(oun) and several projecting functional heads on top, e.g., Num(ber) and D(eterminer), see more on specific nominal heads in the next section. The key principle underlying this framework is that the lexical head and its associated functional projections share the same category, differing only in their lexical vs. functional status (Grimshaw 2000).

It is important to keep in mind that complement-head relations can hold between elements of different categories, as well as between elements of the same extended projection. Consider the verbal structure in Figure 5, adapted from (Grimshaw 2000). The head I takes a VP complement just as the head V takes a DP complement. However, while I and VP share the category [verbal], V and DP belong to different categories — V is [verbal], and DP is [nominal].

Figure 5: *Extended projection of the verb with a nominal complement* (Grimshaw 2000: 118)[3]



The structure of an extended projection ensures that only lexical heads {F0} select for complements outside their own extended projection. In Figure 5, this is clearly illustrated by the lexical verb (V) taking a DP complement, which belongs to a different category. I would like to point out that a complement that belongs to a different extended projection does not necessarily belong to a different category. For example, a V head could take an IP complement, and both would be [verbal]. However, lexical heads are always the lowest elements of extended projections, and they cannot extend projections of their complements. As a result, the complement of a lexical head is always a different extended projection.

One of the major advantages of extended projection is that it addresses problems with selection. Since some verbs have selectional restrictions on the animacy or number of their complements,

[3] One might disagree with specific label choices in the tree—such as using IP instead of TP, or omitting a vP projection — but these details are not crucial. The concept of the extended projection does not hinge on the precise labels, and this nuance can therefore be set aside.

like in (48), there is a concern how those nominal features are seen through D. Selection is supposed to be local, but no verbs select for D features. Verbs never select for a noun phrase with a particular article, modifier or a particular definiteness/specificity value, see discussion in (Baltin 1989; Salzmann 2020: 29–32). In the configuration provided by extended projection, features are shared across the nominal spine and are available on the highest functional head. This ensures that verbs can select for an N-based projection without needing to specifically target D or other functional layers.

(48) ENGLISH

a. *John built his house.* b. #*John built his son.*

Another advantage of extended projection is that it derives the observed pattern of complement-taking. In Chapter 3, I argue that certain heads can take nominal complements smaller than DP. This means that selection should be modeled in such a way that a head can select for a full structure, as well as for a structure that lacks one or more upper functional layers. Extended projection allows for this, because all functional heads share the same category and all features of lower heads are available on the higher head. Crucially, if a nominal structure is reduced, it must lack all projections above the missing layer — intermediate layers cannot be selectively omitted. Figure 6 shows a full DP (a), a well-formed small nominal (b), which lacks all projections higher than NumP, and an ill-formed one (c), where NumP has been cut out from the middle.

Figure 6: *Well-formed and ill-formed extended nominal projections*

a. NP < nP < FP < NumP < DP b. NP < nP < FP c. *NP < nP < FP < DP

Since functional projections build hierarchically, structures like (c), where a middle layer (NumP) is omitted while higher layers (DP) remain, are ungrammatical. This follows directly from extended projection theory, which states that functional heads place stricter selectional restrictions on their complements than lexical heads do, typically allowing only one specific type. Similarly, Adger (2003: 108) introduces a special Hierarchy of Projections, which ensures that a functional head always takes the same complement. As a result, the theory predicts that functional heads cannot be skipped in an arbitrary fashion.

2.3. Functional projections of the nominal spine

2.3.1. DP as the highest nominal projection

Since the seminal works of Abney (1987) and Szabolcsi (1983), full nominal projections have been widely considered DPs rather than NPs. Some scholars have proposed that case has its own syntactic projection KP on top of DP (Lamontagne & Travis 1986, 1987; Loebel 1994; Bittner & Hale 1996). Sometimes, PP is taken to constitute part of the extended nominal projection, PP being built on top of DP (Grimshaw 1993, a.o.). However, I adopt the mainstream view that DP is the highest nominal projection and will argue in favor of this position at the end of this section.

DP is the projection of the determiner, whose prototypical instantiation is a definite article. Even in articleless languages, this projection is postulated for distributional reasons. The presence of the DP layer has been considered crucial for the ability of nominals to function as arguments rather than predicates (Higginbotham 1983; Stowell 1989; Longobardi 1994; Chierchia 1998, a.o.). The core idea is that nominals (at least in some languages) are predicates and must be embedded under some functional structure to be further inserted as arguments into a larger syntactic structure.

This common view that the DP layer is required for argumenthood conflicts with the assumption adopted in this dissertation that not all arguments must be DPs. To resolve this tension, I propose a weaker formulation of the DP layer's role: rather than being necessary for argumenthood per se, the DP layer introduces features characteristic of certain types of arguments.

Beyond this purely mechanical function, there seems to be no meaning that could be attributed to the DP. Lyons (1999) argues that only definite nominals project DP, making DP the projection of definiteness. However, cross-linguistic evidence shows that determiners encode more than definiteness, often including specificity or other interpretative features (Matthewson 1996). Given this variability, I take D, whether overt (e.g., a determiner) or null, to be the syntactic projection that allows a nominal to function as a full argument.

Abstracted away from semantics and even general argumenthood, DP has crucial structural functions. First, as I already said and as follows from its name, DP hosts determiners (they are considered to be in the D head). Second, DP aggregates and hosts nominal features, making them available for agreement probes. This is reflected in DPs' ability to move and trigger agreement. Finally, it is the D head that gets the case feature, and therefore, it is DPs and not NPs that need case.[4]

Given these considerations, I treat DP as a mostly formal projection important for the external syntax of the nominal. It is relatively easy to tell that DP projection is present if a nominal has an overt determiner, can freely move and trigger agreement, or gets an overt case. The presence of one or more of these properties is a good indicator that the projection DP is present.

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[4] Even though some PNI objects can be marked for case, e.g., Hungarian PNI objects (Kiefer 1990; Bartos 1997; Farkas & de Swart 2003, a.o.), I still assume that SNs do not bear a case feature, because I depart form the assumption that PNI objects are necessarily structurally smaller than DP.

There are also certain modifiers located in Spec,DP, such as demonstratives and possessors. However, research suggests that both elements originate lower in the structure and move to Spec,DP during derivation (Giusti 1997, 2002; Brugè 2002; Brugè & Giusti 1996; Panagiotidis 2000; Grohmann & Panagiotidis 2005; Shlonsky 2004; Elsaadany & Shams 2012) on demonstratives and (Szabolcsi 1983; Horrocks & Stavrou 1987) on possessors. Because these elements do not inherently require a DP projection, their presence does not guarantee that a DP is projected, a point that will be relevant in Section 4.4 and Chapter 6, Section 6.6.

Now, it is time to address the potential nominal projections higher than DP rejected in my dissertation. Some researchers postulated another projection on top of DP. This projection is KP, the projection of case (Lamontagne & Travis 1986, 1987; Loebel 1994; Bittner & Hale 1996). I consider the arguments used to support this introduction not convincing in the theoretical paradigm I am adopting here.

A major argument for KP as a separate syntactic projection comes from languages like Turkish, where case marking alternates with zero marking (49). However, as discussed in Chapter 6, this alternation can be explained without positing KP. Specifically, the presence or absence of case marking can be tied to the presence vs. absence of DP, rather than requiring an additional KP layer. Alternatively, the presence and the absence of morphological marking can be accounted for post-syntactically, with case being a dissociated morpheme (Embick 1997).

(49) TURKISH (Kornfilt 2008: 81)
a. *(ben)* ***bir kitap*** *oku-du-m*
I a book read-PST-1SG
'I read a book.'

b. *(ben)* ***bir kitab-ı*** *oku-du-m*
I a book-ACC read-PST-1SG
'I read a certain book.'

Another argument for KP comes from parallels between case markers and complementizers, since both can be omitted under adjacency. However, this does not necessarily mean that KP parallels CP — DP itself could account for this parallelism just as well, with DP being the nominal counterpart of CP (Szabolcsi 1994; Giusti 2006, a.o.).

A final argument is purely theoretical. DP was split to DP proper and KP so that one projection can be responsible for referential characteristics, while the other hosts a purely syntactic feature — case. I follow Giusti (1995) in treating case as part of DP instead.

Another hypothesis is that Prepositional Phrase (PP) is part of the extended nominal projection, with P built on top of DP (Grimshaw 2000). However, there is no consensus among researchers whether P should be considered a lexical or a functional head, see (Asbury et al. 2008: 6–7) for an overview. The classification of P as functional would justify its inclusion in the extended nominal projection, but the lack of clear empirical cross-linguistic support makes this claim uncertain or at least language specific, see also (Den Dikken 2010) making the case against P being a part of nominal extended projection.

Based on the discussion above, I conclude that DP is the highest projection within the nominal domain, rejecting KP and PP as necessary components of the extended nominal projection. DP serves a primarily structural role, ensuring that nominals can function as arguments, nominal features are available for agreement, and case gets assigned. The rejection of KP and PP simplifies the structure while maintaining empirical adequacy and theoretical elegance.

2.3.2. Eliminating AgrP from the nominal spine

The second-highest projection often assumed within the nominal structure is the Agreement Phrase (AgrP), which is primarily associated with possessive agreement (Alexiadou et al. 2007). In languages with adnominal possessors, such as Italian and Hungarian, possessors are often analyzed as occupying Spec,AgrP, unless they move higher to Spec,DP:

(50) ITALIAN (adapted from Alexiadou et al. 2007: 574)

[$_{DP}$ [$_{D}$	*la*] [$_{AgrP}$	*loro* [$_{NumP}$	*brutale*	*invasione…*]]]
	the	their	brutal	invasion

‘Their brutal invasion’

The argumentation for placing the possessor in a specifier lower than DP comes from the word order in the languages where determiners co-occur with possessor. For instance, in Italian, the definite article *la* precedes the possessor *loro* (50). The same order is observed in Hungarian, where the definite article *a* also precedes the nominative possessor:

(51) HUNGARIAN (adapted from Szabolcsi 1994: 180)

a	*te*	*kalap-ja-i-d*
DEF	2SG	hat-POSS-PL-2SG

‘your hats’

While the low base position of possessors is well supported, I disagree with the need for a dedicated AgrP projection to accommodate them. Alexiadou et al. (2007) discuss an alternative analysis in which possessors occupy Spec,NumP (Alexiadou et al. 2007: 572), see Section 2.3.3 on projection NumP. They reject the Spec,NumP proposal based on Hungarian morphology, where number and possessor agreement are expressed by separate morphemes. In (51), the number of the possessum is encoded by the suffix -*i*-, while the 2nd person possessor agreement is expressed by the suffix -*d*.

However, this morphological fact does not necessitate an additional syntactic projection. The presence of multiple morphemes for different features does not inherently imply the existence of separate syntactic heads. Within DM (Embick 1997), bundles of features can be realized as multiple morphemes post-syntactically.

The assumption of AgrP within the nominal domain is largely historical. The nominal structure has long been assumed to parallel clausal structure (Abney 1987; Szabolcsi 1994; Siloni 1997; Guisti 2006; Laenzlinger 2010, a.o.). Under this analogy AgrP in the noun phrase was intended to mirror AgrP in the clause, which was responsible for subject-predicate agreement (Pollock 1989), see also (Belletti 2001) for an overview. However, AgrP has been eliminated from the clausal structure in minimalist syntax (Chomsky 1995: 348).

Retaining AgrP within nominal syntax while removing it from clausal syntax introduces an asymmetry that is theoretically undesirable. If AgrP has been eliminated from the clausal spine, it should not persist in nominal syntax either. Moreover, AgrP as an optional projection raises additional theoretical concerns. First, its only role is to host agreement morphology. Second, it only appears when there is an element to agree with. Third, the Num head is already a plausible candidate for hosting agreement. Thus, I argue for eliminating AgrP from the nominal domain and assigning its functions to NumP.

Further issues with AgrP arise when comparing possessive constructions to prepositional phrases (PPs). Consider (52) for the examples of a Moksha possessive construction and a PP. If we assume that AgrP is a separate projection, it must be below DP, since DP is considered to be the highest projection of the nominal spine. If we look at the PP in (52)b, on the other hand, we have to postulate that there is an AgrP on top of PP, as the agreement marker attaches to the postposition. This runs into selectional issues[5]. If AgrP is an independent projection, it should be optional, appearing only when agreement is present. However, optional functional projections introduce unwanted theoretical complexity — heads should not optionally select for AgrP depending on whether agreement is required.[6] This would also disrupt parallelism between DP and PP, with AgrP being projected below the maximal projection DP in nominals and above the maximal projection in PPs, which is undesirable.[7]

(52) MOKSHA

a.	*mon'*	*kud-əz'ə-n'*	b.	*mon'*	*lank-sə-n*
	1SG.GEN	house-1SG.AGR.SG-GEN		1SG.GEN	top-IN-AGR.1SG
	'of my house'			'on me'	

The assumption that possessors occupy Spec,NumP, on the other hand, fits nicely into the contemporary view of the grammar (see Section 2.3.3 on the placement of Numerals in a functional projection lower than NumP). First, NumP is the closest projection below DP, making it a natural landing site for possessors. Second, NumP already encodes agreement features, just as TP does in clauses (after the elimination of AgrP from clausal syntax). Third, AgrP is theoretically redundant, adding unnecessary complexity to both nominal and prepositional structures.

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[5] I thank Marcel den Dikken for pointing this out to me.

[6] I acknowledge that, as pointed out for defenders of optional functional projections, case and agreement are indeed often optional. However, I believe that dissociated morphemes added post-syntactically provide enough freedom to account for the messy empirical landscape while keeping the syntactic structure simple (see Section 1.2.1 on post-syntactic heads).

[7] One could resolve this issue by accounting for the placement of possessive agreement markers in post-syntactic terms. However, see Section 4.3 for arguments in favor of a syntactic analysis of this phenomenon.

Thus, I argue that possessors occupy Spec,NumP, and there is a probe associated with the Num head, see Section 2.3.3 for the NumP projection and Section 1.2.2.2 for the mechanics of agreement.

2.3.3. NumP as the nominal projection immediately dominated by DP

Apart from DP, the only other widely accepted functional projection in the nominal domain is NumP, which encodes grammatical number. The projection of number NumP (or #P) was first introduced by Ritter (1991a) to account for word order in Semitic languages. However, as Danon (2011) points out, this initial analysis only necessitates some functional head above NP rather than specifically requiring a number projection.

What Danon (2011) proposes as an independent argument in favor of a separate number projection on top of NP are concord data from Finnish. Consider example (53), in which the adjective preceding the numeral is marked as plural. Adjectives following the numeral, on the other hand, get singular marking, as in (47) above, repeated here as (54). The proposal is that NP itself is not specified for number, and number features are located higher, in a special functional projection. Hence, everything that is located below that projection gets the default singular marking.

(53) FINNISH (Danon 2011: 302; from Brattico 2010)
ne *pilaantune-et* *kaksi* *leipä-ä*
those.PL rotten-ACC.PL two.SG bread-PART.SG
'those two rotten breads'

(54) FINNISH (Danon 2011: 301; from Brattico 2010)
Ne *kaksi* *pien-tä* *auto-a* *seiso-ivat* *tiellä.*
those.PL two.SG small-PART.SG car-PART.SG stand-PAST.3PL road.ADESS
'Those two small cars stood at the road.'

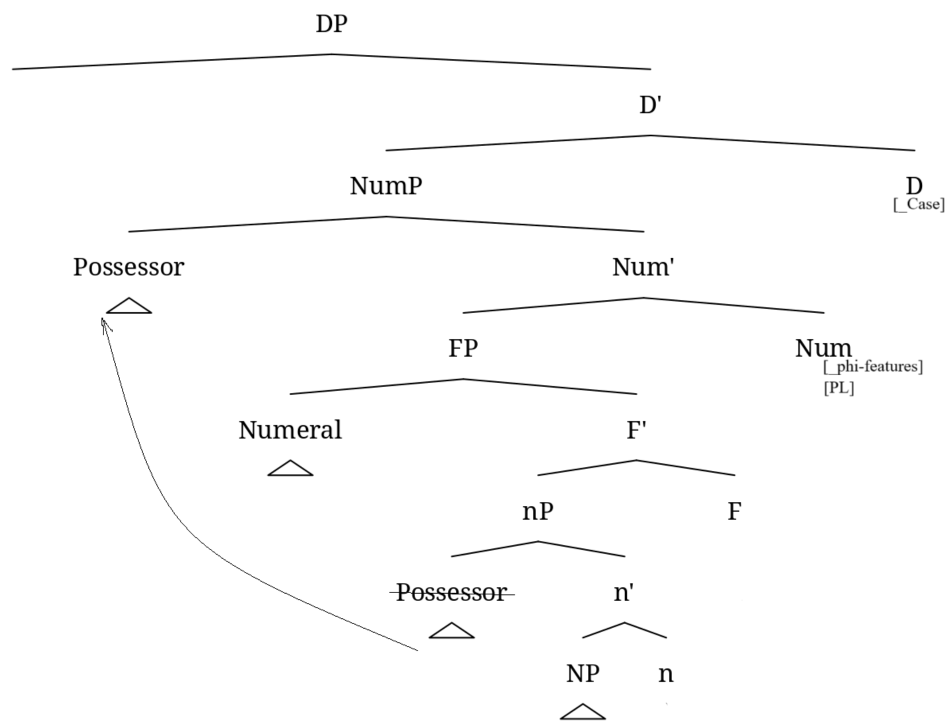
These patterns suggest that NP itself lacks number specification, and that number features are introduced higher in the nominal structure, in a dedicated NumP. This explains why elements below NumP (e.g., post-numeral adjectives) receive default singular marking, while elements above NumP (e.g., pre-numeral adjectives) show plural concord. Thus, NumP hosts number features, and the presence of overt number morphology on the noun head or agreement with the predicate serves as evidence for NumP's existence in the structure.

When it comes to modifiers occupying Spec,NumP, most researchers put numerals in this position (Stavrou & Terzi 2009; Danon 2012, a.o.).[8] This is typically motivated by the need to place numerals above NP but below DP, and NumP naturally provides such a position. However, semantic accounts (Scontras 2014; Martí Martínez 2020) propose that numerals should be placed below NumP, since number must scope over numerals. In this dissertation, I adopt the latter and depart from the assumption that numerals occupy Spec,NumP. In fact, in Chapter 4, Section 4.4, I argue that Moksha provides a syntactic argument in favor of placing numerals in a functional projection below number, see Section 2.3.4 on the functional structure between NumP and NP. If all this is correct, then Spec,NumP must be available for another element.

[8] There are two main approaches to analyzing numerals: as phrases occupying a specifier position (Selkirk 1977; Franks 1994; Ionin & Matushansky 2004; Danon 2012, a.o.) or as heads (Ritter 1991b; Barbiers 1992; Giusti 1997; Danon 2012, a.o.). In this dissertation, I only consider the former view, as it offers broader empirical coverage. The head analysis is more limited in scope and cannot adequately account for complex numerals. Even if some simplex numerals may function as heads in certain languages, all languages must make use of a structure in which the numeral appears in a specifier position.

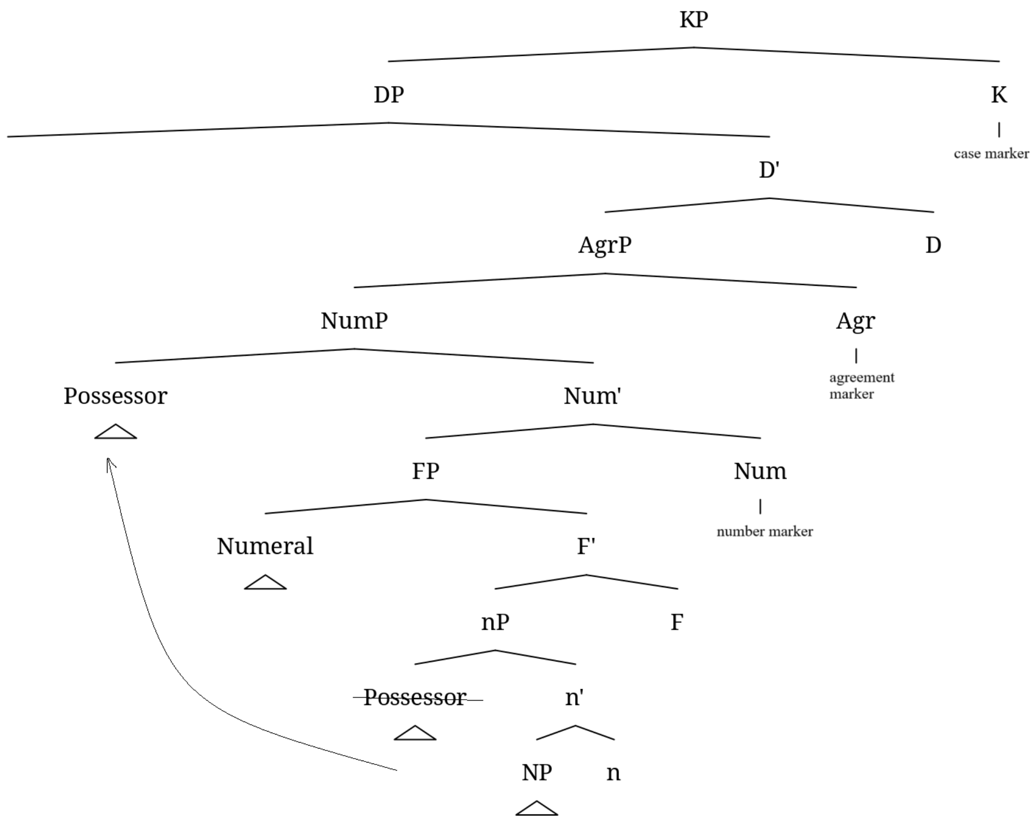
One strong candidate for Spec,NumP is the possessor (or any dependent DP). As outlined in the previous section, if we eliminate AgrP from the narrow syntax, then NumP is the most suitable projection to host possessors (after they move up from their base-generation position, Spec,nP, see Section 2.3.4) and possessive agreement. The exact placements of possessors and agreement probe are outside of the scope of this dissertation, and I leave this as a question for future research. In the meantime, I see no counter-examples against placing possessors in the Spec,NumP and leave it in my model. I also place the nominal agreement probe in Num. Although the morphological realization of agreement requires a post-syntactic addition of an Agr node to be realized as a dissociated morpheme (see Section 1.2.1 on DM and dissociated morphemes), the probing in the narrow syntax is associated with the syntactic Num head. Figure 7 demonstrates possessor agreement and movement within the narrow syntax, and Figure 8 exemplifies the corresponding morphological tree including nodes added post-syntactically for realization of dissociated morphemes, such as nominal agreement and case.

Figure 7: *Narrow syntax: nominal structure with a possessor, a numeral, and an agreement*



probe

Figure 8: *Morphological component: nominal structure with a number marker, a nominal agreement marker, and a case marker*



Despite these possibilities, neither numerals nor possessors serve as definitive indicators of NumP's presence. The only clear diagnostic for NumP remains overt number marking, since numerals might be below NumP, rather than in Spec,NumP; possessors may vary cross-linguistically in their placement; and NumP's presence should not depend on optional elements like numerals or possessors.

Thus, I tentatively adopt the view that possessors occupy Spec,NumP, and the nominal agreement probe is in Num, together with number features. A more precise placement of possessors and agreement probes, as well as possible cross-linguistic variation in this domain, are beyond the scope of this dissertation and remain open questions for future research.[9]

2.3.4. Functional structure between NumP and NP

From the well-established findings discussed so far, we arrive at the following minimal nominal structure:

(55) NP < NumP < DP

With a lexical projection and two functional layers on top, we have sufficient structural space to account for internal movement within the noun phrase (Ritter 1991a). This configuration allows for deriving word order, modeling nominal agreement, and positioning possessors.

However, given the variety of modifiers that need to be generated and placed within a nominal phrase, additional functional layers above NP but below NumP may be required. This is not surprising, considering the parallelism between nominal and clausal structures — the extended projection of the clause includes more than just TP and CP (see Alexiadou et al. 2007: 23–31 for discussion).

One well-motivated functional projection between NP and NumP is nP, which introduces possessors. The proposal that Spec,nP serves as the base position for possessors is inspired by Larson's (1988) VP-shell model, where Spec,vP is the base position for clausal subjects.

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[9] In the languages with optional number marking, number is better analyzed as an adjunct rather than a functional head (Wiltschko 2008).

Although some accounts treat possessors as complements (Ouhalla 1991; Delsing 1993; Horrocks & Stavrou 1987), there is clear syntactic evidence against such an approach (Alexiadou et al. 2007). Specifically, modifiers, unlike complements, can appear in copular constructions (Grimshaw 1990: 97), as shown in (56) and (57) vs. (58).

(56) ENGLISH (Grimshaw 1990: 97)

a. *John's dog*

b. *The dog is John's*.

(57) ENGLISH (Grimshaw 1990: 97)

a. *the book by* / *about* / *on Chomsky*

b. *The book was by* / *about* / *on Chomsky*.

(58) ENGLISH (Grimshaw 1990: 98)

a. *the construction of the building*

b. **The construction was of the building*.

Furthermore, in some languages like Hungarian, possessors trigger agreement on their possessums the same way subjects trigger agreement on predicates. In Hungarian, possessors bear the same case as subjects — nominative case, and trigger agreement morphology on the noun (Szabolcsi 1994):

(59) HUNGARIAN (Szabolcsi 1994: 180)

a	*Mari*	*kalap-ja*
the	Mari-NOM	hat-3SG

'Mary's hat'

The fact that possessors and thematic arguments of nouns occupy the same surface position suggests that possessors do not receive their thematic role in that position. Instead, the thematic relation is assigned in Spec,nP, just as clausal subjects receive their role in Spec,vP. I adopt the following base-generation structure from (Alexiadou et al. 2007):

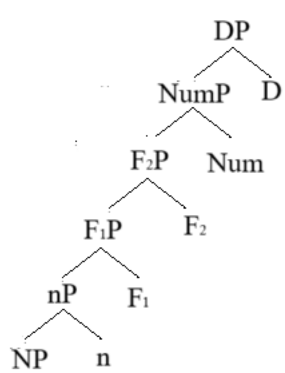
(60) a. *John's book*
b. [nP John's [n bookj] [NP [N tj]]]
(Alexiadou et al. 2007: 562)

While possessors originate in Spec,nP, they do not always remain there. As discussed in Section 2.3.2, possessors move upward to a projection below DP — which I have argued is NumP (see Section 2.3.3). In some cases, possessors can move even higher to Spec,DP (see Section 2.3.1). However, the crucial point is that possessors are generated low in the structure, just above NP.

Some researchers, e.g., Picallo (1991) and Bernstein (1993), propose additional projections above NP but below NumP, such as Gender (GenP) or Word Marker (WMP). These projections are often introduced without reference to nP, so I assume that nP can fulfill their function, making GenP/WMP unnecessary as distinct projections.

The precise nature of other potential projections within the nominal domain remains an open question. However, I leave room for additional functional layers between nP and NumP, depicted as FP (Functional Projection) in Figure 9. In Chapter 4 (section 4.4), I explore how this FP zone is utilized in Moksha nominal syntax. Specifically, I argue that the FP zone is split into two projections accommodating numerals (F_1P) and demonstratives (F_2P).

Figure 9: *Hierarchical model of the noun phrase*



I assume that DP always takes NumP as its complement, NumP takes F_2P as its complement, F_2P takes F_1P, F_1P takes nP, and nP takes NP. A key assumption in this model is that functional projections within the extended nominal structure cannot be skipped. If a higher projection is present, all lower layers must also be present.

2.4. The problem of the NP vs. DP languages debate

As mentioned at the beginning of this chapter, while many researchers argue for the presence of additional functional heads above the nominal projection NP (Abney 1987; Valois 1991; Ritter 1991a; Alexiadou 2001; Brugè 2002; Punske 2014, a.o.) — citing evidence such as case marking in nominalizations, nominal agreement, and the placement of definite determiners — there remains debate over whether the DP is universally present across languages. In this section, I challenge the classification of languages into those that have DPs and those that have NPs, arguing instead that nominal structures of varying sizes can be observed within the same language.

The idea that all argumental nominals are DPs cross-linguistically, regardless of whether they have overt determiners (see Longobardi 1994), is theoretically appealing. It offers a streamlined grammatical architecture, linking argumenthood to the presence of a specific projection—DP. However, the lack of sufficient evidence for DP in certain languages has led to the parametrized DP hypothesis.

This approach posits that DP does not project unless it is manifested by an overt determiner. Specifically, Bošković (2008) argues that languages with articles have DPs, while articleless languages have NPs. His main argument for this parametrization is that languages with articles exhibit different syntactic properties from those without. For example, he claims that languages with articles disallow left-branch extraction, whereas articleless languages permit it. Compare (61)a from Serbo-Croatian, an articleless Slavic language that allows left-branch extraction, with (61)b from Bulgarian, a related Slavic language with articles that disallows it.

(61) a. SERBO-CROATIAN (Bošković 2008: 2)

skupa / *ta*$_i$ *je* *vidio* [t$_i$ *kola*]
expensive that is seen car
'He saw an expensive / this car.'

b. BULGARIAN

**nova-ta*$_i$ *prodade* *petko* [t$_i$ *kola*]
new-the sold Petko car
'The new car, Petko sold.'

Among other distinguishing syntactic properties, Bošković also considers factors such as adjunct extraction, scrambling, negative raising, superiority effects in multiple wh-fronting, clitic doubling, availability of more than one adnominal genitive, superlative readings, island sensitivity of head internal relatives, and polysynthesis. The summary of the properties and their implications for the status of the language with respect to articles is given in Table 2, see (Bošković 2010) for more parameters.

Table 2: *Parameters distinguishing DP and NP languages (languages with and without articles)* (Norris 2014: 29)

CRITERION
Left branch extraction → NP
Adjunct extraction → NP
Double genitives → DP
Majority of 'most' → DP
Superiority in multiple wh-fronting → DP
Negative Raising → DP
Clitic Doubling → DP
Locality in IHRCs → NP
Polysynthesis → NP

A comprehensive examination of Bošković's parameters is beyond the scope of this dissertation, but Norris (2014) and Pleshak (2017a) have shown that these distinctions do not consistently correlate with the presence of articles in a language. For instance, Norris (2014) demonstrates that Estonian (a Finno-Ugric, Uralic language) lacks overt articles but behaves like a DP-language (language with articles) according to Bošković's parameters. In fact, many of the proposed correlations are inconclusive for Estonian, and the only two applicable parameters suggest that Estonian should be classified as a DP-language.

Table 3: *Parameters distinguishing DP and NP languages applied to Estonian* (adapted from Norris 2014: 29)

CRITERION	ESTONIAN	CONCLUSION
Left branch extraction → NP	no Left branch extraction	—
Adjunct extraction → NP	No Adjunct Extraction	—
Double genitives → DP	Yes	DP
Majority of 'most' → DP	Yes/unclear	DP/—
Superiority in multiple wh-fronting → DP	N/A (no MWF)	—
Negative Raising → DP	N/A (no known NPIs)	—
Clitic Doubling → DP	N/A (no clitic doubling)	—
Locality in IHRCs → NP	N/A (no IHRCs)	—
Polysynthesis → NP	N/A (not polysynthetic)	—

Similarly, Moksha (Uralic) presents a challenge to the DP/NP classification. Although Moksha has a definite article (62), as shown in (Pleshak 2017a), and thus should qualify as a DP language, it exhibits properties characteristic of NP-languages (i.e., articleless languages). It permits adjunct extraction (63) and disallows negative raising (64). Moreover, it shows no superiority effects in wh-fronting (65). Moksha also exhibits scrambling (see Chapter 5). All of these are characteristics that Bošković associates with NP-languages.

(62) MOKSHA DEFINITE ARTICLE

a.	*morkš*	b. *morkš-s'*
	table[NOM]	table-DEF.SG[NOM]
	'a table'	'the table'

(63) MOKSHA: ADJUNCT EXTRACTION ALLOWED → NP

a.	*mon*	*luv-in'ə*	*kniga-t'*	***mordovijɛ-t'***	***kolga***
	1SG[NOM]	read-PST.3.O.1SG.S	book-DEF.SG.GEN	Mordovia-DEF.SG.GEN	about

'I read a book about Mordovia.'

b.	***mez'ə-n'***	***kolga***	*luv-it'*	*kn'iga-t'?*
	what-GEN	about	read-PST.3.O.2SG.S	book-DEF.SG.GEN

'What was the book you read about?'

(64) MOKSHA: NEGATIVE RAISING DISALLOWED → NP

a.	*maša*	*dumanda-j,*	*čto*	*šejər'-s'*	*af*
	Masha[NOM]	think-NPST.3SG	that	mouse-DEF.SG[NOM]	NEG
	susk-ənd-i	*fkɛ-vək*	*modamar'*		
	bite-IPFV-NPST.3SG	one-ADD	potato		

b.	**maša*	*af*	*dumanda-j,*	*čto*	*šejər'-s'*
	Masha[NOM]	NEG	think-NPST.3SG	that	mouse-DEF.SG[NOM]
	susk-ənd-i		*fkɛ-vək*	*modamar'*	
	bite-IPFV-NPST.3SG		one-ADD	potato	

'Masha thinks that the mouse isn't going to bite a single potato.'

(65) MOKSHA: NO SUPERIORITY EFFECTS IN WH-FRONTING →NP

a.	*məz'ardə*	*mejama*	*son*	*t'ejə-t*	*mɛr'k-s'?*
	when	what	3SG[NOM]	PRON.DAT-2SG.AGR	say-PST.3SG
b.	OK*mejama*	*məz'ardə*	*son*	*t'ejə-t*	*mɛr'k-s'?*
	what	when	3SG[NOM]	PRON.DAT-2SG.AGR	say-PST.3SG

'When and what did he tell you?'

More strikingly, Moksha demonstrates an internal inconsistency in left-branch extraction, depending on the marking of the nominal from which the extraction is being performed. Based on Bošković's logic, we would expect left-branch extraction to be either possible or impossible in Moksha, regardless of the marking. Yet, in Moksha, left-branch extraction is permitted when the DO lacks an overt article (66), but is blocked when the DO is definitively marked (67), see Chapter 3 for the specifics of the Moksha declension and formal expression of definiteness.

(66) MOKSHA: DO WITH NO ARTICLE → SUBEXTRACTION IS POSSIBLE

a. *mon* *sud'ər'ɛ-n'* ***ravžə*** *traks*
1SG[NOM] pet-PST.1SG black cow

b. OK***ravžə*** *mon* *sud'ər'ɛ-n'* *traks*
black 1SG[NOM] pet-PST.1SG cow
'I pet a black cow.'

(67) MOKSHA: DO WITH AN ARTICLE → SUBEXTRACTION IS IMPOSSIBLE

a. *mon* *sud'ər'ɛ-jn'ə* ***ravžə*** *traks-t'*
1SG[NOM] pet-PST.3.O.1SG.S black cow-DEF.SG.GEN

b. ****ravžə*** *mon* *sud'ər'ɛ-jn'ə* *traks-t'*
black 1SG[NOM] pet-PST.3.O.1SG.S cow-DEF.SG.GEN
'I pet the black cow.'

Another parameter that gives us a split result is the adnominal genitive one. Two adnominal genitives are disallowed when both carry definite marking (68), but they are permitted when at least one lacks definite marking (69).

(68) MOKSHA: TWO DEFINITE GENITIVES
*[[*t'ɛ* *c'ora-**t'***] [*mašina-**t'***] *šari-sə(-nzə)*] *var'ɛ-n'ɛ*
this boy-DEF.SG.GEN car-DEF.SG.GEN wheel-IN-AGR.3SG hole-DIM[NOM]
Int.: 'There is a hole in [this boy's$_{\text{Possessor}}$ wheel$_{\text{Possessum}}$ of the car$_{\text{Possessor}}$].'

(69) MOKSHA

a. TWO INDEFINITE GENITIVES
[*tona* *s't'ər-n'ɛ-**n'***] [*t'ɛ* *ras't'en'ijɛ-**n'***] *kor'ɛn'-c'*
that girl-DIM-GEN this plant-GEN root-DEF.SG
'the root$_{\text{Possessum}}$ of this plant$_{\text{Possessor}}$ that belongs to that girl$_{\text{Possessor}}$'

b. ONE DEFINITE AND ONE INDEFINITE GENITIVE
[[*t'ɛ* *c'ora-**t'***] [*mašina-**n'***] *šari-s'*] *urdaz-u*
this boy-DEF.SG.GEN car-GEN wheel-DEF.SG[NOM] dirt-ATTR
'[This boy's$_{\text{Possessor}}$ wheel$_{\text{Possessum}}$ of the car$_{\text{Possessor}}$] is dirty.'

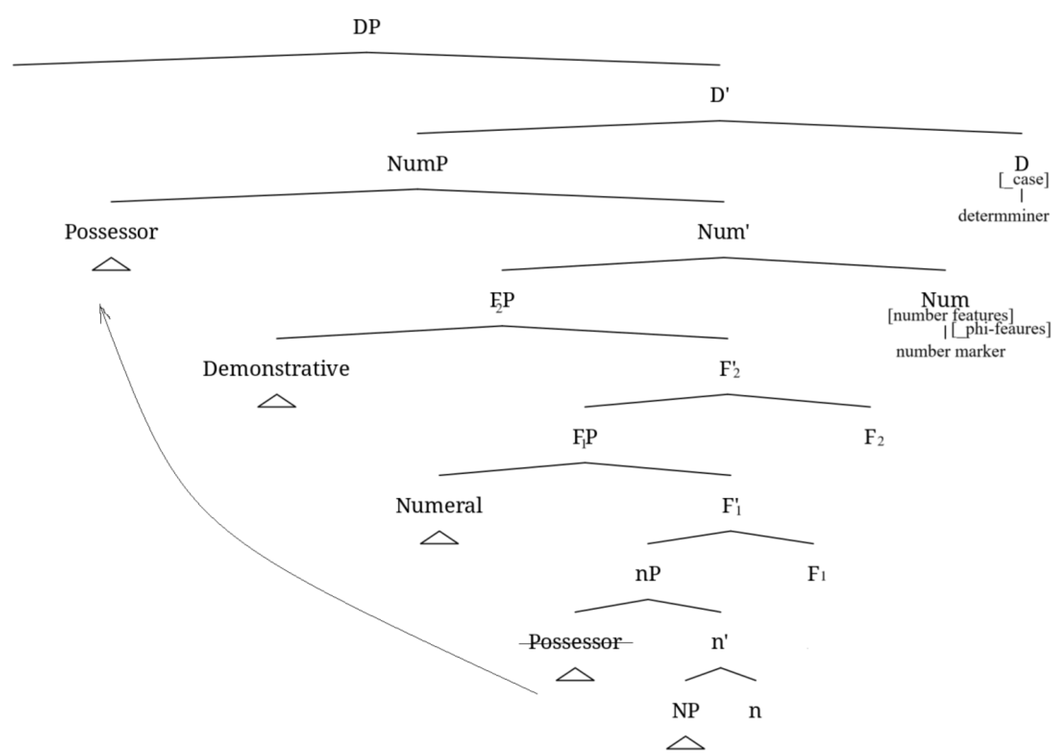
These data from Moksha do not merely challenge the cross-linguistic applicability of Bošković's parameters; they also reveal inconsistencies within a single language. Such findings suggest that the distinction between DP and NP is not a strict cross-linguistic parameter but rather allows language-internal variation. Furthermore, Bošković's analysis assumes a binary DP/NP distinction, ignoring intermediate functional projections. When considering an extended projection approach, such strict parametrization appears untenable.

A more nuanced perspective, which acknowledges the presence of nominals of different sizes within a single language, provides a better framework for understanding cross-linguistic variation in nominal structures. Additional evidence for this claim, specifically within Moksha and Hill Mari, is discussed in Chapters 4, 5 and 6 of this dissertation.

2.5. Interim summary

In this chapter, I provided an overview of the arguments supporting the hierarchical structure of nominals, drawing on evidence from ellipsis, pronominal substitution, coordination, noun phrase-internal movement, and agreement. I also introduced the concept of extended projection (Grimshaw 1993) as a framework for modeling this hierarchy. Additionally, I outlined the exact expectations of how each functional layer contributes to the properties of the whole nominal, see Figure 10.

Figure 10: *Internal structure of the noun phrase*



In particular, I treat DP as a formal projection hosting definite determiners, case features and features relevant for agreement; NumP as a projection hosting number markers and number features, as well as the nominal agreement probe; nP as the projection of base-generation of possessors. I also proposed the presence of two FP projections for cardinal numerals (F_1P) and demonstratives (F_2P). Finally, I argued against the rigid cross-linguistic classification of languages into NP and DP types, presenting evidence from a language where the definiteness marking of the argument affects its syntactic behavior, challenging the validity of such strict parametrization.

Chapter 3: Moksha nominal morphosyntax

Moksha is a Mordvin language of the Finno-Ugric branch of the Uralic family, spoken by approximately 20,000 people[10] in the Republic of Mordovia, Russia. As is typical of Finno-Ugric languages, Moksha exhibits rich morphology. In the nominal domain, four categories — number, possession, definiteness, and case — are expressed through specialized morphological affixes.

The goal of this chapter is to establish a general framework for the Moksha DP, which will serve as a baseline for my analysis of reduced nominal structures in the following chapters. Section 3.1 provides an overview of how nominal categories are morphologically expressed. Section 3.2 examines the basic properties of adnominal modifiers and their interaction with morphological marking. Finally, Section 3.3 maps these data onto the hierarchical DP structure introduced in Chapter 2.

3.1. The expression of case, definiteness, nominal agreement, and number

3.1.1. Case

Descriptions of the Moksha case paradigm include up to 16 cases (Evsevjev 1963; Koljadenkov, Zavodova 1962; Feoktistov 1975; Plaksina 2002; Kholodilova 2018a). As noted by Kholodilova (2018a: 81), some nominal markers lack the typical properties of case markers, at least according to Corbett's (2012) definition. Some researchers analyze markers lacking case-marker properties as derivational morphemes.

[10] According to 2020 census.

Table 4 presents nine markers whose status as cases is least disputed. These include the nominative, which marks subjects; the genitive, which marks direct objects (DOs), possessors and complements of postpositions; the dative, which marks indirect objects; the ablative, which is selected by certain verbs; and several semantic cases that encode spatial relations.

Table 4: *A fragment of the Moksha case paradigm* (adopted from Kholodilova 2018a: 66)

Case	Marker
Nominative	∅
Genitive	*-ən' / -ən'n'ə*
Dative	*-ən'd'i*
Ablative	*-də / -tə / -d'ə / -t'ə*
Inessive	*-sə*
Elative	*-stə*
Illative	*-s*
Lative	*-u / -v / -i*
Prolative	*-əva / -ga / -ka*

Based on their functions, all the cases above can be divided into **structural** and **inherent** cases. The nominative and genitive, as structural cases, mark positions of the core arguments: nominative for subjects and genitive for objects, possessors, and postpositional complements. The remaining cases are associated with specific semantic roles (see Woolford 2006 on the role of theta-roles in structural vs. inherent case distinction).

The nominative case, which is phonologically null in Moksha, marks both intransitive (70)a and transitive (70)b subjects. Indefinite DOs also appear in this phonologically null form (70)c, which may or may not be analyzed as nominative. In Chapter 5, I analyze Moksha bare DOs as bearing a morphologically null object case. In contrast, definite DOs (70)b are marked with the genitive.

(70) a. INTRANSITIVE SUBJECT → NOMINATIVE (Toldova 2018: 552)

maša-n'	***s'ora-c***	*juma-s'*	*paks'a-v*
Masha-GEN	boy-3SG.AGR.SG[NOM]	get.lost-PST.3.SG	field-LAT

‘Masha’s son got lost in the field.’

b. TRANSITIVE SUBJECT → NOMINATIVE; DO → GENITIVE (Toldova 2018: 552)

maša-n'	***s'ora-c***	*juksta-z'ə*	***kaza-t'***[11]	*paks'a-v*
Masha-GEN	boy-3SG.AGR.SG[NOM]	forget-PST.3SG.O.3SG.S	goat-DEF.SG.GEN	field-LAT

‘Masha’s son forgot the goat in the field.’

c. INDEFINITE DO → NOMINATIVE (?) (Toldova 2018: 552)

maša-n'	*s'ora-c*	*ɛxɛr'*	*juksta-j*	***kodamə***	***bəd'ə***	***kaza***	*paks'a-v*
Masha-GEN	boy-3SG.AGR.SG	always	forget-NPST.3.SG	which	INDEF	goat	field-LAT

‘Masha’s son always forgets some goat in the field.’

Beyond marking definite DOs (70)b, the genitive case encodes adnominal dependents, typically possessors (71), as well as complements of all but one postpositions, independent from their semantics (72); see below on the postposition *baška*. This independence of semantics supports the classification of nominative and genitive as bona fide structural cases.

[11] The genitive case in Moksha has several allomorphs, and its phonological realization depends on the nominal categories expressed within the phrase (see Section 3.2.4 below). For instance, *-t'* appears on singular definite genitive phrases, while *-n'* is used in most other contexts. The variation in genitive marking is irrelevant unless discussed explicitly.

(71) POSSESSOR → GENITIVE

ava-t' *sumka-**c*** *pra-s'*
woman-DEF.SG.GEN bag-3SG.AGR.SG fall-PST.3SG
'The woman's bag fell.'

(72) COMPLEMENT OF A POSTPOSITION (Toldova & Pleshak 2018)

vel'ə-n'əkə-n' *esə* *er'ɛ-s'* *koz'ɛ,* *koz'ɛ* *al'ɛ*
village-1PL.AGR-GEN in.IN live-PST.3[SG] rich rich man
'There lived a rich main in our village.'

The dative case marks indirect objects, including recipients (73) and goals (74). It also appears in passives, where it marks external arguments (similar to a by-phrase) (75), as well as in debitive constructions (76). Experiencers with a restricted set of predicates can also be marked with the dative case (77). Because its distribution depends on specific theta roles, the Moksha dative qualifies as an inherent case (see Woolford 2006 for an overview of functional distribution of inherent cases).

(73) RECIPIENT → DATIVE (Kholodilova 2018a: 87)

pet'ɛ *maks-s'* *pan'čf-t* *mon'* *s't'ər'-əz'ə-**n'd'i*** /
Petja[NOM] give-PST.3[SG] flower-PL 1SG.GEN girl-1SG.AGR.SG-DAT
s't'ər'-əzə-n***
girl-ILL-1SG.AGR
'Petja gave flowers to my daughter.'

(74) GOAL → DATIVE (Kholodilova 2018a: 87)

pet'ɛ *suva-s'* *mon'* *kud-əz'ə-**n'd'i*** /
Petja[NOM] enter-PST.3[SG] 1SG.GEN house-1SG.AGR.SG-DAT
*kud-**əzə-n***
house-ILL-1SG.AGR
'Petja entered my house (lit. to my house).'

(75) EXTERNAL ARGUMENT OF A PASSIVE CONSTRUCTION → DATIVE (A. Kozlov et al. 2016: 541)

t'ɛ *karaf-s'* *šuv-əv-s'* ***vas'ɛ-n'd'i***
this pothole-DEF.SG dig-PASS-PST.3[SG] Vasja-DAT
'This pothole has been dug by Vasja.'

(76) EXTERNAL ARGUMENT OF A DEBITIVE CONSTRUCTION → DATIVE (Zakirova 2018: 768)

s'ora-n'ɛ-t'i	*luv-əma*	*t'ɛ*	*kn'iga-s'*	/	*kn'iga-t'*
boy-DIM-DEF.SG.DAT	read-NZR	this	book-DEF.SG		book-DEF.SG.GEN

‘The boy has to read this book.’ (lit. ‘The reading of this book occurs to the boy.’)

(77) EXPERIENCER → DATIVE (Moksha Corpus; from Toldova 2018: 555)

i	*konečno*	***mon'-d'ejə-n***	*ul'-s'*	*viz'ks*
and	of.course.RUS	1SG.OBL-PRON.DAT-1SG.AGR	be-PST.3[SG]	shame

‘And, of course, I was ashamed.’

The ablative case encodes the theme of ingestive verbs (78) (Toldova 2018: 556); in addition, it is selected by certain verbs, for example, *pel'əms* ‘be afraid’ (79), and by the postposition *baška* ‘except’ (80), the only postposition that does not require a genitive complement.

(78) THEME OF INGESTIVE VERBS → ABLATIVE (Stenin 2018: 518)

jar̥ca-n'	*l'ɛm-də*
eat-PST.1SG	soup-ABL

‘I ate soup.’

(79) THE COMPLEMENT OF *PEL'ƏMS* ‘BE AFRAID’ → ABLATIVE (Toldova 2018: 554)

maša-n'	*s'ora-c*	*pel'-i*	*mɛn'*	*jomla*	***pin'ə-də***
Masha-GEN	boy-3SG.AGR.SG	be.afraid-NPST.3[SG]	even	little	dog-ABL

‘Masha's son is afraid of even a small dog.’

(80) THE COMPLEMENT OF *BAŠKA* ‘EXCEPT’ → ABLATIVE

(Muravyeva & Kholodilova 2018: 231)

*pet'a-**də***	*baška*	*s'embə*	*sa-s'-t'*
Peter-ABL	except	all	come-PST.3-PL

‘Except for Peter, everybody came.’

The remaining forms in Table 4 are locative cases, which encode spatial arguments and adjuncts (Kozlov L. 2018) (81)–(82). As I mentioned in Chapter 1, these cases might be syntactically PPs, but here, I follow the traditional descriptions and refer to them as “cases” for consistency, since this does not affect the analysis I am developing below.

(81) *kn'iga-s'* *ašč-i* *karopka-**sə***
book-DEF.SG be.situated-NPST.3[SG] box-IN
'The book is in a box.' (Kozlov L. 2018: 156)

(82) *mon* *povfta-jn'ə* *kart'ina-t'* *s't'ena-**s***
1SG hang-PST.3.O.1SG.S painting-DEF.SG.GEN wall-ILL
'I hang the painting on the wall.' (Kozlov L. 2018: 168)

Beyond spatial relations proper, locative cases may encode some other meanings. For instance, inessive also has instrumental (83) and comitative uses.

(83) *al'ɛ-z'ə* *šuft-t'* *ker'-si* *uz'ər'-**sə***
man-1SG.AGR.SG tree-DEF.SG.GEN cut-NPST.3.O.3SG.S axe-IN
'My father is cutting the tree with an axe.' (Kozlov L. 2018: 157)

A thorough description of functions of oblique cases in Moksha is beyond the scope of this dissertation, and I refer the reader to (Kozlov L. 2018) for details. For the present study, the nominative (unmarked) and genitive cases are the relevant ones, because they are the only ones that participate in the case-marking of the complement of P and DOs — the two syntactic positions I discuss in my dissertation. It is important to keep in mind that Moksha postpositions (almost) uniformly assign a structural case — genitive, not a variety of inherent cases.

3.1.2. Definiteness

Moksha exhibits a set of markers that function as definite determiners. Definiteness is usually expressed together with number and case, and these markers are represented in Moksha grammars in the form of the so-called **definite declension**. This contrasts with the **indefinite declension**, as shown in Table 5. The markers in the indefinite declension do not encode other nominal categories. Table 5 presents the two declensions side by side. Notably, only the first three cases — nominative, genitive and dative — allow for the overt marking of definiteness.

Table 5: *Indefinite and Definite declensions in Moksha* (adopted from Kholodilova 2018a: 66)

Case	Indefinite declension		Definite declension	
	SG	PL	SG	PL
Nominative	∅	*-t* / *-t'*	*-s'*	-(*t'*)*n'ə*
Genitive	*-ən'* / *-ən'n'ə*		*-t'*	-(*t'*)*n'ə-n'*
Dative	*-ən'd'i*		*-t'i*	-(*t'*)*n'ə-n'd'i*
Ablative	*-də* / *-tə* / *-d'ə* / *-t'ə*			
Inessive	*-sə*			
Elative	*-stə*			
Illative	*-s*			
Lative	*-u* / *-v* / *-i*			
Prolative	*-əva* / *-ga* / *-ka*			

The use of the definite declension in Moksha extends beyond canonical definiteness. In addition to expressing uniqueness (84), identifiability (85) and familiarity (86), it is also influenced by the information-structural status of the noun phrase (Kashkin 2018: 123).

(84) UNIQUENESS (Kashkin 2018: 126)

t'ɛči	***ši-s'***	/ ****ši***	*valdəpt-i*	*valctə*
today	sun-DEF.SG[NOM]	sun	give_light-NPST.3SG	bright.EL

'Today, the sun shines bright.'

(85) IDENTIFIABILITY (Kashkin 2018: 126)

t'ɛ	***kuc'***	/ ****kud***	*oc'u*	*i*	*mazi*
this	house.DEF.SG[NOM]	house	big	and	nice

'This house is big and beautiful.'

(86) FAMILIARITY (Kashkin 2018: 127)

mon *mol'-ən'* *ul'c'a-va* *i* *n'ɛj-ən'* *pin'ə,*
1SG[NOM] go-PST.1SG street-PROL and see-PST.1SG dog
i ***pin'ə-s'*** / ****pin'ə*** *uv-əman'*
and dog-DEF.SG[NOM] dog bark-PST.1SG.O.3SG.S
'I walked along the street and saw a dog; and the dog barked at me.'

When a nominal refers to an entity from a known set, the definite determiner may be used, though it is not obligatory. However, if no such set is contextually available, a nominal that is not otherwise definite cannot be marked with a definite determiner:

(87) AN ENTITY FROM A KNOWN SET (Kashkin 2018: 137)

mon *apak* *uč-s'ə-k* *vas'ft-in'ə*
1SG[NOM] NEG.NFIN wait-FREQ-CN.NFIN meet-PST.3.O.1SG.S
fkɛ ***al'ɛ-t'***
one man-DEF.SG.GEN
i. *{I was walking along a very empty street.} 'Suddenly, I met one man…'.
ii. {Yesterday, we were discussing your friends.} 'Suddenly, I met one man (one of them)…'.

Marking an entity from a definite set with a definite determiner is also influenced by the topicality of the noun phrase. As noted by Toldova (2017), definiteness marking is applied when the noun phrase is in the topic position. The contrast in (88) is telling. Given the context in (88)a, the noun phrase *kaftə kal̥t* 'two fish' receives definiteness marking in (88)b, because it is in the topic position. In (88)c, on the other hand, where it is not in the topic, there is no definiteness marking.

(88) a. *d'ɛd'ɛ-z'ə* *pid'-əs'* ***kolmə*** ***kal̥-t***
mother-1SG.AGR.SG[NOM] boil-PST.3SG three fish-PL
'My mother boiled two fish.'

b. ***kaftə*** ***kal̥-n'ə-n'*** *səv-əz'ən'* *katə-s'*
two fish-DEF.PL-GEN eat-PST.3SG.S.3SG.O cat-DEF.SG[NOM]
'Two of the fish were eaten by the cat.'

c. *katə-s'* *səv-s'* ***kaftə*** ***kal̥-t***
cat- DEF.SG[NOM] eat-PST.3SG two fish-PL
'The cat ate two fish {and Vasja ate one.}' (Toldova 2017: 150)

Thus, if a noun is semantically definite, it is marked as such. Otherwise, it must be determined whether it belongs to a relevant discourse set. If it does, then one needs to determine whether the noun phrase is in the topic. If it is topical, then it gets marked as definite.

Definiteness marking obeys special restrictions in three categories: pronouns, proper names and possessed nominals. With exception of *mez'ə* 'what', which behaves like a regular noun (89), items classified in descriptive grammars as different types of pronouns do not get definiteness marking, e.g., the animate interrogative pronoun *ki(jə)* 'who' (90) and the demonstrative *t'ɛ* 'this' (91). Genitive forms of personal pronouns are suppletive and seem to not have an identifiable definiteness marker (92).

(89) INANIMATE INTERROGATIVE PRONOUN
mez'ə-t' / #*mez'ə-n'*[12]
what-DEF.SG.GEN what-GEN
'what?'

(90) ANIMATE INTERROGATIVE PRONOUN
ki-n' / **ki-t'*
who-GEN who-DEF.SG.GEN
'who?'

(91) DEMONSTRATIVE PRONOUN
t'ɛ-n' / **t'ɛ-t'*
this-GEN this-DEF.SG.GEN
'this one'

(92) PERSONAL PRONOUN
a. *mon*
1SG[NOM]
'I'

[12] The form *mez'ə-n'* can appear in the adnominal position, but it exhibits properties distinct from those of a typical referential noun phrase; see Pleshak (2021) for a discussion of *-n'* marking in adnominal contexts in Moksha.

b. *mon'* / **mon-t'*
1SG.GEN 1SG-DEF.SG.GEN
'me'

Despite being semantically definite, by default, animate proper names do not receive definite marking. However, in certain marked contexts, definite markers may appear:

(93) PROPER NAME (Kashkin 2018: 128)
vas'ɛ kel'k-si maša-n' / maša-t'
Vasja like-NPST.3SG.O.3SG.S Masha-GEN Masha-DEF.SG.GEN
'Vasja loves Masha.'

Lastly, possessed nominals are not marked for definiteness because definite affixes and nominal agreement markers are in complementary distribution (see Sections 3.1.3. and 3.1.5 for discussion):

(94) POSSESSED NOMINAL (Kholodilova 2018a: 77)
**katə-z'ə-s' / katə-s'-əz'ə*
cat-1SG.AGR.SG-DEF.SG[NOM] cat-DEF.SG.NOM-1SG.AGR.SG
Int.: 'my cat'

Summing up, Moksha has a definite determiner: definiteness is marked with a suffix and expressed together with case and number. Beyond canonical semantic definiteness, such as familiarity or uniqueness, this marking is influenced by discourse structure, especially topicality. However, pronouns, proper names, and possessed nominals generally do not take definite marking.

3.1.3. Nominal agreement

In Moksha, adnominal DPs trigger agreement on the nominal head. The agreement marker encodes person and number features of the adnominal DP, typically a possessor (95). Traditionally, Moksha grammars refer to this as possessive agreement (see Toldova et al. 2018 and references therein), but I adopt the more general term nominal agreement to better capture the fact that these agreement markers appear both on possessive constructions and postpositions. Furthermore, the DP that triggers agreement is not always a possessor, reinforcing the need for a broader term.

(95) a. *mon'* *morkš-əz'ə*
1SG.GEN table-1SG.AGR.SG[NOM]
'my table'

b. *maša-n'* *morkš-əc*
Masha-GEN table-3SG.AGR.SG[NOM]
'Masha's table'

In the nominative, genitive and dative cases, the agreement marker also encodes the number of the head noun if the adnominal DP is singular. However, if the adnominal DP is plural, the number of the head noun is neutralized[13].

Table 6 presents a fragment of the Moksha case paradigm, illustrating nominal agreement markers. In Moksha grammars, this system is referred to as the **possessive declension** (cf. the indefinite and definite declensions discussed earlier). The first three rows of the table demonstrate the number contrast with a singular possessor and the absence of this contrast with a plural possessor (see also Section 3.1.4 on the number expression in Moksha). The last row represents the behavior of the rest of the Moksha cases, where the number contrast is completely absent.

[13] By neutralization of number, I refer to the form of the fused number and nominal agreement marker. In constructions with a singular possessor, the portmanteau marker encodes both the phi-features of the possessor and the number of the possessum (the head). However, with a plural possessor, the nominal agreement marker appears identical regardless of whether the possessum is singular or plural, and there is no other overt means of expressing the number of the possessum.

Additionally, the shape of the agreement marker in the nominative, genitive, and dative cases differs from that used in the rest (represented by the inessive case).

Table 6. *Nominal agreement markers in the Moksha case paradigm* (Kholodilova 2018a: 67)

Case	**Person and number of the possessor**								
	1SG		**2SG**		**3SG**		**1PL**	**2PL**	**3PL**
	Number of the possessum								
	SG	**PL**	**SG**	**PL**	**SG**	**PL**			
NOM	*-əz’ə*	*-n’ə*	*-c’ə*	*-(t’)n’ə*	*-əc*	*-ənzə*	*-n’(ə)kə*	*-ən’t’ə*	*-snə*
GEN	*-əz’ə-n’*	*-n’ə-n’*	*-c’ə-n’*	*-(t’)n’ə-n’*	*-ənc*	*-ənzə-n*	*-n’(ə)kə-n’*	*-ən’t’ə-n’*	*-snə-n*
DAT	*-əz’ə-n’d’i* / *-z’t’i*	*-n’ə-n’d’i*	*-c’ə-n’d’i* / *-c’t’i*	*-(t’)n’ə-n’d’i*	*-ən cti*	*-ənzə-ndi*	*-n’(ə)kə-n’d’i*	*-ən’t’ə-n’d’i*	*-snə-ndi*
IN	*-sə-n*		*-sə-t*		*-sə-nzə*		*-sə-n(ə)k*	*-sə-nt*	*-sə-st*

The possessive and definite declension markers are in complementary distribution; nominal agreement does not co-occur with a determiner within one noun phrase (see Section 3.1.5 for discussion).

(96) a. **morkš-əz'ə-s'*
table-1SG.AGR.SG-DEF.SG[NOM]
Int.: ‘my table’

b. **morkš-s'-əz'ə*
table-DEF.SG-1SG.AGR.SG

Thus, Moksha employs specialized nominal markers that encode the person and number features of the adnominal DP. Depending on the case, these markers may also be fused with the number marker.

3.1.4. Number

Moksha distinguishes between singular and plural number. In indefinite nominative noun phrases, where no other overt morphological markers are present, singular is expressed by a zero marker, while plural is marked with *-t*:

(97)	a. *vel'ə-**Ø*** village-SG ‘a village’	b.	*vel'ə-**t*** village-PL ‘villages’

However, the overt expression of number depends on the other grammatical categories expressed within the nominal. Specifically, in the absence of a definite determiner or a nominal agreement marker, the plural marker appears only when the noun has a null case marker (typically nominative; see Chapter 5 for my discussion of the null object case in Moksha). In (98), where the nominal is not marked as definite or possessed, no number morphology appears.

(98)	*vat-t,*	*ic'*	*kanfetka-**də***	*jar̥ca-j*
	look-IMP.SG	child.DEF.SG[NOM]	candy-ABL	eat-NPST.3.SG

i. ‘Look, the kid is eating a piece of candy.’
ii. ‘Look, the kid is eating pieces of candy.’

In (99)–(100), on the other hand, the word *vel'ə* ‘village’ combines with a nominal agreement affix encoding the person and number of the possessor. Number is expressed together with nominal agreement, as a fusional part of the same marker. In (99)a and (99)b, the features of the possessor are the same — 1st person singular, but the number of the possessum is different, leading to distinct agreement markers. A similar contrast is observed in (100) with a 3rd person singular possessor.

(99) SINGULAR POSSESSOR → NUMBER OF POSSESSUM EXPRESSED

a. *vel'ə-**z'ə*** village-1SG.AGR.SG ‘my village’	b.	*vel'ə-**n'ə*** village-1SG.AGR.PL ‘my villages’

(100) SINGULAR POSSESSOR → NUMBER OF POSSESSUM EXPRESSED

a. *vel'ə-**c***
village-3SG.AGR.SG
'his/her village'

b. *vel'ə-**nzə***
village-3SG.AGR.PL
'his/her villages'

However, this number distinction is neutralized when the possessor is plural. Unlike (99), where the 1st person singular possessor, triggers a contrast in possessum number, (101) shows that a 1st person plural possessor has the same nominal agreement marker regardless of whether the possessum is singular or plural.

(101) PLURAL POSSESSOR → NUMBER OF POSSESSUM IS NOT EXPRESSED
*vel'ə-**n'əkə***
village-AGR.1PL
'our village/villages'

Similarly, the number distinction disappears when the nominal head receives any case other than nominative, genitive or dative. For example, in the inessive case, number marking is absent:

(102) OBLIQUE CASE → NUMBER OF POSSESSUM IS NOT EXPRESSED
*vel'ə-sə-**n***
village-IN-AGR.1SG
'in my village/villages'

Since number is expressed together with the definite determiner (see Section 3.1.2), it becomes obligatory when definiteness is marked overtly:

(103) OVERT DEFINITENESS → OBLIGATORY NUMBER

a. *vel'ə-s'*
village-DEF.SG[NOM]
'the village / *the villages'

b. *vel'ə-t'n'ə*
village-DEF.PL[NOM]
'the villages / *the village'

Thus, the overt expression of number depends on whether grammatical definiteness or nominal agreement is present. Number is always expressed cumulatively with either the definite determiner or the nominal agreement marker. Only nouns without overt case marking (typically nominative) allow for number to be expressed independently via the marker *-t*. This interdependency of the expression of number and definiteness/nominal agreement will be relevant for the discussion of nominal sizes of Moksha complements, as well as for the account of the morphological exponence of the object case.

3.1.5. On the categorial status of nominal agreement markers

Before I conclude this section, I will discuss the categorial status of Moksha nominal agreement markers. Since nominal agreement markers never co-occur with definite declension markers (104), the question arises as to whether both are exponents of the category D. If this were the case, it would contradict the theoretical assumptions I adopt in this dissertation, as I place nominal agreement in Num. Below, I show that there is no independent evidence supporting the claim that Moksha nominal agreement markers belong to D rather than Num. Instead, nominal agreement markers and determiners are better analyzed as belonging to different categories, and their complementary distribution in Moksha is merely an accidental, post-syntactic phenomenon.

(104) a. **morkš-əz'ə-s'*
table-1SG.AGR.SG-DEF.SG[NOM]
Int.: 'my table'

b. **morkš-s'-əz'ə*
table-DEF.SG-1SG.AGR.SG
Int.: 'my table'

Besides the morphological incompatibility of nominal agreement and definiteness markers, there are two key data points that could shed light on the categorial status of nominal agreement markers: obligatory case marking in the presence of a nominal agreement marker, and obligatory object agreement with nominals that bear definiteness or nominal agreement markers.

The first point does not provide a conclusive answer but is compatible with the view that nominal agreement is below D. The second point, while potentially compatible with both analyses, faces more challenges if nominal agreement is assumed to be in D.

I will start with the point regarding the obligatoriness of case marking in the presence of a nominal agreement marker (105), see Chapter 5 for more details. Case marking is required for full DPs, implying that the presence of a nominal agreement marker signals the presence of the DP layer. One might conclude from this that nominal agreement markers are in D. However, as I argued earlier, the nominal agreement probe is only available in the presence of D, even though the probe itself is located in Num. This means that while all nominals bearing agreement suffixes are necessarily DPs, the suffixes themselves do not need to be in D.

(105) a. *mon'* *mird'ə-z'ə* *ašəz'* *t'ijə* [*min'*
1SG.GEN husband-1SG.AGR.SG[NOM] NEG.PST.3SG do[CN] 1PL.GEN
s't'ər'-n'əkə-n' *fkɛ-vək* ***fotografija-nc***]
girl-AGR.1PL-GEN one-ADD picture-3SG.AGR.SG.GEN
'My husband hasn't taken a single picture of our daughter.'

b. **mon'* *mird'ə-z'ə* *ašəz'* *t'ijə* [*min'*
1SG.GEN husband-1SG.AGR.SG[NOM] NEG.PST.3SG do[CN] we.GEN
s't'ər'-n'əkə-n' *fkɛ-vək* ***fotografijɛ***]
girl-AGR.1PL-GEN one-ADD picture

c. **mon'* *mird'ə-z'ə* *ašəz'* *t'ijə* [*min'*
1SG.GEN husband-1SG.AGR.SG[NOM] NEG.PST.3SG do[CN] we.GEN
s't'ər'-n'əkə-n' *fkɛ-vək* ***fotografija-c***]
girl-AGR.1PL-GEN one-ADD picture-1SG.AGR.SG
Int.: 'My husband hasn't taken a single picture of our daughter.'

The second point concerns object agreement on the predicate. The possibility and obligatoriness of agreement depend on the presence of either a definite marker or a nominal agreement marker. The simplest explanation would be that the object agreement targets a feature in D. However, the pattern is such that while DPs with a definiteness marker trigger obligatory agreement (106), DPs with a nominal agreement marker trigger agreement optionally (107). See also the absence of agreement if none of the markers are present (108). If both definiteness and nominal agreement markers belonged to the same category, it would be unclear why the same D feature results in obligatory agreement in one case and optional agreement in the other.

(106) DP WITH A DETERMINER (DEFINITE DECLENSION) → OBLIGATORY PREDICATE AGREEMENT

a. *min'* *jalga-n'əkə* *n'ɛj-**əẓ'ə*** *t'ɛ* *fotografijɛ-t'*
1PL.GEN friend-AGR.1PL[NOM] see-PST.3SG.O.3SG.S this foto-DEF.SG.GEN

b. **min'* *jalga-n'əkə* *n'ɛj-**s'*** *t'ɛ* *fotografijɛ-t'*
1PL.GEN friend-AGR.1PL[NOM] see-PST.3SG this foto-DEF.SG.GEN
'Our friend saw this picture.'

(107) DP WITH A NOMINAL AGREEMENT MARKER (POSSESSIVE DECLENSION) → OPTIONAL PREDICATE AGREEMENT

a. *min'* *jalga-n'əkə* *n'ɛj-**əẓ'ə*** *min'* *s't'ər'-n'əkə-n'*
1PL.GEN friend-AGR.1PL[NOM] see-PST.3SG.O.3SG.S 1PL.GEN girl-AGR.1PL-GEN
t'ɛ *fotografija-nc*
this foto-AGR.3SG.GEN

b. *min'* *jalga-n'əkə* *n'ɛj-**s'*** *min'* *s't'ər'-n'əkə-n'*
1PL.GEN friend-AGR.1PL[NOM] see-PST.3SG 1PL.GEN girl-AGR.1PL-GEN
t'ɛ *fotografija-nc*
this foto-AGR.3SG.GEN
'Our friend saw this picture of our daughter.'

(108) DP WITH NO OVERT MARKER (INDEFINITE DECLENSION) → NO PREDICATE AGREEMENT
(Toldova 2018: 577)

**vas'ɛ* *ker'-əz'ə* *šuftə*
Vasja[NOM] cut-PST.3SG.O.3SG.S tree
Int.: 'Vasja cut a tree.'

If, on the other hand, nominal agreement is of some different category than definite determiners, the difference in obligatoriness of agreement can be modelled. For instance, one could postulate a separate feature on D that is sensitive to the presence of the nominal agreement on Num. This way we would see agreement depending on the feature specification of the probe.

Thus, despite their incompatibility within a single nominal phrase, definite and agreement markers do not appear to belong to the same category in Moksha. Instead, their complementary distribution is likely due to post-syntactic morphological processes rather than an inherent syntactic constraint.

3.1.6. Interim summary

In Moksha, four nominal categories can be overtly expressed: case, definiteness, nominal agreement, and number. However, due to restrictions on their interaction, not all possible combinations of their respective exponents are grammatical. Definiteness and nominal agreement are mutually exclusive, while definiteness and number must be either expressed or omitted together.

The number feature of the head noun interacts with the number feature of nominal agreement, with a plural feature in nominal agreement neutralizing the head noun's number distinction. Case also interacts with number, as certain cases are incompatible with the overt marking of number.

Additionally, while some cells in the nominal paradigm follow an agglutinative pattern, where each morphological marker corresponds to a single feature, other cells exhibit fusional morphology, where multiple features are bundled together and realized as a single portmanteau morpheme.

Having examined the morpho-syntactic properties of these nominal categories, we now turn to adnominal modifiers, analyzing how they interact with these categories to shape the structure and interpretation of Moksha noun phrases.

3.2. Moksha adnominal modifiers

Moksha noun phrases can contain various adnominal modifiers, some of which impose specific morphological requirements on the head noun, while others do not. The available modifiers include unmarked nouns (109)a, adjectives (109)b, interrogative and indefinite pronouns (109)c, numerals (109)d, possessors (109)e, demonstrative pronouns (109)f, universal quantifiers (109)g, participial relative clauses (109)h, as well as finite relative clauses and some postpositional phrases.

(109)

a. *pona s'ur'ə*
wool thread
'wool thread'

b. *mazi panar*
nice dress
'nice dress'

c. *kat'i kodamə al'ɛ*
INDEF which man
'some man'

d. *kaftə c'ora-t*
two boy-PL
'two boys'

e. *mon' kruška-z'ə*
1SG.GEN mug-1SG.AGR.SG
'my mug'

f. *t'ɛ s't'ər'-n'ɛ-s'*
this girl-DIM-DEF.SG
'this girl'

g. *s'embə jarmak-n'ə*
all money-DEF.PL
'all money'

h. *kas-i šuftə*
grow-PTCP.ACT tree
'growing tree'

Certain modifiers — unmarked nouns, adjectives, relative clauses, postpositional phrases and even interrogative and indefinite pronouns — do not impose any specific morphological marking on the head noun. However, numerals, possessors, demonstrative pronouns, and universal quantifiers do require particular marking on the head. The following sections will examine these morphological requirements in detail, as they provide important insights for identifying the nominal projections present in Moksha noun phrases.

3.2.1. Unmarked nouns and adjectives

In Moksha, adjectives are traditionally distinguished as a separate category from nouns (Kolyadenkov & Zavodova1962, a.o). However, the distinction between these two categories is subtle. Both adjectives and nouns can appear in attributive positions (109), and can take nominal morphology — such as number, nominal agreement, definiteness, and case — when they occur in argument positions (Kholodilova 2018a: 94–95). Moreover, both adjectives (110) and nouns (111) can form comparatives, and both categories combine with the same degree expressions like *pɛk* 'very' (112)–(113).

(110) COMPARATIVE FORMED FROM AN ADJECTIVE (Kholodilova 2018b: 779)

pin'ə-s'	*katə-də*	***s'a-də***	***taza***
dog-DEF.SG[NOM]	cat-ABL	that-ABL	strong

'Dogs are stronger than cats.'

(111) COMPARATIVE FORMED FROM A NOUN (Kholodiliva 2018b: 780)

s'ora-c	***s'a-də***	***durak,***	*čem*	*al'a-c*
boy-3SG.AGR.SG[NOM]	that-ABL	dunce	than	man-3SG.AGR.SG[NOM]

'The son is even more of a dunce than his father.'

(112) ADJECTIVE + PƐK (Moksha Corpus)

a	*vel'ə-sə*	*ul'-s'*	***pɛk***	***mazi***	*s't'ər'-n'ɛ*
and	village-IN	be-PST.3SG	very	beautiful	girl-DIM

'And there was a very beautiful girl in the village.'

(113) NOUN + PƐK (Moksha Corpus)

məz'ardə	*ul'-s'-t'*	***pɛk***	***požar̥-t***
when	be-PST.3-PL	very	fire-PL

'When there were a lot of fires…'

Despite these similarities, the distinction between nouns and adjectives remains relevant. Specifically, nouns, but not adjectives, can be further modified with adjectives (Kolyadenkov & Zavodova1962: 176) (114); and nouns, but not adjectives, can take a special translative marker in the predicative position (Kholodilova 2018a: 95) (115).

(114) ADNOMINAL NOUN MODIFIED BY AN ADJECTIVE

mon'	*ɛvft'-əman'*	***kel'i***	***kopər'***	*al'ɛ*
1SG.GEN	scare-PST.1SG.O.3SG.S	wide	back	man

‘I was scared by a man with a broad back.’

(115) TRANSLATIVE MARKER (Kholodilova 2018a: 95)

a.

son	*ul'-s'*	*užə*	*sir'ə* /	**sir'ə-ks*
3SG[NOM]	be-PST.3SG	already	old	old-TRANS

‘He was already old.’

b.

son	*ul'-s'*	*užə*	*sir'ə*	*loman'* /	*loman'-**ks***
3SG[NOM]	be-PST.3SG	already	old	person	person-TRANS

‘He was already an old person.’

Despite being separate syntactic categories, nouns and adjectives in the attributive position exhibit similar properties. They do not show concord with the head noun and do not take nominal markers, such as a plural marker (116) or a case marker (117).

(116) ADNOMINAL POSITION → NO NUMERAL CONCORD

a.

mar'	/	**mar̥'-t'*	*ket'-t'*
apple		apple-PL	skin-PL

‘apple peels’ (Pleshak & Kholodilova 2018: 286)

b.

mazi	/	**mazij̊-t'*	*panar̥-t*
nice		nice-PL	dress-PL

‘nice dresses’

(117) ADNOMINAL POSITION → NO CASE CONCORD

a.

mar'	/	*mar'-t'	ked'-t'
apple		apple-DEF.SG.GEN	skin-DEF.SG.GEN

‘of the apple peel’

b.

mazi	/	*mazi-t'	panar̥-t'
nice		nice-DEF.SG.GEN	dress-DEF.SG.GEN

‘of the nice dress’

Nouns and adjectives are the adnominal dependents closest to the noun head in a neutral word order:

(118) ADNOMINAL NOUN FOLLOWING ADNOMINAL ADJECTIVE

mazi	***paks'ɛ***	*pan'čf-n'ə*	*pan'č-s'-t'*	*n'i*
beautiful	field	flower-DEF.PL[NOM]	open-PST.3-PL	already

'Beautiful wild flowers already finished blossoming.'

(119) ADJECTIVE FOLLOWING DEMONSTRATIVE

mon	*sud'ər'ɛ-jn'ə*	***t'ɛ***	***ravžə***	*traks-t'*
1SG[NOM]	pet-PST.3.O.1SG.S	this	black	cow-DEF.SG.GEN

'I pet this black cow.'

Adnominal nouns and adjectives do not attach any nominal morphology in concord with the noun head, nor do they impose restrictions on the marking of the head noun. While they share several syntactic and morphological properties, key differences — such as the ability to be modified by adjectives and take translative markers — justify treating them as distinct categories.

3.2.2. Interrogative and indefinite pronouns

Lexical items classified as pronouns in descriptive grammars of Moksha form a heterogeneous class (Kholodilova 2018a: 97). This class includes demonstratives, pronominal possessors, interrogative and indefinite pronouns, all of which can occur in the adnominal position (120). Indefinite pronouns are derived from interrogative pronouns through the addition of indefinite particles. Compare (120)a to (120)b.

(120)

a.

kodamə	*st'ər'*
which	girl

'which girl'

b.

kodamə	*bəd'ə*	*st'ər'*
which	INDEF	girl

'some girl'

Pronouns — especially indefinite pronouns — are primarily distinguished based on their functional distribution. Reference grammars define them by providing a closed list (Bikina 2018 and references therein).

Interrogative pronouns behave similarly to nouns and adjectives in that they do not show concord with the head noun (121). However, indefinite pronouns can take plural marking when modifying a plural noun (122).

(121) *kodamə* / **kodamə-**t*** *panar̥-t* *ton* *mus'kə-t'?*
which which-PL dress-PL 2SG[NOM] wash-PST.2SG
'Which shirts did you wash?'

(122) *mon* *mus'kə-n'* *kodamə* / *kodamə-**t*** *bəd'ə* *panar̥-t*
1SG[NOM] wash-PST.1SG which which-PL INDEF dress-PL
'I washed some shirts.'

This possibility of taking plural marking sets indefinite pronouns apart within the broader pronominal system.

3.2.3. Numerals

Moksha cardinal numerals appear in their unmarked form in the adnominal position (109), similar to nouns, adjectives and interrogative pronouns. Nevertheless, Moksha cardinal numerals should be recognized as a distinct syntactic category due to their unique properties.

One key distinction between numerals and adjectives is that numerals cannot occur in nominal predication (Kholodilova 2018a: 107):

(123) a. *kstij-˚n'ə* *ul'-s'-t'* *tan'c't'ij-˚t'*
berry-DEF.PL[NOM] be-PST.3-PL tasty-PL
'The berries were tasty.'

b. **kstij-˚n'ə* *ul'-s'-t'* *vet'ə* / *vet'ə-t*
berry-DEF.PL[NOM] be-PST.3-PL five five-PL
Int.: 'There were five of the berries.' (Kholodilova 2018a: 107).

Moksha numerals with a numerical value lower than ten differ from nouns, higher numerals exhibit noun-like properties regarding number marking and word order (Sidorova 2018b: 329 and references therein).[14] First, only higher numerals (124)a, not numerals lower than ten (124)b can take a plural marker:

(124) a. *tosə* *er'ɛ-j-˚t'* *lamə* *s'at-t* *loman'*
there live-NPST.3-PL many hundred-PL person

b. **tosə er'ɛ-j-˚t'* *lamə* *kemət'-t'* *loman'*
there live-NPST.3-PL many ten-PL person
'Many tens of people live there.' (Sidorova 2018a: 254-255)

Second, numerals above ten (125), similar to nouns (126), require singular number on the head, while numerals below ten require plural number on the head (127).

(125) NUMERAL > 10 → SG HEAD (Sidorova 2018b: 330)
vel'ə-t' *esə* ***kivet'ijə*** *katə* / **katə-t*
village-DEF.SG.GEN in.IN fifteen cat cat-PL
'There are fifteen cats in the village.'

(126) NOUN → SG HEAD (Kholodilova 2018a: 108)
mon *rama-n'* ***d'es'atka*** *al* / **al-t̥*
1SG[NOM] buy-PST.1SG ten egg egg-PL
'I bought a ten of eggs.'

(127) NUMERAL < 10 → PL HEAD (Sidorova 2018b: 329)
vel'ə-t' *esə* ***vet'ə*** *katə-t* / **katə*
village-DEF.SG.GEN in.IN five cat-PL cat
'There are five cats in the village.'

Third, higher numerals (128) behave like nouns (129) and must follow demonstratives in the adnominal position; meanwhile, lower numerals (130) can appear before or after the demonstrative.

[14] Classifying numerals higher than ten falls outside of the scope of this dissertation, see (Sidorova 2018a) for arguments supporting the distinction between higher numerals and nouns.

(128) DEM + HIGH NUM (Sidorova 2018a: 255)
lavka-v usk-əz'
store-LAT bring-PST.3.O.3PL.S

a. *t'ɛ mil'ion kn'iga-t'n'ə-n'*
this million book-DEF.PL-GEN

b. **mil'ion t'ɛ kn'iga-t'n'ə-n'*
million this book-DEF.PL-GEN
'These million books were brought to the store.'

(129) DEM + NOUN (Sidorova 2018a: 255)
vas'ɛ kand-əz'ən'
Vasja[NOM] carry-PST.3PL.O.3SG.S

a. *t'ɛ sumka kn'iga-t'n'ə-n'*
this bag book-DEF.PL-GEN
'[Vasja brought] this bag of books.'

b. **sumka t'ɛ kn'iga-t'n'ə-n'*
bag this book-DEF.PL-GEN
Int.: 'Vasja brought a bag of these books.'

(130) DEM + LOW NUM (Sidorova 2018a: 255)
lavka-v usk-əz'
store-LAT bring-PST.3.O.3PL.S

a. *t'ɛ kaftə kn'iga-t'n'ə-n'*
this two book-DEF.PL-GEN

b. *kaftə t'ɛ kn'iga-t'n'ə-n'*
two this book-DEF.PL-GEN
'These two books were brought to the store.'

Here, I will focus on the numerals with numerical values lower than 10, as they differ in their properties from nouns and definitely constitute their own category. They are also more frequent in use and are less likely to be replaced by their Russian counterparts compared to higher numerals (Khomchenkova & Pleshak 2021).

The property of influencing number marking within the noun phrase does not only serve to distinguish the class of cardinal numerals. Cardinal numerals with low numerical values induce plural marking on associated nouns, which is useful as a syntactic diagnostic. A noun phrase modified by a numeral below 10 must be marked as plural, regardless of whether it is definite (131) or indefinite (127).

(131) a. *d'ɛd'ɛ-s'* *maks-əz'ən'* *id'-ənzə-ndi* [*t'ɛ* *vet'ə*
mother-DEF.SG[NOM] give-PST.3PL.O.3SG.S child-3SG.AGR.PL-DAT this five
mar̥'-n'ə-n']
apple-DEF.PL-GEN

b. **d'ɛd'ɛ-s'* *maks-əz'ə* *id'-ənzə-ndi* [*t'ɛ* *vet'ə*
mother-DEF.SG[NOM] give-PST.3SG.O.3SG.S child-3SG.AGR.PL-DAT this five
mar'-t']
apple-DEF.SG.GEN
'The mother gave her kids these five apples.'

When overt plural marking is unavailable, such as in indefinite dative forms, the unmarked form is used (132), see Table 5 in Section 3.1.2 for the nominal paradigm and Section 3.1.4 for number neutralization).

(132) *mon* *kaz'-ən'* *pan'čf-t* *kat'i* *kodamə* *kaftə* ***s't'ər'-n'ɛ-n'd'i***
1SG[NOM] give-PST.1SG flower-PL INDEF which two girl-DIM-DAT
'I gave flowers to two girls.' (Sidorova 2018b: 329)

The interaction between the available plural marking and numerals plays a crucial role in the syntax of bare complements (Chapters 4 and 5). Notably, plural marking is always required in bare DOs (Chapter 5), but it is impossible in bare complements of postpositions, even in the presence of low numerals (Chapter 4). I argue that the impossibility of plural marking in the contexts where it would otherwise be required signals the absence of the corresponding functional projection — NumP.

3.2.4 Possessors

In this section, I examine the marking of possessors and look at the nominal agreement with possessor DPs.

3.2.4.1. Genitive marking of possessors

I understand possessors in a broad structural sense, encompassing all specific genitive DPs that modify a noun, regardless of their exact semantic relationship with the noun head they modify. As noted in Section 3.1.1, Moksha genitive marking varies phonologically depending on the presence of a definite determiner in the nominal. This variation should not cause confusion when encountering different genitive markers on various types of nominals. Below, I outline the genitive marking patterns and the factors influencing their selection.

Table 7 provides a fragment of the Moksha case paradigm focusing on genitive markers. As we can see, the marker *-n′* appears throughout the entire genitive paradigm with the exception of one cell — singular of the definite declension. When combined with the singular number and the determiner, the genitive marker looks like *-t′*. The interaction of case, definiteness, nominal agreement, and number within Moksha nominals was discussed in greater detail in Section 3.1.

Table 7: *A fragment of the Moksha case paradigm* (adapted from Kholodilova 2018a)

case	indefinite declension (just case)[15]		definite declension (case + number + determiner)		possessive declension (case + number + nominal agreement)	
	SG	PL	SG	PL	1SG.SG	1SG.PL
Genitive	*-ən'*		*-t'*	*-(t')n'ə-n'*	*-z'ə-n'*	*-n'ə-n'*

Let us look at the distribution of different genitive forms in more detail. Possessor DPs expressed with regular nouns require a determiner if they are not themselves possessed (in which case they get a nominal agreement marker incompatible with marking of definiteness). When a singular possessor is present, it takes the *-t'* marker (133). In plural possessors, the number is encoded together with the determiner, and the case is expressed with the marker *-n'* (134).

(133) REGULAR NOUN, SINGULAR POSSESSOR
s't'ər'-n'ɛ-t' *d'ɛd'a-c*
girl-DIM-DEF.SG.GEN mother-3SG.AGR.SG
'The girl's mother'

(134) REGULAR NOUN, PLURAL POSSESSOR
s't'ər'-n'ɛ-t'n'ə-n' *d'ɛd'a-snə*
girl-DIM-DEF.PL-GEN mother-3PL.AGR.SG
'The girls' mother'

[15] Here, I follow traditional grammatical descriptions in referring to the suffix *-ən'* as a case marker. While I argue in Pleshak (2021) that from a theoretical standpoint, this nominal marker does not encode case, the precise status of -*ən'* is not crucial for the present analysis, and a detailed discussion of its nature falls outside the scope of my dissertation.

When the possessor DP is itself possessed, it carries a nominal agreement marker that also encodes the number of the possessum, and the genitive case is *-n'*:

(135) REGULAR NOUN, SINGULAR POSSESSED POSSESSOR

jalga-z'ə-n'	*d'ɛd'a-c*
friend-1SG.AGR.SG-GEN	mother-3SG.AGR.SG

'my friend's mother'

When it comes to possessors expressed by nominals other than regular nouns, the determiner is not used (see also the end of Section 3.1.2 on the expression of definiteness). Human proper names take the genitive marker without a determiner, so they are marked with *-n'* (136). The same marking strategy is used by non-personal pronouns, e.g., interrogative pronouns (137).

(136) HUMAN PROPER NAME, SINGULAR POSSESSOR

*pet'ɛ-**n'***	*kuc'uv-**əc***
Petja-GEN	spoon-3SG.AGR.SG

'Petja's spoon'

(137) INTERROGATIVE PRONOUN, SINGULAR POSSESSOR

ki-n'	*d'ɛd'a-c*
who-GEN	mother-3SG.AGR.SG

'who's mother'

Finally, personal pronouns have special genitive forms:

(138) PERSONAL PRONOUN, SINGULAR POSSESSOR

mon'	*d'ɛd'ɛ-z'ə*
1SG.GEN	mother-1SG.AGR.SG

'my mother'

In contrast to proper names and pronouns, unmodified regular nouns without a determiner are interpreted as denoting properties rather than entities (139). When a modifier is present, the determiner becomes optional, allowing either the *-t'* or *-n'* genitive marker (140).

(139) ***ava-n'*** *sumka*-***s'***
woman-GEN bag-DEF.SG
'bag for women'
*'the bag of a woman'

(140) a. [***t'ε*** ***ava-t'***] *sumka*-***c***
this woman-DEF.SG.GEN bag-3SG.AGR.SG

b. [***t'ε*** ***ava-n'***] *sumka*-***s'***
this woman-GEN bag-DEF.SG
'this woman's bag'[16]

Thus, the Moksha genitive system exhibits different allomorphs depending on the nominal categories that need to be expressed overtly in a nominal. This variation is crucial for our understanding of definiteness marking in Moksha and has broader implications for the underlying representation of definiteness. This discussion provides a foundation for subsequent sections, where I explore overt definiteness marking and its syntactic representation.

3.2.4.2. Nominal agreement on the possessum

In Moksha, possessors trigger nominal agreement on the possessum. This agreement is obligatory when the possessed nominal appears in the subject or object position, but it remains optional in other cases.

Possessors expressed by regular nouns with a determiner, possessors that bear nominal agreement themselves, as well as those expressed by animate proper names and certain pronouns,

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[16] This noun phrase can also be interpreted as "this bag for women", but that reading corresponds to a different structure:
(i) ***t'ε*** [***ava-n'*** *sumka*]-***s'***
this woman-GEN bag-DEF.SG
'this bag for women'

follow the same agreement pattern, despite differences in the allomorphs of the genitive case. Consider examples of obligatory agreement with personal pronouns as possessors (141), animate possessors expressed by proper names (142), as well as inanimate possessors (143).

(141) *mo**n'*** *kuc'u-**z'ə*** / **kuc'u-s'*
1SG.GEN spoon-1SG.AGR.SG[NOM] spoon-DEF.SG[NOM]
ašč-i *morkš* *lank-sə*
be.situated-NPST.3SG table on-IN
'Petja's spoon is on the table.'

(142) *pet'ɛ-**n'*** *kuc'uv-**əc*** / **kuc'u-s'*
Petja-GEN spoon-3SG.AGR.SG[NOM] spoon-DEF.SG[NOM]
ašč-i *morkš* *lank-sə*
be.situated-NPST.3SG table on-IN
'Petja's spoon is on the table.'

(143) *t'ɛ ras't'en'ijɛ-**t'*** *kor'ɛn'-**əc*** / **kor'ɛn'-c'*
this plant-DEF.SG.GEN root-3SG.AGR.SG[NOM] root-DEF.SG[NOM]
ašč-i *škatulka-sə-n*
be.situated-NPST.3SG box-IN-AGR.1SG
'The root of this plant is in my box.'

However, when noun phrases headed by regular nouns take the genitive without a determiner or nominal agreement, they do not trigger agreement on the noun head, compare (144) with (143).

(144) *t'ɛ ras't'en'ijɛ-**n'*** *kor'ɛn'-**c'*** / **kor'ɛn'-əc*
this plant-**GEN** root-**DEF.SG[NOM]** root-3SG.AGR.SG[NOM]
ašč-i *škatulka-sə-n*
be.situated-NPST.3SG box-IN-AGR.1SG
'The root of this plant is in my box'.

Summing up the section about possessors in Moksha, they receive genitive marking. The genitive case is encoded by *-n'* unless it is expressed together with the definite determiner, in which case it is *-t'*. Possessor DPs require either a determiner or a nominal agreement marker, unless they are pronominal or expressed by animate proper names, in which case they get marked with *-n'* — the genitive marker without a determiner. Obligatory agreement is triggered on the possessed noun when the possessor is marked with a determiner or nominal agreement. In some cases, regular possessors (non-pronominal and not expressed by animate proper names) may lack a determiner even in the absence of a nominal agreement marker. Such possessors must be modified, and they do not trigger agreement on the head noun.

3.2.5. Demonstratives

Like other adnominal dependents in Moksha, demonstratives do not exhibit concord with the head noun. Moksha has a special demonstrative for the plural (Koljadenkov & Zavodova 1962: 234), which is attested in the dialect I have worked with (145). However, its usage is rare, and the demonstrative that is described as singular is used for both singular and plural referents (146).

(145) *n'ɛ* *kal̥-n'ə*
this.PL willow-DEF.PL
'these willows' (Moksha Corpus)

(146) a. *t'ɛ* *c'ora-n'ɛ-t'n'ə*
this boy-DIM-DEF.PL
'these boys'

b. *t'ɛ* *c'oran'ɛ-s'*
this boy-DIM-DEF.SG
'this boy'

Demonstratives require the presence of a definite determiner or a nominal agreement marker if one is available in the paradigm. For instance, in (147), the dative-marked nominal must include definiteness marking because the dative case has a definite form, recall also a nominative DP in (85), Section 3.1.2, repeated here as (148). However, when a nominal agreement marker is present (149), the definite determiner becomes impossible (see Section 3.1.3. on the incompatibility of the definite determiner with nominal agreement markers), but the demonstrative remains grammatical.

(147) *son* *maks-əz'ə* *kajmə-n'ɛ-t'*
3SG[NOM] give-PST.3SG.O.3SG.S shovel-DIM-DEF.SG.GEN
n'av ***s'ɛ*** *s'ora-n'ɛ-**t'i*** / **s'ora-n'ɛ-**n'd'i***
PTCL that boy-DIM-DEF.SG.DAT boy-DIM-DAT
'He gave the shovel to that boy over there.' (Kashkin 2018: 126)

(148) ***t'ɛ*** ***kuc'*** / ****kud*** *oc'u* *i* *mazi*
this house.DEF.SG[NOM] house big and nice
'This house is big and beautiful.' (Kashkin 2018: 126)

(149) *t'ɛ* *mon'* *tabur'etka-**z'ə*** / **tabur'etka-**s'*** *s'a-də* *od,*
this 1SG.GEN chair-1SG.AGR.SG[NOM] chair-DEF.SG[NOM] that-ABL new
čem *tona-s'*
than that-DEF.SG
'This chair of mine is newer than that one.'

Nominals marked with *-n'* form an exception to the rule that demonstratives require definiteness marking if it is available. This applies both when they appear in the adnominal position (150) or as complements of postpositions (151). Even though there is clearly a way to encode the genitive case together with a determiner (marker *-t'*), nominals modified with a demonstrative can simply take the *n'*-suffix, without a definite determiner or nominal agreement marker.

(150) *t'ɛ* *ras't'en'ijɛ-n'* *kor'ɛt'-t'-n'ə*
this plant-GEN root-PL-DEF.PL[NOM]
'the roots of this plant / of these plants'

(151) *s'ɛ* *morkš-ən'* *šir'ə-sə*
that table-GEN side-IN
'near that table'

Unlike nominative, genitive and dative cases, spatial cases do not allow overt definiteness marking. However, nominals in spatial cases can still be modified by demonstratives:

(152) *targa-k* *tr'apka-t'* ***t'ɛ*** ***škaf-t'*** ***ezdə*** /
take_out-IMP.3SG.O.SG.S rag-DEF.SG.GEN this dresser-DEF.SG.GEN in.ABL
škaf-stə
dresser-EL
'Take the rag out of this dresser.' (Kashkin 2018: 132)

In sum, demonstratives do not show any concord with the head noun, just like nouns, adjectives and numerals, but they do have restrictions on the marking on the noun head. In particular, they require definiteness marking when it is available in the paradigm. Genitive-marked nouns are an exception, as can be modified by demonstratives without a definite determiner or nominal agreement marker. In spatial cases, on the other hand, definiteness marking is unavailable, and demonstratives can still be used.

3.2.6. Universal quantifiers

There are several universal quantifiers in Moksha: *s'embə* 'all', *každaj*, *ɛr'* 'each', *(f)s'akaj*, *fs'akajən'* 'any', *l'ubovaj* 'whichever', *marn'ək*, and *optəm* 'entire' (Zakharova 2018: 205). The last two, *marn'ək* and *optəm*, do not function as adnominal modifiers (ibid: 207).

The quantifier *s'embə* only combines with plural nominals (153). The same preference for plural nominals is also observed with *fs'akaj* (154).

(153)	a.	*s'embə*	*lomat'-**t'n'ə***	*kel'k-saz'*	*padarka-t'n'ə-n'*
		all	person-DEF.PL[NOM]	like-NPST.3.O.3PL.S	present-DEF.PL-GEN
	b.	**s'embə*	*loman'-**c'***	*kel'k-sin'ə*	*padarka-t'n'ə-n'*
		all	person-DEF.SG[NOM]	like-NPST.3PL.O.3SG.S	present-DEF.PL-GEN
		'All people like presents.' (Zakharova 2018: 206)			

(154)	a.	*fs'akaj*	*lomat'-**t'n'ə***	*kel'k-saz'*	*padarka-t'n'ə-n'*
		all	person-DEF.PL[NOM]	like-NPST.3.O.3PL.S	present-DEF.PL-GEN
	b.	?*fs'akaj*	*loman'-**c'***	*kel'k-sin'ə*	*padarka-t'n'ə-n'*
		all	person-DEF.PL[NOM]	like-NPST.3PL.O.3SG.S	present-DEF.PL-GEN
		'All people like presents.' (Zakharova 2018: 206)			

In contrast, the quantifiers *každaj* and *ɛr'* require singular nominals:

(155)	a.	*každaj* /	*ɛr'*	*loman'-**c'***	*kel'k-sin'ə*	*padarka-t'n'ə-n'*
		each	each	person-DEF.SG[NOM]	like-NPST.3PL.O.3SG.S	present-DEF.PL-GEN

b.	**každaj /*	*εr'*	*lomat'-**t'n'ə***	*kel'k-saz'*	*padarka-t'n'ə-n'*
	each	each	person-DEF.PL[NOM]	like-NPST.3.O.3PL.S	present-DEF.PL-GEN

‘Every person likes presents.’ (Zakharova 2018: 206)

Finally, *l'ubovaj* is compatible with both singular and plural nominals:

(156)	a.	*l'ubovaj*	*loman'-**c'***	*kel'k-sin'ə*	*padarka-t'n'ə-n'*
		any	person-DEF.SG[NOM]	like-NPST.3PL.O.3SG.S	present-DEF.PL-GEN
	b.	*l'ubovaj*	*lomat'-**t'n'ə***	*kel'k-saz'*	*padarka-t'n'ə-n'*
		any	person-DEF.PL[NOM]	like-NPST.3.O.3PL.S	present-DEF.PL-GEN

‘Every person likes presents.’ (Zakharova 2018: 206)

I will restrict my attention to the universal quantifier *s'embə*, as it is the most semantically versatile — it can be used in different types of quantification (Zakharova 2018: 209-211) — and the most morphosyntactically restrictive — it requires the nominal to be marked as plural and to bear a definite determiner or a nominal agreement marker.

As shown in (153), *s'embə* ‘all’ can only combine with plural nominals. Moreover, plural nominals without a determiner are incompatible with *s'embə*. This is evident in (157) and (158), where the definite determiner is obligatory. However, nominals with a nominal agreement marker can appear with *s'embə*, even without an overt definite determiner (159).

(157)	a.	*s'embə*	*lomat'-**t'n'ə***	*kul-s'-ij-°t'*
		all	person-DEF.PL[NOM]	die-FREQ-NPST.3-PL
	b.	**s'embə*	*lomat'-**t'***	*kul-s'-ij-°t'*
		all	person-PL[NOM]	die-FREQ-NPST.3-PL

‘All people are mortal. Lit.: All people die.’ (Kashkin 2018: 147)

(158)	a.	*mon'*	*d'εd'ε-z'ə*		*kel'k-sin'ə*		
		1SG.GEN	mother-1SG.AGR.SG[NOM]		like-NPST.3PL.O.3SG.S		
		s'embə	*lomat'-**t'n'ə-n'**,*	*i*	*c'ebεr̥'-n'ə-n',*	*i*	*plox-n'ə-n'*
		all	person-DEF.PL-GEN	and	good-DEF.PL-GEN	and	bad-DEF.PL-GEN

b. **mon'* *d'ɛd'ɛ-z'ə* *kel'g-i* / *kel'k-sin'ə*
1SG.GEN mother-1SG.AGR.SG[NOM] like-NPST.3SG like-NPST.3PL.O.3SG.S
s'embə *lomat'**-t'*** *i* *c'ebɛr̜'-n'ə-n',* *i* *plox-n'ə-n'*
all person-PL and good-DEF.PL-GEN and bad-DEF.PL-GEN
'My mother likes all people, both bad and good.' (Kashkin 2018: 147)

(159) *s'embə t'ev-n'ə-n'* *kacajn'ə*
all task-1SG.AGR.PL-GEN leave.NPST.3PL.O.1SG.S
'…I leave all my tasks' (Moksha Corpus)

The quantifier *s'embə* can also appear with nominals that lack a definite determiner:

(160) ?*vojna-t'* *kolga* *soda-j-°t'* ***s'embə vel'ə-sə***
war-PL about know-NPST.3-PL all village-IN
'In all villages, they know about the war.'

In summary, the universal quantifier *s'embə* 'all' combines only with plural nominals marked as definite or bearing a nominal agreement marker unless these nominals bear oblique cases. Similar to demonstratives, *s'embə* is compatible with forms that lack overt definiteness marking only when no definite form exists in the paradigm.

3.3. Moksha DP

Having introduced the nominal categories that can be expressed overtly within the Moksha nominal and the main adnominal modifiers, let me discuss how they all map to the hierarchical model of DP outlined in Chapter 2.

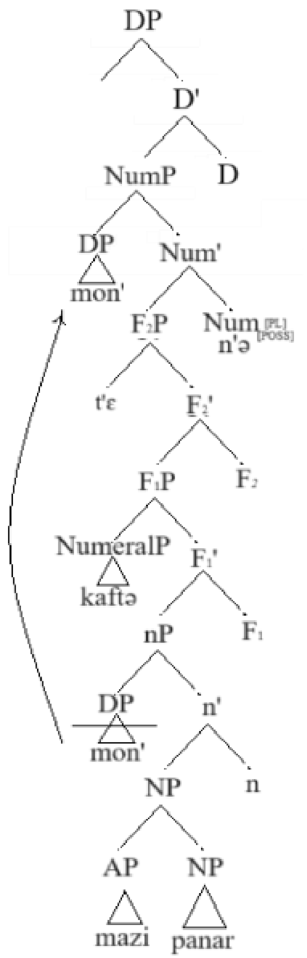
3.3.1. Moksha DPs in the subject and DO positions

In this section, I argue that Moksha subjects and DOs must be full DPs. The argument for definite subjects and DOs being full DPs is straightforward: they bear determiners, which are hosted by the D head. Indefinite and possessed subjects and DOs, however, could potentially be just NumPs. Nevertheless, the obligatory presence of definiteness marking on subjects and DOs when a demonstrative is present — despite its optionality/impossibility in other positions (see Sections 3.3.2 and 3.3.3 below) — suggests that all subjects and DOs in Moksha must be full DPs. In Chapter 5, I further elaborate my analysis of Moksha possessed and bare indefinite DOs as full DPs.

The basic structure of a Moksha nominal (example (161)) is illustrated in Figure 11. This structure accounts for the marking and modification restrictions observed in Moksha while adhering to the theoretical framework outlined in Chapter 2. I consider adjectives to be NP adjuncts, as the closest adnominal dependents. Then, the possessor, which originates in Spec,nP, triggers agreement on the Num head and moves to Spec,NumP. This movement results in the nominal agreement marker appearing on the head noun and the possessor surfacing on the left of other adnominal modifiers. As discussed in Chapter 2, numerals and demonstratives originate in a functional projection below NumP. To avoid multiple specifiers, I proposed two F-projections — F_1P and F_2P, which I further justify in Chapter 4.

(161)	*mon'*	*t'ɛ*	*kaftə*	*mazi*	*panar-n'ə*
	1SG.GEN	this	two	nice	dress
	'these two dresses of mine'				

Figure 11: *The basic structure of a Moksha nominal*



I treat the Moksha definite determiner as a definite article and assume that all nominals marked for definiteness are full DPs. Moksha definite subjects and DOs obligatorily bear the definite determiner, whether their definiteness is derived from a demonstrative (162)–(163) or is purely interpretational, e.g., the subject is unique (164)–(165). From this, I conclude that Moksha definite subjects are always full DPs.

(162) ***t'ɛ*** ***kuc'*** / ****kud*** *oc'u* *i* *mazi*
this house.DEF.SG[NOM] house big and beautiful
'This house is big and beautiful.' (Kashkin 2018: 126)

(163) a. *son* *dasad'ɛ-z'ə* ***t'ɛ*** ***s'ora-n'ɛ-t'***
3SG[NOM] offend-PST.3SG.O.3SG.S this boy-DIM-DEF.SG.GEN

b. **son* *dasad'ɛ-s'* ***t'ɛ*** ***s'ora-n'ɛ***
3SG[NOM] offend-PST.3SG this boy-DIM
'He offended this boy.' (Kashkin 2018: 126)

(164) *t'ɛči* ***ši-s'*** / ****ši*** *valdəpt-i* *valctə*
today sun-DEF.SG[NOM] sun[NOM] shine-NPST.3SG bright.EL
'Today, the sun is shigning bright'. (Kashkin 2018: 126)

(165) a. *ton* *n'ɛj-sak* *mazi* *men'əl'-t'?*
2SG[NOM] see-NPST.3SG.O.2SG.S nice sky-DEF.SG.GEN

b. **ton* *n'ɛj-at* *mazi* *men'əl'?*
2SG[NOM] see-NPST.2SG nice sky
'Do you see the nice sky?' (Kashkin 2018: 126)

As long as a nominal is marked as definite, it must also be marked for number. The nominal in (166) is unambiguously singular, and its plural counterpart looks like (166), with another portmanteau morpheme encoding plural number and formal definiteness.

(166) a. *vel'ə-**s'***
village-DEF.SG
'the village / *the villages'

b. *vel'ə-**t'n'ə***
village-DEF.PL
'the villages / *the village'

The obligatoriness of the expression of number in the presence of the definite article aligns with the assumption that DP projection is built on top of NumP, the number projection, and that functional layers within an extended projection cannot be skipped. Consequently, as long as a DP is present, NumP must also be present.

Turning to Moksha subjects and DOs that are not marked for definiteness, they must be at least NumPs because number marking is obligatory in them. In (167), the noun *jalga* 'friend' combines with a nominal agreement affix. The distinction between singular and plural is preserved, as number is encoded together with agreement in a fusional marker, compare (167)a to plural (167)b. If the nominal agreement probe is located on Num, as argued in Chapter 2, Section 2.3, then it is technically possible that possessive noun phrases are NumPs rather than full DPs, as there is no direct evidence for them to be full DPs. However, as I argue in Chapter 5, Moksha possessive subjects and DOs are better analyzed as full DPs.

(167) a. *mon* *t'er'd'-in'ə* *es'* *jalga-z'ə-n'*
1SG[NOM] call-PST.3.O.1SG.S REFL friend-1SG.AGR.SG-GEN
i. 'I called my friend.'
ii. *'I called my friends.'

b. *mon* *t'er'd'-in'ə* *es'* *jalga-n'ə-n'*
1SG[MON] call-PST.3.O.1SG.S REFL friend-1SG.AGR.PL-GEN
'I called my friends.' (Kholodilova 2018a: 70)

Similarly, number marking is obligatory in indefinite subjects and DOs. Noun phrases without plural marking in these positions can only be interpreted as singular:

(168) a. *morkš* *lank-sə* *mar̥'-t'*
table top-IN apple-PL
'There are apples on the table.'

b. *morkš* *lank-sə* *mar'*
table top-IN apple
i. 'There is an apple on the table'.
ii. *' There are apples on the table.' (Kholodilova 2018a: 70)

Indefinite nominals lack overt marking except for number, making it difficult to determine whether they are full DPs or merely NumPs. However, as discussed in Chapter 5, I conclude that possessive and indefinite DOs are still full DPs.

Subjects and DOs may be modified by all types of adnominal dependents. Importantly, in these positions, nominals get marked with nominative or genitive case, which have full paradigms including cased forms with determiners, number and nominal agreement markers.

In contrast to some other positions where noun phrases modified by low numerals may lack plural marking (Section 3.2.3), subjects and DOs must overtly express plurality. Demonstratives in subject and direct object positions must always appear with a definite determiner or nominal agreement marker on the head noun (169), but this requirement is relaxed in other positions, e.g., with adnominal possessors (see Sections 3.2.4.1, 3.2.5 and 3.3.2) or locative objects (see Section 3.2.5).

(169)	***t'ɛ***	***kuc'***	/	****kud***	*oc'u*	*i*	*mazi*
	this	house.DEF.SG[NOM]		house	big	and	nice

‘This house is big and beautiful.’ (Kashkin 2018: 126)

Given the obligatory formal marking of number and definiteness in the presence of numerals and demonstratives in subjects and DOs, even when it can be skipped in other positions (possessors or locatives), I conclude that Moksha subjects and DOs are always full DPs. In Chapter 4, I examine the restrictions on marking and availability of adnominal modifiers in Moksha bare complements of postpositions, assuming Moksha subject and DO DPs (and the structure in Figure 11) as the baseline for comparison.

3.3.2. Moksha nominals in the positions other than subject and DO

In this section, I argue that while definite nominals in the subject and DO positions are invariably full DPs, only a subset of definite possessors are. Furthermore, nominals in spatial cases are never full DPs in Moksha.

3.3.2.1. Adnominal possessors

The key distinction between definite subjects and DOs, on the one hand, and definite possessors, on the other, is that only the former must be obligatorily marked for definiteness. As mentioned in Section 3.2.5, an adnominal noun phrase can lack a definite determiner even when modified by a demonstrative:

(170) [*t'ɛ* *ava-n'*] *sumka-s'*
this woman-GEN bag
'this woman's bag'

The absence of formal definiteness marking on the head noun, despite its semantic definiteness, suggests that the determiner-hosting head is missing. This indicates that adnominal possessors like the one in (170) are not full DPs. Moreover, number is not overtly expressed either, and -n'-marked adnominal possessors are number neutral (171). Based on this, one can conclude that such possessors lack NumP as well.

(171) ***t'ɛ*** *ras't'en'ijɛ-**n'*** *kor'ɛt'-t'-n'ə* *ašč-ij̊-t'* *mon'*
this plant-**GEN** root-PL-DEF.PL[NOM] be_situated-NPST.3-PL 1SG.GEN
škatulka-sə-n
box-IN-AGR.1SG
'The roots of this plant / of these plants are in my box.'

In further support of analyzing of *-n'*-marked possessors as structures smaller than DP, the universal quantifier *s'embə* 'all', which requires plural and definiteness marking when modifying subjects and DOs, can combine with -n'-marked possessors despite their lack of overt plural or definite marking:

(172) *s'embə* *ava-n'* *panar̥-n'ə*
all woman-GEN dress-DEF.PL
'all women's dresses'

Additionally, only possessors explicitly marked as definite trigger agreement on the possessed head noun:

(173) a. *t'ɛ ras't'en'ijɛ-**t'** kor'ɛn'-**əc*** / **kor'ɛn'-c'*
this plant-DEF.SG.GEN root-3SG.AGR.SG[NOM] root-DEF.SG[NOM]
ašč-i škatulka-sə-n
be_situated-NPST.3SG box-IN-AGR.1SG
'the root of this plant is in my box.'

b. *t'ɛ ras't'en'ijɛ-**n'** kor'ɛn'-**c'*** / **kor'ɛn'-əc*
this plant-GEN root-DEF.SG[NOM] root-3SG.AGR.SG[NOM]
ašč-i škatulka-sə-n
be_situated-NPST.3SG box-IN-AGR.1SG
'The root of this plant is in my box.'

The only problem of the non-DP analysis of -n'-marked nominals is that they are not bare and appear to bear case marking, which contradicts the assumption that only DPs receive case, see Chapter 1, Section 1.2.2.1. However, as suggested in (Pleshak 2021), the -n'-marker may be ambiguous, functioning either as a genitive case marker (following plural definite and nominal agreement markers; see Section 3.2.4.1) or as a marker of an SN, which is not a case marker form a theoretical standpoint.

Two key arguments support the idea that -n'-marked nominals are not genuine genitive DPs without overt definiteness marking. The first argument comes from the cooccurrence of several differently marked genitive phrases within one DP. The second argument comes from the impossibility of -n'-marked forms in the DO position.

I will start with the first argument. According to Alexiadou & Anagnostopoulou (2001), when two DP arguments originate within a VP, one must move out due to restrictions on multiple unchecked Case features. This is supported by the transitivity restriction in English Quotative Inversion, (174). The placement of the quantifier *all* shows that under inversion, the subject is in situ (174)a, as only in situ subjects can appear to the right of floating quantifiers (Sportiche 1988). Furthermore, this subject cannot be moved and placed to the left of the quantifier, (174)b. On the other hand, in absence of inversion, the subject moves and appears to the left of the quantifier, (174)c.

(174) ENGLISH (Alexiadou & Anagnostopoulou 2001: 197)
a. *"We must do this again," the guests all declared to Tony.*
b. *"We must do this again," declared all the guests to Tony.*
c.* *"We must do this again," declared the guests all to Tony.*

Alexiadou and Anagnostopoulou use these data to motivate a generalization whereby "by Spell-Out VP can contain no more than one argument with an unchecked Case feature" (Alexiadou & Anagnostopoulou 2001: 216).[17] Richards (2006) discusses the same phenomenon arguing that this restriction is due to linearization constraints, not case checking. No matter what the reason for this constraint is, we do not expect to see two DPs very close to each other, structurally speaking.[18]

Applying this reasoning to Moksha, we do not expect two DPs marked with the definite-declension genitive (two DPs) to appear within the same noun phrase. However, no such restriction applies to the combination of a "real" genitive nominal and an -n'-marked nominal, or to two -n'-marked nominals. This prediction is borne out. In (175), we see that two *t'*-marked noun phrases

[17] In their analysis, case is assigned under agreement, which differs from the assumptions made in this work. However, the core of the argument remains unaffected by this difference in the case-licensing model.

[18] In both cited works, closeness is defined in terms of phase boundaries: there must be at least one phase boundary separating the two DPs.

are incompatible. Meanwhile, *n'*-marked noun phrases can co-occur within the same noun phrase, (176) and (177).

(175) *[[*t'ɛ* *c'ora-**t'***] [*mašina-**t'***] *šari-sə(-nzə)*] *var'ɛ-n'ɛ*

this	boy-DEF.SG.GEN	car-DEF.SG.GEN	wheel-IN-AGR.3SG	hole-DIM[NOM]

Int.: 'There is a hole in [this boy's$_{\text{Possessor}}$ wheel$_{\text{Possessum}}$ of the car$_{\text{Possessor}}$].'

(176) [*tona* *s't'ər-n'ɛ-**n'***] [*t'ɛ* *ras't'en'ijɛ-**n'***] *kor'ɛn'-**c'***

that	girl-DIM-**GEN**	this	plant-**GEN**	root-**DEF**.SG

'the root$_{\text{Possessum}}$ of this plant$_{\text{Possessor}}$ that belongs to that girl$_{\text{Possessor}}$'

(177) [[*t'ɛ* *c'ora-**t'***] [*mašina-**n'***] *šari-s'*] *urdaz-u*

this	boy-DEF.SG.GEN	car-GEN	wheel-DEF.SG[NOM]	dirt-ATTR

'[This boy's$_{\text{Possessor}}$ wheel$_{\text{Possessum}}$ of the car$_{\text{Possessor}}$] is dirty.'

Now, let us turn to the second argument. If -n' were a true genitive allomorph occurring only in the absence of formal definiteness marking, we would expect it to appear in both adnominal and DO positions. However, in DO position, when a noun phrase lacks a definite determiner, it appears in an unmarked form (compare indefinite unmarked DO in (178)b to the definite genitive DO in (178)a, and -n'-marking is not allowed (179).

(178) a. DEFINITE DO → DEFINITE+GENITIVE

maša-n'	*s'ora-c*	*juksta-zə'*	***kaza-t'***
Masha-GEN	boy-3SG.AGR.SG[NOM]	forget-PST.3SG.O.3SG.S	goat-DEF.SG.GEN
paks'a-v			
field-LAT			

'Masha's son forgot the goat in the field.'

b. INDEFINITE DO → BARE (Toldova 2018: 552)

maša-n'	*s'ora-c*	*ɛxɛr'*	*juksta-j*
Masha-GEN	boy-3SG.AGR.SG[NOM]	always	forget-NPST.3SG
kodamə	*bəd'ə*	***kaza***	*paks'a-v*
some	INDEF	goat	field-LAT

'Masha's son always forgets some goat(s) in the field.'

(179) INDEFINITE DO → *GENITIVE

*vas'ɛ kola-s' / kola-z'ə kodamə bəd'ə tar'elka-n'

Vasja break-PST.3SG break-PST.3SG.O.3SG.S which INDEF plate-GEN

Int.: 'Vasja broke a plate (lit. some plate).'

If -n' were a genuine genitive allomorph in the absence of the formal definiteness marking, we would expect it to occur in both positions. Given this restriction, I conclude that -n'-marked forms do not function as genitive DPs in the same way as fully inflected genitives. When referring to genitive DPs, I consider only regular nominals marked with the genitive of the definite or possessive declension as full DPs. Regular nominals marked with *-n'* (the genitive of the indefinite declension) do not exhibit DP properties and are excluded from my diagnostics.

3.3.2.2. Oblique nominals

The behavior of Moksha oblique nominals aligns more closely with n'-marked possessors than with subjects and DOs. First, oblique cases such as inessive or elative lack markers of the definite declension (see Section 3.1.2) and, consequently, never bear a determiner. Additionally, they are incompatible with the plural marker and exhibit number neutrality (see section 3.1.4). Notably, oblique nominals remain number neutral even when carrying an agreement, suggesting the absence of both DP and NumP:

(180) *vel'ə-sə-n*

village-IN-AGR.1SG

'in my village/in my villages'

Personal pronouns only combine with some oblique cases. Oblique pronouns, if acceptable, resemble postpositional constructions, with the case marker being followed by a nominal agreement marker in addition to the preceding pronominal stem:

(181) *mon'-d'ədə-n*
1SG-ABL-AGR.1SG
'of me'

Since nominals in oblique cases do not exhibit a full set of properties expected from DPs, I do not consider them in discussions regarding DPs in Moksha.

3.3.3. Summary

In this chapter, I provided a brief overview of the Moksha nominal marking system and examined the interaction between common adnominal modifiers and marking on the head noun. Moksha is traditionally described as having up to 16 cases, though some may be more accurately analyzed as postpositional phrases rather than true cases. However, only 9 of these cases are compatible with nominal agreement markers, and just 3 allow for the combination of a plural marker and/or a definite determiner. Number marking is always expressed either in conjunction with nominal agreement or with a definite determiner.

When plural or definite marking is available in the paradigm, certain modifiers require number and/or definiteness to be overtly marked on the head noun. Cardinal numerals below ten necessitate plural marking, demonstratives require definiteness marking, and the universal quantifier *s'embə* 'all' demands both. Possessors, meanwhile, trigger nominal agreement on the possessed noun. However, when plural or definite marking is unavailable in the paradigm, these modifiers can still co-occur with nominals that lack overt definite or plural marking.

A notable asymmetry exists between subjects and DOs, on the one hand, and other syntactic constituents, on the other. Subjects and direct objects always feature the full range of marking options in the paradigm, and these forms are obligatorily used. In contrast, other syntactic positions allow for optionality in such marking, regardless of whether definite and plural forms exist in the paradigm. For example, adnominal possessors are marked with the genitive case, which has all forms available, yet definite and plural marking is not required when such possessors are combined with demonstratives, low numerals, and the universal quantifier *s'embə* ‘all’. Similarly, oblique cases lack these marking options but remain compatible with the same modifiers.

I interpret these marking restrictions as evidence that subjects and DOs are always DPs. Possessors, by contrast, do not behave as DPs if they are regular nouns marked with the genitive of the indefinite declension (*-n'*). Nominals in other syntactic positions consistently fail to exhibit DP properties. For these reasons, talking about DP properties in Moksha, I will only use definite subjects, objects and possessors as DPs.

Chapter 4: Detecting small nominals: a case study of complements of postpositions in Moksha

In this chapter, I analyze the differential marking of complements of spatial postpositions in Moksha exemplified in (182). In (182)a, the complement of the postposition *lank-sə* ‘on top of’ is marked with the genitive case — the case assigned by postpositions in Moksha (see Chapter 3) — and it triggers nominal agreement on the postposition (see Section 4.1.3 on the specifics of nominal agreement). In contrast, in (182)b, the complement of the same postposition remains bare, and there is no agreement on the postposition.

(182)	a.	*st'ər'-t'*	*šir'ə-sə-nzə*	b.	*morkš*	*šir'ə-sə*
		girl-DEF.SG.GEN	side-IN-AGR.3SG		table	side-IN
		‘near the girl’			‘near a table’	

I argue that complements marked with the genitive case are full DPs that receive case marking from the postposition and contain features relevant for the agreement probe. In contrast, complements without genitive marking are caseless small nominals (SNs), lacking higher functional projections — specifically DP and NumP. The absence of nominal morphology follows from the lack of functional heads that would otherwise host the corresponding overt markers. Likewise, the absence of agreement results from the lack of relevant features that would normally be located on these missing heads.

An SN analysis derives the Moksha case marking and agreement pattern in PPs with minimal additional stipulations. By contrast, a purely morphological DM-style alternative would require a large number of additional assumptions and operations.

The chapter proceeds as follows: In Section 4.1, I describe the empirical pattern of postpositional complementation in Moksha. In Section 4.2, I determine the structural size of bare complements of postpositions. Section 4.3 elaborates on the mechanics of the SN analysis of Moksha bare complements of postpositions. Section 4.4 evaluates alternative analyses and demonstrates their shortcomings compared to an SN approach. In Section 4.5. I discuss how my analysis contributes to our understanding of noun phrase structure. Finally, in Section 4.6, I draw conclusions.

4.1. Marking of complements of spatial postpositions in Moksha

In this section, I go through the properties of genitive and bare complements, such as the possibility of expression of number, nominal agreement and definiteness, as well as ability of triggering agreement.

4.1.1. Expression of different nominal categories

In Moksha, four nominal categories can be morphologically expressed: number, nominal agreement, definiteness, and case (see Chapter 3). When analyzing the case-marking of postpositional complements, it is crucial to consider whether other nominal categories are also expressed:

(183) **Overt marking generalization**:

If number, nominal agreement, or definiteness is overtly expressed, the complement of the postposition must be genitive.

Let us look at the marking of formal definiteness first. If the complement of a spatial postposition is explicitly marked for definiteness, it must be genitive. In (184), *morkš-s'* represents an attempt to express definiteness without the genitive case, since the definite nominative is the closest candidate for this form (see Chapter 3 for the structure of the Moksha case paradigm). Non-genitive definite complement is ungrammatical.

(184) *morkš-t'* / **morkš-s'* *šir'ə-sə*
table-DEF.SG.GEN table-DEF.SG[NOM] side-IN
'near the table'

Similarly, nominal agreement requires the genitive case. In (185), *morkš-əc* again represents an attempt to mark nominal agreement without marking the case. Like it was with the definiteness marking, the nominative form is the best candidate.

(185) *morkš-əz'ə-n'* / **morkš-əz'ə* *šir'ə -sə*
table-1SG.AGR.SG-GEN table-1SG.AGR.SG[NOM] side-IN
'near my table'

Finally, let us look at plural complements. Plural marking also correlates with the appearance of the genitive case on the complement. In (186), the form *morkš-n'ə*, which is a definite plural form without the genitive marker, is ungrammatical. However, as I showed above, definiteness marking alone already requires genitive marking, so this form might be ruled out for that reason. What is more telling, the form *morkš-t* with an overt plural marker in the absence of any other nominal morphology is also ungrammatical.

(186) *morkš-n'ə-n'* / **morkš-n'ə* / **morkš-t* *šir'ə -sə*
table-DEF.PL-GEN table-DEF.PL table-PL side-IN
'near (the) tables'

It is also worth noting that as long as both plural and genitive are expressed, the nominal should also be formally marked as definite due to the structure of the Moksha paradigm (see Chapter 3). Recall that there is no plural indefinite genitive form available in Moksha.

So far, I have shown that in Moksha, nominal categories must be expressed in a bundle: the expression of number, nominal agreement of definiteness requires the expression of case. Conversely, the overt expression of genitive case necessitates the marking of number, nominal agreement, or definiteness.

Before I move on, I would like to point out that there are two classes of nominals that in normal case do not bear overt definiteness marking or nominal agreement, and yet must be genitive. These are proper names of animate beings and most pronouns (see Chapter 3 for the overview and Chapter 6 for the discussion of the indefinite inanimate pronoun *mez'ə* 'what' behaving like regular nouns). This is captured by the following generalization:

(187) **Category restriction generalization**:

If the complement of the postposition is a pronoun or an animate proper name, it must be genitive.

Examples (188)-(189) illustrate this:

(188)	a.	*mon'*	*šir'ə-sə-n*	*lijə-nd-i*	*melav-n'ɛ*	
		1SG.GEN	side-IN-AGR.1SG	fly-IPFV-3.SG	butterfly-DIM[NOM]	
	b.	**mon*	*šir'ə-sə-n* /	*šir'ə-sə*	*lijə-nd-i*	*melav-n'ɛ*
		1SG[NOM]	side-IN-AGR.1SG	side-IN	fly-IPFV-3.SG	butterfly-DIM[NOM]

'There is a butterfly flying near me.'

(189)	*melav-n'ɛ-s'*	*lijə-nd-i*	*pet'ɛ-n'* /	**pet'ɛ*	*šir'ə-sə*
	butterfly-DIM-DEF.SG[NOM]	fly-IPFV-3.SG	Petja-GEN	Petja	side-IN

'The butterfly is flying near Petja.'

There are also two types of nominals that always take genitive case due to inherent featural properties. These are nouns denoting humans and inalienable possessive constructions. Examples (191) and (192) illustrate the generalization in (190).

(190) **Featural restriction generalization**:

If the complement of a postposition is a noun denoting a human or an inalienably possessed object, it must be genitive.

(191) *st'ər'-t'* / **st'ər'* *lank-sə*
girl-DEF.SG.GEN girl top-IN
'on a/the girl'

(192) *morkš-t'* *pil'gə-nc* / **pil'gə* *lank-sə(-nzə)*
table-DEF.SG.GEN leg-3SG.AGR.SG.GEN leg on-IN-AGR.3SG
'on the table's leg'

So far, I have shown that as long as number, nominal agreement and definiteness are expressed overtly, whether via specialized morphemes or implied in proper names and pronouns, the complements must be marked with the genitive case. Crucially, the overt expression of possession and plurality is optional in the syntactic position of the complement of the postposition; semantically possessed or plural nominals in the complement of a postposition do not have to bear the corresponding markers. In such cases, they do not get genitive marking either:

(193) **Interpretation-marking mismatch generalization**:

Possessed, plural and definite complements do not get genitive marking as long as the corresponding categories are not expressed overtly.

In (194), the complement of the postposition is a possessed noun. The possessor is present, but there is no nominal agreement on the head of the complement, and the complement must be bare. Recall that the presence of nominal agreement would require genitive marking (185).

(194) [*mon'* *morkš*] *šir'ə-sə-n*
1SG.GEN table side-IN-1SG.AGR
'near my table'

The complement in (195) is semantically plural, since it contains a numeral. However, there is no plural marker on the head, and there is no genitive. If the noun were marked as plural, the genitive marker would have been required (196).

(195)	*kolmə*	*morkš*	*šir'ə-sə*
	three	table	side-IN
	'near three tables'		

(196)	*kolmə*	*morkš-n'ə-n'* /	**morkš-n'ə*	/	**morkš-t*	*šir'ə-sə*
	three	table-DEF.PL-GEN	table-DEF.PL		table-PL	side-IN
	'near the three tables'					

Similarly, bare complements of postpositions can still be semantically definite; consider an example where the referent of the nominal has been introduced in the previous discourse (197)–(198). Consider also an example in which the complement is both possessed and definite (introduced in the previous discourse), yet it remains bare (199).

(197)	***polka-t'***	/	***polka***	***lank-sə***	*ašč-i*	*var'ɛga-c'ə*
	shelf-DEF.SG.GEN		shelf	top-IN	be_situated-NPST.3SG	mitten-2SG.AGR.SG[NOM]
	'{When you enter the room, there is a shelf in the corner.} Your mitten is on the shelf.'					
	(Kashkin 2018: 130)					

(198)	a.	*komnata-sə*	*ašč-ij̊-t'*	***fkɛ***	***morkš***	*i*	*fkɛ*	*stul*
		room-IN	be_situated-NPST.3-PL	one	table	and	one	chair
		'There is a table and a chair in the room.'						
	b.	***morkš***	***lank-sə***	*ašč-i*	*mazi*	*vaza*		
		table	top-IN	be_situated-NPST.3SG	nice	vase		
		'There is a nice vase on the table.'						

(199)	a.	***mon'***	***morkš-əz'ə***	*ašč-i*	*s'a-də*	*mala-sə*
		1SG.GEN	table-1SG.AGR.SG[NOM]	be_situated-NPST.3SG	that-ABL	close-IN
		s'embə	*morkš-n'ə-n'*	*kor'as*		
		all	table-DEF.PLGEN	according		
		'My table is the closest one. (lit.: my table stands the closest among all tables)'				

	b.	*put-k*	*t'ɛ*	*vaza-t'*	***mon'***	***morkš***	*langə-zə-n*
		put-3.O.IMP.SG.O.SG.S	this	vase-DEF.SG.GEN	1SG.GEN	table	top-ILL-AGR.1SG
		'Put this vase on my table.'					

It is worth noting that complements not formally marked for definiteness or number usually remain bare, but they do not have to do so. Indefinite complements can take the marker *-n'* (200), which occupies the indefinite genitive cell of the Moksha case paradigm (see Pleshak 2021 for the formal analysis of the marker *-n'*). Definite complements can also be marked with the genitive without a definiteness marking (201).

(200)

kud-ən'	*šir'ə-sə*	*kas-i*	***kelu-s'***
house-GEN	side-IN	grow-NPST.3SG	birch-DEF.SG[NOM]

'The birch tree grows near a house.'

(201)

ɛr'	*s't'ər'-n' ɛ-s'*	*t'i-s'*	*fkɛ*	*fotografijɛ*
each	girl-DIM-DEF.SG[NOM]	do-PST.3SG	one	photo
t'ɛ	***kelu-n'***	*šir'ə-sə*		
this	birch-GEN	side-IN		

'Every girl made a picture near this birch tree.'

To sum up, the presence of a definite determiner, a plural marker or a nominal agreement marker requires genitive. Proper names and pronouns require genitive as well. Irrespective of semantics, in the absence of a definite determiner and a plural marker on a regular noun, the use of the bare form is preferred. In the absence of a nominal agreement marker on a possessed nominal, the bare form is the only option.

4.1.2. Modification

Genitive complements do not have restrictions on the types of modifiers they can contain. Bare nominals also allow for a wide range of adnominal modifiers, but they are incompatible with demonstratives and universal quantifiers, as stated in (202).

(202) **Modification generalization**:

If a postpositional complement combines with a demonstrative or an adnominal universal quantifier, this complement must be in the genitive case.

The following examples illustrate genitive complements of postpositions modified by various elements, including adjectives (203), possessors (204), numerals (205), relative clauses (206), demonstratives (207), and universal quantifiers (208).

(203) ADJECTIVE (Moksha Corpus)
stud'enčəskaj *apš'ižit'ijat'* *ingəl'ə*
student.ADJ dorms-DEF.SG.GEN front.LOC
'in front of the student dorms'

(204) POSSESSOR (Moksha Corpus)
mon' *al'ε-z'ə-n'* *mar̥tə* *t'εd'ε-z'ə*
1SG.GEN man-1SG.AGR.SG-GEN with mother-1SG.AGR[NOM]
'my mother with my father'

(205) NUMERAL
kolmə *morkš-n'ə-n'* *šir'ə-sə*
three table-DEF.PL-GEN side-IN
'near the three tables'

(206) RELATIVE CLAUSE
mon' *c'ora-z'ə* *ašč-i* *morkš-t'* *vaks-sə,*
1SG.GEN boy-1SG.AGR.SG[NOM] be_situated-NPST.3SG table-DEF.SG.GEN near-IN
[***kona-n'*** ***t'ij-əz̦'ə*** ***mon'*** ***at'ε-z̦'ə***]
which-GEN do-PST.3SG.S.3SG.O 1SG.GEN father-1SG.AGR.SG[NOM]
'My son is standing near the table made by my father.'

(207) DEMONSTRATIVE (Moksha Corpus)
t'ε *c'oran'εt'* *lank-s*
this boy-DIM-DEF.SG.GEN top-ILL
'on top of this boy'

(208) UNIVERSAL QUANTIFIER (Kashkin 2018: 147)
s'embə *vel'ə-t'n'ə-n'* *vaks-sə* *ul'-ij-°t'* *ki-t*
all village-DEF.PL-GEN near-IN be-NPST.3-PL road-PL
'There are roads near all villages.'

The examples below show the contrast between genitival and bare complements of postpositions. The latter also can be modified by adjectives (209), possessors (210), numerals (211) and relative clauses (212), but are incompatible with demonstratives (213) and universal quantifiers (214).

(209) ADJECTIVE (Moksha Corpus)

...i	*pov-s'*	***oc'u***	*kev*	*al-u*
and	occur-PST.3SG	big	stone	bottom-LAT

‘And went under a big rock.’

(210) POSSESSOR

mon'	*morkš*	*lank-sə-n*
1SG.GEN	table	top-IN-AGR.1SG

‘on my table’

(211) NUMERAL (Moksha Corpus)

kazan'εs'	*t'is'*	*skomn'ε*	***kaftə***	*pil'gən'ε*	*lank-sə*
goat-DIM-DEF.SG[NOM]	make-PST.3SG	bench	two	leg-DIM	top-IN

‘The goat made a bench on two legs.’

(212) RELATIVE CLAUSE

mon'	*c'ora-z'ə*	*aščč-i*	*morkš*	*vaks-sə,*
1SG.GEN	boy-1SG.AGR.SG[NOM]	be_situated-NPST.3SG	table	near-IN
[***kona-n'***	***t'ij-əz'ə***	***mon'***	***at'ε-z'ə***]	
which-GEN	do-PST.3SG.S.3SG.O	1SG.GEN	father-1SG.AGR.SG[NOM]	

‘My son is standing near the table made by my father.’

(213) DEMONSTRATIVE

****t'ε***	*morkš*	*lank-sə*
this	table	top-IN

Int.: ‘on this table’

(214) UNIVERSAL QUANTIFIER

****s'embə***	*morkš(-t)*	*lank-sə*
all	table-PL	top-IN

Int.: ‘on all tables’

A further distinction arises between animate and inanimate complements. Inanimate complements may appear in the bare form when unmodified or when modified by certain elements, such as an indefinite pronoun (215). In contrast, animate complements cannot normally appear in the bare form; they are only marginally acceptable when modified by an indefinite pronoun (216).

(215) INANIMATE
(*kodama* *bəd'ə*) *morkš* *šir'ə-sə*
which INDEF table side-IN
'near (some / this) table'

(216) ANIMATE
a. [?]*kodama* *bəd'ə* *st'ər-n'ɛ* *šir'ə-sə*
which INDEF girl-DIM side-IN
'near a girl (lit.: near some girl)'

b. **st'ər-n'ɛ* *šir'ə-sə*
girl-DIM side-IN
Int.: 'near a girl'

In summary, genitive and bare inanimate complements behave similarly, except for the restriction on demonstratives and universal quantifiers. Animate complements, on the other hand, have a further restriction: they must be modified to appear bare.

4.1.3. Nominal agreement

When a complement contains a definite DP — typically a possessor — genitive and bare complements exhibit distinct behaviors concerning the obligatoriness of agreement with their dependent DPs. This contrast is summarized in (217). I have already demonstrated that all complements with nominal agreement must be genitive (183) and that complements without nominal agreement can be bare (194). What (217) further establishes is that complements lacking nominal agreement must be bare, and genitive complements cannot appear without agreement when a possessor is present (218).

(217) **Internal nominal agreement generalization**:
Genitive complements must agree with their possessors; bare complements cannot do so.

(218) **mon'* *morkš-t'* *lank-sə(-n)*
1SG.GEN table-DEF.SG.GEN on-IN-AGR.1SG
Int.: 'on my table'

Nominal agreement appears not only on nominal heads but also on postpositions. However, genitive and bare complements behave differently in terms of the possibility and obligatoriness of triggering agreement on the postposition. Specifically, genitive complements can trigger nominal agreement on the postposition, whereas bare complements never do, as summarized in (219).

(219) **Complement agreement generalization**:

Genitive complements can trigger nominal agreement on the postposition.

a. Personal and reflexive pronouns trigger obligatory agreement
b. Animate nominals (excluding interrogative pronouns) trigger optional agreement
c. Inanimate nominals and interrogative pronouns do not trigger agreement (at least in some idiolects)[19]

Example (220) illustrates that when the complement is a personal pronoun, the postposition must agree with it in person and number. In (221)a, the complement is an animate noun, and agreement is optional, whereas in (221)b, where the complement is an animate interrogative pronoun, agreement is ungrammatical. Complements expressed with an inanimate noun or a demonstrative pronoun, as shown in (222), do not trigger agreement as well, at least in some idiolects.

[19] Among all the examples provided by Muravyeva and Kholodilova (2018), there are two that show nominal agreement with an inanimate complement; this does not align with the judgments I obtained during my own fieldwork. I leave the question of this variation for future research.

(220) PERSONAL PRONOUN

mon'	*šir'ə-sə-**n***	/	******šir'ə-sə*
1SG.GEN	side-IN-AGR.1SG		side-IN

'near me'

(221) a. ANIMATE NOUN

st'ər'-t'	*šir'ə-sə*	/	*šir'ə-sə-**nzə***
girl-DEF.SG.GEN	side-IN		side-IN-AGR.3SG

'near the girl'

b. ANIMATE INTERROGATIVE PRONOUN

ki-n'	*šir'ə-sə*	/	******šir'ə-sə-**nzə***
who-GEN	side-IN		side-IN-AGR.3SG

'near who?'

(222) a. INANIMATE NOUN

morkš-t'	*šir'ə-sə*	/	******šir'ə-sə-**nzə***
table-DEF.SG.GEN	side-IN		side-IN-AGR.3SG

'near the table'

b. DEMONSTRATIVE PRONOUN

t'ɛ-n'	*šir'ə-sə*	/	******šir'ə-sə-**nzə***
this-GEN	side-IN		side-IN-AGR.3SG

'near this'

A crucial aspect of postpositional agreement is the possibility of the postposition agreeing with the possessor of its complement. The ability of the postposition to agree with the possessor is determined by whether the complement is genitive or bare. Specifically, genitive complements block external agreement with their possessor, whereas bare complements allow it. This distinction is outlined in (223).

(223) **Generalization: external agreement with possessor**:

a. Postpositions cannot agree with possessors of genitive complements, regardless of their animacy, but they can agree with possessors of bare complements.

b. The restrictions on possessor agreement in genitive complements mirror those seen in genitive complements themselves.

The restrictions on possessor agreement in genitive complements mirror those seen in genitive complements themselves. Example (224) shows that when the complement is genitive, the postposition cannot agree with the possessor. In (224)a, nominal agreement is already present on the complement head, preventing agreement with the possessor. In (224)b-c, the agreement on the complement head is absent, yet the agreement on the postposition is ruled out.

(224) a. GENITIVE COMPLEMENT WITH A NOMINAL AGREEMENT MARKER

mon'	*morkš-**əz'ə-n'***	*lank-sə* /	**lank-sə-n*
I.GEN	table-1SG.AGR.SG-GEN	on-IN	on-ILL-1SG.AGR

‘on my table’

b. GENITIVE COMPLEMENT WITH A DEFINITENESS MARKER

**mon'*	*morkš-**t'***	*lank-sə-n*
I.GEN	table-DEF.SG.GEN	on-IN-1SG.AGR

Int.: ‘on my table’

c. GENITIVE COMPLEMENT WITH NO OTHER CATEGORIES EXPRESSED

**mon'*	*morkš-**ən'***	*lank-sə-n*
I.GEN	table-GEN	on-IN-1SG.AGR

Int.: ‘on my table’

Now consider (225)–(227), where the complement is bare. In these cases, the postposition can agree with the possessor of its complement. The obligatoriness of agreement follows the same pattern as in (219). If the possessor is a personal pronoun, agreement is obligatory (225). If the possessor is animate, agreement is optional (226). However, determining whether inanimate possessors trigger agreement on the head is difficult because those compatible with spatial postpositions typically occur in part-whole constructions, which cannot be bare complements (227).

(225) POSSESSOR PERSONAL PRONOUN → OBLIGATORY AGREEMENT

mon'	*morkš*	*lank-sə-**n*** /	**lank-sə*
I.GEN	table	top-IN-1SG.AGR	top-IN

‘on my table’

(226) ANIMATE POSSESSOR → OPTIONAL AGREEMENT
maša-n' *morkš* *lank-sə(-**nzə**)*
Masha-GEN table on-IN-AGR.3SG
'on Masha's table'

(227) *INANIMATE POSSESSOR
morkš-t'* *pil'gə* *lank-sə(-nzə**)*
table-DEF.SG.GEN leg on-IN-AGR.3SG
Int.: 'on the table's leg'

Thus, possessors of bare complements of postpositions behave similarly to genitive complements in that they are marked with the genitive case and trigger agreement on the postposition.

4.1.4. Interim summary

In this section, I have described the properties of genitive and bare complements of postpositions. These two types of complements differ in the availability of nominal category marking and have distinct restrictions on modification, as summarized in Table 8. They also follow different patterns with respect to nominal agreement, which is outlined in Table 9.

Table 8: *Marking and modification properties of genitive and bare complements*

	genitive	bare
expression of number	OK	*
expression of nominal agreement	OK (obligatory)	*
expression of definiteness	OK	*
pronouns	OK	only *mez'ə* 'what'
human proper names	OK	*
animate	OK	only if modified by an indefinite pronoun
inalienable possessive constructions	OK	*
modification with universal quantifiers	OK	*
modification with relative clauses	OK	OK

modification with demonstratives	OK	*
modification with cardinal numerals	OK	OK
modification with adjectives	OK	OK

Table 8 shows that genitive complements can freely express nominal features such as number, agreement, and definiteness. They are also unrestricted in modification, allowing demonstratives, universal quantifiers, and relative clauses. In contrast, bare complements lack number marking, definiteness, and agreement, and their modification possibilities are more restricted. For example, they do not permit demonstratives or universal quantifiers and only allow animate nouns when modified by an indefinite pronoun.

Table 9: *Combinations of the instances of nominal agreement within a PP*

	P agreement with its complement	P agreement with the possessor of its complement
no possessor	OK pronominal complement (obligatory) OK animate genitive complement * inanimate genitive complement * bare complement	n/a
Nominal head agreement with its possessor	OK animate genitive complement * inanimate genitive complement n/a bare complement	*
Possessor, no nominal agreement within the complement	* bare complement n/a genitive complement	OK pronominal possessor (obligatory) OK animate possessor * inanimate possessor

Table 9 highlights the interaction between agreement and complement type within a postpositional phrase. Genitive complements generally require agreement, except when they are inanimate. Bare complements, by contrast, never trigger agreement but allow the postposition to agree with their possessors.

From these tables, we observe that genitive complements exhibit no restrictions on marking and modification, and only inanimate genitive complements fail to trigger agreement on the postposition. Bare complements, on the other hand, lack nominal marking, cannot trigger agreement, and uniquely allow the postposition to agree with their possessors. Despite their lack of marking, they are compatible with many adnominal modifiers, except for demonstratives and universal quantifiers. I offer an analysis of these properties in the next section.

4.2. Determining the size of bare complements of postpositions

In this section, I argue that the distinctions between genitive and bare complements reflect differences in their syntactic structure. Specifically, I propose that genitive complements are full DPs, while bare complements are SNs — structures that lack higher functional projections. More precisely, I argue that bare complements of postpositions in Moksha correspond to F_1Ps, which are larger than nP but smaller than NumP.

The analysis aims to account for the following empirical generalizations discussed in Section 4.1:

(228) a. **Overt marking generalization**:
If number, nominal agreement, or definiteness is overtly expressed, the complement of the postposition must be genitive.

b. **Category restriction generalization**:
If the complement of the postposition is a pronoun or an animate proper name, it must be genitive.

c. **Featural restriction generalization**:

If the complement of a postposition is a noun denoting a human or an inalienably possessed object, it must be genitive.

d. **Interpretation-marking mismatch generalization**:

Possessed, plural and definite complements do not get genitive marking as long as the corresponding categories are not expressed overtly.

e. **Modification generalization**:

If a postpositional complement combines with a demonstrative or an adnominal universal quantifier, this complement must be in the genitive case.

f. **Internal nominal agreement generalization**:

Genitive complements must agree with their possessors; bare complements cannot do so.

g. **Complement agreement generalization**:

Genitive complements can trigger nominal agreement on the postposition.

- Personal and reflexive pronouns obligatorily trigger agreement
- Animate nominals (excluding interrogative pronouns) optionally trigger agreement
- Inanimate nominals and interrogative pronouns do not trigger agreement (at least in some idiolects)

h. **Generalization: external agreement with possessor**:

- Postpositions cannot agree with possessors of genitive complements, regardless of their animacy, but they can agree with possessors of bare complements.
- The restrictions on possessor agreement in genitive complements mirror those seen in genitive complements themselves.

I build my argument for the presence and absence of higher nominal projections in genitive and bare complements respectively by taking the hierarchical structure of the noun phrase introduced in Chapter 2 (229). I examine whether the nominals in question exhibit the properties associated with each functional projection.

(229) NP < nP < F_1P < F_2P < NumP < DP

Each layer of this hierarchy is associated with specific syntactic properties: DP hosts definite determiners and interacts with external agreement probes; NumP hosts number marking and is associated with nominal agreement probes; F_1P and F_2P are structural layers between nP and NumP that can host certain adnominal modifiers; and nP base-generates possessors.

In this dissertation, I will look at nominal agreement, which is phi-feature agreement on a non-verbal head, nominal or postpositional head in particular. I also assume that only DPs receive case marking, while SNs remain caseless, even when they appear in syntactic positions where case is typically assigned.

This analysis aligns with the perspective that not all noun phrases in argument positions contain a full DP structure (Pereltsvaig 2006). Under this view, structures that lack DP or other consecutive higher projections are called small nominals. In this framework, NumPs, FPs, nPs, and NPs all qualify as small nominals.

4.2.1. DP layer

I argue that bare complements of postpositions in Moksha lack DP structure. As established in Chapter 2, Section 2.5.3.1, DP serves as the host for formal definiteness marking, case features, and agreement features (see also Chapter 3, Section 3.3 for Moksha DP in particular). The absence of these properties in bare complements suggests that they are structurally smaller than DPs. In contrast, genitive complements exhibit case marking, definiteness marking, and agreement with the postpositional head, indicating the presence of DP, see section 4.1.

I take the impossibility of definiteness marking in the absence of the genitive case to be indicative of the absence of the D head, which would host the determiner and the case feature. In (230), the form *morkšt′* has a *-t′* marker, which is a portmanteau morpheme encoding singular number, definite article, as well as the genitive case, which suggests that the relevant heads (Num and D) are present in the underlying structure. The attempt to mark definiteness without the genitive case, which would be the corresponding singular definite marker for the nominative case[20], results in ungrammaticality. This is expected if both definiteness and case are encoded by D, present in genitive complements and absent in bare complements.

(230)	*morkš-t′*	/	**morkš-s′*	*lank-sə*
	table-DEF.SG.GEN		table-DEF.SG[NOM]	on-IN
	‘on the table’			

Notably, the absence of the definite declension cannot be attributed to semantics, as bare complements of postpositions can be definite, see Section 4.1. In (231)b, the object expressed by a bare noun (*morkš*) has already been introduced in the previous discourse (231)a.

(231)	a.	*komnata-sə*	*ašč-ij̊-t′*	***fkɛ***	***morkš***	*i*	*fkɛ*	*stul*
		room-IN	be_situated-NPST.3-PL	one	table	and	one	chair
		‘There is a table and a chair in the room.’						

	b.	***morkš***	***lank-sə***	*ašč-i*	*mazi*	*vaza*
		table	top-IN	be_situated-NPST.3SG	nice	vase
		‘There is a nice vase on the table.’				

[20] The nominative case itself has no overt realization in Moksha.

Thus, the inability of bare complements to host overt definiteness marking is not due to meaning but rather to syntactic structure — specifically, the absence of D.

Further support for the absence of DP in bare complements comes from their incompatibility with demonstratives and universal quantifiers. Unlike genitive complements, bare complements cannot combine with these elements (232). Given that bare complements have no restriction on semantic definiteness, the explanation should be syntactic. Universal quantifiers are often analyzed as DP adjuncts and are not expected in structures smaller than DP. Demonstrative pronouns, while being generated in a lower functional projection, are usually treated as requiring the presence of DP in the structure, see Chapter 2. However, attributing the ungrammaticality of (232) to the lack of specifically DP would be inconsistent with me treating oblique phrases as SNs, because they do combine with demonstratives and universal quantifiers (this indicates that demonstratives and universal quantifiers in Moksha are lower than D). Instead, I propose that bare complements of postpositions are small enough not to generate demonstratives and universal quantifiers at all, lacking the higher projections of the FP zone.

(232) a. *[***t'ɛ*** *morkš*] *lank-sə*
this table on-IN
Int.: 'on this table'

b. *[***s'embə*** *morkš*] *lank-sə*
all table on-IN
Int.: 'on all tables'

Another key distinction between genitive and bare complements concerns agreement on postpositions. Only genitive complements can trigger agreement on the P head (233). Phrased differently, only genitive complements are visible for agreement probes. This asymmetry is exactly what is predicted if bare complements lack D, which is targeted by agreement probes.

(233) a. *st'ər'-t'* *lank-sə(-nzə)*
girl-DEF.SG.GEN on-IN-AGR.3SG
'on the girl'

b. *kodama bed'ə st'ər' lank-sə / *lank-sə-nzə*
which INDEF girl on-IN on-IN-AGR.3SG
'on a girl (lit. 'on some girl')'

Based on these syntactic properties, I conclude the following. Genitive complements are bona fide DPs, as they exhibit case marking, definiteness marking, compatibility with quantifiers and demonstratives, and agreement on postpositions. Bare complements are small nominals — F_1Ps rather than DPs — as they lack case marking, definiteness marking, and agreement, and cannot host demonstratives or universal quantifiers.

4.2.2. NumP layer

In addition to the absence of DP, bare complements of postpositions in Moksha also lack NumP. As established in Section 4.1, whenever number is overtly expressed, definiteness and case marking become obligatory (234)a (see also Chapter 3, Section 3.1 for a more thorough discussion of Moksha nominal morphology). Conversely, if a definite determiner is present, number must also be expressed (234)b. This follows from the syntactic architecture of the extended nominal projection: the presence of DP entails the presence of NumP, since intermediate projections cannot be skipped.

(234) a. NUMBER → DEFINITENESS+CASE
*morkš-n'ə-n' / *morkš-t lank-sə*
table-DEF.PL-GEN table-PL on-IN
'on (the) tables'

b. DEFINITENESS → NUMBER
morkš-t' *lank-sə*
table-DEF.SG.GEN on-IN
'on the table'
*'on the tables'

The absence of DP does not automatically entail the absence of NumP, but empirical evidence suggests that NumP is also missing in bare complements. Specifically, the lack of overt number marking in bare complements indicates the absence of Num in their structure.

In (234)a, when plural marking is present, definiteness and genitive case are required. In (234)b, the definite determiner *-t'* enforces a singular interpretation, blocking plural reading. These patterns support the idea that number marking requires NumP, which is present in genitive complements but absent in bare complements.

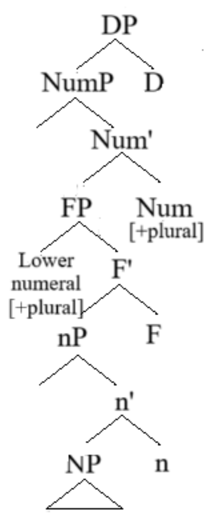
Further evidence for the absence of NumP in bare complements comes from the behavior of cardinal numerals. In Moksha, numerals like *kolmə* 'three' impose specific number marking requirements: in genitive complements, plural marking is obligatory (235)a; in bare complements, plural marking is ruled out (235)b. The obligatoriness of plural marking on the genitive complement in (235)a suggests that only plural Num is compatible with numerals like *kolmə* 'three' in Moksha. The grammaticality of (235)b even without plural marking on the complement suggests that the bare complement lacks NumP entirely. If NumP were present, we would expect a singular-plural mismatch to render the phrase ungrammatical.

(235) a. *kolmə* *morkš-n'ə-n'* / **morkš-t'* *šir'ə-sə*
three table-DEF.PL-GEN table-DEF.SG.GEN side-IN
'near the three tables'

b. *kolmə* *morkš* / **morkš-t* *šir'ə-sə*
three table table-PL side-IN
'near three tables'

My analysis goes as follows. The Num head imposes constraints based on the value of its [±plural] feature. A [+plural] Num head is incompatible with high numerals, and a [+singular] Num head is incompatible with low numerals. In the absence of a Num head, no such restrictions apply — numerals can freely occur without triggering plural marking, because there is no [+singular] feature to block them. This analysis is schematically represented in Figures 12–14.

Figure 12: *Num head with the [+plural] feature combining successfully with a lower numeral*[21]



[21] Here and below I simplify the tree for demonstrational purposes by not splitting FP projection into two when it is not relevant.

Figure 13: *Num head with the [+singular] feature combining unsuccessfully with a lower numeral*

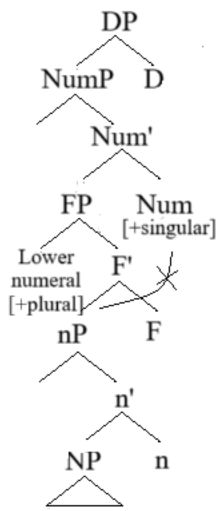
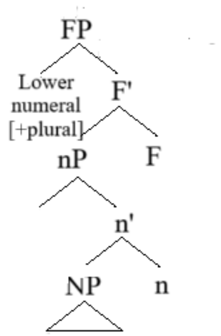


Figure 14: *Lower numeral in the absence of a Num head*



This particular implementation of the number marking-interpretation mismatch diagnostic on Moksha data requires us to assume that numerals are generated lower than NumP. The low position of numerals has been independently motivated in semantic literature (Ionin & Matushansky 2006, 2018; Scontras 2014: 32-35; Marti Martinez 2020). While a full discussion of numeral semantics is beyond the scope of this work, these accounts converge on the idea that number must scope over numerals in the nominal spine, supporting the idea that NumP is higher than the numeral phrase.

Another consequence of the absence of NumP in bare complements is the lack of internal nominal agreement. Given that the internal nominal agreement with the possessor (or any dependent genitive DP) is realized by a probe located somewhere in Num (see Chapter 2 for discussion), we expect bare complements not to show nominal agreement either. This prediction is borne out; the same way as with the definiteness in (230), if the complement is not marked as genitive, it cannot bear nominal agreement (236), see also Section 4.1.

(236)	*morkš-ənc* /	**morkš-əc*	*lank-sə*
	table-3SG.AGR.SG.GEN	table-3SG.AGR.SG[NOM]	on-IN
	'on his table'		

Thus, I have shown that Moksha bare complements of postpositions lack the two most established functional projections of the nominal extended projection — NumP and DP. Due to the lack of well-articulated model of the nominal structure between NP and NumP, it is difficult to tell what other properties are expected from a nominal having those potential lower nominal projections. Nevertheless, I would like to point out that bare complements of postpositions are definitely larger than simple noun heads, and even larger than NPs as they can have some adnominal modifiers, such as possessors (237) and numerals (238). The presence of possessors implies the presence of nP, whereas the presence of numerals implies the presence of an even higher functional projection. Since NP alone does not provide sufficient structure to host different kinds of modifiers, I propose that bare complements are F_1Ps — a category larger than nP but smaller than NumP.

(237)	[*mon'*	*morkš*]	*lank-sə-n*
	1SG.GEN	table	on-IN-1SG.AGR
	'[Put the apples] on my table.'		

(238)	[*kolmə*	*morkš*]	*lank-sə*
	three	table	on-IN
	'on three tables'		

The evidence presented here supports the claim that bare complements of postpositions in Moksha lack both DP and NumP. First, bare complements lack overt number marking, suggesting NumP is absent. Second, numerals impose number restrictions in genitive complements, but not in bare complements, indicating that NumP is missing in the latter. Third, internal nominal agreement, which depends on NumP, is absent in bare complements. Fourth, bare complements can still host possessors and numerals, suggesting a structure larger than nP but smaller than NumP — which I analyze as F_1P. In Section 4.5, I further explore the placement of possessors, numerals, demonstratives, and universal quantifiers within the nominal spine.

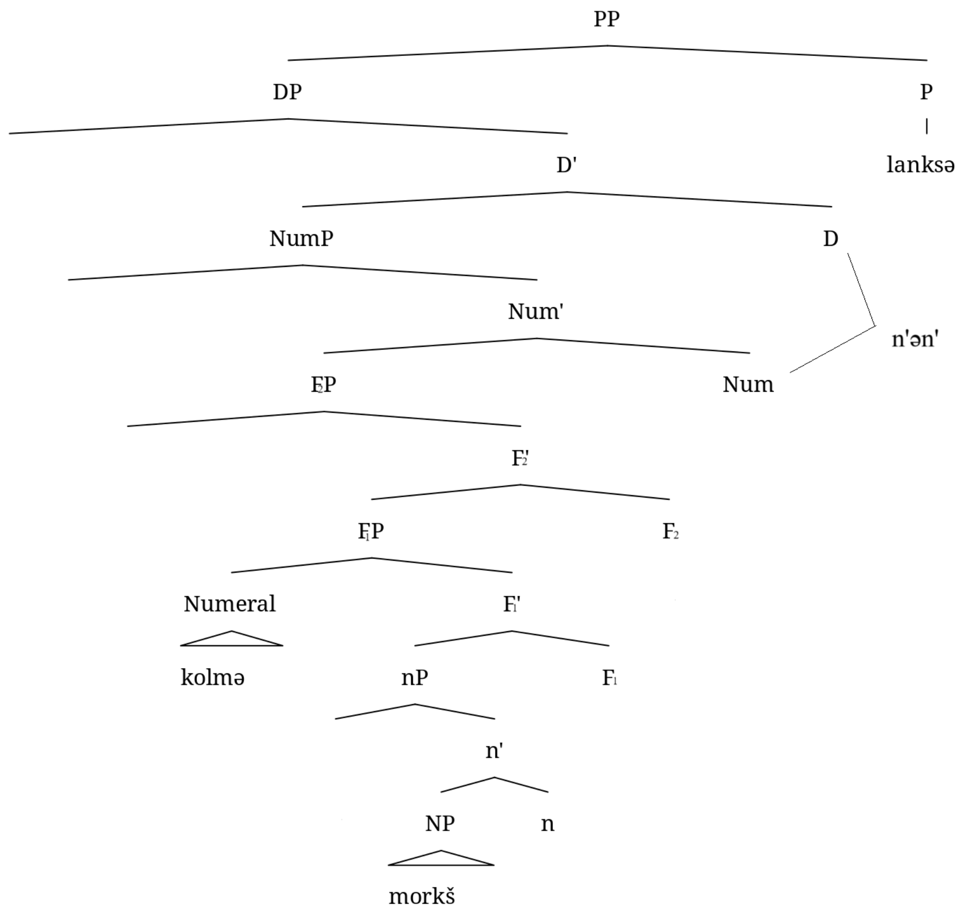
4.3. The mechanics of the small nominal analysis

In Section 4.2, I provided arguments for the structural differences between genitive and bare complements of Moksha spatial postpositions. This section demonstrates how these structural differences derive the agreement patterns described in Section 4.1.

4.3.1. Accounting for case-marking generalizations

The basic structure of a genitive DP complement is illustrated in Figure 15. In this configuration, the P head takes a full DP as its complement, meaning that all functional projections, including NumP and DP, are present within the complement. The genitive case is assigned within the postpositional domain (following the DCT framework, see Chapter 1, Section 1.2.2.1). The presence of NumP and DP results in overt number and definiteness marking, realized here as *-t'*.

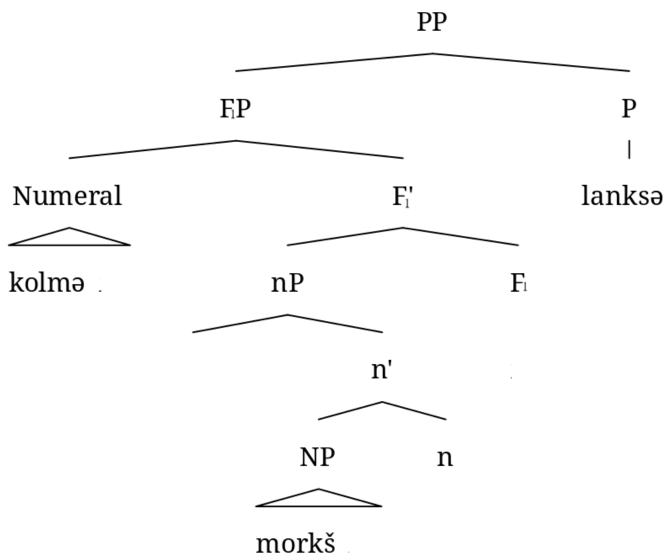
Figure 15: *The structure of a PP with a genitive DP complement*



kolmə *morkš-n'ə-n'* *lank-sə*
three table-DEF.PL-GEN on-IN
'on the three tables'

In contrast, the structure of a bare F_1P complement is shown in Figure 16. Here, P selects an F_1P — a projection below NumP and F_2P— and it is only the functional projections below F_2P that are present within the complement. The complement does not get case due to the absence of D. Since NumP and DP are absent in the structure to begin with, definiteness and number are not overtly realized.

Figure 16: *The structure of a PP with a bare F_1P complement*



kolmə *morkš* *lank-sə*
three table on-IN
'on three tables'

This structural distinction explains the overt marking generalization:

(239) **Overt marking generalization**:

If number, nominal agreement, or definiteness is overtly expressed, the complement of the postposition must be genitive.

Overt realization of definiteness reflects the presence of DP, which requires case; therefore, genitive marking is also present. SNs in Moksha are smaller than NumP, and NumP is always accompanied by DP in Moksha; therefore, overt realization of number necessarily entails definiteness and case marking.

In order to explain the category restriction generalization (240), let me elaborate on the expected nominal structure of pronouns and human proper names and inalienable possessive constructions.

(240) **Category restriction generalization**:

If the complement of the postposition is a pronoun or an animate proper name, it must be genitive.

Pronouns have been argued to be structures of varying sizes (Déchaine & Wiltschko 2002). However, it is reasonable to treat Moksha pronouns that require genitive case as always being full DPs. The pronoun *mez'ə* 'what', which can appear without genitive marking, is assumed to be smaller than DP. A similar explanation applies to human proper names, which appear to be structurally inflexible in Moksha and obligatorily project the DP shell, leading to genitive case assignment.

The featural restriction generalization (241) extends these insights to human-denoting nouns and inalienably possessed objects.

(241) **Featural restriction generalization**:

If the complement of a postposition is a noun denoting a human or an inalienably possessed object, it must be genitive.

I assume that, like pronouns, human nouns in Moksha must project a full DP. This assumption is supported by evidence from binominal compounds, where the non-head noun is treated as being smaller than a DP. In these compounds, the same animacy restriction holds (see Pleshak 2022c on binominal compounds in Moksha). Although part-whole relationships can generally be encoded by a compound, bare noun compounds are ungrammatical when the "whole" is denoted by a human (242). As a result, human-denoting expressions cannot be structurally reduced to functional projections smaller than DP, e.g., F_1Ps. Consequently, all postpositional complements that refer to humans must appear in the genitive.

(242) a. *kelu* *lopa*
birch leaf
'birch leaf'

b. *id'-ən'* / **id'* *kɛd'*
kid-GEN kid hand
'kid's hand'

For inalienable possession, I assume that overt possessor agreement within the noun phrase is required. Agreement within the nominal is on Num, which means that inalienable possessive phrases should be at least NumPs. However, because NumP is always embedded within DP in Moksha, these structures necessarily receive genitive case.

As I argued in Chapter 2, NumP and DP are formal projections, and do not strictly correspond to plural, possessive or definite semantics. This separation allows for plural, possessed, and definite nominals to lack NumP and DP in some cases. When these two highest projections are not present, genitive marking is not expected either, which explains (243).

(243) **Interpretation-marking mismatch generalization**:

Possessed, plural and definite complements do not get genitive marking as long as the corresponding categories are not expressed overtly.

The final case-marking generalization relates to the interaction between demonstratives and universal quantifiers, on the one hand, and nominal structure, on the other:

(244) **Modification generalization**:

If a postpositional complement combines with a demonstrative or an adnominal universal quantifier, this complement must be in the genitive case.

Even though I do not assume that in Moksha demonstratives are located as high as in D, I argue that Moksha bare complements of postpositions are too small to accommodate demonstratives and universal quantifiers. Small caseless nominals are F_1Ps, and cannot combine with them due to the lack of functional projections higher than F_1P, which host these modifiers.

Summing up, these case-marking generalizations follow naturally from the structural distinction between full DP complements (genitive-marked) and bare F_1P complements (caseless), reinforcing the syntactic analysis proposed in Section 4.2. In the following sections, I explore additional morphosyntactic consequences of this structural contrast, including agreement interactions and the placement of quantifiers, demonstratives, and possessors within the nominal spine.

4.3.2. Accounting for agreement generalizations

Beyond the absence of genitive marking, Moksha bare complements of postpositions exhibit an unusual agreement pattern within PPs. This pattern differs from what is observed with genitive-marked counterparts and appears to violate locality restrictions on agreement. As outlined in Section 4.1, my proposal regarding the DP vs. small nominal contrast accurately predicts this agreement behavior. In what follows, I demonstrate how this agreement pattern is derived from the structures in Figures 15 and 16.

Recall that the agreement under discussion is nominal agreement, specifically phi-feature agreement between a non-verbal head and a DP within its search domain. There are two contexts for nominal agreement in Moksha: the noun head agrees with its possessor DP, and the postposition agrees with a DP in its complement. I propose that in Moksha, there are two heads associated with nominal agreement: Num head is the one responsible for the possessor agreement within the noun phrase, and P head is the one responsible for the agreement on the postposition. I also propose that there are two agreement probes associated with P, each targeting a different feature, which results in agreement patterns that differ in optionality with pronominal and non-pronominal arguments (see Section 4.3.2.1 for details).

These two agreement loci yield three instances of nominal agreement in Moksha. First, agreement within possessive constructions — the noun head agrees with a possessor DP, leading to an agreement marker on the noun. Second, agreement between the postposition and its complement — the P head agrees with the complement DP, leading to an agreement marker on the postposition (but only if the complement is animate). Finally, agreement between the postposition and the possessor of its complement — if the possessor of the complement DP is animate, the P head optionally agrees with it, leading to an agreement marker on the postposition. Below, I analyze how genitive and bare complements interact with these three instances of nominal agreement and demonstrate how my structural analysis accounts for the observed patterns.

4.3.2.1. Agreement of a postposition with its complement

In this section, I explain how agreement of postpositions with their complements operates in Moksha and account for the asymmetry between genitive and bare complements.

(245) **Complement agreement generalization**:

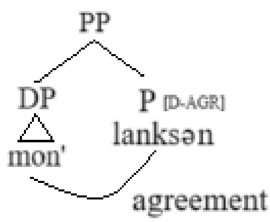
Genitive complements can trigger nominal agreement on the postposition.

- Personal and reflexive pronouns trigger obligatory agreement
- Animate nominals (excluding interrogative pronouns) trigger optional agreement
- Inanimate nominals and interrogative pronouns do not trigger agreement (at least in some idiolects)

Let us start with the situation where the complement of the postposition is a pronoun. Personal and reflexive pronouns trigger obligatory agreement. This follows from my proposal that the agreement probe on P searches for a feature on D (which I will specify below). Since this feature is inherently present on pronominal D heads, the probe always finds a match, making agreement obligatory. The result is an agreement marker on the postposition, encoding person and number, e.g., 1st person singular in (246). The structure for (246) is in Figure 17.

(246) *mon'* *lank-sə-n* / **lank-sə*
1SG.GEN on-IN-AGR.1SG on-IN
'on me'

Figure 17: *Nominal agreement of P with a pronominal complement*



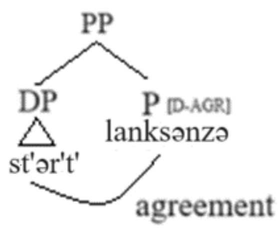
Turning to non-pronominal complements without a possessor, there is an important contrast: animate complements can trigger optional agreement, and inanimate complements never trigger agreement.

The fact that animate complements must be genitive and that all bare complements are inanimate naturally explains why only genitive complements trigger agreement.

What I propose for the agreement of P with animate genitive complements is that the genitive DP complement contains a feature on D, which satisfies the requirements of the probe, and the agreement shows up on the postposition as an AGR.3SG marker, see Figure 18 for (247).

(247) *st'ər'-t'* *lank-sə(-nzə)*
girl-DEF.SG.GEN on-IN-AGR.3SG
'on the girl'

Figure 18: *Nominal agreement of P with an animate DP complement*



By contrast, inanimate complements lack the relevant feature on D, preventing agreement from occurring:

(248) *morkš-**t'*** *lank-sə* / **lank-sə-nzə*
table-DEF.SG.GEN on-IN on-IN-AGR.3SG
'on the table.'

This follows from a well-established animacy hierarchy (249) suggested by Silverstein (1976). I propose that in Moksha, only DPs higher in the hierarchy bear the feature needed for agreement. This results in the absence of agreement with inanimate DPs. I remain agnostic regarding the particular formal feature that results in agreement, but its exact label is not that important for my analysis.

(249) **Animacy Hierarchy** (Silverstein 1976):

1st /2nd person personal pronouns > 3rd person personal pronouns > proper names of humans > kinship terms > human regular nouns > other animate > inanimate > abstract

Now let me finally address the contrast between DPs headed by animate common nouns and pronominal DPs. I propose that the P head always has an agreement probe, and that there are two possible agreement probes: F1, which targets animate DPs and above, and F2, which targets only personal pronouns. Pronouns have both F1 and F2, meaning they always trigger agreement. Animate non-pronominal DPs only have F1, so agreement depends on which probe merges. Inanimate DPs have neither F1 nor F2, so they never trigger agreement. Since either probe can merge in a PP, pronouns always trigger agreement, while animate DPs only trigger it when F1 is present.

Importantly, lack of agreement does not cause ungrammaticality, as argued in Chapter 1, Section 1.2.2.2. When the merged probe does not find a matching feature, agreement fails still yielding a grammatical structure.

Finally, my analysis correctly predicts that bare complements never trigger agreement:

(250) **st'ər'* *lank-sə -nzə*
girl on-IN-AGR.3SG
Int.: 'on the girl'

(251) *morkš* *lank-sə* / **lank-sə-nzə*
table top-IN on-IN-AGR.3SG
'on a table'

Since bare complements are not DPs, they lack D entirely. This means they cannot contain F1 or F2, preventing agreement from occurring.

This contrast is not particularly informative for my analysis, as animate DPs must be genitive anyway, and inanimate DPs do not trigger agreement even when genitive. Nevertheless, this pattern aligns with my SN analysis, which will further support the agreement generalizations discussed in the following sections.

4.3.2.2. Agreement of the nominal head with its possessor

In this section, I examine agreement between a possessor and the head noun within a postpositional complement and derive the generalization in (252).

(252) **Internal nominal agreement generalization**:
Genitive complements must agree with their possessors; bare complements cannot do so.

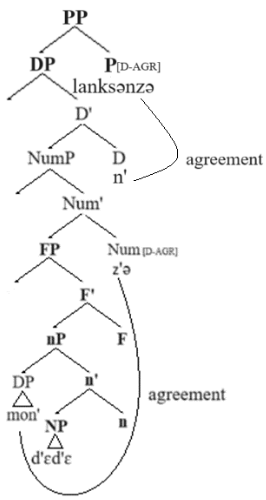
My proposal is based on the assumption that the nominal agreement probe is always associated with Num (see Chapter 2). Unlike P head probes, which differentiate between DP types, the Num probe is not selective — it simply searches for a D-feature. Agreement only fails when no DP is present within the nominal structure.

Since genitive complements are DPs, they necessarily project a NumP. This guarantees that a nominal agreement probe is always present, leading to obligatory possessor agreement whenever a dependent DP is available. In contrast, bare complements lack a NumP, meaning there is no nominal agreement probe — hence, no agreement with a possessor DP.

Example (253) illustrates a genitive DP complement where the following processes take place. First, the Num probe agrees with the possessor DP (realized as a 1st person singular agreement marker on the noun). Then, the P head probe probes for a DP in its domain and finds an animate DP complement (which has F1). Finally, since the P head probe in (253) happens to be one that targets F1, it successfully agrees with the complement, resulting in a 3rd person singular agreement marker on the postposition. The structure for (253) is represented in Figure 19.

(253) *mon'* *t'ɛd'ɛ-z'ə-n'* *lank-sə-nzə*
1SG.GEN mother-1SG.AGR.SG-GEN on-IN-AGR.3SG
'on my mother'

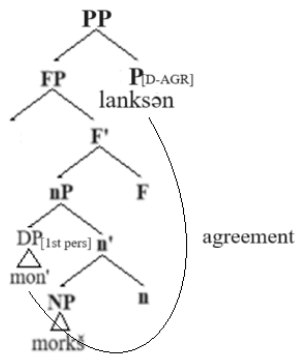
Figure 19: *Nominal agreement of Num with a possessor DP and nominal agreement of P with an animate DP complement*



In bare complements, possessor agreement fails because NumP is absent. This means there is no nominal probe, and thus no agreement with the possessor DP. Example (254) illustrates this. Since NumP is missing, Figure 20 illustrates that no nominal agreement occurs within the complement. The external agreement on P (which I address in the next subsection) remains unaffected by this lack of internal nominal agreement.

(254) *mon'* *morkš* *lank-sə-n*
1SG.GEN table on-IN-1SG.AGR
'on my table'

Figure 20: *Absence of nominal agreement with a possessor DP within a bare complement*



This distinction between genitive and bare complements further supports the SN analysis, which accounts for different patterns of nominal agreement in PPs depending on the case-marking of the complement.

4.3.2.3. Postpositional agreement with the possessor of its complement

In this section, I analyze the conditions under which a postposition can agree with the possessor of its complement and derive the generalization in (255), which highlights a key difference between bare and genitive complements of postpositions.

(255) **Generalization**: **external agreement with possessor**

- Postpositions cannot agree with possessors of genitive complements, regardless of their animacy, but they can agree with possessors of bare complements.
- The restrictions on possessor agreement in genitive complements mirror those seen in genitive complements themselves.

The first observation concerns the impossibility of postpositional agreement with a possessor inside a genitive complement. Since genitive complements are DPs, I propose that an external probe cannot access material inside them because DPs are phases in the sense of Chomsky (2008), where due to the Phase Impenetrability Condition, no elements embedded in a phase are accessible for the external syntax. Hence, the phase boundary blocks agreement, preventing the P head from probing into the DP to target the possessor.

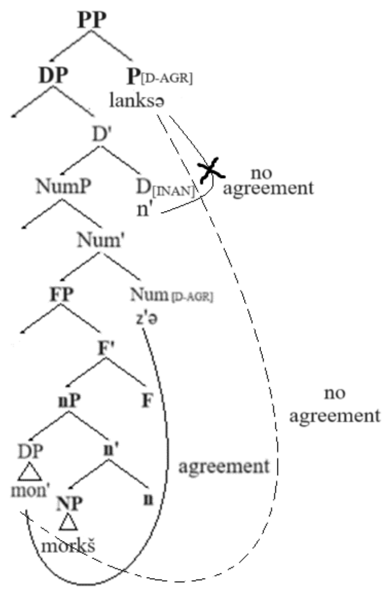
Notably, the blocking effect cannot be attributed to intervention by the genitive DP itself. If that were the case, we would expect possessor agreement to be possible when the complement is inanimate, given that inanimate complements do not trigger agreement on P. However, even inanimate genitive complements block agreement with their possessor, reinforcing the idea that it is the DP phase boundary that prevents external agreement.

Consider (256), where the complement is the inanimate DP *mon' morkšəz'ən'* 'of my table'. The postposition does not agree with the complement because it is inanimate and lacks the F1 feature required for agreement (see Section 4.3.2.1). The agreement probe on P searches for F1, but fails to find it. As we will see in the next section, the probe can keep looking further after the failure. However, after this failure, the probe does not continue searching inside the DP, even though the possessor DP does have F1 (and F2 features). This indicates that the DP phase blocks further probing, preventing agreement with the possessor.

(256)	*mon'*	*morkš-əz'ə-n'*	*lank-sə* /	**lank-sə-n*
	1SG.GEN	table-1SG.AGR.SG-GEN	on-IN	on-IN-1SG.AGR
	'on my table'.			

The ungrammaticality of 1st person singular agreement on the postposition *lanksən* in (256) confirms that a postposition cannot agree with the possessor of a genitive complement when the complement itself is a DP with internal nominal agreement. The structure of (256) is illustrated in Figure 21:

Figure 21: *Nominal agreement of Num with a possessor DP and absence of nominal agreement of P with either the complement or its possessor*

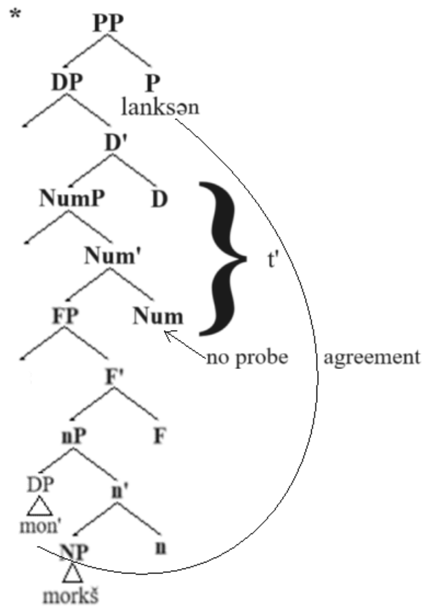


A potential question arises: Is the DP phase the only reason why P cannot agree with the possessor? Another possibility is that the Num probe inside the DP, which first agrees with the possessor, might prevent the P probe from accessing the same possessor. Unfortunately, Moksha does not provide a way to tease apart these possibilities, since the complements that have NumP also have DP. It is also impossible to have a DP with no internal agreement when a possessor is present. As long as DP is case-marked, we see a possessive marker on the head of the complement:

(257) **mon'* *morkš-**t'*** *lank-sə-n*
1SG.GEN table-DEF.SG.GEN on-IN-AGR.1SG
Int.: 'on my table.'

The structure of (257) is illustrated in Figure 22:

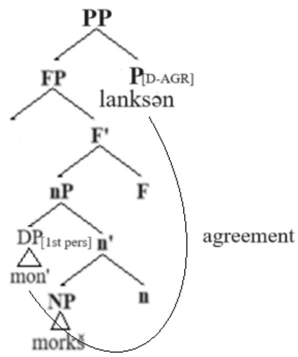
Figure 22: *(Ungrammatical) Absence of nominal agreement of Num with a possessor DP and nominal agreement of P with the possessor of the complement*



A different pattern emerges with bare complements. Because bare complements are not DPs, they do not create a phase boundary, allowing the P probe to continue searching beyond the complement itself. This enables the P head to agree with the possessor DP, as shown in Figure 23 for (258).

(258) *mon'* *morkš* *lank-sə-n*
1SG.GEN table on-IN-1SG.AGR
'on my table'

Figure 23: *Absence of nominal agreement with a possessor DP within an FP complement and nominal agreement of P with the possessor of the complement*



In contrast to PPs with genitive complements, when the complement is an F_1P (bare complement), there is no phase boundary blocking external agreement; there is no Num head competing to agree with the possessor; and the postpositional probe keeps searching and successfully agrees with the possessor DP.

This agreement pattern aligns perfectly with my small nominal analysis. Bare complements lack higher projections and are transparent to external agreement probes. Genitive complements contain DP projections that block external agreement, enforcing a strict boundary.

Before concluding this section, let me point out that the restrictions on postpositional agreement with the possessor of a bare complement are identical to the restrictions on agreement with the complement itself and different from the restrictions on the agreement associated with the Num head. In particular, pronominal possessors trigger obligatory agreement (compare ungrammatical (259) with grammatical (258)); and animate possessors trigger optional agreement (260).

(259) **mon'* *morkš* *lank-sə*
1SG.GEN table on-IN
Int.: 'on my table'

(260) *maša-n'* *morkš* *lank-sə(-nzə)*
Masha-GEN table on-IN-AGR.3SG
'on Masha's table'

Unfortunately, inanimate possessors do not provide clear data, because inanimate possessors occur in whole-part constructions — a type of inalienable possession, which is incompatible with bare complements:

(261) **morkš-t'* *pil'gə* *lank-sə(-nzə)*
table-DEF.SG.GEN leg on-IN-AGR.3SG
Int.: 'on the table's leg'

Thus, the agreement behavior of postpositions with the possessor of a bare complement matches its agreement with the complement itself, while internal nominal agreement on Num follows a different pattern. This distinction supports my analysis, which proposes the following:

- Genitive complements are DPs, while bare complements are F_1Ps.
- Num hosts an obligatory non-discriminating probe looking for a D-feature.
- P has two distinct agreement probes (looking for features F1 and F2), one of which is obligatorily present.
- DPs act as phases, blocking external agreement with their internal possessors.
- F_1Ps lack a phase head, making them transparent for external probes, including agreement with possessors.

The contrast between genitive and bare complements in Moksha follows naturally from their different syntactic sizes. Because genitive complements are full DPs, they block external agreement, while bare complements, as F_1Ps, remain accessible for external agreement probes. The agreement asymmetry between Num and P also supports this structural distinction, with Num hosting a non-selective probe and P hosting feature-specific probes. This analysis thus provides a unified account of postpositional agreement and possessor agreement patterns in Moksha.

An alternative SN-based analysis for deriving agreement of the postposition with the possessor of its bare complement would posit that, in the absence of sufficient nominal structure within the complement itself, possessor raises to a specifier in the extended projection of the postposition, and the agreement of the postposition with the possessor in its specifier is local. This contrasts with my analysis, in which the possessor remains in-situ in Spec,nP. At present, I lack empirical evidence to determine whether the possessor remains within the small nominal argument and is seen by the external probe, or whether it moves to Spec,PP. In any case, I do not consider the distinction between the two options significant for the purposes of my dissertation. Both analyses share a crucial assumption: the complement of the postposition must be smaller than a DP, thereby allowing the probe access the possessor. In my analysis, locality of agreement is ensured by the absence of a phase boundary between the probe and the goal; in the alternative analysis, it results from movement of the possessor into the probe's projection. For now, I stay with the analysis that does not require major restructuring, while leaving open the possibility that the possessor is, in fact, structurally external to the small complement.

4.4. Advantages of a small nominal analysis

As I demonstrated in Section 4.2, bare complements of postpositions lack key properties expected of DPs, which supports their classification as SNs (see Section 4.3.1). In Section 4.3.2, I further strengthened this argument by showing how the agreement patterns with possessors of such complements follow naturally from my analysis. In this section, I evaluate alternative analyses and argue that my small nominal approach provides the most explanatory and consistent account.

Suppose that contrary to my analysis, both genitive (262)a and bare (262)b complements were DPs. Then, an alternative analysis must be morphological. In this view, the bare complement in (262)b is not truly caseless but is instead a DP with a null allomorph of some case marker.

(262)	a.	*st'ər'-t'*	*lank-sə-nzə*	b.	*morkš*	*lank-sə*
		girl-DEF.SG.GEN	top-IN-AGR.3SG		table	top-IN
		'on the girl'			'on a table'	

Under this hypothesis, the null allomorph could be a different realization of the same genitive case, or a distinct case that happens to surface as null. However, any morphological explanation of the case-marking pattern must also account for the agreement pattern. In what follows, I critically assess each of these possibilities and demonstrate their shortcomings. At the end of this section, I will also look at a syntactic analysis of the variable agreement placement proposed for a similar Mongolian pattern (Lim et al. 2023) and at a variation of a small nominal analysis different from mine proposed for similar Meadow Mari data (Burukina 2023).

4.4.1. The absence of the genitive as the null allomorph of the genitive

First, let us consider the possibility that the null allomorph is the same case — the genitive. If both complements in (262) are cased DPs marked as genitive, then the difference is in the realization of genitive morphology. Specifically, the genitive case is overtly realized as *-t'* in (262)a but appears as *-Ø* in (262)b. In other words, there are two allomorphs of the genitive case, with one being null in certain environments.

Several possible explanations could account for the null realization of the genitive case in (262)b. Below, I consider five possible analyses and the challenges each one faces.

First, the absence of overt case morphology can be conditioned by adjacency with P, which allows for the case node to be filled with a null element obeying the Empty Category Principle, in spirit of Lamontagne and Travis (1987). Second, the null allomorph can be the contextual allomorph used in the context of P following D, in line with the DM framework. Third, the null realization of the genitive case may be the result of the joint exponence of case and postposition. Fourth, the null allomorph can be the contextual allomorph used in the context of P having lowered to D, which can also be modeled within DM framework. Finally, the usage of the null allomorph can be due to the absence of some feature present in (262)a, such as animacy, definiteness or number. I will consider each analysis separately discussing the problems it faces.

4.4.1.1. The absence of the genitive as the usage of a null allomorph of the genitive under adjacency: the ECP analysis

An alternative explanation for the alternation between overt and null case realization is that case drop is conditioned by adjacency to a particular head. This approach was originally proposed by Lamontagne and Travis (1987), who analyzed case-drop and complementizer-drop as instances where an overt marker fails to surface due to its proximity to a governing head. Their proposal suggests that an empty category (EC) is permissible when its features can be recovered from an adjacent head, in accordance with the Empty Category Principle (ECP). They look at the cases where the adjacency is to the V-head, but the idea is more general, so it can be easily extended to the adjacency to a postposition (P head).

Adapting this framework to Moksha, we could argue that the contrast in (263) reflects case drop, where the bare complement in (263)b lacks overt case marking because it is adjacent to P.

(263) a. *st'ər'-t'* *lank-sə-nzə*
girl-DEF.SG.GEN top-IN-AGR.3SG
'on the girl'

b. *morkš* *lank-sə*
table top-IN
'on a table'

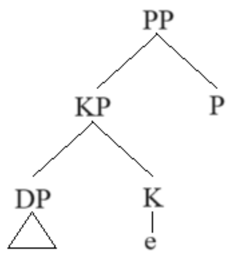
To implement this approach, several theoretical assumptions must be adopted. First, this analysis requires rejecting the post-syntactic addition of KP and instead assuming that K is introduced in narrow syntax. K must be a syntactic functional head that can have overt realization or host an empty category. Second, it is necessary to adopt the ECP, which was a part of the Government and Binding Theory, but whose status in Minimalism is still to be determined. The ECP is formulated in (264). This principle states that all empty categories must be identifiable, either through a local antecedent or by standing in a complement-of relation. Finally, case assignment must be head-driven rather than configurational. Specifically, ECP requires that features be recoverable from the head of the governor, meaning that case cannot be assigned purely based on structural position.

(264) **Empty Category Principle** (Lamontagne & Travis 1987: 177-180):
(i) All empty categories (Xs and XPs) must be identified.
(ii) a. The position of the gap is identified iff
- it has a local antecedent, or
- it (or its maximal projection) must stand in the relation 'complement of'
b. The contents of the gap are identified through feature recoverability (subject to Head Feature Transmission Constraint)

(265) **Head Feature Transmission Constraint**: Head features may only be transmitted from a head to its sister.

The structural configuration for case drop is therefore as in Figure 24:

Figure 24: *The structural configuration for case drop*

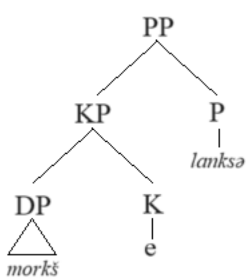


Now, let us apply this analysis to the Moksha data and derive the PPs in (263) repeated here as (266). Under this view, P assigns the genitive case, meaning that the [GEN] feature is present on P. In (266)a, K is overtly realized, while in (266)b, K is empty but satisfies ECP because KP is P's complement, and [GEN] can be recovered on K. This explains why bare complements are possible under adjacency to P, see Figure 25.

(266) a. *st'ər'-t'* *lank-sə-nzə*
girl-DEF.SG.GEN top-IN-AGR.3SG
'on the girl'

b. *morkš* *lank-sə*
table top-IN
'on a table'

Figure 25: *The bare complement of P as an EC*



While this approach accounts for the bare complement pattern in (266)b, it fails to explain the ungrammaticality of certain bare complements, particularly animate ones like (267).

(267) **st'ər'* *lank-sə (-nzə)*
girl on-IN-AGR.3SG
Int.: 'on the girl'

Moreover, this analysis faces three major challenges related to key generalizations in Moksha:

(268) a. **Overt marking generalization**:

If number, nominal agreement, or definiteness is overtly expressed, the complement of the postposition must be genitive.

b. **Category restriction generalization**:

If the complement of the postposition is a pronoun or an animate proper name, it must be genitive.

c. **Featural restriction generalization**:

If the complement of a postposition is a noun denoting a human or an inalienably possessed object, it must be genitive.

In order for the ECP analysis to explain these generalizations, we need to stipulate that the 'complement of' relation in the case of the nominals that must be overtly genitive has been "broken" in some way. The way to achieve this would be stipulating that pronouns, nominals denoting humans and nominals marked as definite or possessed move out from the position of the complement of P, presumably for licensing purposes (e.g., the features [human], [definite], etc. must be licensed by some head located higher in the structure). However, this creates a timing paradox. If movement happens before case assignment, these DPs would have no source of genitive case, contradicting the data. If, on the other hand, case assignment happens before movement, then adjacency should still hold, making case drop possible. This theoretical inconsistency weakens the plausibility of the ECP analysis.

Another problem with the ECP account is that it does not restrict case drop to a specific syntactic position (P-complements in our case), as long as the case features are recoverable from the governor-head. If K can be bare in one position satisfying ECP, it should be able to be bare in any other position satisfying ECP. For instance, the same genitive case is assigned to the DO within the VP, so we expect the complements of V to obey the same restrictions. In particular, a direct object (DO) should be able to appear bare as long as it is not marked as definite or possessed, is not a pronoun or a nominal denoting a human. While Moksha does allow bare DOs, as in (269), their distribution is not tied to structural adjacency to the verb, as they can appear in different clause positions (270).

(269)	*kunc'-əmə*	*kal-n'a-t*	*i*	*pic'ə-mə*	*kal-ən'*	*l'ɛm-n'ɛ*
	catch.FREQ-PST.1PL	fish-DIM-PL	and	boil.FREQ-PST.1PL	fish-GEN	soup-DIM

‘Caught little fish and boiled fish soup.’ (KMT, «Rybalka [Fishing]»)

(270)	*spuskən'ɛ-t'i*	*kaja-sajn'ə,*	*šor'a-sajn'ə*
	braga_pail-DEF.SG.DAT	pour-NPST.3PL.O.1SG.S	mix-NPST.3.O.1SG.S

l'ɛmbə …	***kaj-an***	*dagə*	*lang-əzə-st*	***ravžə***	***počf***
warm …	pour-NPST.1SG	ADD	top-ILL-3PL.AGR	black	flour

‘I pour it into the braga pail, mix with warm… also, put rye flour there.’(KMT, «Recept bragi [Braga recipe]»)

Even if we abandon the theoretical assumptions made in Chapter 1 and adopt those required for an ECP analysis, this approach faces two insurmountable problems: a theoretical and an empirical problem. The theoretical problem with this analysis is the timing of case assignment and movement triggered by the need of licensing features on D. The empirical problem is undergeneration; this analysis predicts that no bare complements can be moved out of their base-generation position, which is not borne out.

My SN analysis, on the other hand, avoids the problem of the movement and case assignment timing. It also makes predictions specifically regarding the position of the complement of a postposition, and the same behavior should not be expected from complements in other positions, see the discussion of the Moksha Differential Object Marking in Chapter 5.

4.4.1.2. The absence of the genitive as the usage of a null allomorph of the genitive under adjacency: the contextual allomorphy analysis

Another possible explanation for the alternation between an overt and a null realization of the genitive case is contextual allomorphy. In this approach, the presence of an adjacent node determines the shape of the morpheme that realizes a given feature. Unlike the ECP analysis, this dependency is established post-syntactically, within the morphological component, without invoking syntactic principles like the ECP. The key idea is that the bare complement appears without an overt case marker because of a morphological rule requiring the genitive to be realized as null when adjacent to a P.

I will model this within the DM framework (see Chapter 1, Section 1.2.1), in which syntactic nodes — bundles of grammatical features — receive phonological realization through vocabulary insertion at a late stage of derivation. Under this proposal, the null genitive allomorph is conditioned by adjacency to the P head.

In DM, the process of assigning morphological forms to structural nodes is called vocabulary insertion. Structural nodes and vocabulary items (morphemes) are each specified for particular sets of features. A vocabulary item is inserted if its features match a subset of those on the structural node. When multiple vocabulary items can realize the same node, the most highly specified item is chosen.

Although I assume that KP is not present in the narrow syntax (see Chapter 2), I propose that by the time of vocabulary insertion, the K node has been added post-syntactically as a dissociated morpheme (Embick 1997) to host the case marker. Additionally, I assume that fusion is available in Moksha, meaning that two adjacent nodes can merge into a single unit containing features from both (Halle & Marantz 1993: 116). The exact conditions that trigger fusion usually are not discussed, but I assume that in Moksha, fusion occurs due to the absence of an independent vocabulary item capable of realizing certain node configurations.

Now, consider the following proposed vocabulary items for the Moksha genitive case, capturing the contextual dependency on P:

(271) Vocabulary items for Moksha genitive:

a. [singular; def*; genitive] <-> /t'/, where [def*] is a formal feature whose distribution goes beyond canonical definiteness; it is present on nominals that bear a definite determiner

b. [inanimate; genitive] <-> -Ø /_P (i.e., null genitive before a postposition)

c. [genitive] <-> /-ən'/ (elsewhere case marker)

This specification of morphemes raises two potential concerns. First, the lack of an unconditional inanimate genitive marker. Second, the paradigm structure is shaped by feature specification: the only feature that is common across the vocabulary items is [genitive], and it may seem that they do not belong to the same paradigm.[22] So, I will address these issues first.

When it comes to the first issue, the vocabulary item in (271)b is conditioned by adjacency to P, but there is no corresponding inanimate genitive marker that appears in other contexts in Moksha. If we introduce such a morpheme, as in (272)c, we miss the generalization that *-ən′* is also the elsewhere genitive form.

(272) Vocabulary items for Moksha genitive: revision 1:

a. [singular; def*; genitive] <-> /t'/

b. [inanimate; genitive] <-> -Ø /_P

---

[22] I thank Maša Bešlin for sharing her concerns with me.

c. [inanimate; genitive] <-> /-ən'/

d. [genitive] <-> /-ən'/

At the same time, if we proposed that the null form is the conditioned counterpart of the elsewhere genitive form, like in (273)b, we would generate ungrammatical examples like (274), where an animate complement is bare.

(273) Vocabulary items for Moksha genitive: revision 2:

a. [singular; def*; genitive] <-> /t'/

b. [genitive] <-> -Ø /_P

c. [genitive] <-> /-ən'/ elsewhere

(274) **st'ər' lank-sə*
girl top-IN
Int.: 'on a girl'

Now let me address the second issue. The vocabulary items in (271) appear to belong to different paradigms — (271)a is specified for number and definiteness, while (271)b is specified for animacy. However, all three items can lexicalize the fused n+Num+D+K node. The features [+/- animate] and [+/- inanimate] originate in n, [+/- singular] and [+/- plural] in Num, [+/- def]* in D, and [+/- genitive] in K. Because the fused node carries all these features, the vocabulary items in (271) all have a subset of the features on this node. vocabulary insertion applies across the paradigm. Some of the morphemes will contain more relevant features, some will contain extra features, and the most highly specified morpheme will win via the Subset Principle (see Chapter 1, Section 1.2.1.1).

Given the vocabulary insertion rules in (271), let us derive the forms in (275), which will result in (276).

(275) a. *st'ər'-t'* *lank-sə-nzə*
girl-DEF.SG.GEN top-IN-AGR.3SG
'on the girl'

b. *morkš* *lank-sə*
table top-IN
'on a table'

(276) a. [root][animate; singular; def*; genitive] → *st'ər'-t'*

b. [root][inanimate; singular; genitive] → *morkš-Ø*

In (276)a, the fused n+Num+D+K node[23] is also specified as animate, but this does not prevent *-t'* from being mapped to it, as it is still the morpheme specified for the largest subset of the needed features. In (276)b, *-t'* does not get inserted, because the complement does not bear the feature [def*]. The morpheme *-n'* is basically the elsewhere option for genitive, and -Ø wins over it as more specified.

This analysis correctly predicts that bare indefinite inanimate nouns appear adjacent to postpositions. However, it faces problems when applied to other syntactic positions, such as DO position. In Moksha, bare indefinite inanimate DOs are also possible, as in (277). Furthermore, even animate DOs can be bare, unlike postpositional complements (278).

(277) *vas'ɛ* *ker'-s'* ***šuftə***
Vasja cut-PST.3[SG] tree
'Vasja cut a tree.' (Toldova 2018: 575)

(278) *mon* *n'ɛj-ən'* *pɛk* *mazi* ***sen'əm*** ***s'el'mə*** ***s't'ər'-n'ɛ***
1SG[NOM] see-PST.1SG very pretty blue eye girl-DIM
'I saw a very beautiful blue-eyed girl.'

---

[23] I omit the discussion of why the nodes are fused in this case, as it would take us too far afield. However, the vocabulary items I propose are such that, proceeding from left to right, no nodes can be lexicalized until they are merged together in the resulting node.

In both (277) and (278), the noun is neither adjacent to a P nor a V. This contradicts the prediction that null genitive is conditioned by adjacency to P, indicating that the analysis undergenerates. I remove the contextual dependency condition and entertain a similar morphological analysis with a different specification of morphemes in Section 4.4.1.5.

The bottom line is that, while the contextual allomorphy account improves upon the ECP-based analysis by avoiding unnecessary syntactic assumptions, it ultimately fails to capture the full range of data. Specifically, it incorrectly predicts that bare nouns should only occur adjacent to postpositions, while bare DOs — both animate and inanimate — are attested in various positions. In contrast, my small nominal analysis (see Chapter 5) avoids these complications altogether by making position-specific predictions, allowing for a more accurate account of genitive marking in Moksha.

4.4.1.3. The absence of the genitive as the result of joint lexicalization of K and P

Adjacency to P can result in what appears to be a null realization of the genitive case through the joint lexicalization of adjacent K and P nodes by a single morpheme — the postposition. This idea was proposed by Dékány (2009) for Hungarian "dressed" postpositions, which take a complement without overt morphological case and bear nominal agreement on the head.

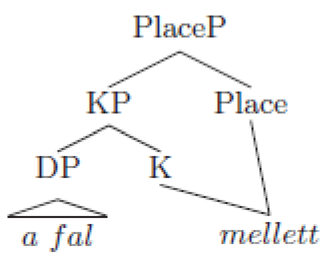
If this analysis were applied to Moksha, the key distinction between genitive and bare complements would be that, in genitive complements, K is lexicalized separately or together with Num and/or D. In contrast, in bare complements, K is lexicalized together with P.

Dékány develops her analysis within the Nanosyntactic framework, which, like Distributed Morphology (DM) (see Chapter 1.2.1), is a non-lexicalist approach to syntax. Both frameworks assume that terminal nodes in the hierarchical structure are abstract, with morpheme insertion occurring late in the derivation. However, unlike DM, Nanosyntax treats each feature as an individual node, though morphemes may still lexicalize multiple structural nodes. For nodes to be lexicalized together, they must be adjacent.

An example of joint lexicalization of K and P in a Hungarian PP with a bare complement (279) is shown in Figure 26.

(279) HUNGARIAN (Dékány 2009: 56)
a *fal* *mellett*
DEF wall next_to
'next to the wall'

Figure 26: *Joint lexicalization of K and P[24] in Hungarian dressed postpositions* (Dékány 2009: 56)

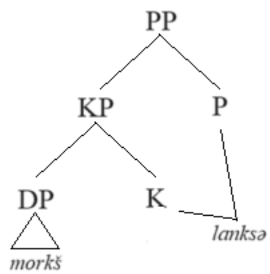


A similar Nanosyntactic structure can be proposed for Moksha PPs with a bare complement (280), as illustrated in Figure 27:

(280) *morkš* *lank-sə*
table top-IN
'on a table'

[24] Dékány assumes that locative postpositions realize the projection PlaceP, which is part of the articulated structure of PP she adopts. Labeling this as either PP or PlaceP has no effect on this analysis or my implementation of it.

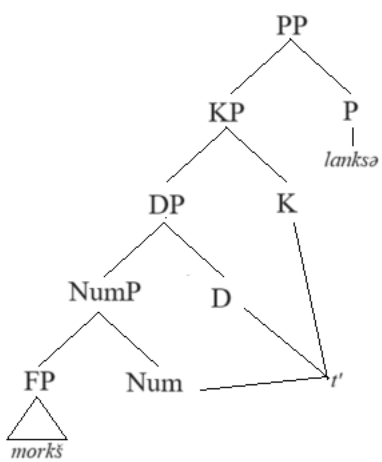
Figure 27: *Joint lexicalization of K and P in Moksha postpositions*



By contrast, when the complement is genitive-marked (281), K and P are lexicalized separately, as illustrated in Figure 28:

(281) *morkš -t'* *lank-sə*
table-DEF.SG.GEN top-IN-AGR.3SG
'on the table'

Figure 28: *Separate lexicalization of K and P in Moksha postpositions*

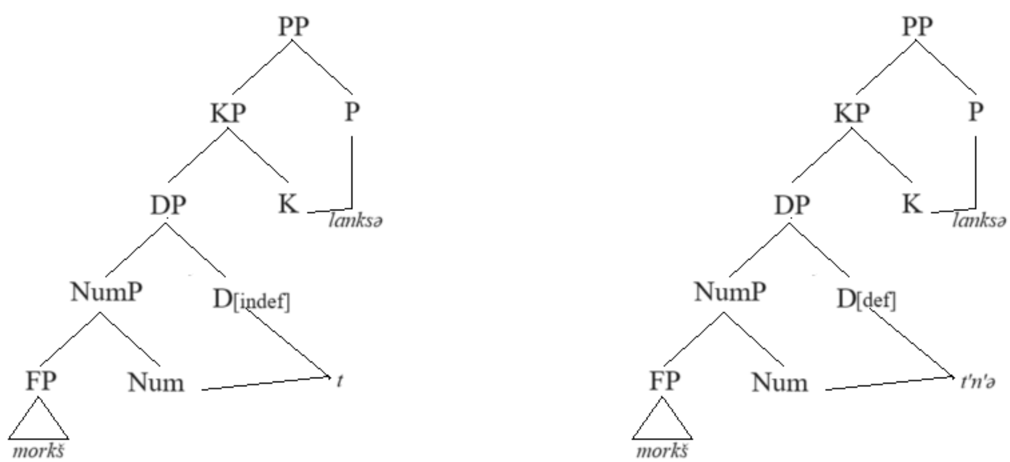


While this approach provides a possible explanation for the null realization of the genitive case, it faces several challenges when applied to Moksha. The first challenge is the lack of a fixed classification of postpositions. In Hungarian, postpositions fall into two distinct classes, one of which takes cased complements, while the other takes caseless complements. In Moksha, however, the same postposition can take either a genitive or a bare complement. Therefore, it is unclear what determines whether K and P are jointly lexicalized in Moksha in some cases but not others. The second challenge is the failure to explain the lack of number marking in bare complements. The nanosyntactic analysis would predict ungrammatical structures like (282), see Figure 29a-b, and therefore, overgenerate.

(282) **morkš-t* / *morkš-t'n'ə* *lank-sə*
table-PL table-DEF.PL top-IN
Int.: 'on (the) tables'

Figure 29: *Number expression in the context of joint lexicalization of K and P*

(a) (b)



The final challenge is the incorrect prediction for agreement placement. Dékány (2009) proposes that in Hungarian, agreement does not have a dedicated projection and cliticizes to the morpheme that lexicalizes K. In cased complements, K is separate, so agreement attaches to K (283)a. In bare complements, K and P are lexicalized together, so agreement attaches to P (283)b.

(283) HUNGARIAN
a. *én-vel-em* *szemben*
I-INSTR-1SG opposite
'opposite to me' (Dékány 2009: 65)

b. *én alatt-am*
I under-1SG
'under me' (Dékány 2009: 48)

In Moksha cased complements, however, agreement markers precede the case marker rather than follow it. This contradicts Dékány's cliticization explanation, making joint lexicalization of K and P inadequate for accounting for Moksha agreement placement.

(284) *morkš-əz'ə-n'* *lank-sə*
table-1SG.AGR.SG-GEN top-IN
'on my table'

Rather than relying on superficial processes like joint lexicalization and cliticization, my SN analysis takes into account structural differences between genitive and bare complements. This approach avoids the need for variable exponence rules; correctly predicts the lack of number marking in bare complements; and explains agreement placement in a way that aligns with Moksha-specific patterns.

By treating bare complements as structurally distinct from genitive-marked complements, my analysis offers a more accurate and consistent account of Moksha postpositional structures.

4.4.1.4. The absence of the genitive as the result of Lowering of P

Another possible explanation for the realization of genitive case as different allomorphs is the intervention of an additional node, which disrupts the adjacency between two other nodes, thereby altering how they map onto phonological forms. Specifically, I explore the post-syntactic downward head movement of P (Lowering), which modifies the structural sequence of nodes bleeding the application of certain post-syntactic operations and preventing the insertion of overt genitive morphemes. Additionally, the Lowering of P results in nominal agreement appearing on the postposition rather than on the head noun of the complement.

I base this analysis of Moksha PPs on McFadden (2004), who proposed a Lowering-based account for nominals in spatial cases (treated as PPs) in both Mordvin languages, Erzya and Moksha. His primary goal was to explain the unusual placement of nominal agreement markers, which follow spatial case markers (285)a rather than preceding them, as in other cases such as genitive or dative (285)b.

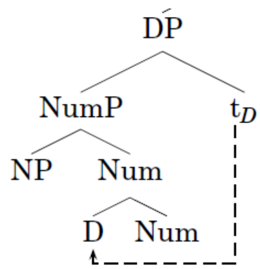
(285) a. *vel'ə-sə-**n*** [village-IN-AGR.1SG] 'in my village / in my villages'

b. *vel'ə-**z'ə**-n'* [village-1SG.AGR.SG-GEN] 'of my village'

If McFadden's analysis accounts for the unexpected placement of agreement after the spatial case marker, it may also explain why agreement appears on the right of the postposition instead of on the left. In addition to accounting for the agreement pattern, this analysis should also be compatible with the observed case alternation.

Lowering is a post-syntactic operation within the DM framework (see Chapter 1, Section 1.2.1). It involves the downward movement of a head in the hierarchical structure derived from narrow syntax, causing the head to adjoin to the left of the head of its complement (Embick & Noyer 2001), as illustrated in Figure 30. Once Lowering has taken place, the structure undergoes linearization, and the nodes are then mapped to their corresponding phonological forms according to insertion rules.

Figure 30: *Lowering* (from Guseva & Weisser 2018:16)

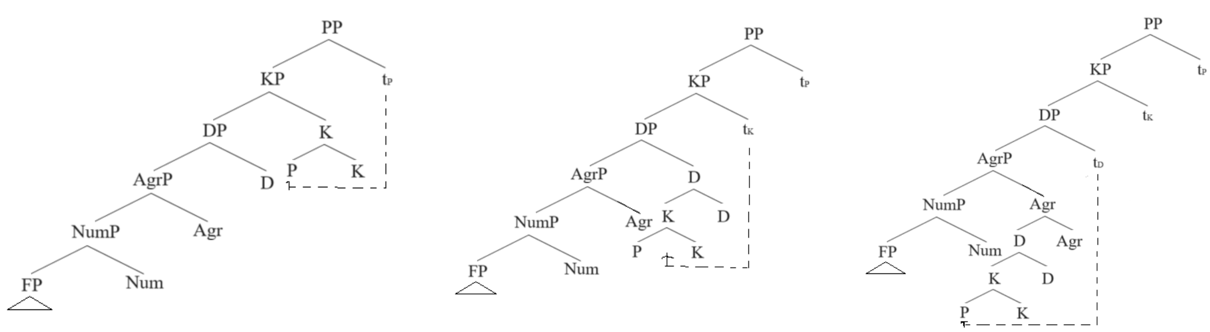


For Moksha PPs, the proposed Lowering process operates as follows. Without Lowering, the sequence of functional nodes available for lexicalization appears as in (286). At this stage of the derivation, the node for morphological agreement (AgrP) and the node for morphological case (KP) have been added (see Chapter 1, Section 1.2.1.2 on the insertion of "ornamental" morphemes). However, if the P head lowers to D, or in possessed nominals, to Agr (which yields the correct agreement placement), the order of nodes is significantly altered, as shown in (286). The step-by-step process of P Lowering to Agr is depicted in Figure 31.

(286) a. Num(+Agr)+D+K+P

b. Num+P+K+D(+Agr)

Figure 31: *Sequential Lowering of P to Agr*



The vocabulary items available for lexicalizing nodes containing genitive case are as follows:

(287) Vocabulary items for the genitive case:

a. [singular; def*; genitive] < → /t'/

b. [genitive] <-> -Ø /_D

c. [genitive] <→ /-ən'/

I assume that Moksha lacks vocabulary items specified exclusively for number (288), which means that Num cannot be lexicalized on its own due to the Subset Principle. This leads to the fusion of Num with adjacent nodes, such as Num+Agr, Num+D, or Num+D+K.

(288) Vocabulary items for number[25]:

a. [singular; def*; nominative] <→ /s'/

b. [singular; def*; genitive] <→ /t'/

c. [plural; def*] <-> /n'ə/

d. [plural; indefinite] <-> /t/

e. [singular; 1sg] <-> /z'ə/

f. [plural; 1sg] <-> /n'ə/

When a sequence like (286) is due exponence, Num fuses with Agr (or D if agreement is absent), after which K is realized separately, followed by the exponence of P. Consider (289). The PP in (289)a enters the derivation as (289)b. Due to the morphological requirements of Moksha, Agr and

[25] The list for the possessive forms is not exhaustive, but the other person and number combinations behave similarly.

K are inserted into the structure (289)c[26]. Num and Agr then fuse and are lexicalized as (288)e, yielding (289)d. Next, D fuses with K, and the combined node D+K is lexicalized as (287)c, producing (289)e. Finally, P is lexicalized as *lanksə*, using the simplified vocabulary item for superessive P in (290) .

(289) a.

mon'	*morkš-əz'ə-n'*	*lank-sə*
1SG.GEN	table-1SG.AGR.SG-GEN	on-IN

‘on my table’

b. $[[[[[[\textit{mon'}]_{DP}\ [\text{root}]_{NP}\ \text{n}]_{nP}\ \text{F}]_{FP}\ \text{Num}_{[SG]}]_{NumP}\ \text{D}_{[GEN]}]_{DP}\ \text{P}_{[SUPER;ESS]}]_{PP}$

c. $[[[[[[[[\textit{mon'}]_{DP}\ [\text{root}]_{NP}\ \text{n}]_{nP}\ \text{F}]_{FP}\ \text{Num}_{[SG]}]_{NumP}\ \text{Agr}]_{AgrP}\ \text{D}]_{DP}\ \text{K}_{[GEN]}]_{KP}\ \text{P}_{[SUPER;ESS]}]_{PP}$

d. $[[[[\textit{mon' morkšəz'ə}]\ \text{D}]_{DP}\ \text{K}_{[GEN]}]_{KP}\ \text{P}_{[SUPER;ESS]}]_{PP}$

e. $[[\textit{mon' morkšəz'ən'}]_{KP}\ \text{P}_{[SUPER;ESS]}]_{PP}$

(290) Vocabulary item for Superessive P in Moksha (simplified):

[super; ess] <-> /lanksə/

Next, consider the derivation of (291)a. The PP enters the morphological component in the same way as (289)a, and the initial steps of the derivation mirror (289). However, the P head then undergoes Lowering, adjoining to the left of K (291)b, followed by a second Lowering step where the combined P+K head adjoins to D (291)c. Finally, this joint head P+K+D undergoes further Lowering and adjoins to the left of the morphological agreement node AgrP (291)d.

(291) a.

mon'	*morkš*	*lank-sə-n*
1SG.GEN	table	on-IN-AGR.1SG

‘on my table’

[26] Even though, in this derivation, Num+Agr and D+K are fused together again—making the insertion of Agr and K appear vacuous — this operation does not seem vacuous in other derivations. I therefore assume it to be an obligatory and automatic step, adopted for analytical uniformity.

b. $[[[[[[[[\mathit{mon'}]_{DP}\ [root]_{NP}\ n]_{nP}\ F]_{FP}\ Num_{[SG]}]_{NumP}\ Agr]_{AgrP}\ D]_{DP}\ P_{[SUPER;ESS]i}+K_{[GEN]}]_{P+KP}\ e_i]_{PP}$

c. $[[[[[[[[\mathit{mon'}]_{DP}\ [root]_{NP}\ n]_{nP}\ F]_{FP}\ Num_{[SG]}]_{NumP}\ Agr]_{AgrP}\ (P_{[SUPER;ESS]i}+K_{[GEN]})_j+D]_{P+K+DP}\ e_j]_{P+KP}\ e_i]_{PP}$

d. $[[[[[[[[\mathit{mon'}]_{DP}\ [root]_{NP}\ n]_{nP}\ F]_{FP}\ Num_{[SG]}]_{NumP}\ ((P_i+K_{[GEN]})_j+D)_l+Agr]_{AgrP}\ e_l]_{P+K+DP}\ e_j]_{P+KP}\ e_i]_{PP}$

After these structural modifications, the derivation proceeds through linearization, where each node, now containing feature bundles, is lexicalized by a vocabulary item that is specified for the corresponding features (or a subset of them).

Since Num cannot get lexicalized on its own due to the lack of corresponding vocabulary items, it gets fused with the following node, which is P. The resulting node gets lexicalized with (290). This item is underspecified for number, which is permitted by the Subset Principle due to the absence of a more specified alternative that suits the node and there are no extra features not present in the node. Next, K is lexicalized as (287)b, influenced by the presence of D in its immediate right context. Finally, D fuses with Agr and is realized as (292).

(292) Vocabulary item for 1SG agreement in Moksha:
[1sg] <-> /n/

Thus, the Lowering analysis accounts for genitive allomorphy by altering the sequence of nodes, thereby modifying the order of exponence and triggering different morphemes. It also explains agreement marker allomorphy: *-z'ə* appears when 1st person features are bundled with number, while *-n* is inserted otherwise. Finally, this analysis derives the differing placements of agreement markers.

Despite its strengths, the Lowering analysis struggles with both undergeneration and overgeneration, particularly in the obligatoriness of agreement when it appears on the postposition.

I begin with the problem of undergeneration. If agreement with a third-person possessor DP is obligatory, as seen in (293)a, the lowering analysis fails to explain why it becomes optional when agreement appears on the postposition, as in (293)b. Under this analysis, structures like (293)b without possessor agreement would not be generated.

(293) a.

Maša-n'	*morkš-ənc*	/	**morkš-t'*	*lank-sə*
Masha-GEN	table-3SG.AGR.SG.GEN		table-DEF.SG.GEN	top-IN

b.

Maša-n'	*morkš*	*lank-sə* /	*lank-sə-nzə*
Masha-GEN	table	top-IN	top-IN-AGR.3SG

'on Masha's table'

At the same time, the lowering analysis leads to overgeneration by incorrectly predicting the acceptability of PPs like (294)a, which are ungrammatical in Moksha. The issue arises because, in the absence of lowering, agreement with inanimate possessors is obligatory — making (294) parallel to (293)a. However, unlike (293)b, PPs such as (294)b are ungrammatical, and the lowering analysis does not provide a principled reason why they should not be generated.

(294) a.

morkš-t'	*pil'gə-nc*	/	**pil'g-t'*	*lank-sə*
table-DEF.SG.GEN	leg-3SG.AGR.SG.GEN		leg-DEF.SG.GEN	top-IN

b.

**morkš-t'*	*pil'gə*	*lank-sə* /	*lank-sə-nzə*
table-DEF.SG.GEN	leg	top-IN	top-IN-AGR.3SG

'on the table's leg'

Thus, while the lowering analysis successfully derives the basic agreement pattern, it fails to account for the asymmetry between agreement marking on the noun head and the postposition. In contrast, my SN analysis resolves this issue by positing two distinct probes responsible for the different agreement placements. Specifically, the agreement marker preceding the case marker reflects successful agreement of the probe on Num, while the agreement marker following the case marker reflects agreement of the probe on P.

4.4.1.5. The absence of the genitive as the usage of a null allomorph of the genitive: different feature specification analysis

In this section, I explore an approach where the null allomorph arises due to the absence of certain nominal features. Specifically, in Moksha, the morpheme specified for inanimate and genitive is phonologically null, unlike other genitive allomorphs, which are conditioned by different feature sets. Following the DM framework (see Chapter 1, Section 1.2.1), I assume late insertion of morphemes, with phonological realizations listed in the vocabulary alongside the abstract features they lexicalize. According to the Subset Principle, the morpheme specified for the largest subset of relevant features — without additional, incompatible ones — gets inserted.

Given the available genitive allomorphs in the Moksha case paradigm, the following vocabulary items for genitive can be proposed[27]:

(295) Vocabulary items for the genitive case:

a. [singular; def*; genitive] < → /t'/
b. [inanimate; genitive] <-> /Ø/
c. [genitive] <→ /-ən'/

With these specifications, the following derivations (297) account for the complements in (296):

[27] For concerns regarding the vocabulary items in (295) being specified for disparate features and not forming a coherent paradigm, see the discussion following (374) in Section 4.4.1.2 above.

(296)	a.	*st'ər'-t'*	*lank-sə-nzə*	b.	*morkš*	*lank-sə*
		girl-DEF.SG.GEN	top-IN-AGR.3SG		table	top-IN
		'on the girl'			'on a table'	

(297) a. [root][animate; singular; def*; genitive] → *st'ər'-t'*

b. [root][inanimate; singular; indefinite; genitive] → *morkš-Ø*

The morpheme in (295)a is underspecified for animacy but contains the largest subset of features matching the node in (297)a. In (297)b, (295)a is ruled out due to its extra feature [def*], making (295)b the best fit under the Subset Principle, even though it lacks number and indefiniteness specification.

The derivation of a regular possessive phrase with agreement on the noun head and genitive case in (298) follows from the vocabulary items in (295) and the number and agreement morpheme in (299). Number gets realized together with agreement, and then the genitive case gets realized with the elsewhere option for genitive — /n'/.

(298) [root][inanimate; singular; 1sg][genitive] → *morkš-əz'ə-n'*

(299) Vocabulary item for number and agreement:

[singular; 1sg] <-> /z'ə/

However, this analysis overgenerates when dealing with plural nominals. Consider the vocabulary items for plural, which align with those proposed in Section 4.4.1.4:

(300) Vocabulary items for number:

a. [singular; def*; nominative] <⟶ /s'/

b. [singular; def*; genitive] <⟶ /t'/

c. [plural; def*] <-> /n'ə/

d. [plural; indefinite] <-> /t/

e. [singular; 1sg] <-> /z'ə/

f. [plural; 1sg] <-> /n'ə/

When an indefinite plural complement undergoes lexicalization, this analysis incorrectly predicts the following ungrammatical form:

(301) [root][inanimate; plural; indefinite] [genitive] → **morkš-t-ən'*

Additionally, bare complements of postpositions are generally incompatible with plural morphology. However, since plural marking is structurally and linearly too far from the P head, it cannot be conditioned by it (see Section 4.4.1.4 for an attempt to resolve this via Lowering). A more generalized modification of plural vocabulary items would cause undergeneration of plural forms in positions other than the complement of a postposition.

In contrast, my SN analysis naturally accounts for the absence of plural morphology on bare complements of postpositions, while preserving plural marking in other syntactic environments. This follows from the structural absence of the number feature in bare complements of postpositions, eliminating the need for additional stipulations.

4.4.2. The absence of genitive as the null allomorph of some other case

Now, let us consider the possibility that bare complements of postpositions bear a null allomorph of a case other than genitive. For the purposes of this discussion, identifying the exact label of this case — whether it corresponds to a well-known case in the Moksha nominal paradigm (e.g., nominative) — is unnecessary. Instead, I will refer to it as the complement case.

Under this analysis, the complement in (302)a is assigned genitive case, while the complement in (302)b is assigned complement case:

(302)	a.	*st'ər'-t'*	*lank-sə-nzə*	b.	*morkš* *lank-sə*
		girl-DEF.SG.GEN	top-IN-AGR.3SG		table top-IN
		'on the girl'			'on a table'

I assume a strict version of the theory of case, where case assignment by a particular head or in a given configuration does not allow free variation. Consequently, for variable case assignment to be possible, one must assume either two distinct case assignment positions within the PP, or that the complement case arises as a default, assigned when genitive case fails to be assigned[28].

4.4.2.1. Two case assignment positions within PP

I will outline the former analysis first. The idea of having two case assignment positions within the PP parallels the object shift analysis within the clause. In object shift, certain features trigger movement, affecting case assignment. Applying this idea to Moksha, bare complements of postpositions remain in situ and receive complement case, while genitive complements move and receive genitive case.

For movement to be possible, PP must have an articulated structure, as illustrated in Figure 32. I adopt Svenonius' (2006) Axial Part model, to reflect the lexical content of the postposition and represent the lexical head in the extended projection of P (see Chapter 2, Section 2.2 for extended projection). Additionally, I introduce pP, analogous to vP, to serve as a phase head in the P domain. While I remain agnostic about other functional projections (e.g., P_1, P_2), their presence does not affect the core analysis.

[28] I thank Alec Marantz for pointing out this logical possibility to me.

Under this architecture, the complement in (302)a moves to Spec,pP and receives genitive case (Figure 33), whereas the complement in (302)b stays in situ (Figure 32) and receives complement case:

Figure 32: *The articulated structure of PP with a DP complement in-situ*

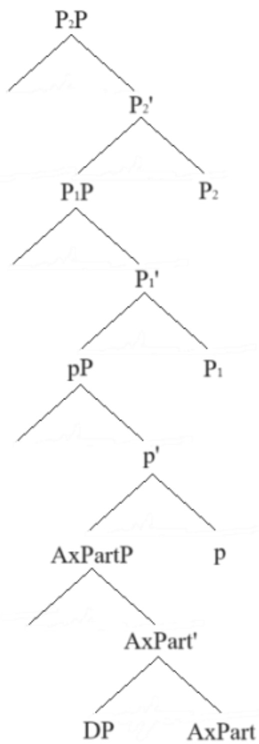
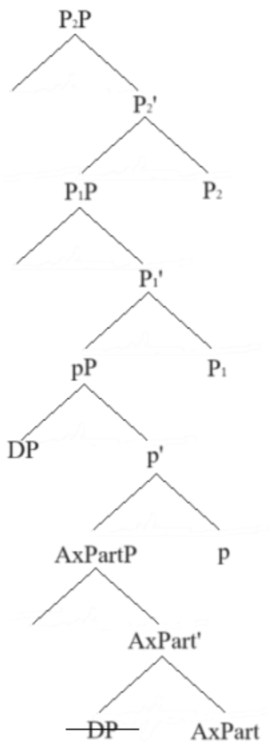


Figure 33: *The articulated structure of PP with a DP complement moved to Spec,pP*



Although as a head-final language Moksha provides no data that would allow us to make an argument for such articulated structure, A split PP structure is not a novel proposal (see Svenonius 2006; Den Dikken 2010; Koopman 2010). So let us assume that we still can adopt the new architecture, and the trees in Figure 32 and Figure 33 represent the structure for PPs with bare and genitive complements respectively. Now, we have to explain why some DPs move to Spec,pP. We could propose that there is a feature on some DPs that gets attracted by p. Ideally, this feature would constitute a natural class of DPs that always get genitive marking (303).

If we adopt the movement-based analysis, we must explain why some DPs move to Spec,pP. One possibility is that p attracts a specific feature present in DPs that require genitive marking.

Ideally, this feature would constitute a natural class of DPs that always get genitive marking, as this aligns with the following empirical generalizations:

(303) a. **Overt marking generalization**:

If number, nominal agreement, or definiteness is overtly expressed, the complement of the postposition must be genitive.

b. **Category restriction generalization**:

If the complement of the postposition is a pronoun or an animate proper name, it must be genitive.

c. **Featural restriction generalization**:

If the complement of a postposition is a noun denoting a human or an inalienably possessed object, it must be genitive.

A simple definiteness feature is insufficient, because possessive phrases can be definite and indefinite, but in either case always get genitive marking. as possessive phrases can be definite or indefinite, yet always take genitive marking. To account for this, we might posit a purely formal feature [def*], akin to Danon's (2001) formal definiteness feature in Hebrew, which regulates syntactic processes rather than semantic definiteness. Let us assume that there is some mechanism that is responsible for the marking of DPs as [def*].

A mechanism that will ensure that a DP has the feature [def*] is needed for the declension choice anyway. However, this feature-based approach encounters a problem: animate nouns require genitive case, but there is no clear explanation for why they must also bear [def*]. If [def]* were required for animate nouns, they should always appear with definite determiners or nominal agreement in all syntactic positions — yet in direct object positions, they do not (compare (304) to (305) below).

(304)	*st'ər'-t'* /	**st'ər'*	*lank-sə (-nzə)*
	girl-DEF.SG.GEN	girl	on-IN-AGR.3SG
	'on the girl'		

In my account, the stipulation is that animate nouns require the full extended projection up to DP, which will result in the marking appropriate for the DP in the position of the complement of a spatial postposition, which is genitive marking. In this morphological account, on the other hand, the stipulation should be that animate nouns must be also marked as [def*]. The prediction this account makes is that animate nouns must always bear a definite determiner or a nominal agreement marker, including other syntactic positions. Since animate nouns in DO position can remain bare, the [def*] feature would make incorrect predictions about where genitive marking is obligatory. Thus, a formal analysis with a particular feature is more problematic than the SN analysis, which requires the difference in the nominal size in a particular syntactic position.

(305)	*mon*	*n'ɛj-ən'*	*pɛk*	*mazi*	*sen'əm*	*s'el'mə*	*s't'ər'-n'ɛ*
	1SG[NOM]	see-PST.1SG	very	pretty	blue	eye	girl-DIM
	'I saw a very beautiful blue-eyed girl.'						

Moreover, an analysis that ties genitive marking to the presence of an agreeing probe that attracts the DP faces further problems when considering morphological agreement within the PP. If genitive complements move to Spec,pP due to an agreement probe, it is reasonable to assume that the morphological agreement is the result of the same agreement operation. Then we expect a one-to-one correspondence between case and morphological agreement: all genitive DPs must trigger agreement, and all non-genitive DPs should lack agreement.

However, Moksha does not exhibit this pattern. Animate genitive DPs optionally trigger agreement, and inanimate genitive DPs do not trigger agreement at all.

To salvage the movement account, we would need another probe targeting a different formal feature (F1), similar to the feature used in my SN analysis (see Section 4.3). The derivation for the grammatical PP in (304) would then involve multiple steps. The complement DP comes with the set of features in (306)a. Then the probe on p probes for [def*], attracting the DP to Spec,pP, where it receives genitive case (306)b. A second probe (possibly in another functional head) searches for F1, triggering morphological agreement.

(306) a. Initial feature set: [animate; singular; def*; F1; case_]
b. Case assignment after movement: [animate; singular; def*; F1; genitive]

This results in an unnecessarily complex system with two separate formal features [def*] and F1 and two different agreement probes targeting different features both of which lack independent empirical motivation. Despite this complexity, the movement analysis still fails to explain why animate nouns always require the genitive case.

In contrast, my SN analysis offers a more parsimonious explanation. First, case assignment depends on nominal size, not movement. Second, agreement is determined by D-features, naturally accounting for why not all genitive DPs trigger agreement. Third, the obligatory genitive marking of animate nouns follows directly from the syntactic size of their nominal projection rather than an arbitrary feature like [def*]. Thus, the SN analysis provides a simpler, empirically superior account of Moksha PP complements and their case marking and agreement patterns.

4.4.2.2. Null case — default case

Now, I will move on to the other option — null allomorph as an allomorph of the default case. This approach allows for a simpler PP structure while still distinguishing two syntactic configurations:

- Genitive case assignment succeeds → DP receives genitive marking.
- Genitive case assignment fails → DP surfaces with default case, which is realized as a null morpheme.

Under this view, Moksha bare complements of postpositions arise in structures where no source for genitive case is present. The asymmetry in genitive marking can be captured by positing two types of postpositions: (i) **Case-assigning P**: Merges and assigns genitive case (whether via Agree or by defining a case-assigning domain, see Chapter 1 for assumptions on case theory). (ii) **Non-case-assigning P**: Merges but does not assign case, causing the complement DP to take default case, which surfaces as Ø.

Taking our example repeated here again as (307), (307)a involves a case-assigning P, resulting in genitive marking, while (307)b involves a non-case-assigning P, causing the complement DP to surface in default case (Ø).

(307)	a.	*st'ər'-t'*	*lank-sə-nzə*	b.	*morkš*	*lank-sə*
		girl-DEF.SG.GEN	top-IN-AGR.3SG		table	top-IN
		'on the girl'			'on a table'	

Despite its simplicity, this analysis encounters significant empirical and theoretical issues. A major drawback is its failure to explain why certain DPs must receive genitive case, such as pronouns, human nouns including proper names, inalienably possessed objects, and DPs marked as definite or bearing nominal agreement.

In my SN analysis, I account for this by assuming that these categories are full DPs, which necessarily require case, realized as genitive. Smaller nominal projections, in contrast, can remain caseless and thus surface with default case (Ø). For the default case analysis, we would instead need to stipulate that these categories cannot receive default case — a highly problematic assumption. From a theoretical standpoint, it is generally not assumed that a DP can impose restrictions on the type of syntactic case it receives. Case assignment is typically determined by syntactic structure, not by an inherent requirement of the DP itself.

Another major issue with the default case hypothesis is that it fails to account for the agreement asymmetry between genitive and bare complements: postpositions can agree with the possessor of a bare complement; postpositions cannot agree with the possessor of a genitive complement. My SN analysis explains this by linking agreement to the structure of the nominal complement: bare complements involve smaller nominal structures, where possessors remain accessible for agreement; genitive complements involve full DPs, where possessors are structurally embedded and inaccessible for agreement. The default case analysis, however, does not provide a mechanism to derive this asymmetry, making it theoretically weaker.

Thus, the default case hypothesis faces theoretical and empirical challenges. First, it requires an unexplained restriction preventing certain DPs from receiving default case. Second, it fails to account for possessor agreement asymmetries within PP. Third, it cannot naturally explain why animate DPs must receive genitive case, unlike my SN analysis, where this follows directly from nominal size. Given these issues, the SN analysis remains the more explanatory and parsimonious approach.

4.4.3. Analysis of the transparent agreement with the possessor

In the previous subsections, I evaluated alternative analyses primarily in terms of their ability to explain differential case marking. In this section, I am looking at an additional possible account of the differential placement of the agreement marker. This account derives the agreement placement variation via movement and antilocality constraints, and it was proposed in (Lim et al. 2023) for Mongolian.

The core of this analysis is that genitive complements are KPs, and both K and P heads have agreement probes that attract the possessor. However, movement from K to P would be too short, violating antilocality, so the possessor stops at K, and there is no agreement on P. In contrast, bare complements are caseless DPs, and since there is no KP layer, the possessor moves directly to PP without violating antilocality, leading to agreement on P. Consider Mongolian examples in (308) and (309) and their derivations in Figure 34 and Figure 35.

(308) KHALKHA MONGOLIAN (Lim et al. 2023: slide 13)
a. *naraa tolgoin deer=min malgai taiv-san*
Naraa head on=1SG hat put-PST

b. **naraa tolgoin=min deer malgai taiv-san*
Naraa head=1SG on hat put-PST
'Naraa put a pat on my head.'

(309) KHALKHA MONGOLIAN (Lim et al. 2023: slide 17)
a. *naraa tolgoin-ii=min deer malgai taiv-san*
Naraa head-GEN=1SG on hat put-PST

b. **naraa tolgoin-ii deer=min malgai taiv-san*
Naraa head-GEN on=1SG hat put-PST
'Naraa put a pat on my head.'

Figure 34: *Nominal agreement on the postposition in Khalkha Mongolian PP* (adapted from Lim et al. 2023: slide 33)

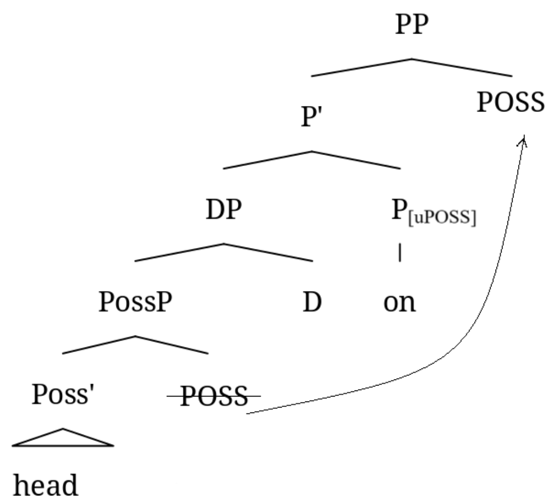
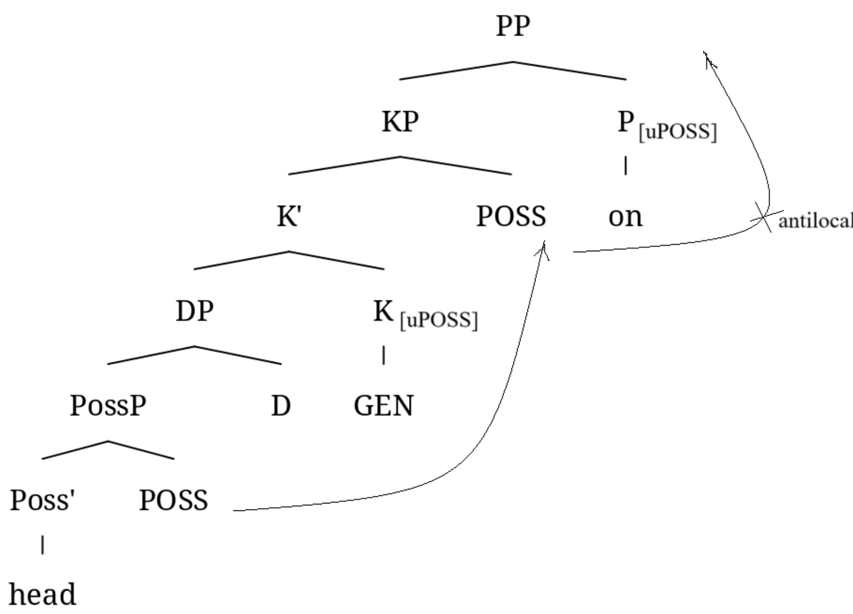


Figure 35: *Nominal agreement on the complement in Khalkha Mongolian PP* (adapted from Lim et al. 2023: slide 39)



The agreement modelled in this analysis is through m-merger that follows phrasal movement (Matushansky 2006; Kramer 2014; Harizanov 2014, a.o.). The agreeing probe on a functional head attracts the possessor DP to the specifier of its head, the possessor DP moves there and then gets remerged as a part of a complex head, surfacing as an agreement clitic. In Figure 34, the probe on the P head attracts the possessor to its specifier, and we see the agreement clitic on the P head, as the possessor remerged as a part of the complex P head. In structures with bare complements, P can still probe and find material below DP, because the authors stipulate that DP in Mongolian PPs is not phasal, which is not unprecedented, see e.g., (Bošković 2014).

The crucial part that accounts for the difference in the placement of agreement is the antilocality part. Spec-to-Spec Antilocality constraint (Erlewine 2016, 2020; Toquero-Pérez 2022) is formulated in (310). Since both KP and PP have an agreement probe, the possessor DP would have moved from KP to PP, crossing only KP and therefore violating antilocality. In PPs with bare complements, possessor crosses the projection of its generation and at least DP, so there is no violation.

(310) **Spec-to-Spec Anti-Locality**: (from Erlewine 2016: 431, as revised in Deal 2019: 408)

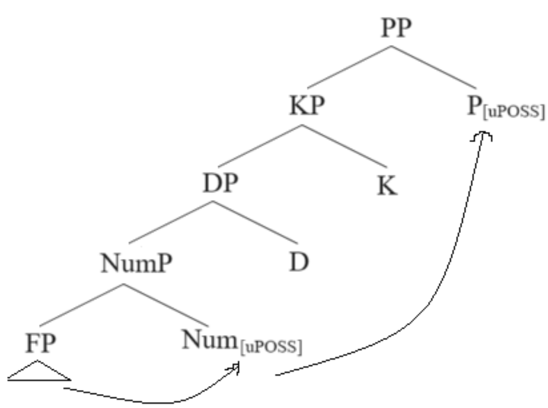
Movement of a phrase from the Specifier of XP must cross a maximal projection other than XP.

While Moksha and Mongolian exhibit similar agreement placement asymmetries, crucial structural differences prevent the antilocality account from extending to Moksha. The first issue is the agreement probe position in Moksha. In Mongolian, agreement marking appears after the case marker, suggesting that the agreement probe is in K. However, in Moksha, the agreement marker precedes the case marker, indicating that the agreement probe is not in K. The second issue is the failure to explain lack of number marking in bare complements.

Consider Moksha examples once again (311). The nominal agreement marker precedes the case marker. In Chapter 2, I argued for the possessive probe to be located on Num. Moving from Spec,NumP to Spec,PP, the possessor DP would have crossed multiple maximal projections, see Figure 36. Even if agreement were on D, movement would still cross KP, avoiding antilocality violations. Thus, the core assumption of the Mongolian analysis — that agreement fails due to movement restrictions — does not hold for Moksha.

(311) *mon'* *morkš-əz'ə-n'* *lank-sə*
1SG.GEN table-1SG.AGR.SG-GEN on-IN
'on my table'

Figure 36: *Nominal agreement in Moksha*[29]



The issue with the number neutrality of bare complements is also challenging for the antilocality analysis. In Moksha, NumP carries number features, and it is present in both DPs and KPs. Even if number features must be checked by D, D is present in both structures, meaning that the DP vs.

[29] Even though the movement is technically Spec-to-Spec, I track the positions of the head, as in this analysis, the possessor in the specifier is remerged together with the head.

KP distinction does not predict number omission. The antilocality analysis simply assumes that some complements are caseless DPs, but it does not derive why these DPs must also lack number marking.

In contrast, my SN analysis predicts the observed morpheme orders in Moksha, as well as naturally accounts for why bare complements lack number (NumP is absent in smaller nominals). Another advantage of the SN analysis is that it also derives the presence vs. absence of case marking without stipulating caseless DPs. Thus, the antilocality analysis is not on the right track for Moksha, and my SN approach remains the superior alternative.

4.4.4. Genitive complements as possessors

Before concluding this section, I would like to discuss Burukina's (2023) analysis of differential P-complement marking in Meadow Mari. In Meadow Mari, pronominal complements receive genitive marking, while non-pronominal ones remain bare. While Burukina's analysis differs from mine in some details and could be considered an alternative, it is fundamentally based on the same core idea that I advocate in this dissertation — namely, that smaller nominal structures appear in the complement of P. Thus, rather than viewing it as a competing analysis, I consider it a variation of the small nominal approach. Below, I outline the key differences between her account and mine and demonstrate that my version is simpler.

Burukina's analysis combines two existing approaches to the structural position of the Ground (i.e., the complement of the postposition) in PPs. In one approach, the Ground is merged as the complement of P, as traditionally assumed. In the other, it is merged as the specifier of a relational noun within the complement of P. According to her analysis, the former structure results in bare complements, while the latter accounts for genitive-marked ones. Applying this to Moksha, genitive-marked complements would be specifiers of relational nouns, whereas bare complements would merge directly as complements of P.

For this analysis to work, one needs to add the assumption (embraced in my analysis as well) that some nominals can be smaller than DP. Specifically, she proposes that pronouns, which receive genitive marking, must project DP in Meadow Mari, whereas non-pronominal arguments are structurally smaller. Another assumption necessary for this analysis is flexibility of the merge position of the postposition. Particularly, the postposition (which may also be analyzed as a relational noun) gets generated as the head of a nominal complement of P or as the exponent of P, depending on the context.[30]

This results in two structural configurations: pronominal genitive complements are possessors within a nominal projection, while non-pronominal complements are NPs or PossPs directly selected by P, as illustrated in Figure 37a and Figure 37b for (312)a and (312)b respectively (Burukina 2023: 159-160). Burukina argues that locational Ps have a requirement on the size of their complement: it should be an NP or a PossP, not a DP and also have a [loc] feature on the complement. The Mari case-marking pattern is derived if we assume that pronouns cannot be reduced to NPs, lack the [loc] feature, and must be merged as possessors of locational nouns. These

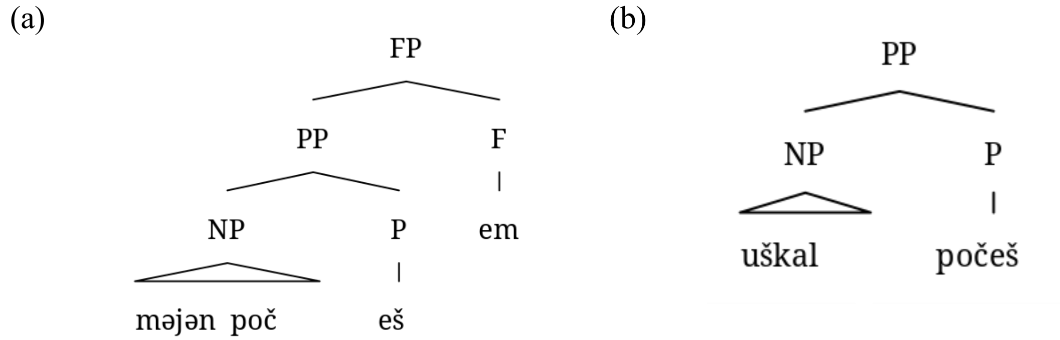
[30] Burukina adopts the split-P model of PP, with separate projections of PlaceP and PathP instead of a single PP (see also Dékány 2009). This split is not significant for the current analysis, and I use P for the simplicity of presentation and better comparability with my analysis.

locational nouns do not project up to DP. Non-pronominal complements merge as NPs[31] (or PossPs) right away. In the latter case, the overt element corresponding to the locational noun is actually merged in P.

(312) MEADOW MARI (adapted from Burukina 2023: 159–160)

a.	*məj-ən*	*počeš(-em)*	b.	*uškal*	*počeš*
	1SG-GEN	after-AGR.1SG		cow	after
	'after me'			'after a cow'	

Figure 37: *Structures for Meadow Mari PPs with genitive and bare "complements"* (adapted from Burukina 2023: 159-160)



The case-marking contrast follows from this structure: since pronouns function as possessors, they receive genitive case within the nominal domain (see Chapter 1, Section 1.2.2.1), whereas non-pronominal complements remain outside the nominal domain and, consequently, caseless. Burukina probably also assumes that there should be no licensing problems, since such caseless Grounds are not DPs but NPs.

[31] Burukina analyzes locative relations between the relational noun and its complement as instances of inalienable possession, with possessors originating in Spec,NP. Therefore, her NPs roughly correspond to nPs in my analysis.

Optional agreement occurring with pronouns is analyzed as an agreement head (F) on top of PP, while the absence of agreement with non-pronominal complements reflects the absence of possessive structure.

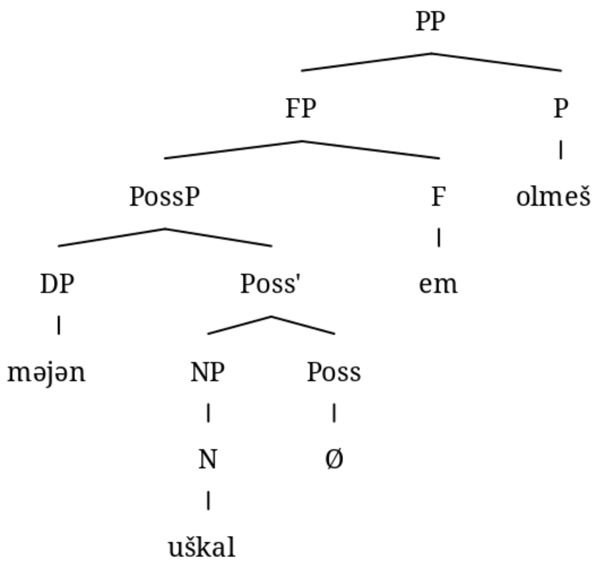
Possessed nominals are merged as PossPs, which are smaller than DPs, see (313) with its structure in Figure 38. The various placement of possessive agreement is analyzed as a post-syntactic phenomenon, following (Guseva & Weisser 2018). When the agreement with the possessor of the complement surfaces on the Place head, it is the result of a post-syntactic operation of Metathesis (314)–(315).

(313) MEADOW MARI (adapted from Burukina 2023: 178)

məj-ən	*uškal-em*	*olmeš*	/	*uškal*	*olmeš-em*
1SG-GEN	cow-AGR.1SG	instead.of		cow	instead.of-AGR.1SG

‘instead of my cow’

Figure 38: *The structure of a PP with a possessed complement* (adapted from Burukina 2023: 178)



(314) Metathesis of Poss and P_{loc}: definition (Burukina 2023: 178)

1. Structural description $[_{PP}$ Possessor NP F X P_{loc}
2. Structural change:
   i. Insert ⟦ to the immediate left of F and ⟧ to the immediate right of P_{loc}.
   ii. Insert ⟩⟨ to the immediate right of F.

(315) Metathesis of Poss and P_{loc}: procedure (Burukina 2023: 178)

NP F (Place) P_{loc} ⇒ NP ⟦ F ⟩⟨ (Place) P_{loc} ⟧ ⇒

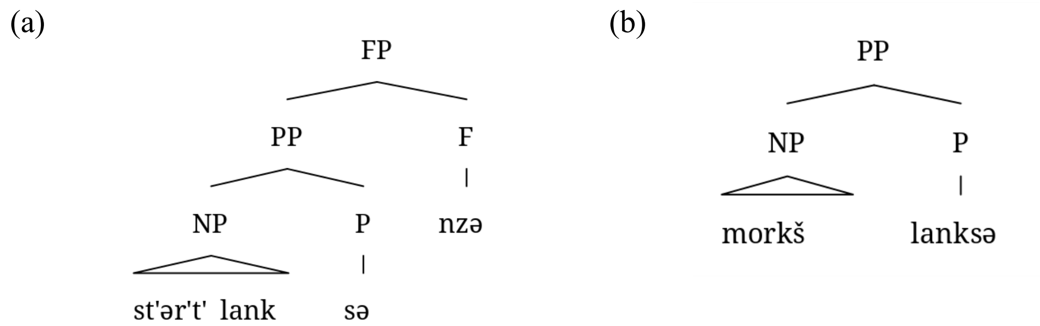
NP – ~~F~~ (Place) P_{loc}- F ~~(Place) P_{loc}~~ ⇒ NP (Place) P_{loc} F

Applying this analysis to Moksha requires several modifications. First, in addition to pronouns, nominals denoting humans and inalienably possessed items must also project DP and/or lack the [loc] feature, ensuring that they merge as specifiers of relational nouns and receive genitive marking. Second, inanimate nominals, unlike in Mari, may project DP but are not required to do so, explaining the variation in case marking. The structures for a genitive (316)a and a bare (316)b complements in Moksha are given in Figure 39.

(316) a. *st'ər'-t'* *lank-sə(-nzə)*
girl-DEF.SG.GEN on-IN-AGR.3SG
'on the girl'

b. *morkš* *lank-sə*
table on-IN
'on a table'

Figure 39: *Structures for Moksha PPs with genitive and bare "complements"*

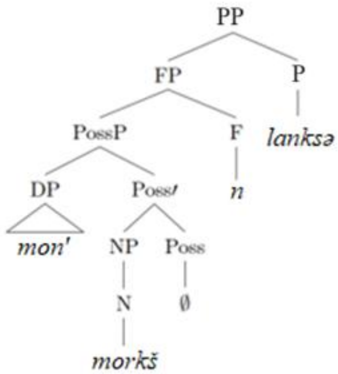


Finally, one should stipulate a cross-linguistic parametrization of the obligatoriness of metathesis, because in Moksha possessive agreement can never appear on the head of the complement itself.[32]

(317) *mon'* *morkš* *lank-sə-n* / **mon'* *morkš-əz'ə* *lank-sə*
1SG.GEN table on-IN-AGR.1SG 1SG.GEN table-1SG.AGR.SG on-IN
'on my table'

[32] The allomorphy of possessive affixes should not pose a problem here, as metathesis operates after linearization but before vocabulary insertion. Contextual rules can account for the variation in affix forms and may also explain the absence of an overt plural morpheme — see my analysis of Lowering in section 4.4.1.3, which produces similar effects.

Figure 40: *The structure of a Moksha PP with a possessed complement*



Although this variation of SN analysis derives the case-marking pattern, it comes with several theoretical and empirical challenges. The first problem is the inconsistent merge position of relational nouns. Burukina's analysis requires the same relational noun to merge as the head of the nominal complement in PPs with pronouns (and human/inalienably possessed nouns in Moksha) but as a P head in other contexts. It is also worth noting that with inanimate non-pronominal complements both merging options are available for some reason. The flexibility in positioning, particularly the optional merging of inanimate non-pronominal complements, lacks a clear motivation.

The second problem is the misalignment with the morphophonological process of n'-omission in Moksha. If a given lexical item can occupy different structural positions due to reanalysis, as suggested by Burukina (2023), we would expect this variation to align with a morphophonological process that tracks the level of grammaticalization of the item. However, this is not the case. According to Muravyeva and Kholodilova (2018), elements functioning as postpositions in Moksha exhibit different degrees of grammaticalization, which manifest in their morphophonological and morphosyntactic properties (see also Pleshak 2021 for a theoretical discussion). One relevant property is that the most grammaticalized postpositions, when they begin with a sonorant, allow for a phonological process on the boundary between the P head and its complement: the final /n'/ of the complement can be omitted (Muravyeva & Kholodilova 2018: 216–218).[33]

(318)

mon	*saj-an*	*jalga-z'ə-n'*	/	*jalga-z'ə*	***mar̥tə***
1SG	come-NPST.1SG	friend-1SG.AGR.SG-GEN		friend-1SG.AGR.SG[GEN]	with

'I'll come with my friend'.

This property is best illustrated by minimal pairs involving the same lexical element. Muravyeva and Kholodilova (2018) argue that certain P-like elements in Moksha can function either as relational nouns or as postpositions, depending on their distribution and intended meaning. For example, in (319) *langə* 'on' functions as a relational noun when expressed as *lanksnə*, appearing in an argument (subject) position and prohibiting *n'*-omission. Conversely, in (320), the same element *langə* when expressed as *lanksə* functions as a postposition, allowing *n'*-omission.

(319) RELATIONAL NOUN → -N' OMISSION IS IMPOSSIBLE (Muravyeva & Kholodilova 2018: 218)

morkš-n'ə-n'	/	**morkš-n'ə*	*lank-snə*	*vad'a-f-t*
table-DEF.PL-GEN		table-DEF.PL	on-3PL.AGR	grease-PTCP.RES-PL

'The surfaces of the tables are dirty.'

(320) GRAMMATICALIZED RELATIONAL NOUN → -N' OMISSION IS POSSIBLE (Muravyeva & Kholodilova 2018: 217)

čɛjn'ək-t	*ašč-ij-°t'*	*morkš-n'ə-n'*	/	*morkš-n'ə*	*lank-sə*
kettle-PL	be.situated-NPST.3-PL	table-DEF.PL-GEN		table-DEF.PL[GEN]	top-IN

'The kettles are on the tables.'

[33] In this dissertation, I do not address why this is treated as a phonological process rather than NOM marking of the complement. See (Muravyeva & Kholodilova 2018: 216–218) for empirical support for this approach.

The contrast between a non-grammaticalized relational noun and a relational noun grammaticalized into a postposition is further confirmed by agreement data. In (320), where *n'*-omission is possible, there is no possessive agreement on the postpositional element. In contrast, in (321), where possessive agreement is indexed on the head, *n'*-omission is blocked.

(321) RELATIONAL NOUN → -N' OMISSION IS IMPOSSIBLE (Muravyeva & Kholodilova 2018: 218)

morkš-n'ə-n' /	**morkš-n'ə*	*lank-sə-st*	*ašč-ij-˚t'*	*čɛjn'ək-t*
table-DEF.PL-GEN	table-DEF.PL	top-IN-3PL.AGR	be.situated-NPST.3-PL	kettle-PL

‘There are kettles on the tables.’

If elements like *langə* prohibit *n'*-omission when they serve as relational nouns, then in structures such as Figure 39a, where they are merged as NP heads, we would expect the same prohibition. This would predict that *n'*-omission should not occur with genitive complements in Moksha. However, this prediction is clearly incorrect: as shown in (320), it is precisely in the context of genitive-marked complements that *n'*-omission is observed. This discrepancy suggests that Burukina’s structural reanalysis does not adequately capture the morphophonological properties of P-complements in Moksha.

The third problem is the unclear licensing of caseless complements. Non-pronominal complements of postpositions are not always inherently locational, making it unclear why all of them should satisfy the [loc] requirement imposed on locational Ps.

The fourth problem is the definition of the nominal domain. Following Baker (2015), the nominal domain corresponds to the spell-out of a phasal D head. However, in Burukina’s structure, there is no D in P-complements, raising the question of how the genitive case is assigned at all.

Finally, Burukina’s approach has undesirable implications for agreement and SP structure. Burukina’s approach requires nominal agreement to be positioned lower than NumP, conflicting with the widely assumed universal DP hierarchy (see Chapter 2). Furthermore, the post-syntactic

Metathesis operation, which shifts possessive agreement from the complement to P, does not work for Moksha. In Section 4.4.1.4, I argued against a post-syntactic Lowering analysis, and similar issues arise with Metathesis due to differences in probe properties.

In contrast, my SN analysis avoids these stipulations while capturing the observed patterns more straightforwardly. First, In the SN analysis, each morpheme has a fixed structural position, eliminating the need for ad hoc reanalysis of relational nouns. Second, my analysis does not require redefining the nominal case-assigning domain. Third, the hierarchical structure of DP remains intact, without requiring unusual assumptions about agreement placement. Finally, my analysis accounts for the variation in agreement placement without relying on post-syntactic operations, which are problematic given Moksha's restrictions on agreement.

Overall, while Burukina's (2023) approach shares the fundamental insight that P-complements vary in size, my analysis achieves the same explanatory power with fewer theoretical complications.

4.4.5. Interim summary

I have demonstrated that all potential alternative analyses encounter both theoretical and empirical challenges. Contextual allomorphy analyses based on adjacency to P overgeneralize the pattern to other syntactic positions, leading to incorrect predictions. Other variations of contextual allomorphy fail to account for the absence of number marking in bare complements. Analyses that assume bare complements carry a case other than genitive are problematic due to category-specific restrictions on case marking. Post-syntactic approaches attempting to explain the placement of agreement with the possessor of the complement on the postposition fall short because they fail to account for the distinct syntactic behavior of internal and external agreement probes. Similarly, the antilocality-based account of variable agreement proves unsuitable due to the ordering of nominal projections.

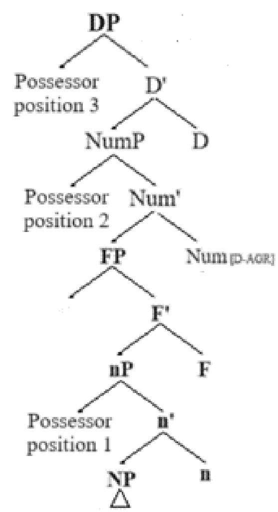
I conclude that some version of the SN analysis, which allows for variation in the size of the complement of P, is necessary. My proposed analysis introduces the minimal number of unmotivated stipulations and is the only one that successfully captures the observed agreement pattern.

4.5. Placing possessors and numerals in the nominal spine

In the previous sections, I have demonstrated that bare complements of postpositions in Moksha are SNs. The first part of my argument relied on fundamental and widely accepted assumptions about the hierarchical structure of noun phrases. Now that I have established that these nominals are smaller than NumP, I can use their properties to gain insight into the nominal structure below NumP. In particular, while such small structures cannot contain demonstratives and universal quantifiers, they can contain possessors and numerals. This suggests that possessors and numerals occupy positions below NumP. This challenges the common assumption that possessors are associated with PossP (between DP and NumP) and that numerals are directly tied to NumP. In this section, I argue that possessors and numerals originate in lower projections within the nominal spine and do not require movement to higher positions.

I begin with the placement of adnominal possessors. There are three potential specifier positions within the nominal structure that can host possessors: Spec,nP, where possessors are potentially base-generated; Spec,NumP, where possessors may move to satisfy agreement probes; and Spec,DP, which can serve as an escape hatch from DP (see Chapter 2, Section 2.3 for details):

Figure 41: *Various positions for possessors*



However, the existence of the lowest position is often overlooked. Possessors are still typically treated as left-peripheral modifiers, and their absence is sometimes taken as evidence for the absence of higher DP layers (Pereltsvaig 2021). Moksha bare complements of postpositions serve as a counterexample, suggesting that possessors can remain in their base-generated position or be targeted by heads other than higher heads in the extended nominal projection. Since possessors occur in bare complements of postpositions (322) — structures smaller than NumP — it is clear that noun phrase's left periphery is not the only place to host possessor. In Figure 42, the possessor remains in Spec,nP. In Figure 43, the possessor moves to a specifier in the extended projection of the postposition.

(322) *maša-n'* *morkš* *lank-sə*
Masha-GEN table on-IN
'on Masha's table'

Figure 42: *Possessor remains in Spec,nP*

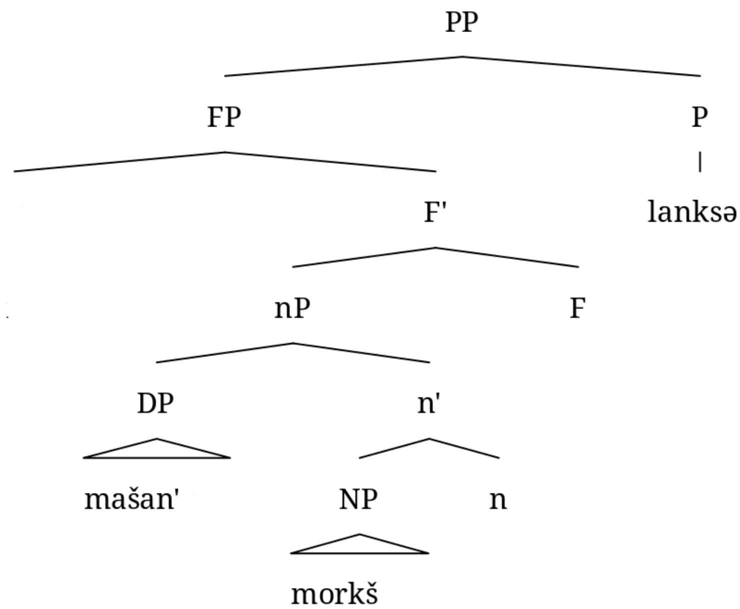
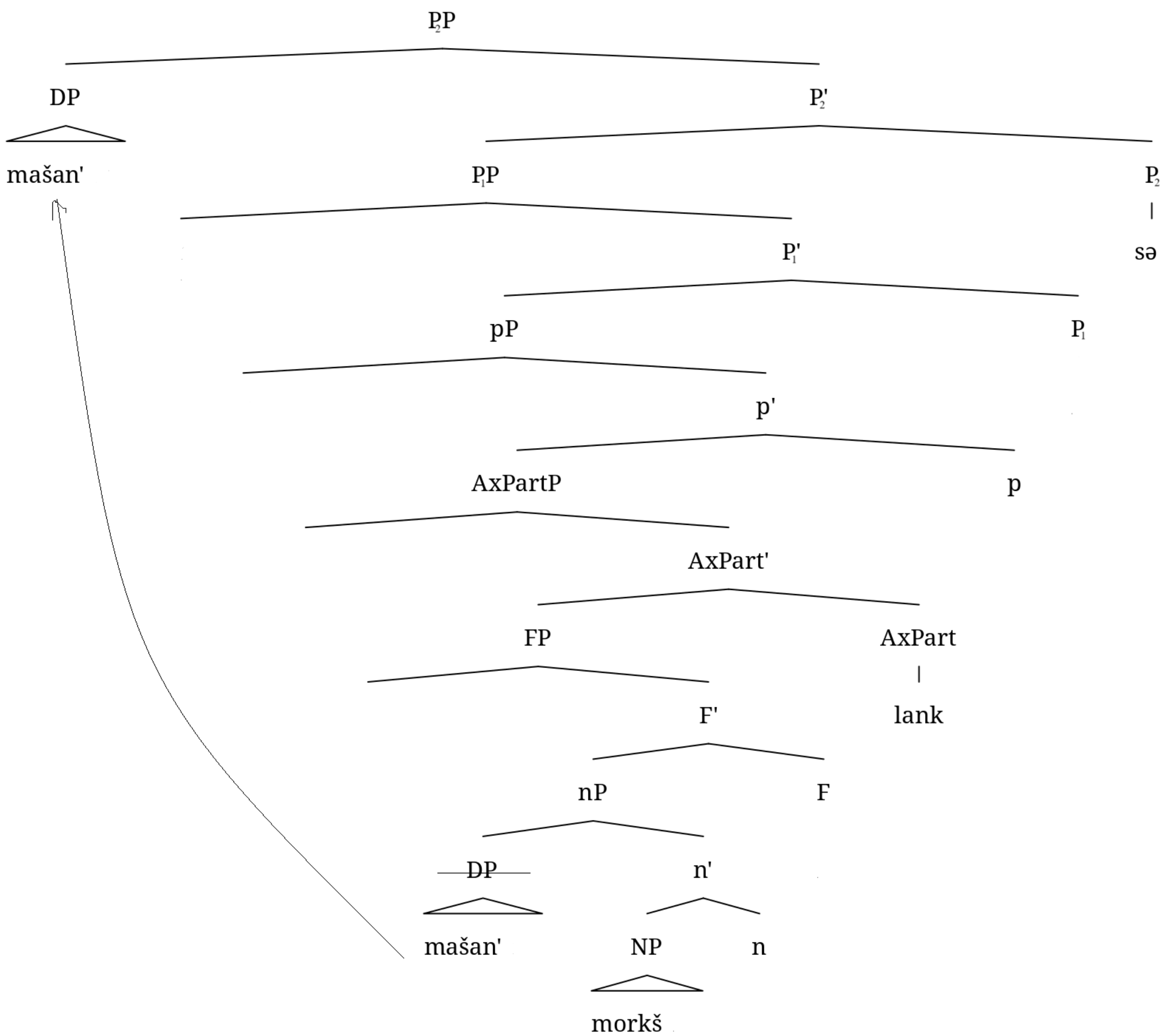


Figure 43: *Possessor moves to Spec,P_1P*



Examples like (322) confirm that possessors do not require Num or D for licensing and can remain in situ (Figure 42) or move to higher specifiers other than nominal (Figure 43). Consequently, the presence of a possessor does not necessarily indicate the presence of NumP or DP, and the absence of a possessor cannot be used as definitive evidence for their absence.

Next, I discuss the position of cardinal numerals within the noun phrase. Although numerals are often linked to the NumP projection (Stavrou & Terzi 2009; Miechowicz-Mathiasen 2011; Danon 2012; Di Sciullo 2022, a.o.), there are at least two arguments for their lower placement. First, numeral phrases do not inherently trigger plural agreement (see (47) repeated here as (323)), suggesting that nominals modified by numerals do not necessarily carry a number feature and thus may lack NumP. Second, in certain contexts, number marking is ungrammatical while numeral modification remains possible. This pattern is observed in Moksha bare complements of postpositions: while numerals can appear in these structures (324), overt number marking is ungrammatical (325), see my analysis of this pattern in Section 4.2.2.

(323) FINNISH (Danon 2011: 301; from Brattico 2010)
Kolme auto-a aja-a tiellä.
three car-PART.SG drive-SG road
'Three cars drive on the road.'

(324) [*kolmə morkš*] *lank-sə*
three table on-in
'on three tables'

(325) **morkš-t lank-sə*
table-PL on-IN
Int.: 'on tables'

A similar phenomenon occurs in Hill Mari DOM. In Hill Mari, direct objects of participial clauses can appear as bare nouns rather than taking accusative case. While such objects allow numeral modification (albeit slightly degraded, possibly due to semantic factors) (326)a, number marking is entirely ungrammatical (326)b (Pleshak & Sirotina 2023), see more on Hill Mari DOM in Chapter 6.

(326) HILL MARI

a. | *ˀkok* | *kol* | *kǝ̂čǝ̂-šǝ̂* | *ǝ̂rvezäš* | *ängǝ̂r-ǝ̂š* | *ke-n* | *vaz-ǝ̂n* |
|---|---|---|---|---|---|---|
| two | fish | catch-PTCP.ACT | boy | creek-ILL | go-CVB | lay-PRET |

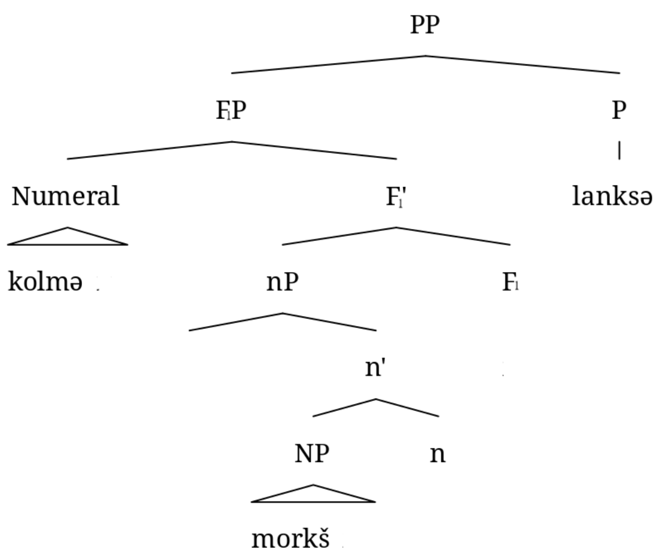
‘The boy who was catching two fish fell in the water.’

b. | *toma* | / | **toma-vlä* | *vǝ̂žalǝ̂-šǝ̂* | *edem* |
|---|---|---|---|---|
| house | | house-PL | sell-PTCP.ACT | man |

‘the man who sells houses’

Thus, I conclude that numerals are base-generated within a lower functional projection (F_1P), which is later selected by NumP in the derivation, see Figure 44 for the complement in (324).

Figure 44: *Moksha numeral in Spec,F_1P*



In conclusion, possessors and cardinal numerals not only originate in lower projections of the nominal structure but can also remain there.

4.6. Summary

In this chapter, I have shown that bare complements of postpositions in Moksha lack genitive marking due to their smaller syntactic structure. The absence of definiteness, number marking, and overt morphological case suggests that these complements are smaller than NumP. This analysis is further supported by the possessive agreement pattern observed in Moksha PPs with bare complements. While a morphological explanation for the absence of marking is possible, it would require numerous ad hoc assumptions. Moreover, no alternative approach successfully accounts for the agreement pattern, which follows naturally from my SN analysis. Crucially, this analysis has broader theoretical implications: if Moksha bare complements of postpositions are indeed smaller than NumP, they provide strong evidence for the low base generation of adnominal modifiers — such as possessors, and numerals— and for the possibility of these modifiers remaining in lower structural positions.

Chapter 5: Identifying imposter small nominals: a case study of DOM in Moksha

In this chapter, I discuss Moksha bare direct objects (DOs) in comparison to bare complements of spatial postpositions. Despite surface similarities between the Differential Object Marking (DOM) pattern observed here (327) and the differential marking of postpositional complements analyzed in Chapter 4, I argue that they require distinct analyses.

The DOM pattern in Moksha resembles the marking of spatial postpositional complements. In (327)a, the DO is marked with the genitive case and triggers agreement on the verb. In (327)b, however, the DO is bare, and there is no object agreement on the verb.

(327) a. *vas'ɛ* *ker'-əz'ə* *šuft-t'*
Vasja[NOM] cut-PST.3SG.O.3SG.S tree-DEF.SG.GEN
'Vasja cut the tree.'

b. *vas'ɛ* *ker'-s'* *šuftə*
Vasja[NOM] cut-PST.3[SG] tree
'Vasja cut a tree.' (Toldova 2018: 575)

A similar contrast is seen in the marking of spatial postpositional complements, as described in Chapter 4 and repeated in (328). In (328)a, the complement is marked with the genitive case and triggers agreement on the postpositional head. In (328)b, the complement is bare and does not trigger agreement. Given this similarity, it is tempting to analyze bare DOs and bare complements of postpositions the same way.

(328) a. *st'ər'-t'* *šir'ə-sə-nzə*
girl-DEF.SG.GEN side-IN-AGR.3SG
'near the girl'

b. *morkš* *šir'ə-sə*
table side-IN
'near a table'

In Chapter 4, I argued that only genitive-marked complements are full DPs, while bare complements lack higher functional projections. The absence of these projections accounts for the lack of nominal morphology and the failure to trigger agreement. Extending this analysis to bare DOs, one might argue that the absence of genitive case in (327)b results from the absence of DP in its structure. Since the features relevant for verbal agreement are located on D, a DO smaller than DP would not be able to trigger agreement.

However, I argue that, unlike bare complements of postpositions, bare indefinite DOs in Moksha are full DPs, not SNs, and that the absence of overt case marking should be explained morphologically rather than syntactically. The argument against a common analysis of bare DOs and bare complements of postpositions comes from two properties that separate the former from the latter. First, bare DOs must be indefinite, whereas bare complements of postpositions need not be. Second, bare DOs can, and even must, be marked for number, unlike bare postpositional complements. This suggests that these two kinds of bare complements cannot receive a common account. If the incompatibility of bare DOs with demonstratives were due to their structural size, they would have to be at least as small as bare postpositional complements (i.e., smaller than NumP); yet the presence of number marking shows that bare DOs are at least as large as NumP. This suggests that this restriction on modification comes from semantics, and the absence of formal definiteness marking does not necessarily indicate the absence of a DP. Moreover, the behavior of bare DOs in causative constructions suggests that they are indeed full DPs.

These differences indicate that not all nominals lacking overt case marking should be analyzed as small nominals, even in a language that allows structures smaller than DP in argument positions. Furthermore, even though some properties of bare nominals in a language seem to align with the properties of SNs found in other languages, one needs to take a closer look at how these properties are derived within the system of this particular language, see my discussion of "universal" SN diagnostics in Chapter 6.

This chapter is structured as follows. Section 5.1 presents the empirical patterns of DOM in Moksha. Section 5.2 examines the structural size of Moksha bare DOs. Section 5.3 argues against the SN analysis of Moksha bare DOs. Section 5.4 develops a "weak case" analysis to account for the observed patterns. Section 5.5 summarizes the findings and concludes the chapter.

5.1. DOM in Moksha

In this section, I present the Moksha DOM pattern through five empirical generalizations. First, bare DOs must be indefinite; I refer to this as the Definiteness Generalization. Second, they cannot bear nominal agreement markers (Nominal Agreement Generalization). Third, predicates cannot agree with bare DOs, while they obligatorily agree with definite-marked DOs and optionally with possessed ones (Predicate Agreement Generalization). Fourth, pronouns and animate proper names cannot appear as bare DOs (Category Restriction Generalization). Fifth, not all modifiers are allowed in bare DOs (Modification Restriction Generalization).

5.1.1. DOM, definiteness and nominal agreement

One of the key factors influencing the marking of Moksha DOs is their referential status. Specifically, only indefinite and non-specific DOs may remain bare:

(329) **Definiteness generalization**:
Only indefinite DOs may be bare; definite DOs must be marked.

The functions of the definite declension in Moksha extend beyond prototypical definiteness (see Chapter 3). However, as long as a DO is eligible for definite declension, it must be marked accordingly and receive the genitive case. All other DOs remain bare. Crucially, no DO that qualifies as definite and is marked as such can remain bare.

I begin by listing the contexts in which a DO receives definite marking. In DO position, all nominals in contexts classified as definite according to Hawkins (1978) receive definite marking. These include anaphoric familiarity (330), uniqueness (331)–(332), and contextual definiteness (333), which is a less prototypical definiteness context (Toldova 2016: 138).

(330) FAMILIARITY (Toldova 2016: 139; from Moksha Corpus)

ves't'	*kir̥ks-s'*	*mu-s'*	*makə-n'*	*vid'm'ə-n'ɛ.*
once	pidgin-DEF.SG[NOM]	find-PST.3SG	poppy-GEN	seed-DIM
… da	*i*	*səv-əz'ə*	*optəm*	*vid'mə-n'ɛ-t'*
and	and	eat-PST.3SG.S.3SG.O	whole	seed-DIM-DEF.GEN

‘Once a pidgin found a poppy seed. And he ate the whole seed.’

(331) UNIQUENESS (GLOBAL) (Toldova 2016: 139)

son	*n'ɛj-əz'ə*	*ši-t'*
3SG[NOM]	see-PST-3SG.S.3SG.O	sun-DEF.GEN

‘He saw the sun.’

(332) UNIQUENESS (LOCAL) (Toldova 2016: 139; from Moksha Corpus)

vet'ə-mbə	*s't'ər-n'ɛ-s'*	*ušt-əz'ə*
night-TMPR	girl-DIM-DEF.SG[NOM]	heat-PST.3SG.S.3SG.O
pɛn'ɛkud-t'		
Russian_stove-DEF.SG.GEN		

‘In the night, the girl heated the Russian stove.’

(333) CONTEXTUAL (Toldova 2016: 140)

min'	*ul'-s'*	*kolmə*	*traks-tə-nək*	
1PL.GEN	EX-PST.3SG	three	cow-ABL-1PL.AGR[NOM]	
fkɛ	*traks-t'*	*al'ɛ-z'ə*	*mi-z'ə*	
one	cow-DEF.SG.GEN	father-1SG.AGR.SG[NOM]	sell-PST.3SG.S.3SG.O	

‘We had three cows. My dad sold one cow (one of the cows).’

Toldova refers to the last context (333) as partitive specific, following (Enç 1991: 5). In addition to specific nominals bound by prior discourse, non-specific nominals receive definite marking when inferred from context or common knowledge. Generic noun phrases, for instance, may be analyzed as referring to kinds opposed to sets of alternative, contextually relevant kinds (334).

(334) GENERIC (Toldova 2016: 146)

modamar̥'-n'ə-n'	*s'oks'e-nda*	*taroks'e-saz'*
potato-DEF.PL-GEN	autumn-TMPR	dig-NPST.3SG.S.3PL.O

'Potatoes are harvested in the fall.'

She also points out that not all partitive relations trigger definite marking. Information structure plays a crucial role in determining definiteness marking in discourse-bound nominals. Specifically, such nominals receive marking only when in a topical position. This is evident in (335), where *kaftə kal̥t* 'two fish' is marked as definite in (335)b when in topic position but remains unmarked in (335)c, where it is not topical.

(335) a.

d'ɛd'ɛ-z'ə	*pid'-əs'*	*kolmə*	*kal̥-t.*
mother-1SG.AGR.SG[NOM]	boil-PST.3SG	three	fish-PL

'My mom boiled three fish.'

b.

kaftə	*kal̥-n'ə-n'*	*səv-əz'ən'*	*katə-s'*
two	fish-DEF.PL-GEN	eat-PST.3SG.S.3SG.O	cat-DEF.SG[NOM]

'Two fish were eaten by the cat.'

c.

katə-s'	*səv-s'*	*kaftə*	*kal̥-t*
cat-DEF.SG[NOM]	eat-PST.3SG	two	fish-PL

'The cat ate two fish {, and Vasja ate one fish}.' (Toldova 2016: 150)

Summarizing Toldova's observations, definiteness marking in Moksha DOs follows a structured hierarchy:

- If a noun is semantically definite, it receives definite object marking.
- If not, one must determine whether it stands in a partitive relation with a relevant discourse set.

- If it does not, the DO surfaces as indefinite.
- If it does, its topical status must be assessed:
  - If topical, it receives definite marking.
  - If non-topical, it remains bare.

As a result, only non-topical partitive and non-partitive indefinite DOs may remain bare, while definite DOs must always be marked:

(336) a. *ton* *n'ɛj-sak* *mazi* *men'əl'-t'?*
2SG[NOM] see-NPST.3SG.O.2SG.S nice sky-DEF.SG.GEN

b. **ton* *n'ɛj-at* *mazi* *men'əl'?*
2SG[NOM] see-NPST.2SG nice sky
'Do you see the nice sky?' (Kashkin 2018: 126)

The obligatory definite marking of semantically-definite DOs distinguishes them from the complements of spatial postpositions (Chapter 4).

There is, however, one exception: possessed nominals. Regardless of their semantics, possessed DOs must bear both nominal agreement markers and case marking:

(337) **Nominal Agreement generalization**:
Possessed nominals must bear nominal agreement and case marking.

This is evident in (338), where even a non-specific indefinite DO must be marked as possessed and receive case marking. Example (338)b shows that the absence of marking leads to ungrammaticality, while example (338)c specifies that not only the absence of genitive is ungrammatical, but also the absence of nominal agreement.

(338) a. *mon'* *mird'ə-z'ə* *ašəz'* *t'ijə* *min'*
1SG.GEN husband-1SG.AGR.SG[NOM] NEG.PST.3SG do[CN] 1PL.GEN
s't'ər'-n'əkə-n' *fkɛ-vək* *fotografija-nc*
girl-AGR.1PL-GEN one-ADD picture-3SG.AGR.SG.GEN

b. **mon'* *mird'ə-z'ə* *ašəz'* *t'ijə* *min'*
1SG.GEN husband-1SG.AGR.SG[NOM] NEG.PST.3SG do[CN] we.GEN
s't'ər'-n'əkə-n' *fkɛ-vək* *fotografija-c*
girl-AGR.1PL-GEN one-ADD picture-3SG.AGR.SG

c. **mon'* *mird'ə-z'ə* *ašəz'* *t'ijə* *min'*
1SG.GEN husband-1SG.AGR.SG[NOM] NEG.PST.3SG do[CN] we.GEN
s't'ər'-n'əkə-n' *fkɛ-vək* *fotografijɛ*
girl-AGR.1PL-GEN one-ADD picture
'My husband hasn't taken a single picture of our daughter.'

This requirement differs from the agreement-related generalizations for complements of postpositions discussed in Chapter 4. There, case marking is required only when agreement is present, and possessed phrases have the option of remaining unmarked for agreement. In contrast, all possessed DOs must be overtly marked.

5.1.2. Category restriction

Another prominent property of bare DOs is that they cannot be filled by most pronouns (339) or human proper names (340). This mirrors the behavior of bare complements of postpositions. In this respect, both bare complement positions exhibit the same properties, including the exception of the inanimate interrogative pronoun *mez'ə* 'what' (341), which receives genitive marking optionally, unlike other pronouns (342) in all syntactic positions, see Chapter 6.

(339) PERSONAL PRONOUN

a. *mon* *ton'* *muj-t'ɛ*
1SG[NOM] 2SG.GEN find-NPST.2SG.O.1SG.S

b. **mon* *ton* *muj-an*
1SG[NOM] 2SG[NOM] find-NPST.1SG
'I will find you.'

(340) HUMAN PROPER NAME (Toldova 2016: 130)
mon *n'ɛj-in'ə* *maša-*(n')*
1SG[NOM] see-PST-1SG.S.3.SG.O Masha-GEN
'I saw Masha.'

(341) INANIMATE INTERROGATIVE
ton *naj* *mez'ə* / *mez'ə-t'* *uč-at?*
2SG[NOM] always what what-DEF.SG.GEN wait-NPST.2SG
'What do you always wait for?'

(342) ANIMATE INTERROGATIVE
ton *naj* *ki-n'* / **ki(jə)* *uč-at*
2SG[NOM] always who-GEN who wait-NPST.2SG
'Who do you always wait for?'

From these observations, we can generalize the restriction on pronouns and animate personal names:

(343) **Category Restriction generalization**:
If the DO is a pronoun, or an animate proper name, it must be genitive.

While the restriction on personal (and demonstrative) pronouns could be attributed to their inherent definiteness, the fact that animate interrogative pronouns cannot be bare does not follow from this explanation. Furthermore, non-pronominal animate DOs can remain bare as long as they meet the indefiniteness condition:

(344) *mon* *n'ɛj-ən'* *pɛk* *mazi* *sen'əm* *s'el'mə* *s't'ər'-n'ɛ*
1SG[NOM] see-PST.1SG very beautiful blue eye girl-DIM
'I saw a very beautiful blue-eyed girl.'

Thus, the requirement for most pronouns and animate personal names to receive genitive marking cannot be reduced to a single semantic feature such as definiteness. Instead, this restriction appears to be categorical in nature.

5.1.3. Modification

Genitive DOs do not have restrictions on the types of modifiers they can contain. Bare DOs allow for a wide range of adnominal modifiers, except for possessors, demonstratives and universal quantifiers:

(345) **Modification Restriction generalization**:

In the presence of a possessor, a demonstrative or a universal quantifier, the complement must be genitive.

The following examples illustrate that genitive DOs occur with adjectives (346), possessors (347), numerals (348), relative clauses (349), demonstratives (348)–(350), and universal quantifiers (351).

(346) ADJECTIVE

škola-s'	*veš-ənci*	***od***	*d'ir'ektər-t'*
scool-DEF.SG[NOM]	look_for-FREQ.NPST.3SG.O.3SG.S	new	headmaster-DEF.SG.GEN

‘The school is looking for a new headmaster.’

(347) POSSESSOR (Toldova 2018: 601)

a.

pet'ɛ	***s'ora-t'***	*morə-**nc***	/	**morə*
Petja[NOM]	boy-DEF.SG.GEN	song-3SG.AGR.GEN		song
*tona-f-n'ə-**z'ə***				
learn-CAUS-FREQ-PST.3SG.O.3SG.S				

b.

**pet'ɛ*	***s'ora-t'***	*morə-**nc***	*tona-f-n'ə-s'*
Petja[NOM]	boy-DEF.SG.GEN	song-3SG.AGR.GEN	learn-CAUS-FREQ-PST.3[SG]

‘Petja learned the boy’s songs.’

(348) DEMONSTRATIVE + NUMERAL (Sidorova 2018b: 331)

a.

d'ɛd'ɛ-s'	*maks-əz'ən'*	*id'-ənzə-ndi*	***t'ɛ***
мать-DEF.SG[NOM]	give-PST.3PL.O.3SG.S	child-3SG.AGR.PL-DAT	this
vet'ə	*maŗ'-n'ə-n'*		
five	apple-DEF.PL-GEN		

b. ******d'ɛd'ɛ-s'* *maks-əz'ə* *id'-ənzə-ndi* ***t'ɛ***
mother-DEF.SG[NOM] give-PST.3SG.O.3SG.S child-3SG.AGR.PL-DAT this
vet'ə *mar'-t'*
five apple-DEF.SG.GEN
'The mother gave her kids these five apples.'

(349) DEMONSTRATIVE + RELATIVE CLAUSE (Privizentseva 2018: 716)
n'ɛft'ə-k ***s'ɛ*** *s'ora-n'ɛ-t',* ***kona-n'd'i*** ***er'av-i***
show-IMP.3SG.O.SG.S that boy-DIM-DEF.SG.GEN which-DAT be_needed-NPST.3SG
maks-əm-s ***kn'iga***
give-INF-ILL book
'Show me the boy, to who I need to give a book.'

(350) DEMONSTRATIVE (Kashkin 2018: 126)
son *dasad'ɛ-z'ə* ***t'ɛ*** *s'ora-n'ɛ-t'*
3SG[NOM] offend-PST.3SG.O.3SG.S this boy-DIM-DEF.SG.GEN
'He offended this boy.'

(351) UNIVERSAL QUANTIFIER (Toldova 2016: 135)
son ***s'embə*** *morə-t'n'ə-n'* *tona-f-n'ə-z'n'e*
3SG[NOM] all song-DEF.PL-GEN learn-CAUS-IPFV-PST.3.S.PL.O
'He learned all the songs.'

Bare DOs also can be modified by adjectives (352), numerals (353), and relative clauses (354). However, they cannot be modified by possessors (355), demonstratives (356), or universal quantifiers (357).

(352) ADJECTIVE (Toldova 2016: 145)
mon *rama-n'* *vas'ə* ***jakst'ər'*** *panar* *a* *s'ɛnger'ɛ* *odo*
1SG[NOM] buy-PST.3SG earlier red shirt and green after
'I bought a red shirt first, and then a green one.'

(353) NUMERAL (Sidorova 2018b: 335)
son *t'er'd'-i* ***kolmə*** *lomat'-t'* / **loman'*
3SG.NOM call-NPST.3SG thee person-PL person
'He's calling three people'.

(354) RELATIVE CLAUSE (Privizentseva 2018: 716)
mon *n'ɛj-an* *kat'i* *kodamə* *loman'* ***kona*** ***sa-s'*** ***is'ak***
1SG[NOM] see-NPST.1SG INDEF what person which come-PST.3SG yesterday
'I see some man, who came yesterday.'

(355) POSSESSOR
mon' mird'ə-z'ə ašəz' t'ijə* ***min'*
1SG.GEN husband-1SG.AGR.SG[NOM] NEG.PST.3SG do[CN] we.GEN
s't'ər'-n'əkə-n' *fkɛ-vək fotografijɛ*
girl-AGR.1PL-GEN one-ADD picture
Int.: 'My husband hasn't taken a single picture of our daughter.'

(356) DEMONSTRATIVE (Kashkin 2018: 126)
son dasad'ɛ-s'* ***t'ɛ* *s'ora-n'ɛ*
3SG[NOM] offend-PST.3SG this boy-DIM
Int.: 'He offended this boy.'

(357) UNIVERSAL QUANTIFIER (Toldova 2016: 135)
son* ***s'embə* *morə-t tona-f-n'ə-s'-t'*
3SG[NOM] all song-PL learn-CAUS-IPFV-PST.3-PL
Int.: 'He learned all songs.'

This restriction on modification closely resembles the one observed with bare complements of postpositions. The primary difference is that DOs have an additional restriction on possessors.

5.1.4. Predicate agreement

Moksha has object agreement, but not all DO can trigger it:

(358) **Predicate Agreement generalization**:

a. Bare DOs cannot trigger predicate agreement;
b. Genitive DOs marked as definite trigger predicate agreement obligatorily;
c. Genitive DOs bearing nominal agreement trigger predicate agreement optionally.

The inability of bare DOs to trigger agreement is demonstrated in (359). The obligatory agreement with genitive DOs depends on the type of declension used to mark the DO (see Chapter 3 for declension types). Object agreement is obligatory with genitive nominals in the definite declension (360). However, it is optional with cased nominals marked with the possessive declension, even if they are semantically definite (361).

(359) BARE DO → NO AGREEMENT (Toldova 2018: 577)

**vas'ɛ ker'-əz'ə šuftə*
Vasja[NOM] cut-PST.3SG.O.3SG.S tree
Int.: 'Vasja cut a tree.'

(360) DO MARKED AS DEFINITE → OBLIGATORY AGREEMENT

a. *min' jalga-n'əkə n'ɛj-əz'ə t'ɛ fotografijɛ-t'*
1PL.GEN friend-AGR.1PL[NOM] see-PST.3SG.O.3SG.S this picture-DEF.SG.GEN

b. **min' jalga-n'əkə n'ɛj-s' t'ɛ fotografijɛ-t'*
1PL.GEN friend-AGR.1PL[NOM] see-PST.3SG this picture-DEF.SG.GEN
'Our friend saw this picture.'

(361) DO MARKED AS POSSESSED → OPTIONAL AGREEMENT

a. *min' jalga-n'əkə n'ɛj-əz'ə min' s't'ər'-n'əkə-n'*
1PL.GEN friend-AGR.1PL[NOM] see-PST.3SG.O.3SG.S 1PL.GEN girl-AGR.1PL-GEN
t'ɛ fotografija-nc
this foto-AGR.3SG.GEN

b. *min' jalga-n'əkə n'ɛj-s' min' s't'ər'-n'əkə-n'*
1PL.GEN friend-AGR.1PL[NOM] see-PST.3SG 1PL.GEN girl-AGR.1PL-GEN
t'ɛ fotografijanc
this foto-AGR.3SG.GEN
'Our friend saw this picture of our daughter.'

Thus, while the presence of predicate agreement in Moksha may be linked to the (in)definiteness of the DO, the obligatoriness of this agreement is determined by the declension type rather than by a purely semantic or categorial feature. The sensitivity of the verbal agreement probe to declension type is unique to verbal agreement. As shown in Chapters 3 and 4, neither the internal nominal agreement probe nor the probe on postpositions is sensitive to the declension type of the goal. Instead, the postpositional probe is sensitive to the animacy of the complement — an aspect that does not affect the verbal predicate agreement probe. For more difference between agreement patterns with complements of postpositions and DOs, including the impossibility of external agreement with the possessor of bare DOs, see Section 5.3.2.

5.1.5. Interim summary

In this section, I outlined the key properties of Moksha DOM. I demonstrated that case marking on DOs is influenced by their specificity and definiteness, their categorial status, and the declension type that is used to mark them. The summary is presented in Table 10.

Table 10: *Properties of genitive and bare DOs*

properties	genitive DO	bare DO
definite interpretation	OK	*
possessed interpretation	OK	*
Indefinite interpretation	*	OK
definite/possessive marking	OK	*
pronouns, animate proper names	OK	*
demonstratives and universal quantifiers as modifiers	OK	*
predicate agreement	OK	*

In particular, all definite DOs are marked with the definite declension and carry a genitive case marker. Pronouns and animate proper names must also be genitive. Indefinite DOs can be possessed, in which case they take a nominal agreement marker and bear a genitive marker. Non-possessed indefinite DOs remain bare, without additional marking. Demonstratives and universal quantifiers require definite declension and are incompatible with bare DOs. Possessors require nominal agreement and, like demonstratives and quantifiers, cannot occur with bare DOs. Crucially, only genitive DOs trigger object agreement on the predicate, and only DOs marked as definite obligatorily trigger agreement.

5.2. Determining the size of the DOs

In this section, I investigate whether the absence of overt case marking on bare DOs can be attributed to the absence of DP in their syntactic structure. This explanation successfully accounts for bare complements of spatial postpositions, which exhibit a similar pattern, so it is worth considering a similar analysis for DOs. Here, I show that as opposed to bare complements of postpositions, there is no evidence of the absence of DP in bare DOs.

Any structural analysis must account for the following empirical generalizations:

(362) a. **Definiteness generalization**:
Only indefinite DOs may be bare; definite DOs must be marked.

b. **Nominal Agreement generalization**:
Possessed nominals must bear nominal agreement and case marking.

c. **Category Restriction generalization**:
If the DO is a pronoun, or an animate proper name, it must be genitive.

d. **Modification Restriction generalization**:
In the presence of a possessor, a demonstrative or a universal quantifier, the complement must be genitive.

e. **Predicate Agreement generalization**:
- Bare DOs cannot trigger predicate agreement;
- Genitive DOs marked as definite trigger predicate agreement obligatorily;
- Genitive DOs bearing nominal agreement trigger predicate agreement optionally.

To explore the structure of bare vs. genitive DOs, I adopt the hierarchical noun phrase structure introduced in Chapter 2:

(363) $NP < nP < F_1P < F_2P < NumP < DP$

Since each functional projection is associated with particular properties expected from the nominal that has it, I should be able to tell which projections are present in the nominal.

As a reminder, my assumption is that not all arguments in a language must be DPs. Small nominals lack the highest layers and may be NumPs, F_2Ps, F_1Ps, nPs or NPs. Within the generalized noun phrase, DP hosts definite determiners, and NumP — number marking and the nominal agreement probe. The nP projection is where possessors are base generated. Only DPs require case and trigger agreement. I assume that small nominals must be caseless even in the environment where there is a source of case marking.

5.2.1. DP layer

For a structural analysis to account for case absence in bare DOs, it would suffice to demonstrate that bare complements lack the DP layer. However, in this section, I show that there is not enough independent evidence to support this claim.

The idea that bare indefinite DOs are SNs — lacking the DP layer — stems from the same reasoning applied to bare complements of postpositions: they lack formal marking for definiteness and case; they do not trigger agreement on the predicate; and they contrast with genitive DOs, which are marked for definiteness and trigger object agreement.

Consider (364), where attempting to mark definiteness without genitive case results in an ill-formed sentence. In contrast, (365)a and (365)b are grammatical, showing that case and definiteness marking are interdependent, with case-marked DOs triggering agreement and bare DOs remaining unmarked.

(364) **Vas'a* *ker'-s'* / *ker'-əz'ə* *šufc'*
Vasja[NOM] cut-PST.3SG cut-PST.3SG.O.3SG.S tree.DEF.SG
Int.: 'Vasja cut the tree.'

(365) a. *vas'ε* *ker'-əz'ə* *šuft-t'*
Vasja[NOM] cut-PST.3SG.O.3SG.S tree-DEF.SG.GEN
'Vasja cut the tree.'
b. *vas'ε* *ker'-**s'*** *šuftə*
Vasja[NOM] cut-PST.3[SG] tree
'Vasja cut a tree.' (Toldova 2018: 575)

At first glance, the lack of a determiner in bare DOs might suggest the absence of a D head. However, an alternative explanation is that bare DOs are obligatorily indefinite. Then the absence of a definite determiner simply follows from the restriction on bare DOs being indefinite (see Section 5.1 for the restriction on the definiteness of the DO), and predicate agreement may track definiteness, rather than the presence of DP. In Section 5.3, I show that an SN analysis indeed fails to account for all properties of Moksha bare DOs.

Thus, while there is no evidence that the DP layer is present, there is also no conclusive evidence that it is absent. The absence of overt marking does not necessarily imply a structural deficiency but may instead be a consequence of indefiniteness constraints on bare DOs.

5.2.2. NumP layer

Moving onto number marking, bare DOs in Moksha must be marked for number. First, plural marking is obligatory for plural interpretation (366). Second, without explicit plural marking, a bare DO is interpreted as singular (367). Third, cardinal numerals with low numerical values (e.g., 'three') require plural marking on DOs (368). In contrast, the same numerals can co-occur with singular nouns in adnominal dependents (369).

(366) PLURAL DO → PLURAL MARKING (Toldova 2018: 581)

pet'ɛ	*kanc'*	[***kolmə***	***gl'ɛn'c'ək-ən'***	*butilka-t*]
Petja[NOM]	carry.PST.3SG	three	glass-GEN	bottle-PL

'Petja brought three glass bottles.'

(367) NO PLURAL MARKING → SINGULAR INTERPRETATION (Toldova 2018: 580)

son	*mi-s'*	*akšə*	***alaša***
3SG[NOM]	sell-PST.3SG	white	horse

'He sold a white horse.'
*'He sold white horses.'

(368) LOW NUMERAL → PLURAL MARKING OF DO

d'ɛd'ɛ-z'ə	*kanc'*	*kolmə*	*mar̥'-t'* /	**mar'*
mother-1SG.AGR.SG[NOM]	bring.PST.3SG	three	apple-PL	apple

'My mother brought three apples.'

(369) LOW NUMERAL →NO PLURAL MARKING OF AN ADNOMINAL DEPENDENT

kolmə	*brad-ən'*	*kuc'*	*ašč-i*	*ber'ɛk*	*tona*	*bok-sə*
three	brother-GEN	house.DEF.SG[NOM]	be_situated-NPST.3SG	shore	that	side-IN

'Three brothers' house is on that side of the shore.'

The fact that plural marking in the context of low numerals cannot be skipped when the nominal is in the DO position suggests that NumP must always be present in DOs. The logic is the following. If plural marking were simply imposed by the numeral, we would expect it to be obligatory in all contexts. This is not borne out, because in Moksha, some syntactic contexts allow for phrases with low numerals not marked for plural (369). Then it should be the Num head, which, when valued as plural, makes sure that the numeral has a low numerical value. Crucially, in the absence of Num, like in the adnominal dependent in (369), no conditions are violated, and a numeral with any value can be present (see my analysis of Moksha numerical constructions in Chapter 4, Section 4.2.2). The fact that in the DO position the conditions imposed by Num cannot be violated is telling; it means that DOs are always at least NumPs.

If bare DOs must be at least NumPs, and if the nominal agreement probe is located in NumP, we might expect bare DOs to allow nominal agreement. However, this is not the case. Bare DOs cannot bear nominal agreement markers (370). Moreover, possessed nominals cannot be bare DOs, whether they bear nominal agreement or not (371).

(370) POSSESSED DO WITH NOMINAL AGREEMENT → *BARE
**mon'* *mird'ə-z'ə* *ašəz'* *t'ijə* *min'*
1SG.GEN husband-1SG.AGR.SG[NOM] NEG.PST.3SG do[CN] 1PL.GEN
s't'ər'-n'əkə-n' *fkɛ-vək* *fotografija-c*
girl-AGR.1PL-GEN one-ADD picture-1SG.AGR.SG
Int.: 'My husband hasn't taken a single picture of our daughter.'

(371) POSSESSED DO WITHOUT NOMINAL AGREEMENT → *BARE
**mon'* *mird'ə-z'ə* *ašəz'* *t'ijə* *min'*
1SG.GEN husband-1SG.AGR.SG[NOM] NEG.PST.3SG do[CN] 1PL.GEN
s't'ər'-n'əkə-n' *fkɛ-vək* *fotografijɛ*
girl-AGR.1PL-GEN one-ADD picture
Int.: 'My husband hasn't taken a single picture of our daughter.'

The presence of NumP in bare DOs is evident from obligatory plural marking. While bare DOs are at least as large as NumP, there is no conclusive evidence that they are not full DPs. In the next section, I demonstrate why the SN analysis, which successfully accounts for bare complements of postpositions, does not extend to bare DOs.

5.3. The failure of the SN analysis for Moksha bare DOs

At the beginning of this chapter, I proposed that the case-marking and agreement patterns of DOs and complements of postpositions (Ps) are similar. Indeed, in both contexts case-marking alternates between the presence of the genitive case and the absence of it, and only genitive complements can trigger agreement.

However, a closer examination reveals that bare DOs and bare complements of postpositions exhibit fundamentally different properties, with the exception of restrictions on definiteness marking, which may stem from distinct underlying mechanisms.

The properties of bare complements of postpositions described in Chapter 4 together with the properties of DOs described in sections 5.1 and 5.2 are summarized in Table 11.

Table 11: *Properties of bare complements of postpositions and bare DOs*

	Bare complements of Ps	Bare DOs
definiteness marking	NO	NO
number marking	NO	YES
demonstratives as modifiers	YES	NO
possessors as modifiers	YES	NO

These differences suggest that bare DOs and bare complements of postpositions cannot receive the same structural analysis. The data indicate that bare DOs are at least NumPs, given their obligatory number marking; and bare complements of postpositions are smaller, likely FPs, given the absence of number marking.

Technically, both could still be SNs, just of different size. Small nominals of different sizes do not have to have identical properties. However, below, I argue that the observed sets of properties contradict each other and cannot be captured under a single, unified analysis. Instead, their structural differences necessitate separate explanations.

5.3.1. The absence of the definite determiner and structural size

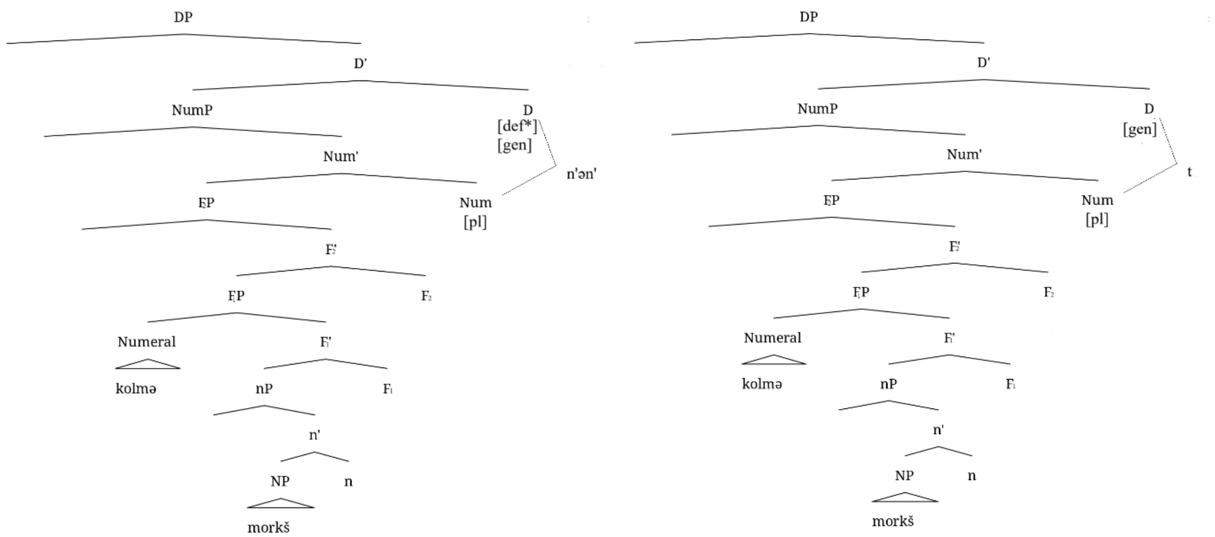
I will start with the absence of definiteness marking and the availability of definite interpretation. One key factor in my SN analysis of bare complements of postpositions was the mismatch between definiteness marking and interpretation — bare complements of postpositions can be interpreted as definite despite lacking overt definiteness marking. Therefore, the absence of the definite declension on bare complements of postpositions cannot be explained semantically. Definite declension does not seem to be optional in Moksha, as definiteness marking is required for noun phrases in the subject and DO positions. Thus, the very contrast with DOs, where definiteness marking is obligatory for definite interpretations, suggests that the absence of definiteness marking in postpositional complements must be structural.

If the absence of definiteness marking in bare DOs were also structural, we would expect to find definite DOs without overt definiteness marking, which we do not. Instead, I propose a different explanation. Bare DOs are full DPs but lack the definiteness feature that would otherwise trigger the definite declension. This absence of the definiteness feature is semantically driven, rather than due to a reduced structure, because marking (or its absence) correlates with interpretation. The difference between definite and indefinite DOs in (372) is illustrated in Figure 45: Notice that both nominals have all projections up to DP, and the difference lies in the presence vs. absence of the definiteness feature in D, not in the structural size of the nominal (presence of D itself).

(372) a. *kolmə morkš-n'ə-n'*
three table-DEF.PL-GEN
'[I see] the three tables'

b. *kolmə morkš-t*
three table-PL
'[I see] three tables'

Figure 45: *The structure of a definite DO and an indefinite DO*



5.3.2. Possessors, nominal agreement and object agreement

If the analysis of bare complements of postpositions from chapter 4 is correct, then possessors, which are generated in Spec,nP (as discussed in chapter 2, section 2.5), can remain low in Moksha and appear in phrases as small as F_1P. Since bare DOs are NumPs, which are structurally larger than F_1Ps, they should in principle also accommodate possessors. Furthermore, given that nominal agreement with possessors is hosted in NumP, we would expect bare DOs to bear nominal agreement markers and allow possessor movement to Spec,NumP.

However, as demonstrated in sections 5.1.1 and 5.2.2, bare DOs do not bear nominal agreement markers, and possessors are never found in bare DOs. From bare complements of postpositions, we know that possessors are allowed with no possessive marking, which suggests a structural explanation of the absence of nominal agreement marking in complements of Moksha postpositions. Given this situation with possessed bare complements of postpositions, the impossibility of possessors in bare DOs should receive an alternative explanation. The absence of agreement follows from the absence of a possessor, rather than structural constraints on agreement itself.

Before I move on, let me discuss an additional issue that arises because of the absence of nominal agreement on bare DOs. In PPs, when nominal agreement is absent from the complement itself, the possessor can trigger agreement on the P head (see chapter 4, section 4.2). If bare DOs were structurally parallel to bare complements of postpositions, we would expect a similar agreement pattern — specifically, the possessor of a bare DO should be able to trigger object agreement on the verb. However, this expectation is not borne out, as demonstrated in (373). Here, the 1st person possessor should trigger 1st person object agreement on the verb if the structure behaved like PPs. Indeed, if the verbal probe looks for a DP goal in its c-commanding domain and does not see the object as it is an SN, it can keep looking and find the possessor DP, as the P probe did. However, this does not happen, further differentiating bare DOs from bare complements of postpositions.

(373)	**Vas'a*	*ker-əma-n'*	[*mon'*	*šuftə*]
	Vasja[NOM]	cut-PST.1.O-SG.O.3SG.S	I.GEN	tree
	Int.: 'Vasja cut my tree.'			

The differences between bare DOs and bare complements of postpositions are not limited to case-marking and agreement patterns; they also extend to possessor behavior. First, bare DOs do not allow possessors without possessive marking, whereas bare complements of postpositions do. Second, possessors in bare DOs do not trigger verbal agreement shift, whereas possessors in PPs can trigger agreement on the P head. As long as the SN analysis holds for bare complements of postpositions, which seems to be the case it cannot account for bare DOs. This supports the claim that bare DOs must have a distinct structural analysis from bare complements of postpositions.

5.3.3. Bare DOs and transitivity of the clause

So far, we have established that the restrictions on modification and interpretation in bare DOs cannot be explained simply by the absence of relevant projections. The only structural information we have about bare DOs is the obligatoriness of number marking, which implies that they are at least as large as NumP. However, Moksha provides no direct evidence against the presence of DP in these constructions. In this section, I present an argument *in favor* of the presence of DP in bare DOs, based on causative clauses.

Cross-linguistically, causees typically receive object case unless this case has already been assigned to the direct object (DO) (e.g., Comrie 1989; Dixon 2000). If a DO is present, it receives the object case, forcing the causee into a postpositional construction.

Under any theory of case, the explanation of these facts proceeds roughly as follows. In a causative construction, there is one source of the object case. It gets assigned if there is an argument that can receive it. Once the case is assigned to one argument, the source is exhausted, and no other argument can bear that case.

Now, let me outline the expectations regarding the behavior of SNs in causative constructions. SNs are nominal phrases that cannot receive case because case is a feature on D. For this reason, we expect them not to exhaust the source of the object case. Consequently, in causative constructions with bare indefinite DOs, we would expect the causee to receive the object case. However, this prediction is not borne out. Instead, causees are forced into a postpositional construction, just as they are when a fully definite DP object is present. This strongly suggests that bare DOs do receive case, which is only possible if they are DPs. The remainder of this section presents the empirical basis for this argument.

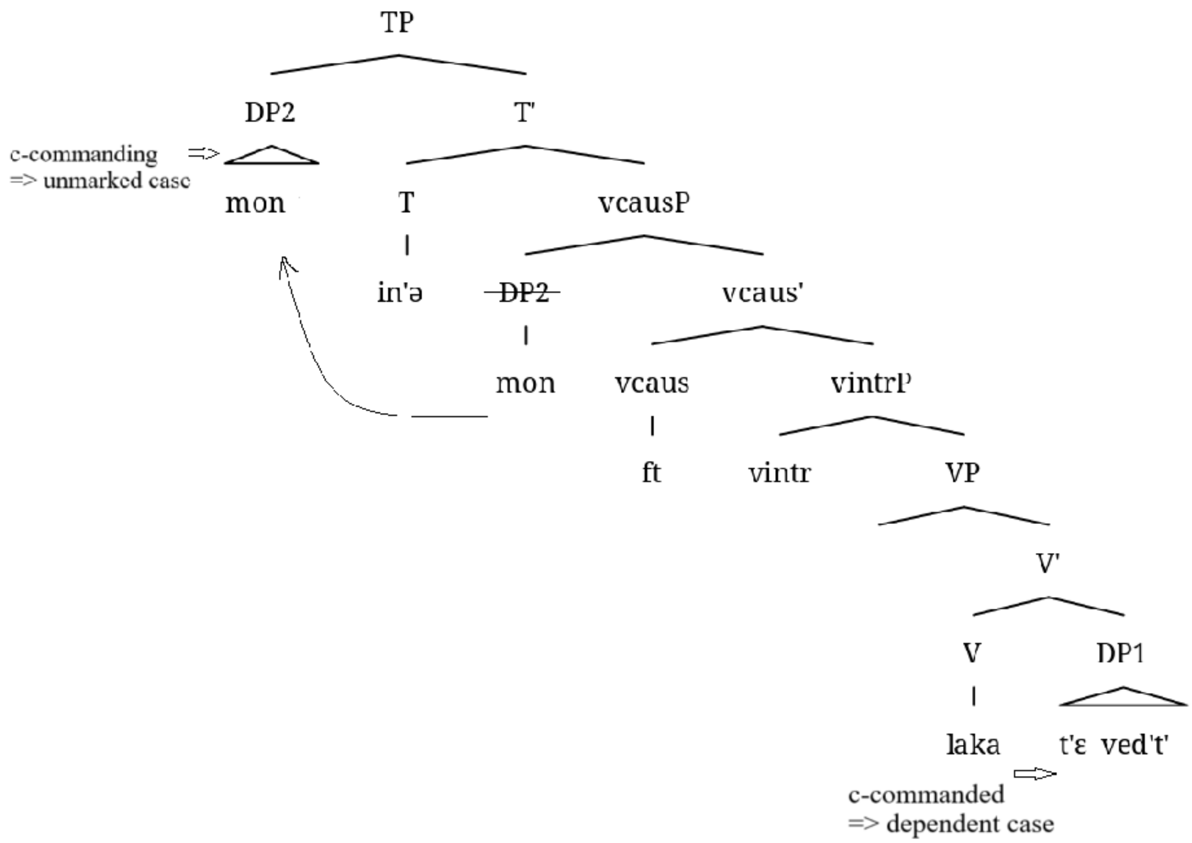
Let me start with the introduction of the case marking pattern in causative constructions with different types of initial predicates, as described in (Nikiforova 2018). When the base predicate is intransitive, the causee receives object marking because there is no DO competing for case assignment. Consider (374)–(375) with the initial intransitive predicate *lakams* 'boil' taking the only argument — what is being boiled (374). In (374), the only argument *vec'* 'water' receives nominative case, as expected. When this predicate is causativized, the causee gets object marking, which is the genitive case in Moksha (375).

(374) INTRANSITIVE BASE PREDICATE
vec' *laka-j*
water.DEF.SG[NOM] boil-NPST.3SG
'The water is boiling.'

(375) CAUSATIVIZED INTRANSITIVE PREDICATE (Nikiforova 2018: 475)
mon *laka-ft-in'ə* *t'ɛ* *ved'-t'*
1SG[NOM] boil-CAUS-PST.3.O.1SG.S this water-DEF.SG.GEN
'I boiled this water.'

Under DCT, the only argument of the predicate in (374) receives nominative as the only DP within the domain. In (375), on the other hand, the causee is the lower DP c-commanded by the subject DP and has to receive the dependent case, which is the object genitive in Moksha. The clausal structure with the case assignment mechanism described above is in Figure 46.

Figure 46: *Case assignment in a clause with a causativized intransitive predicate*



A similar pattern appears when a predicate has two arguments, but the object is marked with an inherent case, such as 'drink'. In (376), we see a two-place predicate whose object is inherently marked as ablative, while the subject is in the nominative. In (377), where the predicate has been causativized, the object remains ablative, while the causee gets the object case.

(376) TWO-PLACE BASE PREDICATE WITH AN INHERENT CASE-MARKED OBJECT (Nikiforova 2018: 475)

mon	*s'im-ən'*	*čɛj-də*
1SG[NOM]	drink-PST.1SG	tea-ABL

'I drank (some) tea.'

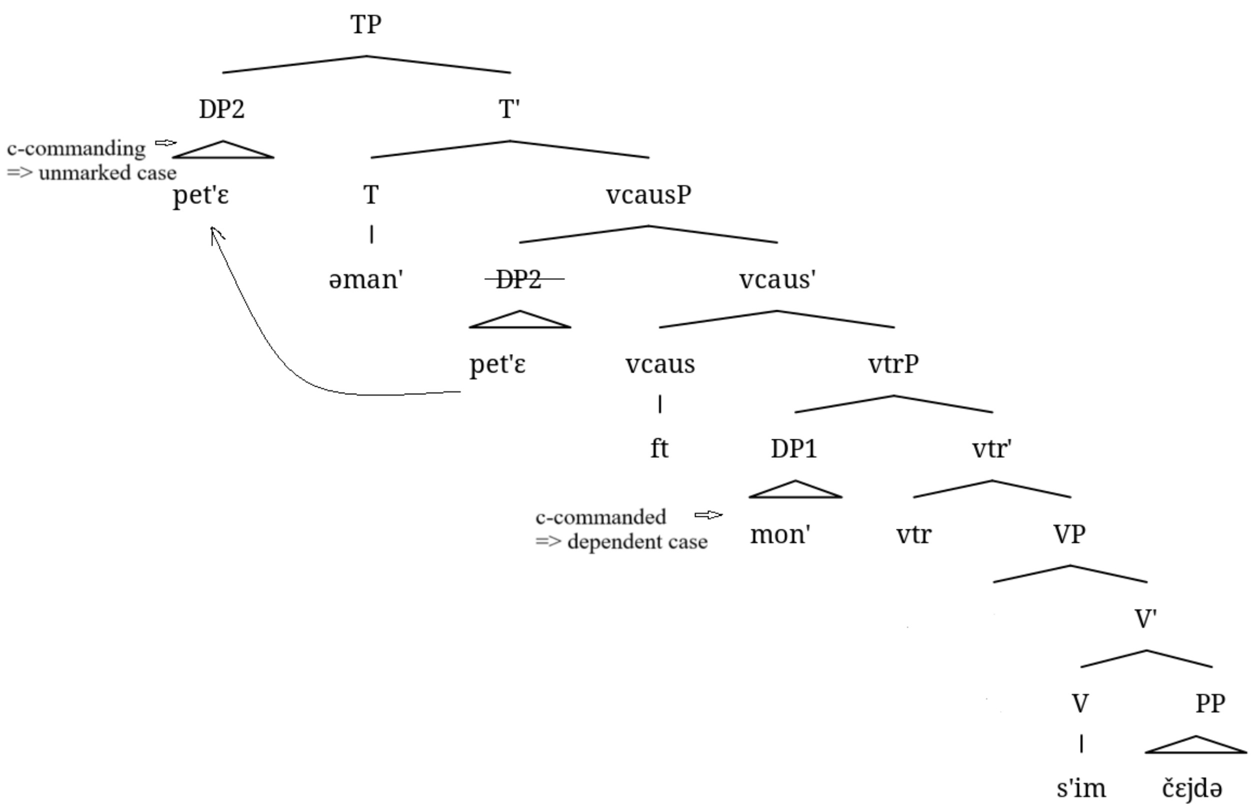
(377) CAUSATIVIZED TWO-PLACE BASE PREDICATE WITH AN INHERENT CASE-MARKED OBJECT (Nikiforova 2018: 475)

pet'ɛ	*s'im-ft-əman'*	*mon'*	*čɛj-də*
Petja[NOM]	drink-CAUS-PST.1SG.O.3SG.S	1SG.GEN	tea-ABL

'Petja gave me (some) tea to drink [lit.: Petja made me drink some tea].'

Again, under DCT, it is easy to model the genitive case assignment in (377). The object gets an inherent case and does not compete for a structural case. The causee, on the other hand, is now c-commanded by the new subject, so it gets the dependent case, see the structure in Figure 47.

Figure 47: *Case assignment in a clause with a causativized two-place base predicate with an inherent case-marked object*



Now, let us look at a transitive clause with a bona-fide definite DO. When a transitive predicate is causativized, the DO preserves its object case, and the causee is forced into a postpositional construction, as in (378)–(379). Specifically, in Moksha, the causee gets embedded in a PP headed by the postposition *kɛctə*. It is worth noting that the causee is marked with the genitive case, which might be confusing at first. Crucially, the genitive on the *al'ɛz'ən'* in (379) is not the genitive of the DO, but the genitive that is assigned within a PP. Therefore, despite being technically marked as genitive, the causee has to receive alternative marking — to be embedded under *kɛctə*.

(378) TRANSITIVE BASE PREDICATE

al'ɛ-z'ə	*ar̥t-əz'ə*	*kut-t'*
man-1SG.AGR.SG[NOM]	paint-PST.3SG.O.3SG.S	house-DEF.SG.GEN

'My father painted the house.'

(379) CAUSATIVIZED TRANSITIVE PREDICATE (Nikiforova 2018: 476)

d'ɛd'ɛ-z'ə	*ar̥t-ft-əz'ə*
mother-1SG.AGR.SG[NOM]	paint-CAUS-PST.3SG.O.3SG.S

kut-t'	[*al'ɛ-z'ə-n'*	*kɛctə*]
house-DEF.SG.GEN	man-1SG.AGR.SG-GEN	hand.EL

'My mother made my father paint the house.'

The mechanics of dependent case assignment in Moksha causative clauses is beyond the scope of my dissertation, and there is no existing analysis readily available to apply to Moksha data. However, the basic idea should be the following. The object DP does not get an inherent case and participates in the dependent-case assignment. It is the lowest DP, and it is c-commanded by another DP, so it receives the dependent case, which is genitive. The new subject of the causative predicate is the highest DP in the domain, which c-commands all the other DPs, so it gets the unmarked nominative case. The tricky part here is the causee, which is in the middle. It is still c-commanded by the subject DP, so it is not immediately obvious why it does not get the dependent case. My argument does not depend on the solution to this problem, and I leave this for future research ("!?" in the figure reflects the uncertainty of the theory regarding the case that should be assigned to a DP that is c-commanding and c-commanded at the same time). For now, I just assume that there is some constraint disallowing dependent case assignment to two DPs in Moksha.

Because of this constraint, the DP that would be both c-commanding and c-commanded has to receive alternative marking. This results in Figures 48-49. Figure 48 shows the case assignment conflict with three DPs present in the domain, and Figure 46 shows the structure with the alternative marking.

Figure 48: *Case assignment in a clause with a causativized transitive predicate; three DPs*

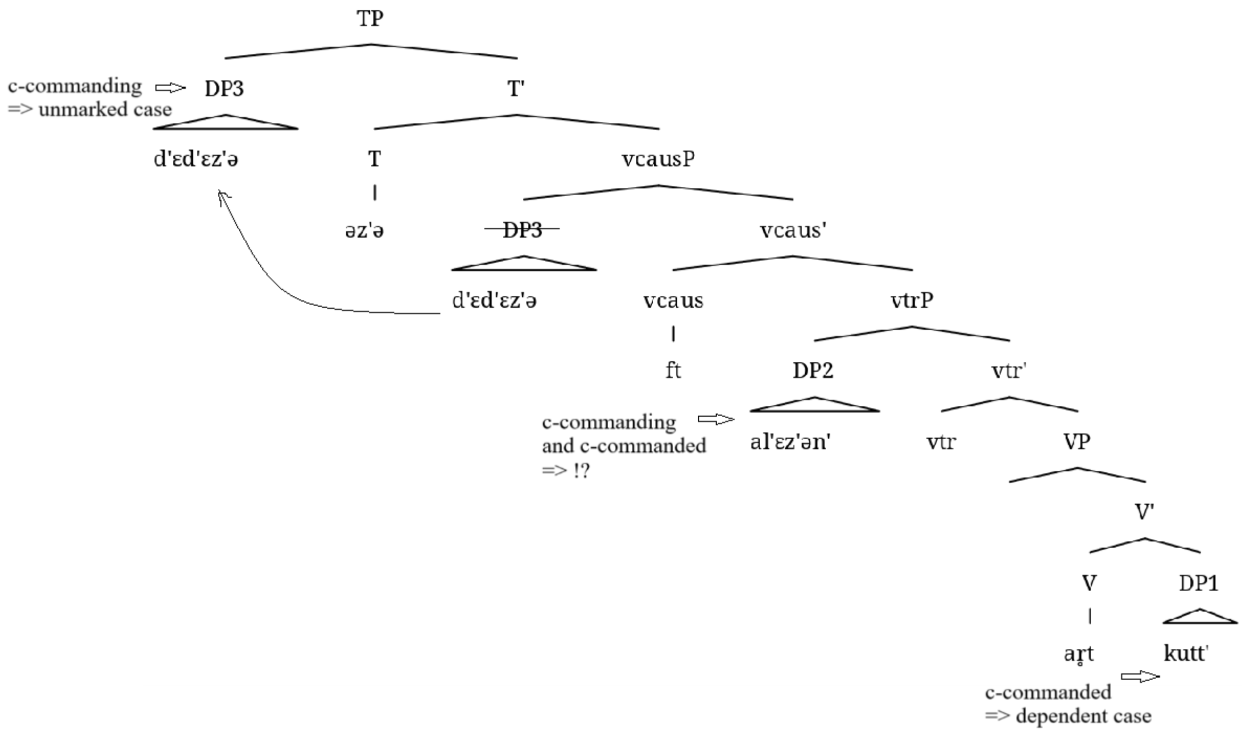
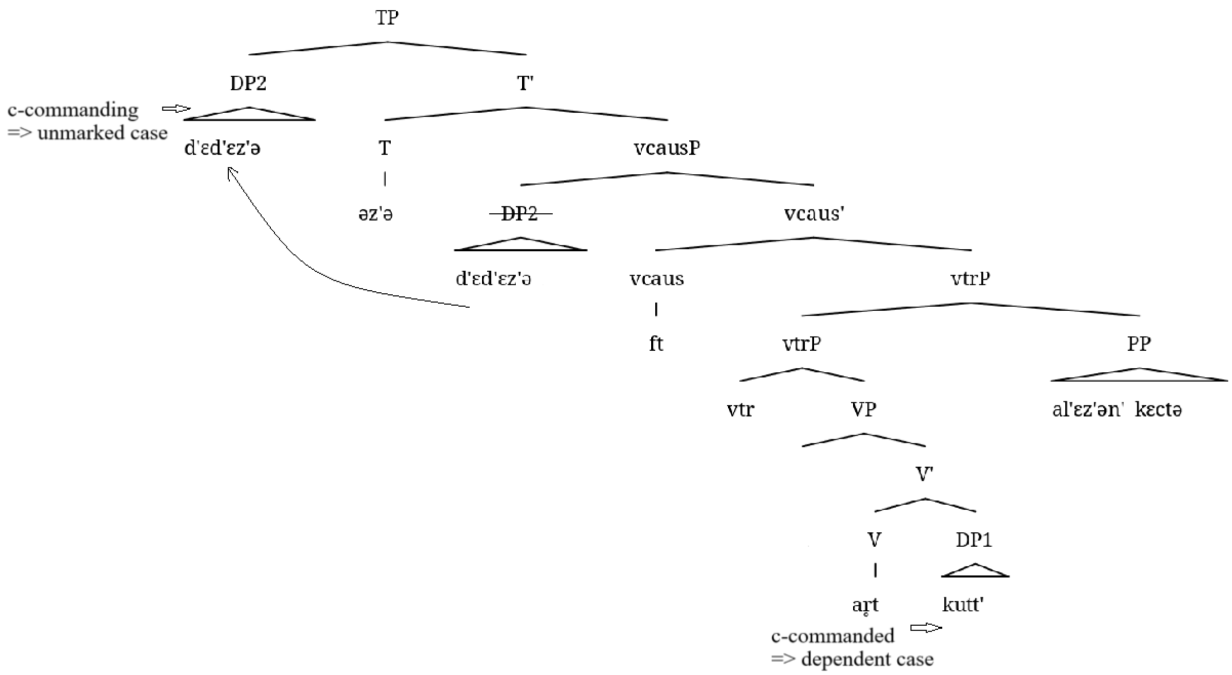


Figure 49: *Case assignment in a clause with a causativized transitive predicate; two DPs and a PP*



The crucial piece is the data concerning the causativization of transitive clauses with bare indefinite DOs, as in (380). If bare DOs were SNs, which do not get case and do not contribute to the transitivity of the clause reflected in the argument alignment, then we would expect a pattern similar to the one observed in bivalent intransitive clauses like (376)–(377): the bare DO remains bare, and the causee as the lower of the two DPs gets the object genitive case. This is not what we see in (380) though. As (380) shows, the DO indeed remains bare, but the causee receives an alternative marking; it gets embedded in a PP with the postposition *kɛctə*, the same way it was in (378)–(379) with a definite genitive DO. An attempt to mark the causee just with the genitive case results in ungrammaticality of the sentence, see (380)b.

(380)	a.	*maša*	*narda-fc'*	***morkš***	*s't'ər'-n'ɛ-t'*	***kɛctə***
		Masha[NOM]	wipe-CAUS.PST.3.SG	table	girl-DIM-DEF.SG.GEN	hand.EL

b. *_maša_ _s't'ər'-n'ɛ-t'_ _narda-ft-əz'ə_ **_morkš_**
Masha[NOM] girl-DIM-DEF.SG.GEN wipe-CAUS-PST.3SG.O.3SG.S table
‘Masha made the girl wipe the table.’

Under an SN analysis of bare indefinite DOs, there is no way to explain why the causee does not get the genitive case. Under the DCT analysis, a bare DO is not a case competitor, which results in two DPs within the domain, and the lower gets assigned the dependent case, while the upper receives the unmarked case, see a hypothetical structure in Figure 50. The fact that the causee has to be embedded in a PP suggests another analysis, the one shown in Figures 51-52. The DO is a full DP, and it gets the dependent case as the lowest DP in the domain. Due to the constraint discussed above regarding (378)–(379) with a definite DO, the causee has to receive alternative marking.

Figure 50: *Case assignment in a clause with a causativized transitive predicate with a bare DO; two DPs and an SN*

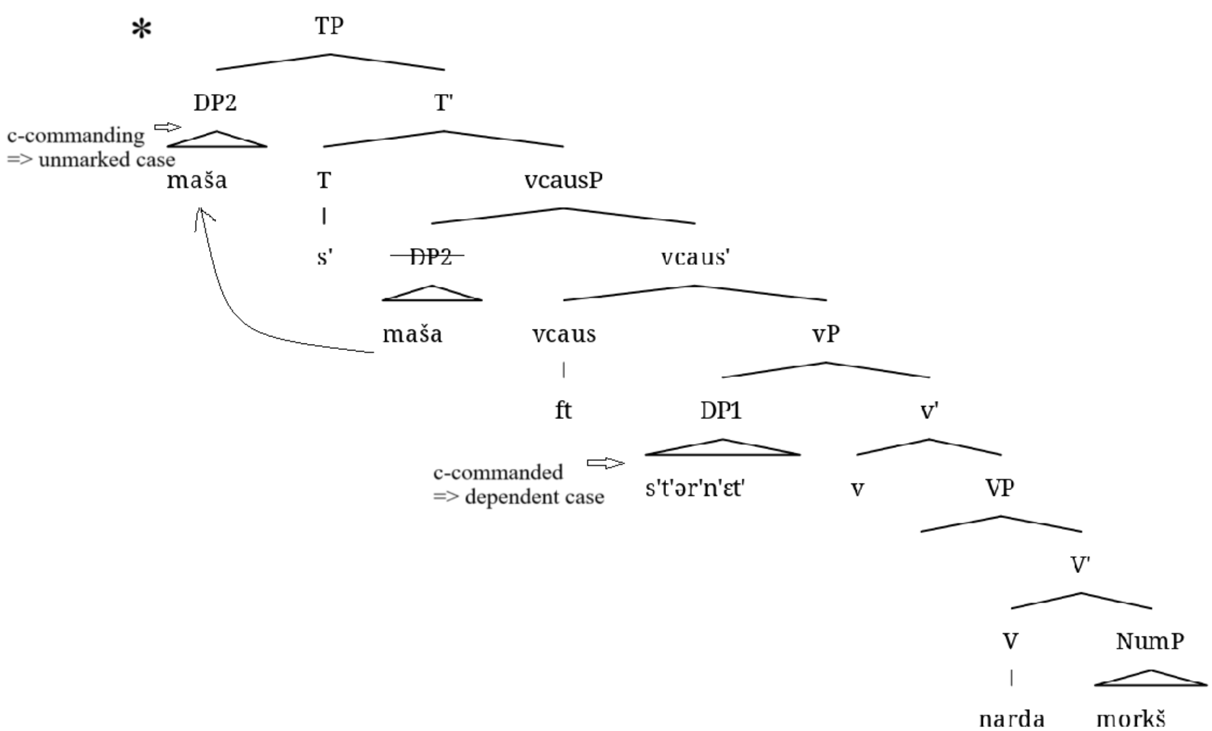


Figure 51: *Case assignment in a clause with a causativized transitive predicate with a bare DO; three DPs*

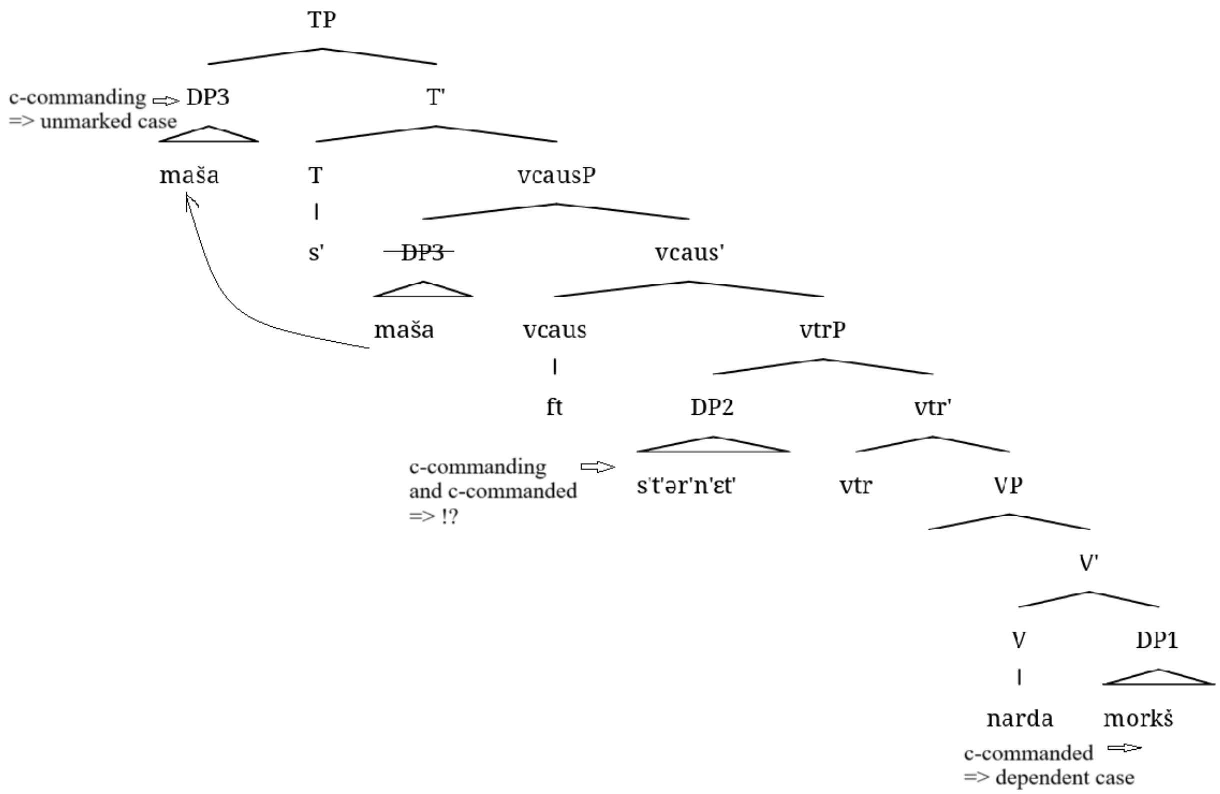
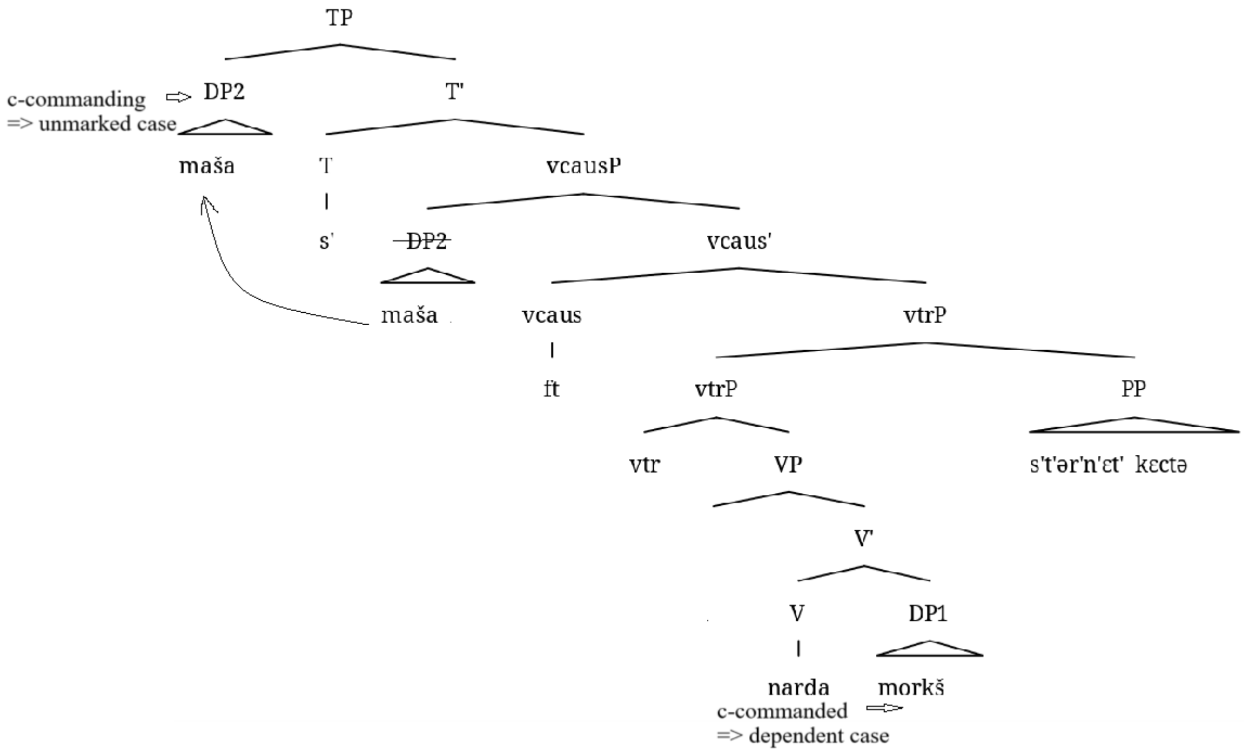


Figure 52: *Case assignment in a clause with a causativized transitive predicate with a bare DO; two DPs and a PP*



Thus, bare indefinite DOs behave syntactically as full DPs that get assigned the genitive case, not like SNs. From this, I conclude that the absence of an overt morphological genitive marker is a superficial phenomenon, which has nothing to do with the absence of corresponding structure. Instead, I propose that the zero morpheme is just the realization of the genitive case in the absence of features that also result in definite or possessive marking. This conclusion is very similar to the one made by Kornfilt (1990; 1996; 2003) for Turkish. She uses the term "weak case" taking it from (de Hoop 1992) to refer to a structural case that has no phonological realization due to the absence of a specificity feature on the nominal, see Section 5.4 below for the discussion of a weak case analysis for Moksha bare DOs.

Summing up, in this section, I provided empirical evidence that bare indefinite DOs in Moksha are full DPs, based on their behavior in causative constructions. Regardless of the specific case framework adopted, the fact that bare DOs pattern with definite genitive DOs — rather than with indirect objects of intransitive predicates — supports the conclusion that they are cased DPs, not SNs.

5.4. A "Weak case" analysis of Moksha bare DOs

In the previous sections, I argued that, unlike bare complements of postpositions, bare DOs do not exhibit evidence of lacking a DP layer. On the contrary, case assignment in causative constructions suggests that bare DOs are full DPs. This finding rules out a small nominal analysis of bare DOs. In this section, I explore an alternative morphological explanation. Specifically, I consider the possibility that Moksha indefinite DOs appear bare due to the absence of certain features. When these features are missing, the object case in Moksha is realized as a null morpheme.

I will begin with a brief introduction to the concept of "weak case", followed by an overview of Kornfilt's (2003) analysis of Turkish. Finally, I will outline how a similar approach can account for the Moksha data.

5.4.1. Weak case

The idea of weak case proposed by de Hoop (1992) comes from the observation that some languages exhibit more than one objective case depending on the (in)definiteness of the object. This distinction may be morphologically realized, as in Finnish (381), or appear as a zero marker, as in Turkish (382). In Finnish, indefinite objects take the so-called partitive case (381)a, whereas definite objects receive accusative marking (381)b. In Turkish, indefinite objects remain bare (382)a, while definite objects are marked with the accusative case (382)b.

(381) FINNISH (adapted from de Hoop 1992: 71)

a. *ostin* *leipää*
buy.PST.1SG bread.PART
'I bought (some) bread.'

b. *ostin* *leivän*
buy.PST.1SG bread.ACC
'I bought the bread.'

(382) TURKISH (adapted from de Hoop 1992: 71)

a. *ali kitap-Ø* *okudu*
Ali book-ACC2 read.PST.3SG
'Ali read a book.'

b. *ali kitab-ɨ* *okudu*
Ali book-ACC1 read.PST.3SG
'Ali read the book.'

The weak vs. strong distinction in case marking is generally linked to the definiteness of the object, as demonstrated in (381)–(382). However, the role of interpretation in distinguishing weak from strong case is less straightforward. De Hoop notes that even definite noun phrases can be marked with a weak case in irresultative predicates (383). Based on this, she concludes that weak and strong case distinctions arise from different semantic types of the object noun phrases and their different relationship to the predicate head. While I abstract away from the semantic details of these object types, I adopt the general idea that accusative case correlates with a greater degree of independence between the object and the predicate.

(383) FINNISH (adapted from de Hoop 1992: 72)

puhumme *suomea*
speak.PRS.1PL Finnish.PART
'We speak Finnish'

To define the strong vs. weak distinction, de Hoop (1992: 76) takes as a starting point the distinction between syntactic (structural) and lexical (oblique) cases, arguing that structural case can be either strong or weak. This aligns with the assumptions I make in this dissertation (see Chapter 3, Section 3.1.1) regarding case taxonomy. Although weak and strong objects exhibit interpretational differences, these differences are not linked to theta-role assignment, which challenges an explanation based on the structural vs. inherent/lexical case distinction, according to which inherent case is associated with a specific theta-role (see Woolford 2006 for discussion).

In addition to providing examples of languages where DOs exhibit differential marking that cannot be explained through the structural vs. inherent case distinction, De Hoop defines weak vs. strong case as follows:

"Weak Case, then, is a structural default Case licensed at D-structure in a certain configuration. For instance, government by a transitive verb can license weak Case; some transitive verbs can only license weak Case, whereas other transitive verbs can only license strong Case. Most transitive verbs seem to be able to alternate between licensing either weak or strong Case. Weak Case is not related to a specific theta-role, which distinguishes it from lexical or oblique Case. Strong Case differs from weak Case in that it is licensed at S-structure." (de Hoop 1992: 80)

However, this definition presents challenges both for the current theoretical framework and for the assumptions made in this dissertation, as the D-structure vs. S-structure distinction no longer holds, and case is not licensed under government. In her account of Turkish case, Kornfilt (2003) defines weak structural case as morphologically null structural case on nominals marked as [-specific], while strong structural case is an overtly realized case on nominals marked as [+specific]. As mentioned earlier, I remain open to different ways in which weak and strong interpretations can manifest, beyond just definiteness and specificity. In my analysis of Moksha DOs, I assume that structural case is assigned configurationally under DCT (see Chapter 1, Section 1.2.2.1), and its overt realization depends on an additional feature on D, which I label [+/-F1]. This feature can be specificity or some other feature relevant for interpretation, movement out of the predicate etc.

In addition to accounting for the differential marking per se, the weak vs. strong case distinction analysis has been designed to account for the adjacency effects observed, e.g., in Turkish. Specifically, marked objects can undergo movement (384), whereas morphologically unmarked objects must remain adjacent to the verb (385). Kornfilt's weak case proposal for Turkish bare DOs posits that when the case head is empty, as in [-specific] nouns, the noun head can move to the case head (K) and then to the verbal head (V) (see Figure 53). This movement is blocked when K contains overt phonological material.

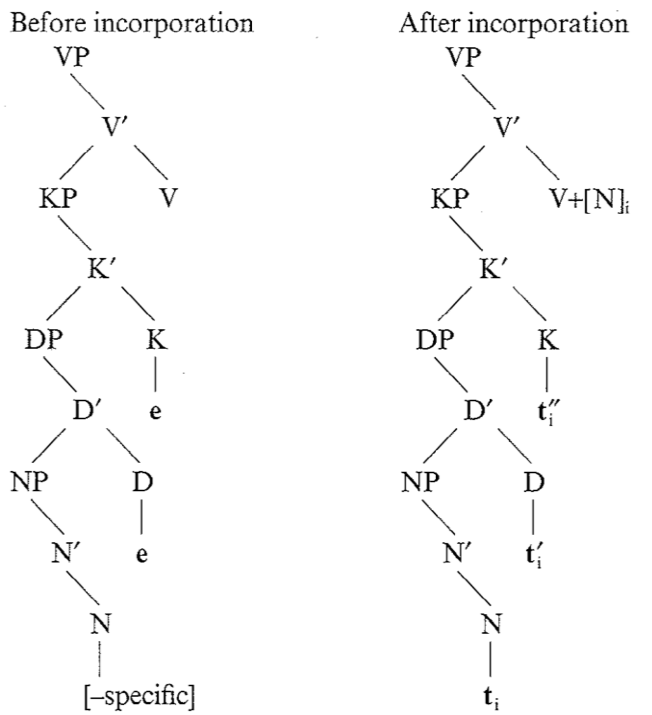
(384) TURKISH (Kornfilt 2003: 132-133)
a. *bir* *daha* [[*sen-in* *gibi*] *bir terzi*]-*yi* *bul-a-ma-m*
one time you-GEN like a tailor-ACC find-ABIL-NEG-1SG

b. [[*sen-in* *gibi*] *bir terzi*]-*yi* *bir* *daha* *bul-a-ma-m*
you-GEN like a tailor-ACC one time find-ABIL-NEG-1SG
'I won't ever be able to find a tailor [+specific] like you again.'

(385) TURKISH (Kornfilt 2003: 132-133)
a. *bir* *daha* [[*sen-in* *gibi*] *bir terzi*] *bul-a-ma-m*
one time you-GEN like a tailor find-ABIL-NEG-1SG

b. *[[*sen-in* *gibi*] *bir terzi*] *bir* *daha* *bul-a-ma-m*
you-GEN like a tailor one time find-ABIL-NEG-1SG
'I won't ever be able to find a tailor [-specific] like you again.'

Figure 53: *Incorporation of the N head into the V head*



As I will demonstrate in Section 5.4.2, Moksha data pose challenges for an incorporation-based analysis, as bare DOs can appear in various positions within the clause. Instead, I propose a featural analysis of the surface realization of the object case that is more flexible than incorporation while still accounting for adjacency effects in languages where such constraints apply.

5.4.2. "Weak case" analysis of Moksha bare DOs

In this section, I present a weak case analysis of Moksha bare DOs. I argue that in (327), repeated here as (386), the genitive-marked DO in (386)a bears a strong case, while the bare DO in (386)b bears a weak case. Crucially, both DOs are full DPs that receive structural case. The difference in their morphological realization arises from the presence of the feature [+def*] on the D head of the DO in (386)a, which not only accounts for the interpretational differences between (386)a and (386)b but also triggers object agreement on the verb. Below, I elaborate on this analysis.

(386) a. *vas'ɛ* *ker'-əz'ə* *šuft-t'*
Vasja[NOM] cut-PST.3SG.O.3SG.S tree-DEF.SG.GEN
'Vasja cut the tree.'

b. *vas'ɛ* *ker'-**s'*** *šuftə*
Vasja[NOM] cut-PST.3[SG] tree
'Vasja cut a tree.' (Toldova 2018: 575)

The analysis I provide here explains why the overt object case marker appears in (386)a but is absent in (386)b, while maintaining the assumption that both objects are DPs and neither is caseless. Additionally, this analysis must account for the interpretational difference between the two: only non-specific and non-possessed indefinite DOs appear as bare. Finally, it must explain why object agreement is present with overtly marked DOs but absent with bare ones.

Given these empirical patterns, the weak vs. strong case analysis provides a suitable framework. This approach has been widely used to explain DOM, including asymmetrical DOM, in which some objects receive overt case marking while others remain bare. Furthermore, weak and strong case marking generally correspond to weak and strong interpretational readings.

As I already said, I do not define weak and strong readings in a strict way and do not claim that they correspond to a definiteness or specificity feature in particular, but rather reflect the status of the object within the predicate more generally. In Moksha, the use of the definite declension does not perfectly align with a specificity feature (see Chapter 3). However, scope distinctions provide evidence for a weak vs. strong contrast. Specifically, bare DOs take narrow scope, while genitive DOs take wide scope, as shown in (387):

(387) a. *ɛr'* *s't'ər'-n'ɛ-s'* *sɛv-əz'ə* *mar'-t'*
every girl-DIM-DEF.SG take-PST.3SG.O.3SG.S apple-DEF.SG.GEN
'Every girl took a [common] apple [one girl after another].'
*'Every girl took a [different] apple.'

b. *ɛr'* *s't'ər'-n'ɛ-s'* *sɛv-s'* *mar'*
every girl-DIM-DEF.SG take-PST.3[SG] apple
'Every girl took a [different] apple.'
*'Every girl took a [common] apple.' (Toldova 2018: 604)

To adapt the weak case analysis to Moksha bare DOs, I make two key modifications. First, unlike de Hoop (1992) and Kornfilt (2003), who assume that structural case is assigned under government, I propose that in Moksha, the DO receives case by virtue of being c-commanded by another still caseless DP within the clause — namely, the subject DP. However, the exact mechanism of case assignment does not play a crucial role in my argument. Second, I discard the D- vs. S-structure distinction assumed by de Hoop (which was discarded from the current theory in general) and adopt Kornfilt's approach, which relies on feature-based morphological realization of the syntactic structure.

At the same time, I do not fully adopt Kornfilt's view that [+specific] nominals receive overt case marking while [-specific] nominals remain bare. Instead, I introduce a formal feature, [+def*], which governs the overt realization of object case.

Recall that the use of the Moksha definite declension is not solely determined by definiteness or specificity (see Chapter 3). In Moksha, noun phrases get formal definiteness marking when they are definite or topical and belong to a definite set. I propose that it is the feature [+def*] that results in the definiteness marking. If we examine the DOM pattern, we observe that DOs marked as definite must bear overt morphological case, whereas bare DOs cannot be marked as definite. Therefore, I argue that in (386)a, the object case is realized as -*t'* due to the presence of [+def*] alongside the case feature. In contrast, when [+def*] is absent, as in (386)b, the object case is realized as Ø.

The introduction of [+def*] also explains why object agreement is obligatory with genitive-marked DOs. If the agreement probe on the verbal head searches for the feature [+def*], and the definite declension results from the presence of [+def*] on the D head, the analysis predicts that all DOs marked with the genitive of the definite declension will trigger object agreement.

Now let us turn to bare DOs. Unlike definite DOs, indefinite DOs are marked as [-def*] (or are not marked as [+def*]). In this context, the exponent of the object case is -Ø rather than *-t′*. Crucially, the agreement probe on the verb, which looks for [+def*], does not see such indefinite DOs, because they are not marked as [+def*].

In the languages where adjacency requirement holds for such bare objects, this adjacency can be explained by restriction on scrambling; if all movement must be associated with specific features, such as [+def*], then the absence of those features necessarily leads to impossibility to move.

One remaining question concerns the *-n′* marking found on plural definites, proper names and possessed DOs with most nominal agreement markers (see Chapter 3 for genitive allomorphs). All plural definites and proper names in Moksha must be marked with the genitive case and trigger object agreement, as shown in (388) and (389). Similarly, all possessed DOs must bear a nominal agreement marker, be marked with an overt genitive case, and optionally trigger object agreement (390).

(388) PLURAL+DEFINITE → OJECT AGREEMENT (Toldova 2018: 607)

son	*kan'-n'ə-**sin'ə***	*panar̥-**n'ə-n'***	*n'ur̥kɛn'a-stə*
3SG[NOM]	carry-FREQ-NPST.3PL.O.3SG.S	shirt-DEF.PL-GEN	short-EL

'He wears shirts short.'

(389) PROPER NAME → OBJECT AGREEMENT (Toldova 2018: 597 from Moksha Corpus)

nu	*məz'ardə*	*aldun'ɛ*	*n'ɛj-**əz'ə***	*jaku-n'ɛ-**n'**,*
PTCL	when	Dunja[NOM]	видеть-PST.3SG.O.3SG.S	Jacob-DIM-GEN
son	*ɛvəc'*		*ezdə-nzə*	
3SG[NOM]	be_afraid.PST.3[SG]		IN.ABL-3SG.AGR	

'When Dunja saw Jacob, she became afraid of him.'

(390) POSSESSED DO → OBJECT AGREEMENT (Moksha Corpus)

i	*mon*	*s'ora-z'ə-**n'***	*kɛl'gən'd'-**in'ən'***
and	1SG[NOM]	šy-1SG.AGR.SG-GEN	trick-PST.3.O.1SG.S

'I tricked my son as well.'

Notably, in all three cases, the genitive marker is not the same as the definite declension marker -*t'*; instead, it appears as -*n'*. I believe this can be modelled in post-syntactic terms, with -*t'* being the realization of case together with number and [def*], as opposed to realization of case (-*n'*) separately from number and definiteness. I leave the details of this analysis for future research.

5.5. The takeaway from the Moksha asymmetrical complement marking

Chapters 4 and 5 together constitute a case study of nominal complements in Moksha, focusing on two syntactic contexts that allow for variable case marking: postpositional complements and DOs. In both contexts, nominals can either appear in the genitive case and trigger agreement on the head or lack overt morphological case marking and fail to trigger agreement. While these two structures may seem superficially similar, I have demonstrated that they have distinct underlying representations. Specifically, I showed that postpositional complements without overt marking are structurally reduced SNs, and their lack of case marking follows from this reduced structure. In contrast, bare DOs are full DPs whose case is realized as null rather than being truly caseless. This means that the absence of a genitive marker on DOs does not indicate a smaller structural size or a lack of case.

This case study has two key implications for syntactic theory in the context of the DP hypothesis. First, not all argument nominals in a language must be full DPs — some may be structurally smaller and genuinely caseless. This holds even in languages with overt definite determiners, which are generally not considered NP-languages (Bošković 2008), as discussed in Chapter 2, Section 2.4. Second, the absence of overt morphological case does not necessarily imply that a nominal is a small, caseless structure, even in languages where such structures exist in certain contexts. As I have demonstrated for Moksha, bare DOs, despite appearing caseless and superficially resembling SNs, are in fact full DPs that carry a morphologically null case. Crucially, these co-exist alongside true SNs in postpositional complements. This variation in the analyses of what is widely discussed as DOM, in turn, shows that DOM is a descriptive term rather than an analysis, as I also discuss in Chapter 7, Section 7.2 below.

Thus, while it may be tempting to attribute the absence of morphological case to a reduced syntactic structure, this explanation is not always valid, even when other properties — such as the inability to trigger agreement or accommodate certain adnominal modifiers — suggest syntactic "deficiency". Properties commonly associated with SNs must be carefully evaluated within the specific grammatical system of a given language. In Chapter 7, I outline a set of diagnostics for identifying SNs cross-linguistically, emphasizing that these diagnostics are conditional and must be assessed in conjunction with other language-specific properties.

Chapter 6: Small nominals of different sizes: a case study of Hill Mari DOM

This chapter examines Differential Object Marking (DOM) in Hill Mari, a Finno-Ugric (Uralic) language. While all DOs in finite clauses must be marked with the accusative case (391), non-finite clauses allow for differential marking: direct objects (DOs) can appear either in the accusative or as bare (392), where "bare" refers to the absence of overt case morphology.

(391) FINITE CLAUSE → ACCUSATIVE DO
šə̂žar-em *každə̂j* *kečə̈-n* ***ə̂škal-ə̂m*** / ****ə̂škal*** *šə̂pš-əl-ə̂n*
younger_sister-AGR.1SG[NOM] every day-GEN cow-ACC cow pull-ITER-PRET
'My younger sister milked the/a cow every day.'[34]

(392) NON-FINITE CLAUSE → ACCUSATIVE / BARE DO
mӛn' ***jäm-ə̈m*** / ***jäm*** *kapajə̂-maš-eš* *jangə̂l-en-äm*
1SG[NOM] hole-ACC hole dig-NMLZ2-LAT tire-PRET-1SG
'I'm tired of digging a hole.'

Despite having only two surface forms — accusative-marked DOs and bare DOs — Hill Mari exhibits multiple types of bare DOs, each with distinct syntactic properties. For instance, bare DOs

[34] The examples used in this chapter are drawn from a database created jointly with my colleague Anastasiia Sirotina. I do not explicitly mark her authorship for each individual example drawn from this source.

in nominalizations can take plural marking (393), whereas those in converbial constructions cannot (394).

(393) *maša-n* ***təgər-vlä*** *ərgə-mə-žə-m* *už-Ø-əm*
Masha-GEN shirt-PL sew-NMLZ-AGR.3SG-ACC see-AOR-1SG
'I saw Masha sew shirts.'

(394) *maša* *posudə-m,* ***mərə*** / ****mərə-vlä*** *mər-en,* *məšk-ən*
Masha[NOM] dishes-ACC song song-PL sing-CVB wash-PRET
'Singing songs, Masha washed the dishes.'

I argue that bare DOs in Hill Mari are small nominals, meaning they lack a DP layer, and that their distinct syntactic behaviors reflect variations in their size. For instance, bare DOs in nominalizations lack only the DP layer and are as large as NumP, while bare DOs in converbial clauses are even smaller, falling below NumP. Unlike the Moksha examples discussed in Chapters 4 and 5 — where I demonstrated that not all bare nominals should be analyzed as SNs — this chapter highlights the internal diversity of SN structures within a single language.

This chapter is structured as follows. Section 6.1 provides an overview of Hill Mari nominal morphosyntax, covering nominal categories (6.1.1), adnominal modifiers (6.1.2), DP structure (6.1.3), and nominal agreement (6.1.4). Section 6.2 examines non-finite clause structures in Hill Mari, as DOM occurs specifically in these contexts. Section 6.3 describes the DOM pattern in Hill Mari. Section 6.4 presents my analysis of Hill Mari DOM, focusing on the syntactic structure of different bare DOs. Section 6.5 discusses four alternative analyses and explains why they fail to fully account for the data. Section 6.6 contains some implications of my SN analysis for noun phrase structure. Section 6.7 summarizes the findings and draws conclusions.

6.1. Hill Mari nominal morphosyntax

Hill Mari is a Mari language within the Finno-Ugric branch of the Uralic family, spoken by approximately 15,000 people[35] in the Mari El Republic (Russia). In the nominal domain, three morphological categories are marked: number, nominal agreement, and case (cf. Moksha, which expresses four, see Chapter 3).

In this section, I provide a brief overview of how these nominal categories are expressed, outline the basic properties of the main adnominal modifiers, and propose a structural analysis of the Hill Mari DP in line with the theoretical framework adopted in Chapter 2. I conclude with a discussion of nominal agreement, which, while not central, plays a role in my analysis of bare DOs.

6.1.1. Expression of nominal categories

The Hill Mari case paradigm consists of 10 cases (Alhoniemi 1993: 45–46; Tuzharov 1987: 73–74; Pleshak & Kashkin 2023: 13–14), see Table 12. These include nominative of subjects, accusative of DOs, genitive of possessors, dative of indirect objects, and several other cases[36] expressing different spatial and other peripheral relations.

[35] According to 2020 census.

[36] I refer to inherent cases as "cases" in line with traditional grammatical descriptions. As mentioned in Chapters 1 and 3, however, these forms may be analyzed syntactically as P.

Table 12: *Hill Mari case paradigm* (from Pleshak & Kashkin 2023: 13)

Case	Marker
Nominative	∅
Accusative	*-(ə̂m / -(ə̈)m*
Genitive	*-(ə̂)n / -(ə̈)n*
Dative	*-lan / -län*
Inessive	*-(ə̂)štə̂ / -(ə̈)štə̈*
Illative	*-(ə̂)š(kə̂) / -(ə̈)š(kə̈)*
Lative	*-eš*
Similative	*-la / -lä*
Comitative	*-ge*
Caritive	*-de*

Hill Mari adnominal DPs (usually possessors) trigger agreement on the head nominal reflecting the person and number features of the adnominal DP (395). However, except for certain kinship terms (see Chapter 6, Section 6.4.2.2), nominal agreement in Hill Mari is optional (Pleshak 2023b: 170).

(395) a. *mən'-ən pört-em-ən / pört-ən*
1SG-GEN house-AGR.1SG-GEN house-GEN
'of my house'

b. *maša-n pört-šə̈-n / pört-ən*
Masha-GEN house-AGR.3SG-GEN house-GEN
'of Masha's house'

The ordering of agreement markers relative to case markers varies: in the genitive, accusative, comitative, and caritive cases, the agreement marker precedes the case marker (396)–(397); in spatial cases, it follows (398); while in the dative and similative cases, both orders are possible (399).

(396) GEN → AGR > CASE
pört-em-ən / **pört-ən-em*
house-AGR.1SG-GEN house-GEN-AGR.1SG
'of my house'

(397) CAR → AGR > CASE
pört-em-de / **pört-te-em*
house-AGR.1SG-CAR house- CAR-AGR.1SG
'without my house'

(398) IN → CASE > AGR
pört-əšt-em / **pört-em-əštə*
house-IN-AGR.1SG house- AGR.1SG-IN
'in my house'

(399) DAT → AGR > CASE / CASE > AGR
pört-em-län / *pört-län-em*
house-AGR.1SG-DAT house-DAT-AGR.1SG
'for my house' 'for my house'

Hill Mari distinguishes singular and plural number. The singular form has no overt marker, while the plural is marked with the suffix *-vlä*:

(400) a. *pört*
house[SG]
'house'

b. *pört-vlä*
house-PL
'houses'

The plural marker always precedes the case marker, regardless of case type (401). However, it can either precede or follow the possessive affix (402)–(403). In spatial cases, the plural marker can precede possessive affixes only if they follow the case marker (403).

(401) a. ACCUSATIVE
pört-vlä-m / * *pört-əm-vlä*
house-PL-ACC house-ACC-PL
'[I see] houses'

b. INESSIVE
pört-vlä-štə / * *pört-əštə-vlä*
house-PL-IN house-IN-PL
'in houses'

(402) ACCUSATIVE
pört-em-vlä-m / *pört-vlä-em-əm*
house-AGR.1SG-PL-ACC house-PL-AGR.1SG-ACC
'[I see] my houses'

(403) INESSIVE
a. *pört-em-vlä-štə* / *pört-vlä-št-em*
house-AGR.1SG-PL-IN house-PL-IN-AGR.1SG

b. **pört-vlä-em-əštə*
house-PL-AGR.1SG-IN
Int.: 'in my houses'

In contrast to Moksha (see Chapter 3), Hill Mari nominal affixes follow a strictly agglutinative pattern. They are combined in a fixed order (sometimes, there is more than one possible order, but the ordering is restricted, not arbitrary), with no restrictions on which categories can co-occur within a single nominal form.

6.1.2. Adnominal modifiers

Hill Mari features a range of adnominal modifiers, including bare nouns (404)a, adjectives (404)b, noun phrases with attributivizers (404)c, interrogative and indefinite pronouns (404)d, numerals (404)e, possessors (404)f, demonstrative pronouns (404)g, universal quantifiers (404)h, and participial relative clauses (404)i. Additionally, finite relative clauses, infinitive clauses, and certain postpositional phrases can function as modifiers.

(404) a. BARE NOUN
püšängə ukš
tree branch
'tree branch'

b. ADJECTIVES
izi ukš
little branch
'little branch'

c. NOUN PHRASE WITH -ATTR
ələštäš-än ukš
leaf-PROP branch
'branch with leaves'

d. INDEFINITE PRONOUN
ta-maxan' ukš
INDEF-what branch
'some branch'

e. NUMERAL
kok ukš
two branch
'two branches'

f. POSSESSOR
mən'ə-n ukš-em
1SG-GEN branch-AGR.1SG
'my branch'

g. DEMONSTRATIVE		h. UNIVERSAL QUANTIFIER		i. PARTICIPIAL REL. CLAUSE		
ti	*ukš*	*cilä*	*ukš*	*tengečə̑*	*vac-šə̂*	*ukš*
this	branch	all	branch	yesterday	fall-PTCP.ACT	branch
'this branch'		'all branches'		'branch that fell yesterday'		

Most adnominal modifiers do not affect the morphological marking of the head noun. However, cardinal numerals and universal quantifiers impose number marking restrictions, and possessors trigger optional nominal agreement on the head, as I have already shown.

Cardinal numerals require singular form of the head noun (405), unless there is a long string of additional adnominal modifiers intervening between the numeral and the head, in which case the noun can bear a plural marker (406), see (Winkler 2023) for discussion.

(405) *stöl* *väl-nä* *kə̂m* *cäškä* / **cäškä-vlä* *šə̈nz-ät*
table top-IN2 three cup[NOM] cup-PL[NOM] sit-NPST.3PL
'There are three cups on the table.' (Winkler 2023: 221)

(406) *stöl* *väl-nä* *kə̂m* ***cever*** ***klovoj*** *cäškä* / *cäškä-vlä* *šə̈nz-ät*
table top-IN2 three beautiful blue cup[NOM] cup-PL[NOM] sit-NPST.3PL
'There are three beautiful blue cups on the table.' (Winkler 2023: 221)

The quantifier *cilä* follows a pattern similar to cardinal numerals. It typically combines with singular-marked nouns (407), but when separated from the noun by intervening material (linearly or structurally), plural marking is allowed (408).

(407) a. ***cilä*** *mən'-ən* *äkä-m* *momoca-š* *ke-n*
all 1SG-GEN elder_sister-AGR.1SG[NOM] sauna-ILL go-PRET
[One or all your sisters went to the sauna?] 'All my elder sisters went to the sauna.'

b. *mən'-ən* ***cilä*** *äkä-m* *momoca-š* *ke-n*
1SG-GEN all elder_sister-AGR.1SG[NOM] sauna-ILL go-PRET
[One or all your sisters went to the sauna?] 'All my elder sisters went to the sauna.
(Solovyeva 2023: 245)

(408) a. ***cilä*** *mən'-ən* *äkä-m-vlä* *momoca-š* *ke-n-ət*
all 1SG-GEN elder_sister-AGR.1SG-PL[NOM] sauna-ILL go-PRET-3PL
'All my elder sisters went to the sauna.'

b. ******mən'-ən* ***cilä*** *äkä-m-vlä* *momoca-š* *ke-n-ət*
1SG-GEN all elder_sister-AGR.1SG-PL[NOM] sauna-ILL go-PRET-3PL
'All my elder sisters went to the sauna.' (Solovyeva 2023: 245)

Adnominal DPs (typically possessors) trigger optional agreement on the head noun, regardless of their phi-features:

(409) a. 1ST PERSON POSSESSOR → OPTIONAL AGREEMENT
mən'-ən *kid-em* / *kid*
1SG-GEN hand-AGR.1SG hand
'my hand'

b. 3RD PERSON POSSESSOR → OPTIONAL AGREEMENT
vas'a-n *kit-šə* / *kid*
Vasja-GEN hand-AGR.3SG hand
'Vasja's hand'

Thus, most adnominal dependents in Hill Mari precede the head noun, and show no concord or other interaction with the nominal head. The exceptions are cardinal numerals and universal quantifiers (I am only discussing the most general universal quantifier *cilä* 'all' here), which influence the number marking of the head, as well as possessors, which trigger optional nominal agreement.

6.1.3. Hill Mari DPs

Since Hill Mari is an articleless language, determining whether a nominal is a full DP relies on indirect theoretical considerations. As stated in Chapter 2, I adopt the view that DPs exist in all languages, though not all nominals within a language must be DPs. I also adopted the assumption that only DPs can trigger agreement and receive structural case. Therefore, if a noun phrase triggers agreement on the verb or nominal head and/or is marked with a structural case, I consider it a full DP.[37]

In Hill Mari, there is no object agreement, but subject agreement in finite clauses is obligatory. The verb must agree with the subject in person and number features (410). In case of 3rd person subjects, number agreement reveals the presence of agreement, while person agreement in 3rd person is null (411). Based on the obligatoriness of subject agreement, I conclude that Hill Mari subjects are DPs.

(410) *mən′ už-Ø-âm, tən′ maša-lan* ***palšə-š-âc*** / ****palšə-š***
1SG[NOM] see-AOR-1SG 2SG[NOM] Masha-DAT help-AOR-2SG help-AOR
'I saw that you helped Masha.' (Pleshak & Kashkin 2023b: 557)

(411) *mən′ už-Ø-âm, ədər-vlä maša-lan* ***palš-evä*** / ****palšə-š***
1SG[NOM] see-AOR-1SG girl-PL[NOM] Masha-DAT help-AOR.3PL help-AOR
'I saw that the girls helped Masha.'

When it comes to objects, there is no agreement to show that they are DPs. However, the overt accusative marking suggests that they are DPs, since only DPs receive structural case. In finite clauses, DOs must be marked with the accusative case (412), which supports the conclusion that Hill Mari direct objects in finite clauses are DPs.

(412) *šəžar-em každâj kečə-n* ***əškal-âm*** / ****əškal*** *šəpš-əl-ân*
younger_sister-AGR.1SG[NOM] every day-GEN cow-ACC cow pull-ITER-PRET
'My younger sister milked the/a cow every day.'

---

[37] I emphasize that the reasoning here does not presuppose a dependency between case and agreement. The only claim being made is that both case and agreement are expected in DPs, but not in SNs.

Having established what prototypical DPs in Hill Mari are, I can identify properties that are a DP should have.

First, all non-generic plural DPs without numerals must carry a plural marker, otherwise they are interpreted as singular (413) (Winkler & Kashkin 2023). The obligatoriness of number marking is expected in DPs given that NumP is below DP.

(413)

stöl	*vəl-nə*	*ta-maxan'*	***tärelkä***	*ki-ä*
table	top-IN2	INDEF-what	plate	lay-NPST.3SG

‘There is a plate on the table (lit.: on the table, some plate lays).’
*‘There are some plates on the table.’ (Bochkova 2018: 64)

Second, nominal agreement is optional within DPs, and its absence cannot serve as the sign of a smaller size. Subjects trigger agreement on the verb whether or not an agreement marker is present (414). Objects marked with the accusative case, which suggests that they are DPs, may lack nominal agreement (415). Thus, nominals that are full DPs by other diagnostics, clearly show optionality of agreement.

(414)

mən'-ən	*plat'-em-vlä*	/	*plat'ə̂-vlä*	*keč-ä-t*	*škap-ə̂štə̂*
1SG-GEN	dress-AGR.1SG-PL[NOM]		dress-PL	hang-NPST.3-PL	dresser-IN

‘My dresses are hanging in the dresser.’

(415)

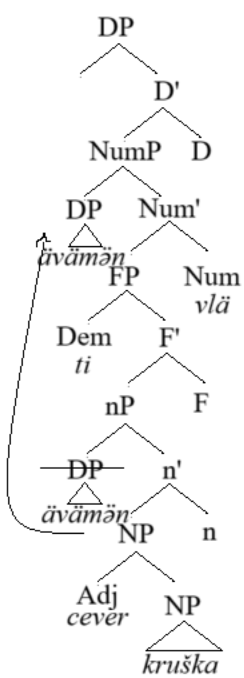
nənə	*vas'a-n*	*ät'ä-žə-n*	*mašinä-(žə-)m*	*pə̂də̂rt-en-ət*
3PL[NOM]	Vasja-GEN	father-AGR.3SG-GEN	car-AGR.3SG-ACC	break-PRET-3PL

‘They broke Vasja’s father’s car.’

The basic structure of a Hill Mari nominal without a cardinal numeral (416) is illustrated in Figure 54. I treat adjectives as NP adjuncts, as the closest adnominal dependents. The possessor originates in Spec,nP, but, as the word order shows (see discussion below), it undergoes movement. At this point, I remain agnostic regarding the exact motivation for this movement. It may be related to phi-feature checking, given that the nominal agreement probe is associated with Num. However, the nature of the movement in cases where overt agreement is absent remains to be determined. In Figure 54, it moves to Spec,NumP, as it is one of the possible landing sites for possessors (see Chapter 2 for the basic hierarchical structure of nominals). However, I do not rule out the possibility that possessors may move even higher in certain derivations. Demonstratives originate in a functional projection below NumP. Like in the other chapters, I label the corresponding projection F_2P (see also Chapter 3 about Moksha). Due to variation in number agreement with noun phrases containing cardinal numerals, as well as the unclear status of Hill Mari numerals themselves (see Winkler 2023 for a tentative analysis of Hill Mari low numerals as heads), I exclude numerical constructions from the discussion and leave them for future research.

(416) *ävä-m-ən* *ti* *cever* *kruška-vlä*
mother-AGR.1SG-GEN this beautiful mug-PL
'these beautiful mugs of my mother's'

Figure 54: *The basic structure of a Hill Mari nominal*



The movement of the possessor to Spec,NumP is suggested by word order facts. Possessors are generally preferred in a position preceding demonstratives (417), whereas lower dependents such as adjectives must follow demonstratives (418).

(417)	***ävä-m-ən***	***ti***	/	?***ti***	***ävä-m-ən***	*sumka-žə̂*
	mother-AGR.1SG-GEN	this		this	mother-AGR.1SG-GEN	table-AGR.3SG
	‘this bag of my mother’s’					

(418)	*ti*	*jakšar*	/	??*jakšar*	*ti*	*tə̂gə̂r*
	this	red		red	this	shirt
	‘this red shirt’					

Further evidence for Spec,NumP as a landing site for possessors comes from extraction data. Lower adnominal modifiers, such as adjectives and demonstratives, must bear a plural marker when extracted from a plural nominal (419)–(420). In contrast, possessors’ plural marking in extraction contexts is degraded (421).

(419)	***izi-vlä-m***	/	****izi-m***	*mən'*	*cäškä-vlä-m*	*pə̂də̂r-t-en-äm*
	small-PL-ACC		small-ACC	1SG[NOM]	cup-PL-ACC	break-CAUS-PRET-1SG
	‘It’s small cups that I broke.’					

(420)	***tidə-vlä-m***	/	****tidə-m***	*mən'*	*cäškä-vlä-m*	*pə̂də̂r-t-en-äm*
	this-PL-ACC		this-ACC	1SG[NOM]	cup-PL-ACC	break-CAUS-PRET-1SG
	‘It’s these cups that I broke.’					

(421)	***ävä-m-ən-əm***	/	??***ävä-m-ən-vlä-m***	*mən'*	*cäškä-vlä-m*
	mother-AGR.1SG-GEN-ACC		mother-AGR.1SG-GEN-PL-ACC	1SG[NOM]	cup-PL-ACC
	pə̂də̂r-t-en-äm				
	break-CAUS-PRET-1SG				
	‘It’s mother’s cups that I broke.’				

This contrast is explained if possessors are extracted from a position above NumP, while lower modifiers originate under Num. When adjectives and demonstratives move out of the noun phrase, they “cross” the Num head and inherit plural marking (Figure 55). Possessors, on the other hand, are extracted from a higher position and do not pass through NumP, thus avoiding plural marking (Figure 56). I remain agnostic regarding the exact nature of the inheritance of the features in the process of extraction. However, it is worth noting that all modifiers cross the DP when extracted, and all of them show up with case-marking, which is in line with my idea that crossing a functional layer somehow triggers the inheritance of the features contained in it.

Figure 55: *Adjective extraction from the noun phrase*

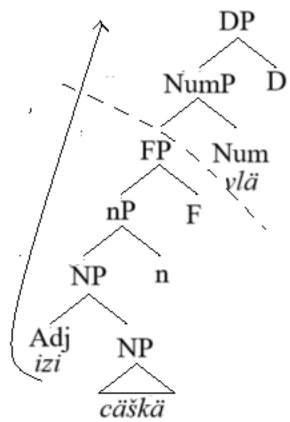
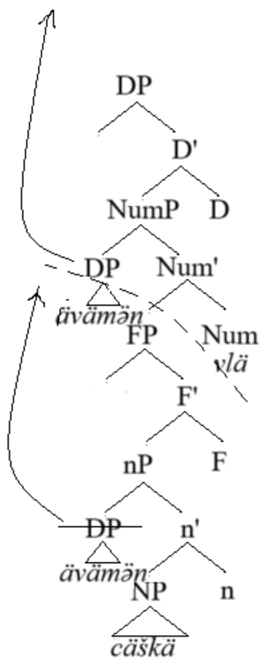


Figure 56: *Possessor extraction from the noun phrase*



6.1.4. Nominal agreement

So far, I have avoided discussing nominal agreement in Hill Mari. Following the universal hierarchical structure outlined in Chapter 2, I assume that the agreement probe in Hill Mari is located in Num. The optionality of nominal agreement, as well as the possibility of an agreement marker being entirely absent rather than defaulting to a specific form, suggests that Hill Mari nominal agreement functions as clitic doubling in terms of (Anagnostopoulou 2007; Preminger 2014). A thorough examination of the nature of Hill Mari agreement falls outside the scope of this work and is not essential to the analysis of DOM. However, since I consider the presence and absence of nominal agreement as part of my diagnostics, it is necessary to establish a representational model for how it operates.

I propose that the agreement marker is hosted by a morphological node generated post-syntactically, positioned to the right of the maximal projection head — D, in this case. Figures 57 and 58 illustrate the derivations of an accusative possessed nominal and an inessive possessed nominal, respectively, showing two different orderings of nominal agreement and case markers.

In an accusative nominal *tən'-ən cäškä-vlä-et-əm* [2SG-GEN cup-PL-AGR.2SG-ACC] 'your cups', the probe on Num becomes active as soon as the D head is merged. The possessor moves to Spec,NumP. The clitic that reflects the possessor's features is hosted by the Agr head, which is inserted post-syntactically on top of D. Following this, another post-syntactic node, K, is introduced to host the case marker (see Figure 57).

In an inessive nominal *tən'-ən cäškä-vlä-št-et* [2SG-GEN cup-PL-IN-AGR.2SG] 'in your cups' the syntactic derivation proceeds similarly. However, in this case, the D head lacks a case feature and instead moves to P forming a complex D+P head. Since I consider locative cases to be syntactically P heads rather than case features on D (as is the case for structural cases), the post-syntactic component behaves differently. Here, the Agr node attaches to the complex D+P node, and the K node is not generated due to the absence of a case feature on D. As a result, the nominal agreement marker follows the inessive marker rather than preceding it (see Figure 58).

Figure 57: *Nominal agreement in an accusative nominal*

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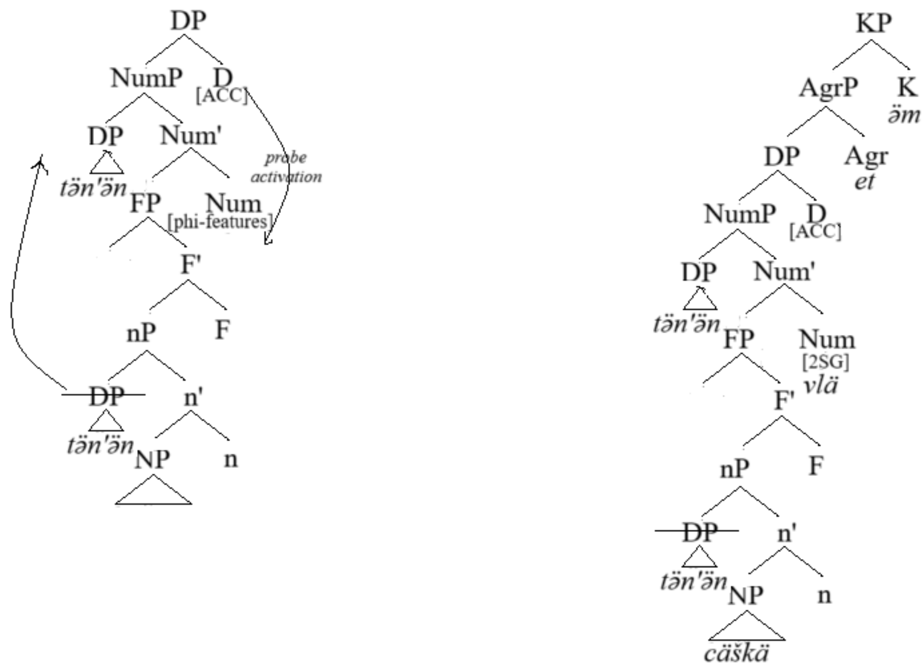
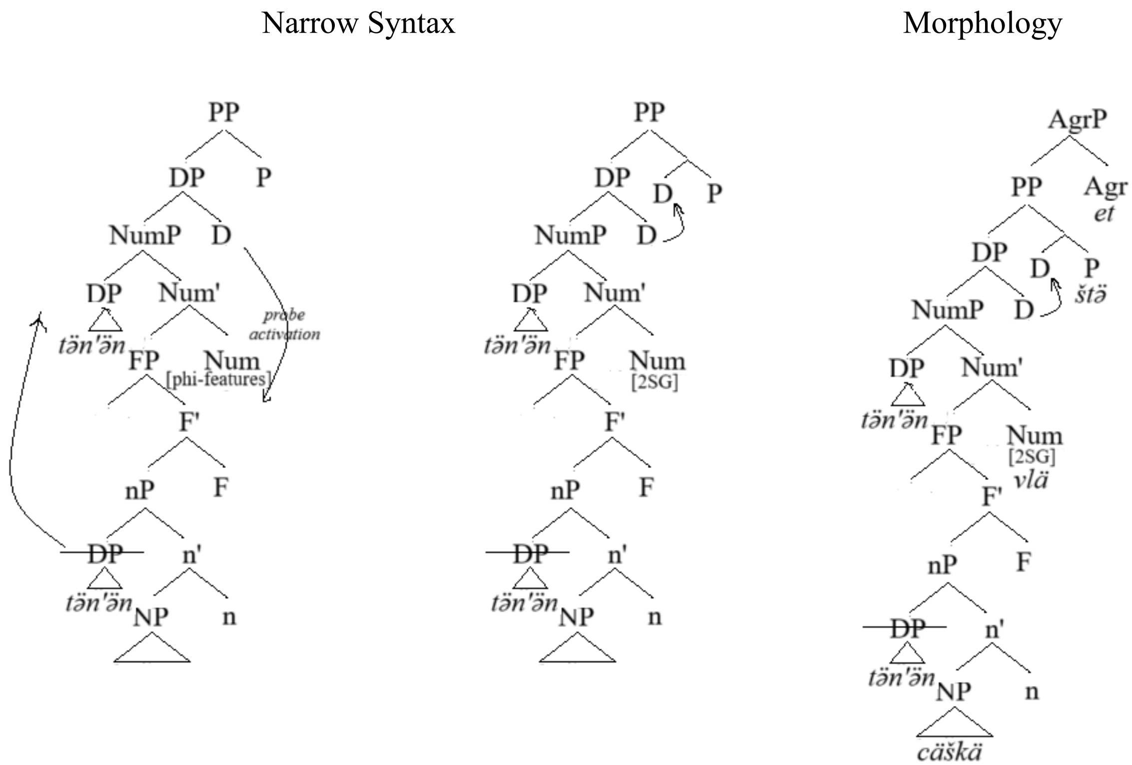


Figure 58: *Nominal agreement in an inessive nominal*



I set aside forms where the nominal agreement marker precedes the number marker, assuming for now that these result from post-syntactic reordering processes, similar to what is proposed by McFadden (2004) and Guseva and Weisser (2018) for related languages.

6.2. Hill Mari non-finite structures

Since DOM is present in constructions with non-finite predicates, I will first introduce the non-finite forms present in Hill Mari and their syntactic distribution. I will begin with nominalizations, then move on to infinitives and converbs, and conclude with participles.

6.2.1. Nominalizations

Hill Mari has two forms that are considered nominalizations: one ending in *-mə̂* (422)a and the other in *-maš* (422)b. The latter is usually considered a derivational morpheme that forms nouns from verbs (Savatkova 2002: 120). In this dissertation, I focus on the former, as it exhibits more clausal properties than the *-maš* nominalization and can function as the head of a sentential argument (Pleshak et al. 2023: 726).

(422) a. *ävä-žə* *vas'a-n* *mən'-əm* ***obižäjə-mə-m*** *už-ən*
mother-AGR.3SG[NOM] Vasja-GEN 1SG-ACC offend-NMLZ-ACC see-PRET

b. *ävä-žə* *vas'a-n* *mən'-əm* ***obižäjə-mäš-əm*** *už-ən*
mother-AGR.3SG[NOM] Vasja-GEN 1SG-ACC offend-NMLZ2-ACC see-PRET
'His mother saw how Vasja offended me.' (Pleshak et al. 2023: 726)

Nominalizations serve as sentential arguments for certain types of predicates, including emotive verbs (423), manipulative predicates (424), speech verbs (425), perception verbs (426) and psych verbs (427).

(423) EMOTIVE VERB (Pleshak et al. 2023: 743)
ävä-žə *van'a* *šərgə-štə* ***jam-mə̂*** *gəc*
mother-AGR.3SG[NOM] Vanja[NOM] forest-IN disappear-NMLZ EL
lüd-eš
be_afraid-NPST.3SG
'His mother is afraid that Vanja is going to be lost in the forest.'

(424) MANIPULATIVE PREDICATE (Pleshak et al. 2023: 745)

maša	*pet'a-m*	*pičə-m*	***tör-lə-mə***	*gišän*	*šüd-en*
Masha[NOM]	Petja-ACC	fence-ACC	level-DENOM-NMLZ	about	order-PRET

‘Masha called Petja to fix the fence.’

(425) SPEECH VERB (Pleshak et al. 2023: 747)

pet'a	*maša-lan*	*zabor*	***tör-lə-mə-m***	*keles-en*
Petja[NOM]	Masha-DAT	fence	level-DENOM-NMLZ-ACC	talk-PRET

‘Petja told Masha that the fence has been fixed.’

(426) PERCEPTION VERB (Pleshak et al. 2023: 749)

mən'	*maša-n*	***rovotajə̑-mə̑-m***	*už-ə̑n-am*
2SG[NOM]	Masha-GEN	work-NMLZ-ACC	see-PRET-1SG

‘I saw that Masha worked.’

(427) PSYCH VERB (Pleshak et al. 2023: 749)

tädə	*päl-ä*	*mən'-ən*	***rovotajə̑-m-em-əm***
3SG[NOM]	know-NPST.3SG	1SG-GEN	work-NMLZ-AGR.1SG-ACC

‘He knows that I work.’

The *-mə̑* nominalization can attach to a full verbal stem, including derivational affixes such as mediopassive etc. (Pleshak et al. 2023: 726). However, there are no tense morphemes for nominalizations in Hill Mari, and the tense of the nominalized clause fully depends on the context, e.g., a temporal adverbial modifying the clause:

(428)

mən'ə	[***tengečə***	*tən'-ən*	*zont'ik-əm*	***jam-də̑-m-et-əm***]
1SG[NOM]	yesterday	2SG-GEN	umbrella-ACC	disappear-CAUS-NMLZ-AGR.2SG-ACC
päl-em				
know-NPST.1SG				

‘I know that you lost the umbrella yesterday.’ (Pleshak et al. 2023: 727)

Example (428) also demonstrates that *-mə̑* nominalizations can be modified by adverbials rather than adnominal modifiers, distinguishing them from *-maš* nominalizations attaching an adnominal modifier:

(429) a. *-MƏ̂* NOMINALIZATION → ADVERBIAL (Pleshak et al. 2023: 727)

mən' ***ti-štə*** / ****ti-štə-šə*** ***ələ-mə-m*** *päl-em*
1SG[NOM] this-IN this-IN-ATTR live-NMLZ-ACC know-NPST.1SG
'I know that they live here.'

b. *-MAŠ* NOMINALIZATION → ADNOMINAL MODIFIER (Pleshak et al. 2023: 727)

mən' ***ti-štə-šə*** / ****ti-štə*** ***ələ-mäš-əm*** *päl-em*
1SG[NOM] this-IN-ATTR this-IN live-NMLZ2-ACC know-NPST.1SG
'I know the life here (the life that is here).'

When it comes to nominal properties of *-mə̂* nominalizations, most speakers allow for them to bear a plural marker (430), and all speakers accept optional nominal agreement on the nominalized head (431). Finally, depending on the properties of the matrix predicate, nominalizations get case-marking, see the accusative marker in (430)–(431). Unlike nominal agreement, case-marking is obligatory.

(430) ?*vas'a* *pušängə* ***ke-n*** ***vac-mə̂-vlä-m*** *kol-eš*
Vasja[NOM] tree go-CVB lie-NMLZ-PL-ACC hear-NPST.3SG
'Vasja hears trees fall.' (Pleshak et al. 2023: 728)

(431) *ävä-žə* *už-ə̂n* ***tən'-ən*** *mən'-əm* ***obižäjə-m(-et)-əm***
mother-AGR.3SG[NOM] see-PRET 2SG-GEN 1SG-ACC offend-NMLZ-AGR.2SG-ACC
'His mother saw that you offended me.' (Pleshak et al. 2023: 728)

Nominal agreement on the nominalized head reflects the phi-features of the subject. Like in finite clauses, object agreement is absent in nominalizations:

(432) **ävä-žə* *už-ə̂n* *tən'-ən* ***mən'-əm*** ***obižäjə-m-em-əm***.
mother-AGR.3SG[NOM] see-PRET 2SG-GEN 1SG-ACC offend-NMLZ-AGR.1SG-ACC
Int: 'His mother saw that you offended me.' (Pleshak et al. 2023: 728)

Subjects of nominalizations can appear in either the nominative (like in finite clauses) or genitive case:

(433) *ävä-žə* ***vas'a-n*** / ***vas'a*** ***päšä-lə-mə-m*** *pop-en*
mother-AGR.3SG[NOM] Vasja-GEN Vasja[NOM] work-DENOM-NMLZ-ACC say-PRET
'His mother said that Vasja worked.' (Pleshak et al. 2023: 730)

Unlike Meadow Mari, where nominative and genitive subjects behave differently (Serdobolskaya 2005, 2008), Hill Mari subjects exhibit similar syntactic behavior (Pleshak et al. 2023), with the only notable difference being their effect on nominal agreement. Nominative subjects trigger agreement only when pronominal, compare (434) to (435), while genitive subjects allow agreement regardless of their categorial status (436)–(437).

(434) PRONOMINAL NOMINATIVE SUBJECT → OPTIONAL AGREEMENT (Pleshak et al. 2023: 728)
nənə *səkər-əm* ***näl-mə-štə-m*** / ***näl-mə-m***
3PL[NOM] bread-ACC take-NMLZ-AGR.3PL-ACC take-NMLZ-ACC
ävä-m *äšt-ä*
mother-AGR.1SG[NOM] remember-NPST.3SG
'My mother remembers that they bought bread.'

(435) NON-PRONOMINAL NOMINATIVE SUBJECT → NO AGREEMENT (Pleshak et al. 2023: 728)
t'et'ä ***kek-vlä*** ***väsə-mə-m*** / ****väsə- mə-štə-m*** *päl-ä*
child[NOM] bird-PL fly-NMLZ-ACC fly-NMLZ-AGR.3PL-ACC know-NPST.3SG
'The child knows that birds fly.'

(436) PRONOMINAL GENITIVE SUBJECT → OPTIONAL AGREEMENT (Pleshak et al. 2023: 729)
nənə-n *səkər-əm* ***näl-mə-štə-m*** / ***näl-mə-m***
3PL-GEN bread-ACC take-NMLZ-AGR.3PL-ACC take-NMLZ-ACC
ävä-m *äšt-ä*
mother-AGR.1SG[NOM] remember-NPST.3SG
'My mother remembers that they bought bread.'

(437) NON-PRONOMINAL GENITIVE SUBJECT → OPTIONAL AGREEMENT (Pleshak et al. 2023: 729)

t'et'ä	***kek-vlä-n***	***väsə-mə-m***	/	***väsə-mə-štə-m***	*päl-ä.*
child[NOM]	bird-PL-GEN	fly-NMLZ-ACC		fly-NMLZ-AGR.3PL-ACC	know-NPST.3SG

'The child knows that birds fly.'

Subject marking and the presence of nominal agreement on the head do not influence DOM in Hill Mari, and I will not go into any further details, see (Pleshak et al. 2023) for more data and a preliminary analysis of subjects in Hill Mari non-finite clauses. The central focus of the discussion here is that direct objects in *-mə̂* nominalizations can be either accusative, as in finite clauses, or bare (438). The marking of other arguments and adjuncts remains consistent with finite clauses (439).

(438)

mən'ə	*päl-em*	*maša-n*	*každə̂j*	*kečə-n*	***sam-ə̂m***	/	***sam***
1SG[NOM]	know-NPST.1SG	Masha-GEN	each	day-GEN	weed-ACC		weed

kər-mə-žə-m
tear-NMLZ-AGR.3SG-ACC
'I know that Masha pulls weeds every day.'

(439) a.

mən'	***ül-əkə-lä***	*valə̂-m-em-əm*	*päl-en-äm*
1SG[NOM]	down-ILL2-SIM	descend-NMLZ-AGR.1SG-ACC	know-PRET-1SG

b.

mən'	*päl-en-äm,*	*što*	***ül-əkə-lä***	*val-em*
1SG[NOM]	know-PRET-1SG	that	down-ILL2-SIM	descend-NPST.1SG

'I knew I was going down.' (Pleshak et al. 2023: 750-751)

Summing up, this section has provided an overview of the main properties of Hill Mari *-mə̂* nominalizations. These nominalizations exhibit clausal structure, as evidenced by their ability to host adverbials and maintain argument marking. Although they lack an overt tense marker, the presence of temporal adverbials and nominative subjects suggests that TP is present (see also Voznesenskaya 2019 for similar conclusions based on another dialect). At the same time, *-mə̂* nominalizations display nominal characteristics, such as case-marking, nominal agreement, and plural markers. One of the defining features of Hill Mari *-mə̂* nominalizations is the differential marking of subjects and direct objects: subjects may appear in nominative or genitive case, while direct objects may be marked accusative or remain bare.

6.2.2. Infinitives

In Hill Mari, infinitives are formed using the affix *-aš*. They serve various functions, including as arguments of phasal (440), modal (441), evaluative (442), emotive (443), and manipulative (444) predicates. Additionally, infinitives function as adjuncts expressing purpose or goal in both clauses (445) and nominal constructions (446).

(440) PHASAL PREDICATE (Pleshak et al. 2023: 735)

tən'	*rokolma-m*	***šol-t-aš***	*təngäl-ät*
2SG[NOM]	potato-ACC	boil-CAUS-INF	start-NPST.2SG

'You start boiling potatoes.'

(441) MODAL PREDICATE (Pleshak et al. 2023: 738)

mə-läm	*sedərä-m*	***mə̂šk-aš***	*vär-ešt-ən*
1SG-DAT.AGR.1SG	floor-ACC	wash-INF	place-DENOM2-PRET

'I had to wash the floor.'

(442) EVALUATIVE PREDICATE (Pleshak et al. 2023: 741)

mə-läm	*ti-štə*	***tə̂men'-äš***	*jažo*
1SG-DAT.AGR.1SG	this-IN	study-INF	good

'It's good for me to study here.'

(443) EMOTIVE PREDICATE (Pleshak et al. 2023: 743)

mən'	***päšä-l-äš***	*jarat-em*
1SG[NOM]	work-DENOM-INF	like-NPST.1SG

'I like working.'

(444) MANIPULATIVE PREDICATE (Pleshak et al. 2023: 745)

maša	*pet'a-m*	*kol-ə̂m*	***kə̂č-aš***	*jad-ə̂n*
Masha[NOM]	Petja-ACC	fish-ACC	catch-INF	ask-PRET

'Masha asked Petja to catch fish.'

(445) PURPOSE/GOAL CLAUSAL ADJUNCT

a.

kol-ə̂m /	***kol***	*kə̂č-aš*	*man-ə̂n,*	*pu-da*	*nənə-län*	*ängərvaštə̂r-ə̂m*
fish-ACC	fish	catch-INF	say-CVB	give-IMP.2PL	3PL-DAT	rod-ACC

'For them to fish, give them a rod.'

b.

ərvezə	***olma-m*** /	***olma***	*kačk-aš*	*kə̂rgə̂ž-eš*
boy[NOM]	apple-ACC	apple	eat-INF	run-NPST.3SG

'The boy is running to eat an apple.'

(446) PURPOSE/GOAL NOMINAL ADJUNCT (Hill Mari Corpus)

pərcə	*opt-aš*	*pura*
grain	put-INF	granary

'box to store grain'

The infinitive marker in Hill Mari attaches to the full verbal stem, including derivational affixes, but they cannot attach tense markers. The tense of argumental infinitives is fully dependent on the matrix predicate (447), whereas the tense of adjunct infinitives can additionally be influenced by temporal adverbials (448). Temporal adverbials are ruled out in argumental infinitives (449).

(447) a. PRESENT MATRIX PREDICATE → PRESENT INTERPRETATION

mə-läm	*stöl-vlä-m*	***šək-äl-äš***	*nelə*
1SG-DAT.AGR.1SG	table-PL-ACC	push-ATT-INF	hard

'It's hard for me to push tables.'

b. PAST MATRIX PREDICATE → PAST INTERPRETATION

mə-läm	*stöl-vlä-m*	***šək-äl-äš***	*nelə*	*ə̂l-ə̂n*
1SG-DAT.AGR.1SG	table-PL-ACC	push-ATT-INF	hard	be-PRET

'It was hard for me to push tables.'

Infinitival adjunct clauses can contain temporal adverbials (448), whereas argumental infinitives cannot (449).

(448) *mən'* *tengeč=ok* [*tagačə̑* *kagə̑l'-ə̑m* ***ə̑št-äš***] *jämdə-l-en-äm*
1SG[NOM] yesterday=EMPH today pie-ACC do-INF ready-DENOM-PRET-1SG
'I prepared everything yesterday to make the pie today.' (Pleshak et al. 2023: 725)

(449) **mən'* *tengeč=ok* [*tagačə̑* *kagə̑l'-ə̑m* ***sträpäj-äš***] *təngäl-ən-äm*
1SG yesterday=EMPH today pie-ACC cook-INF start-PRET-1SG
Int: 'Already yesterday, I started today's baking of the pie.' (Pleshak et al. 2023: 725)

Infinitives exhibit limited nominal properties. They cannot take plural markers (450) or case markers (451)–(452), regardless of whether they serve as arguments or adjuncts.

(450) a. ARGUMENT CALUSE → NO NUMBER (Pleshak et al. 2023: 723)
mə-läm *stöl-vlä-m* ***šək-äl-äš*** / ****šək-äl-äš-vlä*** *nelə*
1SG-DAT.AGR.1SG table-PL-ACC push-ATT-INF push-ATT-INF-PL hard
'It's hard for me to push tables.'

b. ADJUNCT CLAUSE → NO NUMBER
mən' *stöl-vlä-m* ***šək-äl-äš*** / ****šək-äl-äš-vlä*** *tol-am*
1SG[NOM] table-PL-ACC push-ATT-INF push-ATT-INF-PL come-NPST.1SG
'I come to move the tables.'

(451) ARGUMENT CLAUSE → NO CASE (Pleshak et al. 2023: 723)
pet'a *pirə* *vä-k* ***lü-äš*** / ****lü-äš-əm*** *təngäl-ən*
Petja[NOM] wolf up-ILL2 shoot-INF shoot-INF-ACC start-PRET
'Petja started shooting at the wolf.'

(452) ADJUNCT CLAUSE → NO CASE
tədə *tol-ə̑n* *mən'* *don-em* ***šajə̑št-al-aš*** / ****šajə̑št-al-aš-lan***
3SG[NOM] come-PRET 1SG with-AGR.1SG speak-ATT-INF speak-ATT-INF-DAT
'He came to talk to me.'

Nominal agreement is permitted in adjunct infinitival clauses but generally ruled out in argumental clauses, with the exception of evaluative predicates. Consider example (453) in which the presence of a nominal agreement marker triggers obligatory purpose/goal interpretation; modal reading is only available in the absence of the nominal agreement marker. In (454) with an evaluative predicate, however, agreement is licit.

(453) *mə-län-em*$_i$ [Ø$_i$ *jäm-əm* ***kapaj-aš*** / #***kapaj-aš-em***][38] *kel-eš*
1SG-DAT-AGR.1SG hole-ACC dig-INF dig-INF-AGR.1SG be_needed-NPST.3SG
'I need to dig a hole.'
#'It (shovel) is needed for me to dig a hole.' (Pleshak et al. 2023: 724)

(454) *tä-län-dä* *stöl-vlä-m* ***tärvät-äš-dä*** *nelə*
2PL-DAT-AGR.2PL table-PL-ACC shift-INF-AGR.2PL hard
'It's hard for you to move tables.' (Pleshak et al. 2023: 723)

Infinitival clauses lack overt subjects; the null subject must be coreferential with an argument in the matrix clause. For example, in evaluative constructions, it is the agent in the subject position (454), and in manipulative constructions, it is the causee in the object position (455).

(455) *maša* *pet'a-m* [*kol-əm* ***kəč-aš***] *jad-ən*
Masha[NOM] Petja-ACC fish-ACC catch-INF ask-PRET
'Masha asked Petja to catch fish.' (Pleshak et al. 2023: 745)

DOs in infinitival clauses can appear in the accusative or in bare form (456). Other arguments, such as indirect objects and adjuncts, retain the same marking as in finite clauses (457).

(456) *každəj* *kečə-n* ***sam-əm*** / ***sam*** *sam-l-aš* *kel-eš*
each day-GEN weed-ACC weed weed-DENOM-INF be_needed-NPST.3SG
'One must pull weeds every day.'

[38] Here, "#" in the transcription line indicates that the phrase is grammatical, but not with the target meaning. The changed meaning is given under "#" in the translation line.

(457) a. INFINITIVAL CLAUSE (Pleshak et al. 2023: 743)
ədəräš ***šərgə-štə*** *jam-aš* *lüd-eš*
girl[NOM] forest-IN disappear-INF be_afraid-NPST.3SG
'The girl is afraid of getting lost in the forest.'

b. FINITE CLAUSE (Pleshak et al. 2023: 733)
ävä-žə *lüd-eš,* *što* *van'a* ***šərgə-štə***
mother-AGR.3SG[NOM] be_afraid-NPST.3SG that Vanja[NOM] forest-IN
jam-eš
disappear-NPST.3SG
'His mother is afraid that Vanja is going to get lost in the forest.'

In summary, Hill Mari infinitives vary in their structural behavior depending on the construction they occur in. Some allow temporal adverbials and subject agreement. The realization of agreement is the same as we see in the nominal domain, not in the clausal domain. However, beyond agreement, they do not exhibit nominal properties, as they cannot take case or plural markers. Direct objects may appear in the accusative or in bare form, and subjects cannot be overtly realized, while other arguments retain the forms found in finite clauses.

6.2.3. Converbs

Hill Mari has several affixes that form converbs from verbal stems. The most general converbs of manner (Andreeva 2008) are formed with the affix *-n* (Pengitov et al. (eds.) 1961: 252–253; Savatkova 2002: 234–235). Some stems also have an alternative form with a null marker (458). For stems that allow for the null marker, the choice of the marker depends on the construction (Danilova & Kashkin 2023). Other converbial forms include mə̂kə̂-converbs (459), which express precedence, and meškə̈-converbs (460), which indicate posteriority (Muravyev 2023).

(458) N/Ø-CONVERBS (adapted from Danilova & Kashkin 2023: 769)
kek-vlä ***čongešt-əl-ən*** / ***čongešt-əl-Ø*** *juk-lan-en-ət*
bird-PL[NOM] fly-ITER-CVB fly-ITER-CVB sound-MAN-PRET-3PL
'Birds were screaming while flying.'

(459) MƏ̂KƏ̂-CONVERB (Muravyev 2023: 775)
maša *to-kə̂-žə̂* ***tol-mə̂kə̂*** *amal-aš* *vaz-ə̂n*
Masha[NOM] home-ILL2-AGR.3SG come-CVB.ANT sleep-INF lie-PRET
'When Masha came home, she went to bed.'

(460) MEŠKƏ̈-CONVERB (Muravyev 2023: 775)
maša *kečə̈* ***läk-meškə̈*** *amal-en.*
Masha[NOM] sun exit-CVB.LIM sleep-PRET
'Masha slept until the sunrise.'

All converbs occur in temporal adjunct clauses (458)–(460), but n/Ø-converbs can also be arguments of certain modal and phasal verbs (461)–(462). Lexical verbs in constructions with light verbs must also be in their converbial forms (463). The latter cases are excluded from consideration as they do not allow for DOM (Pleshak & Sirotina 2023: 576), as seen in (464).

(461) MODAL MATRIX PREDICATE
pört-əšt=ok ***äl-en*** *kerd-eš* *tədə*
house-IN=EMPH live-CVB can-NPST.3SG 3SG[NOM]
'It [the cat] can only live in a home.' (Hill Mari Corpus)

(462) PHASAL MATRIX PREDICATE (Pleshak et al. 2023: 736)
t'et'ä-vlä *sir-mäš-əm* ***sir-en*** *pət-är-en-ət*
child-PL[NOM] write-NMLZ2-ACC write-CVB finish-CAUS2-PRET-3PL
'The children finished writing a letter'.

(463) CONSTRUCTION WITH A LIGHT VERB (Hill Mari Corpus)
kot'i *patə̂l-ə̂m* ***kačk-ə̂n*** ***kolt-en***
cat[NOM] sour_cream-ACC eat-CVB send-PRET
'The cat ate the sour cream.'

(464) CONSTRUCTION WITH A LIGHT VERB (Pleshak et al. 2023: 576)
vas'a ***pandə̂-m*** / ****pandə̂*** *ro-en* *šu-en*
Vasja[NOM] stick-ACC stick cut-CVB through-PRET
'Vasja cut the stick (in half)'.

The temporal interpretation of n/Ø-converbs in adjunct clauses (simultaneity or posteriority) depends on the aspectual characteristics of the verb (Danilova 2023: 763), and the interpretation of *məkə̑*- and *meškə̈*-converbs is usually determined by the affix itself. These clauses cannot be modified by independent temporal adverbs (465)–(466). In argumental clauses, temporal interpretation fully depends on the matrix predicate, and temporal adverbial modification of the converbial clause is not permitted either (467).

(465) ADJUNCT CLAUSE
**tengečə̈ šarə̑k-ə̑m tə̈red-ən, tagačə̑ miž-əm və̑žal-en-äm*
yesterday sheep-ACC shave-CVB today wool-ACC sell-PRET-1SG
Int.: 'Having shaved the sheep, I sold the wool today.'

(466) ADJUNCT CLAUSE
**vas'a tengečə̈ čeboksarə̑-š ke-məkə̑, tagačə̑ tol-ə̑n*
Vasja[NOM] yesterday Cheboksary-ILL go-CVB.ANT today come-PRET
Int.: 'Having gone to Cheboksary yesterday, Vasja came back today.'

(467) ARGUMENTAL CLAUSE (Pleshak et al. 2023: 732)
män' tagačə̑ [tengečə̈ lem-əm* ***šol-t-en**] pə̈t-är-en-äm*
1SG[NOM] today yesterday soup-ACC boil-CAUS-CVB finish-CAUS2-PRET-1SG
Int.: 'Today I finished cooking the soup started yesterday.'

Converbs cannot bear affixes except for nominal agreement markers (Muravyev et al. 2023: 69). Specifically, n/Ø-converbs do not bear subject agreement, whether occurring as adjuncts (468) or complements (469)–(470). In contrast, mə̑kə̑- (471) and meškə̈-converbs (472) allow subject agreement, which is expressed with a non-finite agreement affix.

(468) N/Ø-CONVERB (ADJUNCT)
**amasa-m čüč-ən-em, män' tocə̑n-na asked-ən-äm*
door-ACC close-PRET-1SG 1SG[NOM] home.EL-AGR.1PL go_out-PRET-1SG
Int.: 'Having closed the door, I left the house.'

(469) N/Ø-CONVERB (ARGUMENT) (Pleshak et al. 2023: 732)

******tä*	*stöl-vlä-m*	***tärvät-en-dä***	*pət-är-en-dä*
2PL[NOM]	table-PL-ACC	shift-CVB-AGR.2PL	finish-CAUS2-PRET-2PL

Int: ‘You finished moving the tables.’

(470) N/Ø-CONVERB (ARGUMENT) (Pleshak et al. 2023: 732)

**tä*	*stöl-vlä-m*	***tärvät-en-dä***	*kerd-ədä*	*ma?*
2PL[NOM]	table-PL-ACC	shift-CVB-AGR.2PL	can-NPST.2PL	what

Int.: ‘Can you move tables?’

(471) MƏ̑KƏ̑-CONVERB (Muravyev 2023: 775)

kən'əl-mək-em	*maša*	*lem-əm*	*kačk-eš*	*əl'ə̂*
get_up-CVB.ANT-AGR.1SG	Masha[NOM]	soup-ACC	eat-NPST.3SG	RETR1

‘When I got up, [I saw that] Masha was eating soup.’

(472) MEŠKƏ̈-CONVERB (Muravyev 2023: 785)

tol-mešk-em	*maša*	*uže*	*tənäm=ok*	*pət-är-en*
come-CVB.LIM-AGR.1SG	Masha[NOM]	already	then=EMPH	finish-CAUS2-PRET
škol-ə̂m				
school-ACC				

‘By the time I came, Masha had graduated from school.’

The subject of a clause headed by an n/Ø-converb must be coreferential with the subject of the matrix clause and cannot be overt (Danilova 2023: 759), as shown in (473)–(474) for adjunct clauses and (475) for an argument clause. In contrast, clauses with mə̂kə̂- and meškə̈-converbs can have overt nominative subjects, as illustrated in (476)–(477).

(473) N/Ø-CONVERB (ADJUNCT) (Danilova 2023: 759)

šarə̂k-ə̂m	***türed-ən***,	*miž-əm*	*və̂žal-en-äm*
sheep-ACC	shave-CVB	wool-ACC	sell-PRET-1SG

‘After I shaved the sheep (букв.: having shaved the sheep), I sold the wool.’

(474) N/Ø-CONVERB (ADJUNCT) (Danilova 2023: 759)

**t'et'ä-vlä*	*šarə̂k-ə̂m*	***näl-ən**,*	*nənə*	*to-kə̂-štə̂*	*ke-vä.*
child-PL[NOM]	sheep-ACC	take-CVB	3PL[NOM]	house-ILL2-AGR.3PL	go-AOR.3PL

Int.: ‘Having bought a goat, the children went home.’

(475) N/Ø-CONVERB (ARGUMENT) (Pleshak et al. 2023: 732)
vas'a *(*tədə)* *toma* ***stroj-en*** *pət-är-en*
Vasja[NOM] 3SG[NOM] house build-CVB finish-CAUS2-PRET
'Vasja finished building a house.'

(476) MƏ̂KƏ̂-CONVERB (Muravyev 2023: 777)
vas'a ***tol-mə̂kə̂*** *šar* *pə̂dešt-ən*
Vasja[NOM] come-CVB.ANT balloon[NOM] pop-PRET
'When Vasja came, the balloon popped.'

(477) MEŠKƏ̈-CONVERB (Muravyev 2023: 785)
pet'a ***tol-meškə̈*** *mən'* *mə̂rə̂-m* *mə̂r-en-äm*
Petja[NOM] come-CVB.LIM 1SG[NOM] song-ACC sing-PRET-1SG
'Before Petja came, I was singing a song.'

DOs in different adjunct clauses can appear in either the accusative or bare form (478)–(480).

Some speakers also allow DOM in the arguments of phasal predicates (481). However, DOs in the arguments of modal predicates must be in the accusative case (482).

(478) ADJUNCT CLAUSE → ACCUSATIVE / BARE DO
sam-ə̂m / *sam* *kər-ən,* *mä* *jərän-vlä-m* *itəräj-en-nä*
weed-ACC weed pull-CVB 1PL[NOM] patch-PL-ACC clean-PRET-1PL
'Having pulled weeds, we cleaned the patches.'

(479) ADJUNCT CLAUSE → ACCUSATIVE / BARE DO
jäm-əm / ***jäm*** *kapajə̂-mə̂kə̂* *mən'* *kačk-aš* *ke-š-əm*
hole-ACC hole dig-CVB.ANT 1SG[NOM] eat-INF go-AOR-1SG
'Having dug a hole, I went to eat.'

(480) ADJUNCT CLAUSE → ACCUSATIVE / BARE DO
ərvezäš *ti* ***olma*** / ***olma-m*** *kač-meškə̈* *mad-aš* *kej-en.*
boy[NOM] this apple apple-ACC eat-CVB.LIM play-INF go-PRET
'Having eaten this apple, the bey went to play.'

(481) ARGUMENT OF A PHASAL PREDICATE → ACCUSATIVE DO PREFERRED
ərvezäš ***kol-ə̂m*** / ?***kol*** *kə̂č-en* *pət-är-en*
boy[NOM] fish-ACC fish catch-CVB finish-CAUS2-PRET
'The boy finished fishing.'

(482) ARGUMENT OF A MODAL PREDICATE → ACCUSATIVE DO

ərvezə	***olma-m***	/	****olma***	*kačk-ən*	*kerd-eš*
boy[NOM]	apple-ACC		apple	eat-CVB	can-NPST.3SG

‘The boy can eat an apple.’

Other clausal arguments and adjuncts carry the same markers in both converbial and finite clauses. This can be seen in an indirect object functioning as a recipient in the dative case (483) and an oblique object in the inessive case (484).

(483) a.

tä	*mə-läm*	*sir-mäš-əm*	***sir-en***	*kerd-ədä*
2SG[NOM]	1SG-DAT.AGR.1SG	write-NMLZ2-ACC	write-CVB	can-NPST.2PL

b.

**tä*	*mə-läm*	*sir-mäš-əm*	***sir-äš***	*kerd-ədä*
2SG[NOM]	1SG-DAT.AGR.1SG	write-NMLZ2-ACC	write-INF	can-NPST.2PL

‘You can write me a letter.’ (Pleshak et al. 2023: 740)

(484)

d'ivan	*ti*	*vär-əštə*	***šənz-en***	*kerd-eš*
couch[NOM]	this	place-IN	sit-CVB	can-NPST.3SG

‘The couch can stand (fits) here.’ (Pleshak et al. 2023: 740)

Thus, converbial clauses do not bear temporal markers and cannot contain temporal adverbials. Like with infinitives, their subject agreement looks like nominal agreement rather than verbal agreement, see (471)–(472) above. However, the realization of agreement is the only nominal-like property of Hill Mari converbs. They do not take case or plural markers, and only some adjunct converbial clauses permit overt nominative subjects. Depending on the construction, DOs in certain converbial clauses may be marked with accusative or remain bare. Other arguments follow the same marking patterns as in finite clauses.

6.2.4. Participles

The two main affixes that form participles in Hill Mari are *-šə̑* and *-mə̑*. These affixes are used in relative clauses, with *-šə̑* relativizing subjects (485) and *-mə̑* relativizing direct (486), indirect (487), and oblique (488) objects.

(485) SUBJECT RELATIVIZATION (adapted from Pleshak & Demina 2023: 701)

urok-ə̑m	***əštə-šə***	*ərvezäš*
lesson-ACC	do-PTCP.ACT	boy

‘the boy doing his homework’

(486) DO RELATIVIZATION (adapted from Pleshak & Demina 2023: 701)

ərvezäš-ən	***lə̂d-mə̂***	*kn'igä-žə*
boy-GEN	read-PTCP.PASS	book-AGR.3SG

‘the book that the boy reads’

(487) INDIRECT OBJECT RELATIVIZATION (Pleshak & Demina 2023: 702)

kid-əm	***mašə̂-mə̂***	*paškudə̂-n*	*ərvezäš*	*mə-län-em*	*jər-ältə-š*
hand-ACC	wave-PTCP.PASS	neighbor-GEN	boy[NOM]	1SG-DAT-AGR.1SG	smile-PUN-AOR

‘The neighbor-boy who I waved at smiled at me.’

(488) OBLIQUE OBJECT RELATIVIZATION (Pleshak & Demina 2023: 703)

ävä-m	*paj-ə̂m*	***päč-mä***	*käzä*	*nəšk-em-Ø*	*šənz-ən*
mother-AGR.1SG	meat-ACC	cut-PTCP.PASS	knife	dull-INCH-CVB	sit-PRET

‘The knife with which my mother cut meat became dull.’

The temporal interpretation of participial clauses depends on the surrounding context, and they can contain temporal adverbials:

(489)

tengečə	*urok-ə̂m*	***əštə-šə***	*ərvezäš*	*mən'-ən*	*šol'a-em*	*ə̂l-eš.*
yesterday	lesson-ACC	do-PTCP.ACT	boy[NOM]	1SG-GEN	younger_brother-AGR.1SG	be-NPST.3SG

‘The boy who did his homework yesterday is my younger brother.’

(Pleshak & Demina 2023: 702)

Like other adnominal dependents in Hill Mari, participles do not show nominal concord — they do not double case or number affixes present on the head noun. However, they can take nominal morphological markers such as number, agreement (490), and case (491) when the nominal head is absent (adnominal dependents in postposition behave the same way as in nominal ellipsis).

(490) *…ves* *komandə̂-n* ***mad-šə̂-vlä-ǯə̈*** *sv'ečkə̈-m* *xvat'-at…*
other team-GEN play-PTCP.ACT-PL-AGR.3SG candle-ACC catch-NPST.3PL
'…the players from the other team grab the candle…' (Hill Mari Corpus)

(491) *ät'ä-m* *kandə̂-š* *šə̈rgə̈* *gə̈c* *mə̈-län-em* *kostenec-ə̈m*
father-AGR.1SG bring-AOR forest EL 1SG-DAT-AGR.1SG gift-ACC
kə̈lmə̈-šə̈-m *iziš*
freeze-PTCP.ACT-ACC a_little
'My father brought me a slightly frozen present from the woods.' (Hill Mari Corpus)

In colloquial speech, some speakers accept 1^{st} and 2^{nd} person nominal agreement markers on participles in the adnominal position when the nominal head is present (492). In these cases nominal agreement is triggered by the subject of the participial clause.

(492) ?***tə̈n'(-ə̈n)*** *ro-m-**et*** *püšängə̈-m* *mä* *už-ə̂n-na*
2SG-GEN cut-PTCP.PASS-AGR.2SG tree-ACC 1PL[NOM] see-PRET-1PL
'We saw the tree that you had cut.'

Like in nominalizations (see Section 6.2.1), subjects in participles (those that do not relativize the subject position) can be nominative or genitive (493). Typically, participial clause subjects, both nominative and genitive, trigger agreement on the nominal head (494). Agreement with genitive subjects is optional (494), while the agreement with nominative subjects is strongly preferred

(494), see (Pleshak 2022a) for further details and a structural analysis of subject agreement in Hill Mari participial clauses.

(493)	***učen'ik-vlä-n*** /	***učen'ik-vlä***	*irok*	*kač-mâ*	*tə*	*âžar*	*olma-vlä*
	student-PL-GEN	student-PL	morning	eat-PTCP.PASS	that	green	apple-PL[NOM]
	šapâ	*âl-ân-ât*					
	sour	be-PRET-3PL					

‘Those green apples eaten by the students in the morning were tart.’ (Pleshak & Demina 2023: 707)

(494) a.	***tən'-ən***	*ro-mâ*	*püšäng-**et**-əm* /	*püšängə-m*	*mä*	*už-ân-na.*
	2SG-GEN	cut-PTCP.PASS	tree-AGR.2SG-ACC	tree-ACC	1PL[NOM]	see-PRET-1PL
b.	***tən'***	*ro-mâ*	*püšäng-**et**-əm* /	?*püšängə-m*	*mä*	*už-ân-na.*
	2SG[NOM]	cut-PTCP.PASS	tree-AGR.2SG-ACC	tree-ACC	1PL[NOM]	see-PRET-1PL

‘We saw the tree that you had cut.’

DOs of participial clauses can be either accusative or bare:

(495)	*kesə-m* /	*kesə*	*näl-šə*	*edem*
	goat-ACC	goat	take-PTCP.ACT	man

‘the man buying a goat’

Other clausal arguments and adjuncts follow the same marking patterns in participial and finite clauses:

(496)	[*ti-štə*	*kuštâ-šâ*]	*ədəräš*
	this-IN	dance-PTCP.ACT	girl

‘the girl dancing here’

(497)	*mən'*	*kušt-em*	*ti-štə*
	1SG[NOM]	dance-NPST.1SG	this-IN

‘I dance here.’

In this chapter, I will focus on *šâ*-participles, as they frequently appear in transitive clauses with overt DOs, unlike *-mâ* participles, which primarily relativize DOs.

To sum up, participial clauses in Hill Mari have the external syntax of adnominal dependents but constitute full clauses that may include overt subjects and temporal adverbials, although they do not overtly mark tense. The realization of subject agreement is nominal, as in other non-finite forms. Subjects can be either genitive or nominative, and DOs can be accusative or bare.

6.2.5. Interim summary

In this section, I gave an overview of the key properties of Hill Mari non-finite forms. The main distinction between finite and non-finite clauses lies in the latter's inability to carry tense markers, which is inherent to their definition.[39] Additionally, these two clause types differ in how they express subject agreement: while finite clauses use verbal agreement markers, non-finite forms combine with nominal agreement markers.

Unlike finite clauses, non-finite clauses permit Differential Object Marking (DOM), though its realization in infinitives and converbs depends on the type of construction in which they appear. Other clausal arguments and adjuncts, however, are marked identically in both finite and non-finite clauses. Table 13 summarizes the properties of all non-finite forms.

Table 13: *Properties of Hill Mari non-finite forms*

	nominalizations	participles	n/Ø-converbs	*mə̂kə̂*- & *meškə̈*-converbs	adjunct infinitives	Argumental infinitives
temporal adverbials	OK	OK	*	*	OK	*

[39] As mentioned previously, the inability to bear an overt tense marker does not necessarily imply the absence of a TP in the clausal spine. Here, I assume that the overt realization of tense reflects the distinction between finite and non-finite TPs.

Combining with number and case markers	OK	OK in the absence of the nominal head	*	*	*	*
overt subject	OK (NOM/GEN)	OK (NOM/GEN)	*	OK (NOM)	*	*
nominal agreement	OK	?[40]	*	OK	OK	depending on the matrix predicate
DOM	OK	OK	OK	OK	OK	OK

Among non-finite forms, nominalizations and participial clauses most closely resemble finite clauses, as they allow for temporal adverbials and the overt realization of subjects. Converbial clauses, by contrast, cannot contain temporal adverbials, though some constructions permit overt subjects.

Nominalizations exhibit the strongest nominal properties. In addition to allowing nominative subjects, they also permit genitive subjects, similar to adnominal possessors. Furthermore, nominalizations can bear number markers and must be marked for case. Participles, as adnominal modifiers, also display certain nominal properties, such as the possibility of genitive subjects and the ability to take nominal morphological markers when the nominal head is absent. Conversely, converbs and infinitives lack nominal characteristics aside from the form of the agreement markers they attach.

[40] "?" indicates the degraded status of agreement markers on the non-finite form itself.

6.3. DOM pattern in Hill Mari

As noted at the beginning of this chapter, Hill Mari DOM is characterized by three main facts. First, it applies exclusively to DOs in non-finite clauses and not in finite clauses. Second, there are no contexts in which a DO must be bare. Third, the properties of bare DOs vary depending on the specific construction in which they appear. In this section, I first outline the general properties of bare DOs in Hill Mari, followed by a discussion of construction-specific variations.

6.3.1. General properties of Hill Mari bare DOs[41]

Hill Mari bare DOs cannot be marked with a nominal agreement affix (498). As long as nominal agreement is present, the DO must be marked with the accusative case. This is illustrated in examples (499)–(501) across different types of non-finite clauses.

(498) **Overt Marking generalization**:
If nominal agreement is expressed overtly, the DO must be accusative.

(499) NOMINALIZATION

mən'	*lüd-äm*	*vas'a-n*	***škal-na-m*** /	****škal-na***
1SG[NOM]	be_afraid-NPST.1SG	Vasja-GEN	cow-AGR.1PL-ACC	cow-AGR.1PL
šəšk-əl-mə-žə		*gəc-ən*		
slaughter-ITER-NMLZ-AGR.3SG		EL-FULL		

'I'm afraid of Vasja slaughtering our cow.'

[41] This section is based on a chapter of a Hill Mari grammatical description written by me, drawing on my own field data and data collected collaboratively with my colleague A. Sirotina (Pleshak & Sirotina 2023). In what follows, I omit individual citations of that chapter.

(500) INFINITIVE

ədər	*mäm-nä-n*	***šarə̂k-na-m*** /	****šarə̂k-na***	*näl-äš*
girl[NOM]	1PL-AGR.1PL-GEN	sheep-AGR.1PL-ACC	sheep-AGR.1PL	take-INF
tol-ə̂n				
come-PRET				

'The girl came to buy our sheep.'

(501) PARTICIPLE

mänmän	***kesə̈-nä-m*** /	****kesə̈-nä***	*näl-šə̈*	*edem*
1PL.GEN	goat-AGR.1PL-ACC	goat-AGR.1PL	take-PTCP.ACT	man

'the man that bought our goat'

Hill Mari bare DOs are more flexible in terms of number marking. The possibility of bearing a plural marker depends on the specific construction, which will be discussed in Section 6.4.

Bare DOs are further restricted when it comes to certain nominal categories. Alongside with possessed nominals, personal pronouns cannot appear as bare DOs (502). This is exemplified in (503).[42]

(502) **Category Restriction generalization**:
If the DO is a personal pronoun, it must be accusative.

[42] As shown in Section 6.3.2, bare direct objects in nominalizations and purpose/goal infinitives exhibit the fewest restrictions. Therefore, when illustrating general constraints on these constructions, I do not enumerate all possible contexts each time. Constructions that are ungrammatical in nominalizations and purpose/goal infinitives are ungrammatical in all other contexts as well. An implicational hierarchy of constructions is provided in the beginning of Section 6.3.2.

(503) a. *mən'* **tən'-əm** / ***tən'(ə)** *kə̂mə̂l-ang-d-aš* *tol-ə̂n-am*
1SG[NOM] 2SG-ACC 2SG mood-INCH2-CAUS-INF come-PRET-1SG
'I came to greet you.'

b. *mən'* **tədə-m** / ***tədə** *kə̂mə̂l-ang-d-aš* *tol-ə̂n-am*
1SG[NOM] 3SG-ACC 3SG mood-INCH2-CAUS-INF come-PRET-1SG
'I came to greet him.'

It is also worth noting that proper names and kinship terms are allowed as bare DOs only in some idiolects:

(504) **Featural restriction generalization** (effective only in some idiolects):
If the DO is a proper name or a kinship term, it must be accusative.

(505) *ävä-m* **vas'a-m** / ?***vas'a*** *pukš-aš* *ke-n*
mother-AGR.1SG[NOM] Vasja-ACC Vasja feed-INF go-PRET
'My mother went to feed Vasja.'

(506) *ävä-žə* **ergə-m** / ?***ergə*** *pukš-aš* *ke-n*
mother-AGR.3SG[NOM] son-ACC son feed-INF go-PRET
'The mother went to feed the son.'

In contrast to proper names and kinship terms, noun phrases headed by regular nouns can be bare in all idiolects, whether they are animate (507) or inanimate (508).

(507) ANIMATE REGULAR NOUN
ävä-m **t'et'ä-m** / **t'et'ä** *pukš-aš* *ke-n*
mother-AGR.1SG[NOM] child-ACC child feed-INF go-PRET
'My mother went to feed the child.'

(508) INANIMATE REGULAR NOUN
ərvezə **olma-m** / **olma** *kačk-aš* *kə̂rgə̂ž-eš*
boy[NOM] apple-ACC apple eat-INF run-NPST.3SG
'The boy is running to eat an apple.'

When it comes to other classes of pronouns, interrogative pronouns can be bare only in some idiolects.[43] Bare DOs expressed by inanimate interrogative pronouns (509) are more likely to be accepted than those expressed by animate interrogative pronouns (510).

(509) INANIMATE INTERROGATIVE PRONOUN

tən'	***ma-m*** /	?***ma***	*näl-äš*	*kej-et?*	
2SG[NOM]	what-ACC	what	take-INF	go-NPST.2SG	

'What are you going to buy?'

(510) ANIMATE INTERROGATIVE PRONOUN

tən'	***kü-m*** /	??***kü***	*kəməl-ang-d-aš*	*tol-ən-at?*
2SG[NOM]	who-ACC	who	mood-INCH2-CAUS-INF	come-PRET-2SG

'Who did you come to greet?'

Bare DOs cannot combine with universal quantifiers that require a definite interpretation (511). In (512), the universal quantifier *každəj*, which requires a definite interpretation, can only be interpreted as modifying the entire nominalization; it cannot be interpreted as modifying just the DO. Similarly, the universal quantifier *cilä* cannot be interpreted in its DEF-quantification function (in the terms of Tatevosov 2002), quantification of definite sets (513). However, when the same lexical item is used as a free-choice quantifier, it can combine with a bare DO (514).

(511) **Modification generalization**:
In the presence of an adnominal universal quantifier, the DO must be accusative.

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[43] Here, "?" and "??" indicate the number of idiolects in which the phenomenon is accepted, rather than degrees of acceptability within a single idiolect. I avoid using the more common "%" symbol for idiolectal restrictions, since I interpret the number of idiolects as a gradient measure of the overall acceptability of a given syntactic configuration in the language.

(512) *#ävä-m-ən* ***każdəj*** ***əškal*** *šəpš-əl-mə-žə-m*
mother-AGR.1SG-GEN each cow pull-ITER-NMLZ-AGR.3SG-ACC
mən' *päl-em*
1SG[NOM] know-NPST.1SG
'I know about my mother's every milking of the cow.'
*'I know that my mother milked every cow.'

(513) **ävä-m-ən* ***cilä*** ***əškal*** *šəpš-əl-mə-m* *mən'* *päl-em*
mother-AGR.1SG-GEN all cow pull-ITER-NMLZ-ACC 1SG know-NPST.1SG
Int.: 'I know that my mother milked all the cows.'

(514) *mən'* *ävä-m-ən* *xot'* *maxan'/* ***cilä*** ***əškal*** *jaratə-mə-m*
1SG[NOM] mother-AGR.1SG-GEN PTCL what all cow like-NMLZ-ACC
päl-em
know-NPST.1SG
'I know that my mother likes whatever/all cows.'

Restrictions on modification with other adnominal dependents vary depending on the construction. I discuss these restrictions in Section 6.4

Finally, bare DOs must be adjacent to the predicate (515). If an element, such as a temporal adverb (516) or a focus particle (517), intervenes between the DO and the predicate, the DO must take the accusative case. However, if the adjacency requirement is met, bare DOs can still be focused or contrasted (518).

(515) **Adjacency generalization**:
If the DO is not adjacent to the predicate, it must be accusative.

(516) TEMPORAL ADVERB
a. *ävä-m* ***tengečə*** *səkər-əm* / *səkər* *näl-äš* *kašt-ə*
mother-AGR.1SG[NOM] yesterday bread-ACC bread take-INF walk-AOR.3SG
b. *ävä-m* *səkər-əm* / **səkər* ***tengečə*** *näl-äš* *kašt-ə*
mother-AGR.1SG[NOM] bread-ACC bread yesterday take-INF walk-AOR.3SG
'My mother went to buy bread yesterday.'

(517) FOCUS PARTICLE

mən'ə	*päl-em*	*tən'-ən*	*əškal-əm=**ok*** /	**əškal=**ok***
1SG[NOM]	know-NPST.1SG	2SG-GEN	cow-ACC=EMPH	cow=EMPH
näl-m-et-əm				
take-NMLZ-AGR.2SG-ACC				

'I know that it was a cow that you bought.'

(518)

ävä-m	***kesə-m*** /	****kesə***	*agəl,*	*a*	***əškal***	*näl-äš*	*kej-en*
mother-AGR.1SG[NOM]	goat-ACC	goat	NEG	but	cow	take-INF	go-PRET

'My mother went to buy a cow, not a goat.'

In this section, I described the properties of bare DOs as opposed to the baseline accusative DOs. The two types of DOs differ in availability of nominal agreement marking category restrictions, available modifiers and adjacency requirements, see Table 14.

Table 14: *General properties of accusative and bare DOs*

	accusative	bare
expression of nominal agreement	OK	*
personal pronouns	OK	*
interrogative pronouns	OK	?
proper names	OK	?
kinship terms	OK	?
modification by universal quantifiers	OK	*

Accusative direct objects are always possible, with no restrictions on their nominal category, allowing any type of adnominal modification and free movement within the clause. In contrast, bare DOs cannot bear nominal agreement markers, cannot be expressed by personal pronouns, cannot be modified by universal quantifiers, and must remain adjacent to the predicate. In the next section, I will examine the construction-specific properties of bare DOs in further detail.

6.3.2. Construction-specific properties of bare DOs

While bare DOs cannot bear a nominal agreement marker or be modified with a universal quantifier, they can, in some cases, take a plural marker or a demonstrative. The possibility of such marking or modification depends on the specific construction in which the DO occurs. Below, I examine the following types of constructions: nominalized clauses, adjunct purpose/goal infinitival clauses, infinitival arguments of emotive predicates, adjunct converbial clauses, converbial arguments of phasal verbs, and participial relative clauses. These constructions form an implicational hierarchy (519): if a DO in a particular construction is subject to certain restrictions, the same restrictions apply to DOs in the constructions further to the right.

(519) IMPLICATIVE HIERARCHY OF HILL MARI NON-FINITE CONSTRUCTIONS WITH RESPECT TO THE PROPERTIES OF BARE DOS

nominalizations & goal / purpose infinitives > participles > infinitival arguments of evaluative predicates & converbial adjuncts > converbial arguments of phasal verbs and infinitival arguments of phasal and modal verbs

6.3.2.1. Nominalized clauses

Bare DOs in nominalized clauses appear to have the fewest restrictions. Like all bare DOs in Hill Mari, they cannot take a nominal agreement marker (520). However, a plural marker is acceptable in most idiolects (521).

(520)

mən'	*lüd-äm*	*vas'a-n*	*škal-**na**-m*	/	**škal-**na***
1SG[NOM]	be_afraid-NPST.1SG	Vasja-GEN	cow-AGR.1PL-ACC		cow-AGR.1PL
šəšk-əl-mə-žə		*gəc-ən*			
slaughter-ITER-NMLZ-AGR.3SG		EL-FULL			

'I'm afraid of Vasja's slaughtering our cow.'

(521)

$^{?}$*maša-n*	*tə̑gə̑r-**vlä***	*ə̑rgə̑-mə̑-žə̑-m*	*už-Ø-ə̑m*
Masha-GEN	shirt-PL	sew-NMLZ-AGR.3SG-ACC	see-AOR-1SG

'I saw Masha sew shirts.'

Bare DOs in nominalized clauses allow a variety of adnominal modifiers. They can occur with bare nouns (522) and adjectives (523) and also permit possessors (524), numerals (525), and even demonstratives (526).

(522) BARE NOUN

ävi-n	***cəvə***	*lem*	*šol-tə̑-mə̑-m*	*mən'*	*päl-em*
mother.KIN-GEN	chicken	soup	boil-CAUS-NMLZ-ACC	1SG[NOM]	know-NPST.1SG

'I know that my mom is making chicken soup.'

(523) ADJECTIVE

mən'	*už-ə̑n-am*	*tən'-ən*	***u***	*tə̑gə̑r*	*ə̑rgə̑-m-et-əm*
1SG[NOM]	see-PRET-1SG	2SG-GEN	new	shirt	sew-NMLZ-AGR.2SG-ACC

'I saw you sewing a new shirt.'

(524) POSSESSOR

ti	*ədər*	***mämnän***	*koti*	*pukš-ə̑mə̑-m*	*mən'*	*už-ə̑n-am*
this	girl[NOM]	1PL.GEN	cat	feed-NMLZ-ACC	1SG[NOM]	see-PRET-1SG

'I saw that this girl fed our cat.'

(525) NUMERAL

mən'	*päl-em*	*každə̑j*	*ərvezäš-ən*	***kok***	*kn'igä*
1SG[NOM]	know-NPST.1SG	each	boy-GEN	two	book
lə̑d-mə̑-štə̑-m					
read-NMLZ-AGR.3PL-ACC					

'I know that every boy read two books.'

(526) DEMONSTRATIVE

kə̑m	*ərvezäš-ən*	***ti***	*kn'igä*	*lə̑d-mə̑-štə̑-m*	*mən'*	*päl-em*
three	boy-GEN	this	book	read-NMLZ-AGR.3PL-ACC	1SG[NOM]	know-NPST.1SG

'I know that three boys read this book.'

6.3.2.2. Adjunct purpose/goal infinitival clauses

Bare DOs in adjunct purpose/goal infinitival clauses exhibit similar properties to those in nominalized clauses. They allow number marking in most idiolects (527) and can also be modified by bare nouns (528), adjectives (529), possessors (530), numerals (531), and demonstratives (532).

(527) PLURAL MARKER
?*maša* *škal-**vlä*** *šəpš-əl-aš* *ke-n*
Masha[NOM] cow-PL pull-ITER-INF go-PRET
'Masha went to milk cows.'

(528) BARE NOUN
?*ävi* ***cəvä*** *lem* *šol-t-aš* *ke-n*
mother.KIN[NOM] chicken soup boil-CAUS-INF go-PRET
'My mom went to make chicken soup.'

(529) ADJECTIVE
?*ergə-žə* ***u*** *kol'mə̑* *näl-äš* *kej-en*
son-AGR.3SG[NOM] new shovel take-INF go-PRET
'The son went to buy a new shovel.'

(530) POSSESSOR
?*ədər* ***mäm-nä-n*** *šarə̑k* *näl-äš* *tol-ə̑n*
girl[NOM] 1PL-AGR.1PL-GEN sheep take-INF come-PRET
'The girl came to buy our sheep.'

(531) NUMERAL
?*maša* ***kok*** *škal* *šə̑pš-əl-aš* *ke-n*
Masha[NOM] two cow pull-ITER-INF go-PRET
'Masha went to milk two cows.'

(532) DEMONSTRATIVE
?*ədəräš* ***ti*** *ləpə* *kə̑č-aš* *kə̑rgə̑ž-ə̑*
girl[NOM] this butterfly catch-INF run-AOR.3SG
'The girl ran to catch this butterfly.'

The only difference between bare DOs in nominalized clauses and adjunct purpose/goal infinitival clauses is that, while modification of the former is accepted by all speakers, modification of the latter is accepted by fewer speakers.

6.3.2.3. Participial relative clauses

Bare DOs in participial relative clauses cannot be marked as plural (533). However, they allow for a variety of adnominal modifiers, including bare nouns (534), adjectives (535), possessors (536), numerals (537) and demonstratives (538) in most idiolects.

(533) PLURAL MARKER

toma /	**toma-**vlä***	*vəžalə-šə*	*edem*
house	house-PL	sell-PTCP.ACT	man

‘the man selling houses’

(534) BARE NOUN

?***cəvə***	*lem*	*šol-tə-šə*	*ədərəməš*	*maša*	*gan'ə*	*əl-eš*
chicken	soup	boil-CAUS-PTCP.ACT	woman[NOM]	Masha	EQU	be-NPST.3SG

‘The woman making chicken soup looks similar to Masha.’

(535) ADJECTIVE

?***kogo***	*kol*	*kəčə-šə*	*ərvezəš*	*ängər-əš*	*ke-n*	*vaz-ən*
big	fish	catch-PTCP.ACT	boy[NOM]	creek-ILL	go-CVB	lay-PRET

‘The boy catching a big fish fell in the water.’

(536) POSSESSOR

?***mäm-nä-n***	*pört*	*ubiräjə-šə*	*ədərəməš*	*kogo-n*	*itərä*
1PL-AGR.1PL-GEN	house	clean-PTCP.ACT	woman[NOM]	big-ADV	clean

‘The woman that cleans our house is very tidy.’

(537) NUMERAL

?***kok***	*kol*	*kəčə-šə*	*ərvezəš*	*ängər-əš*	*ke-n*	*vaz-ən*
two	fish	catch-PTCP.ACT	boy[NOM]	creek-ILL	go-CVB	lay-PRET

‘The boy catching two fish fell in the water.’

(538) DEMONSTRATIVE

?***ti***	*kol*	*kə̂čə̂-šə̂*	*ərvezäš*	*ängər-əš*	*ke-n*	*vaz-ən*
this	fish	catch-PTCP.ACT	boy[NOM]	creek-ILL	go-CVB	lay-PRET

‘The boy catching this fish fell in the water.’

6.3.2.3. Infinitival arguments of emotive predicates

Unlike in two previous constructions, and similar to bare DOs in participial clauses, plural marking on bare DOs in infinitival arguments of emotive predicates is accepted by very few speakers (539), and I consider it to be marginal in Hill Mari.

(539)

??*mən'*	*kn'igä-**vlä***	*lə̂d-aš*	*jarat-em*
1SG[NOM]	book-PL	read-INF	like-NPST.1SG

‘I like reading books.’

Bare DOs in infinitival arguments of emotive predicates face more restrictions on modification than those in any of the constructions discussed so far. While most speakers accept modification with bare nouns (540) and adjectives (541), possessors (542) and numerals (543) are incompatible with bare DOs in this construction. Demonstratives, on the other hand, are accepted by only a few speakers (544).

(540) BARE NOUN

?*ävi*	***cəvä***	*lem*	*šol-t-aš*	*jarat-a*
mother.KIN[NOM]	chicken	soup	boil-CAUS-INF	like-NPST.3SG

‘My mom likes making chicken soup.’

(541) ADJECTIVE
[?]*mən'* ***mü-än***[44] *olma* *kačk-aš* *jarat-em*
1SG[NOM] honey-PROP apple eat-INF like-NPST.1SG
'I like eating sweet apples.'

(542) POSSESSOR
mən'* ***tən'-ən* *lem* *šolt-aš* *jarat-em*
1SG[NOM] 2SG-GEN soup boil-INF like-NPST.1SG
Int.: 'I like making your soup.'

(543) NUMERAL
mən'* *irok* ***kok* *mə̂nə̂* *kačk-aš* *jarat-em*
1SG[NOM] morning two egg eat-INF like-NPST.1SG
Int.: 'I like eating two eggs in the morning.'

(544) DEMONSTRATIVE
[??]*mən'* ***ti*** ***kn'igä*** *lə̂d-aš* *jarat-em*
1SG[NOM] this book read-INF like-NPST.1SG
'I like reading this book.'

6.3.2.4. Adjunct converbial clauses[45]

In adjunct converbial clauses, plural marking on bare DOs is entirely ruled out (545). However, lower adnominal dependents, such as bare nouns (546) and adjectives (547), can modify bare DOs. In contrast, higher adnominal dependents, such as possessors (548), numerals (549), and demonstratives (550), are ungrammatical in this construction.

[44] Despite exhibiting an internal structure characteristic of a noun phrase with an attributivizer, this word is sufficiently lexicalized to be considered an adjective (Savatkova 2002).

[45] With regard to the properties of bare direct objects, there appear to be no differences among various types of converbs (Pleshak & Sirotina 2023). In this section, I illustrate these properties using *n*-converbs exclusively.

(545) PLURAL MARKER

maša	*posudə̂-m,*	*mə̂rə̂ / ***mə̂rə̂-vlä***		*mə̂r-en,*	*mə̂šk-ə̂n*
Masha[NOM]	dishes-ACC	song	song-PL	sing-CVB	wash-PRET

'Masha washed dishes singing songs.'

(546) BARE NOUN

kesə̈	*šə̈šer*	*jü-n,*	*maša*	*urok-ə̂m*	*ə̈št-ä*
goat	milk	drink-CVB	Masha[NOM]	lesson-ACC	do-NPST.3SG

'Masha is doing her homework sipping on goat milk.'

(547) ADJECTIVE

u	*mə̂rə̂*	*mə̂r-en,*	*ät'ä-m*	*rokolma-m*	*itəräj-ä*
new	song	sing-CVB	father-AGR.1SG[NOM]	potato-ACC	clean-NPST.3SG

'My dad is peeling potatoes singing a new song.'

(548) POSSESSOR

**maša*	*posudə̂-m,*	***tən'-ən***	*mə̂rə̂*	*mə̂r-en,*	*mə̂šk-ə̂n*
Masha[NOM]	dishes-ACC	2SG-GEN	song	sing-CVB	wash-PRET

Int.: 'Masha is washing dishes singing your song.'

(549) NUMERAL

**so,*	***kə̂m***	*mə̂rə̂*	*mə̂r-en,*	*ävä-m*	*ätə̈der-ə̈m*
always	three	song	sing-CVB	mother-AGR.1SG[NOM]	dishes-ACC

mə̂šk-eš
wash-NPST.3SG

Int.: 'My mother usually washes dishes singing three songs.'

(550) DEMONSTRATIVE

****ti***	*mə̂rə̂*	*mə̂r-en,*	*ät'ä-m*	*rokolma-m*	*itəräj-ä*
this	song	sing-CVB	father-AGR.1SG	potato-ACC	clean-NPST.3SG

Int.: 'My father peels potatoes singing this song.'

Thus, abstracting away from inter-speaker variation, bare DOs in adjunct converbial clauses exhibit the same restrictions as infinitival arguments of emotive predicates. They do not allow for plural marking or modification by higher adnominal dependents but do permit modification by lower adnominal dependents.

6.3.2.5. Converbial and infinitival arguments of phasal verbs and infinitival arguments of modal verbs

Bare DOs in converbial and infinitival arguments of phasal verbs, as well as infinitival arguments of modal verbs, are significantly more restricted than any of the previously discussed bare DOs. Not all speakers accept bare DOs in these constructions (551)–(553), and no one allows for plural marking of a bare DO (554)–(556).

(551) CONVERBIAL ARGUMENT OF A PHASAL PREDICATE ; UNMODIFIED DO
[?]*ävi* ***lem*** *šol-t-en* *pət-är-en*
mother.KIN soup boil-CAUS-CVB finish-CAUS2-PRET
'My mom finished making soup.'

(552) INFINITIVAL ARGUMENT OF A MODAL PREDICATE ; UNMODIFIED DO
[?]*mə-läm* ***škal*** *šəpš-əl-aš* *kel-eš*
1SG-DAT.AGR.1SG cow pull-ITER-INF be_needed-NPST.3SG
'I must milk the/a cow.'

(553) INFINITIVAL ARGUMENT OF A PHASAL PREDICATE ; UNMODIFIED DO
[?]***škal*** *šəpš-əl-aš* *təngäl-ən-äm*
cow pull-ITER-INF start-PRET-1SG
'I started milking the/a cow.'

(554) CONVERBIAL ARGUMENT OF A PHASAL PREDICATE ; PLURAL DO
mən'** ***škal-vlä *šə̂pš-əl-Ø* *pət-är-en-äm*
1SG[NOM] cow-PL pull-ITER-CVB finish-CAUS2-PRET-1SG
Int.: 'I finished milking cows.'

(555) INFINITIVAL ARGUMENT OF A MODAL PREDICATE ; PLURAL DO
mə-län-em ***tə̂gə̂r*** / ****tə̂gə̂r-vlä*** *mə̂šk-aš* *kel-eš*
1SG-DAT-AGR.1SG shirt shirt-PL wash-INF be_needed-NPST.3SG
'In mush wash shirts.'

(556) INFINITIVAL ARGUMENT OF A PHASAL PREDICATE ; PLURAL DO
***škal-vlä** *šə̂pš-ə̂l-aš* *təngäl-ən-nä*
cow-PL pull-ITER-INF start-PRET-1PL
Int.: 'We started milking cows.'

Some speakers who accept bare DOs in these constructions also allow for their modification with a bare noun (557)–(559). However, other types of dependents, including adjectives (560)–(562), numerals (563)–(565), and demonstratives (566)–(568) are consistently ruled out.

(557) CONVERBIAL ARGUMENT OF A PHASAL PREDICATE ; DO MODIFIED BY A BARE NOUN
??*ävi* ***cə̈vä*** ***lem*** *šol-t-en* *pə̈t-är-en*
mother.KIN[NOM] chicken soup boil-CAUS-CVB finish-CAUS2-PRET
'My mom finished making chicken soup.'

(558) INFINITIVAL ARGUMENT OF A MODAL PREDICATE ; DO MODIFIED BY A BARE NOUN
??*ävi-län* ***cə̈vä*** ***lem*** *šol-t-aš* *kel-eš*
mother.KIN-DAT chicken soup boil-CAUS-INF be_needed-NPST.3SG
'My mom must make chicken soup.'

(559) INFINITIVAL ARGUMENT OF A MODAL PREDICATE ; DO MODIFIED BY A BARE NOUN
??*ävi* ***cə̈vä*** ***lem*** *šol-t-aš* *təngäl-ən*
mother.KIN[NOM] chicken soup boil-CAUS-INF start-PRET
'My mom started making chicken soup.'

(560) CONVERBIAL ARGUMENT OF A PHASAL PREDICATE ; DO MODIFIED BY AN ADJECTIVE
mən'* ***kərän* ***škal*** *šə̂pš-ə̂l-Ø* *pə̈t-är-en-äm*
1SG[NOM] brown cow pull-ITER-CVB finish-CAUS2-PRET-1SG
Int.: 'I finished milking a brown cow.'

(561) INFINITIVAL ARGUMENT OF A MODAL PREDICATE ; DO MODIFIED BY AN ADJECTIVE
kərän** ***škal *šə̂pš-ə̂l-aš* *kel-eš*
brown cow pull-ITER-INF be_needed-NPST.3SG
Int.: 'It's necessary to milk the/a brown cow.'

(562) INFINITIVAL ARGUMENT OF A PHASAL PREDICATE ; DO MODIFIED BY AN ADJECTIVE

**maša*	***kərän***	***škal***	*šə̂pš-ə̂l-aš*	*təngäl-ən*
Masha[NOM]	brown	cow	pull-ITER-INF	start-PRET

Int.: ‘Masha started milking the/a brown cow.’

(563) CONVERBIAL ARGUMENT OF A PHASAL PREDICATE ; DO MODIFIED BY A NUMERAL

**mən'*	***kok***	***škal***	*šə̂pš-ə̂l-Ø*	*pət-är-en-äm*
1SG[NOM]	two	cow	pull-ITER-CVB	finish-CAUS2-PRET-1SG

Int.: ‘I finished milking two cows.’

(564) INFINITIVAL ARGUMENT OF A MODAL PREDICATE ; DO MODIFIED BY A NUMERAL

**mə-läm*	***kok***	***škal***	*šə̂pš-ə̂l-aš*	*kel-eš*
1SG-DAT.AGR.1SG	two	cow	pull-ITER-INF	be_needed-NPST.3SG

Int.: ‘I must milk two cows.’

(565) INFINITIVAL ARGUMENT OF A PHASAL PREDICATE ; DO MODIFIED BY A NUMERAL

**maša*	***ik***	***škal***	*šə̂pš-ə̂l-aš*	*təngäl-ən*
Masha[NOM]	one	cow	pull-ITER-INF	start-PRET

Int.: ‘Masha started milking one cow.’

(566) CONVERBIAL ARGUMENT OF A PHASAL PREDICATE ; DO MODIFIED BY A DEMONSTRATIVE

**ävi*	***ti***	***lem***	*šol-t-en*	*pət-är-en*
mother.KIN[NOM]	this	soup	boil-CAUS-CVB	finish-CAUS2-PRET

Int.: ‘My mom finished making this soup.’

(567) INFINITIVAL ARGUMENT OF A MODAL PREDICATE ; DO MODIFIED BY A DEMONSTRATIVE

**mə-läm*	***ti***	***škal***	*šə̂pš-ə̂l-aš*	*kel-eš*
1SG-DAT.AGR.1SG	this	cow	pull-ITER-INF	be_needed-NPST.3SG

Int.: ‘I must milk this cow.’

(568) INFINITIVAL ARGUMENT OF A PHASAL PREDICATE ; DO MODIFIED BY A DEMONSTRATIVE
ti škal *šəpš-əl-aš* *təngäl-ən-äm*
this cow pull-ITER-INF start-PRET-1SG
Int.: 'I started milking this cow.'

6.3.2.6. Summary

As demonstrated in this section, the properties of bare DOs vary depending on the specific construction in which they appear. These properties are summarized in Table 15.

Table 15: *Properties of bare DOs by construction*

	No dependents	Bare nouns	Adjectives	Possessors, numerals, demonstratives	Plural marking
nominalizations	OK	OK	OK	OK	?
goal / purpose infinitives	OK	?	?	?	?
Participles	OK	?	?	?	*
Infinitival arguments of evaluative predicates	OK	?	?	*	??
Converbial adjuncts	OK	OK	OK	*	*
Converbial arguments of phasal verbs and infinitival arguments of phasal and modal verbs	?	??	*	*	*

For the analysis presented in the next section, I abstract away from inter-speaker variation, treating "?" as acceptable and "??" as unacceptable. This yields an implicational hierarchy of constructions, ranked from those in which bare DOs exhibit the full range of properties to those where they are most restricted:

(569) nominalizations & goal / purpose infinitives > participles > infinitival arguments of evaluative predicates & converbial adjuncts > converbial arguments of phasal verbs & infinitival arguments of phasal and modal verbs

In the following section, I will provide an analysis of Hill Mari differential object marking (DOM) that not only accounts for the distinction between marked and bare DOs but also explains the variation observed among different types of bare DOs.

6.4. Analysis

In this section, I argue that the variation in the properties of bare DOs across different constructions reflects differences in the size of the nominal phrases occupying the DO position. Specifically, accusative DOs are full DPs, while bare DOs are SNs with reduced syntactic structure. This analysis not only explains the generalizations in (570) but also accounts for the observed variation among bare DOs in different constructions.

(570) a. **Overt marking generalization**:
If nominal agreement is expressed overtly, the DO must be accusative.

b. **Category restriction generalization**:
If the DO is a personal pronoun, it must be accusative.

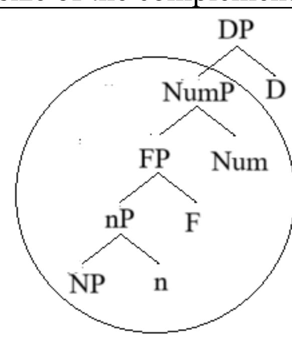
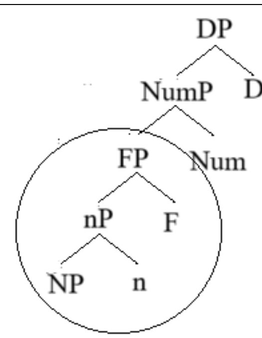
c. **Featural restriction generalization** (effective only in some idiolects):
If the DO is a proper name or a kinship term, it must be accusative.

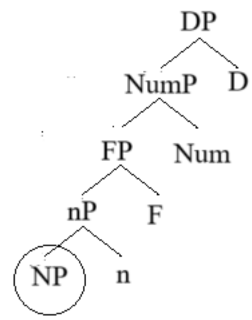
d. **Modification generalization**:
In the presence of an adnominal universal quantifier, the DO must be accusative.

e. **Adjacency generalization**:
If the DO is not adjacent to the predicate, it must be accusative.

Unlike accusative DOs, which are full DPs, bare DOs lack the DP projection and vary in size depending on the construction they occur in. The degree of syntactic deficiency correlates with the restrictions observed across different constructions. In particular, bare DOs in nominalized and goal / purpose infinitival clauses are as large as NumPs, lacking only the DP layer; bare DOs in participial clauses are F_2Ps, which lack NumP in addition to DP; bare DOs in infinitival arguments of evaluative predicates and converbial adjuncts are as small as NPs, lacking all functional structure above the lexical projection; and bare DOs in converbial arguments of phasal verbs and infinitival arguments of phasal and modal verbs are non-projecting heads, making them the most restricted type:

Table 16: *Non-finite structures with corresponding nominal sizes*

<table>
<tr><td>Syntactic environment</td><td>Nominal size of the complement</td></tr>
<tr><td>nominalized clauses</td><td rowspan="2">
</td></tr>
<tr><td>goal / purpose infinitival clauses</td></tr>
<tr><td>participial clauses</td><td>
</td></tr>
<tr><td>infinitival arguments of evaluative predicates</td><td></td></tr>
</table>

converbial adjuncts	
converbial arguments of phasal verbs	N
infinitival arguments of phasal and modal verbs	

This analysis effectively derives the observed distinctions among different types of bare DOs. In the following sections, I will first demonstrate how this approach accounts for variation in DO behavior across constructions, and then I will show how the generalizations in (570) follow naturally from this analysis.

6.4.1. Determining the size of the bare DO in each construction

6.4.1.1. Bare DOs in nominalized and goal / purpose infinitival clauses

As demonstrated in Sections 6.3.2.1 and 6.3.2.2, bare DOs in nominalized and goal/purpose infinitival clauses are the least restricted among bare DOs, exhibiting nearly the full set of properties associated with accusative DOs. A comprehensive analysis must explain both the generalizations in (570), which establish when accusative marking is required, and the possibility of plural marking and broad modification, which are unique to these bare DOs. I argue that bare DOs in these constructions are NumPs — SNs that lack only the DP layer.

The argument in favor of the presence of NumP is straightforward. The presence of plural marking provides direct evidence for the presence of NumP. The plural marker, as seen in (571)–(572), is hosted by the Num head, confirming that these bare DOs contain NumP:

(571)	?*maša-n*	***tâgâr-vlä***	*ərgə̂-mə̂-žə̂-m*	*už-Ø-əm*
	Masha-GEN	shirt-PL	sew-NMLZ-AGR.3SG-ACC	see-AOR-1SG
	'I saw Mahsa sew shirts.'			

(572)	?*maša*	***škal-vlä***	*šə̂pš-əl-aš*	*ke-n*
	Masha[NOM]	cow-PL	pull-ITER-INF	go-PRET
	'Masha went to milk cows.'			

A more challenging task is to demonstrate that these bare DOs do not project a full DP. Like in the previous chapters, I adopt the following assumptions about DPs vs. SNs (see Chapter 1, Section 1.2.2 on case and agreement and Chapter 2 on the extended projection of the noun phrase):

- DPs, but not small nominals, are case-marked.
- D hosts definite determiners.
- Only DPs trigger agreement.
- Nominal agreement requires the phase head D.

Given that the caseless/cased status of bare DOs is rather a question than a fact, since Hill Mari lacks definite determiners and object agreement, the best diagnostic for DP status is nominal agreement.

A key empirical observation about Hill Mari bare DOs is that while possessors can appear in bare DOs, nominal agreement forces accusative marking:

(573) a. *ti* *ədər* *mäm-nä-n* ***koti-m*** / ***koti*** *pukš-əmə̑-m*
this girl[NOM] 1PL-AGR.1PL-GEN cat-ACC cat feed-NMLZ-ACC
män' *už-ə̑n-am*
1SG[NOM] see-PRET-1SG

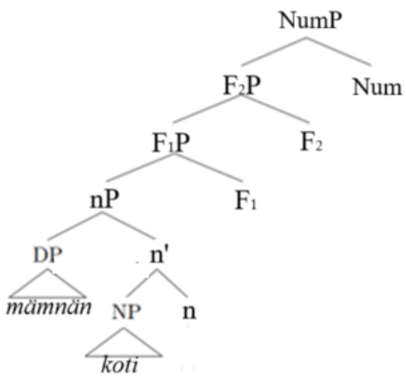
b. *ti* *ədər* *mäm-nä-n* ***koti-nä-m*** / ****koti-nä***
this girl[NOM] 1PL-AGR.1PL-GEN cat-AGR.1PL-ACC cat-AGR.1PL
pukš-əmə̑-m *män'* *už-ə̑n-am*
feed-NMLZ-ACC 1SG[NOM] see-PRET-1SG
'I saw that this girl fed our cat.'

Since agreement is only possible in the presence of accusative case, and since case is typically assigned at DP level, this suggests that bare DOs lack DP projection.

Even though I do not assume that D is the actual host of nominal agreement, I extend the model of feature inheritance (Chomsky 2008) from the clausal domain to the nominal domain. The idea is that in parallel with the clause, where the T head in the complement of the phase head C inherits its features and probes only in the presence of C, the Num head in the complement of the phase head D inherits the features of D, and probes only if D is present. What it gives us is the impossibility of Num head to be the agreement probe in the absence of D. Consequently, in bare DOs, since D is absent, Num cannot act as an agreement probe.

What we see in Hill Mari is exactly what is predicted by the model outlined above. Even though we can see bare DOs marked as plural (571)–(572), indicating that they have a Num head, the agreement is impossible as long as the case is absent (indicating the absence of D) (573). In contrast to some other languages, where possessed nominals cannot be bare, e.g., Moksha (Chapter 5) or Tatar (Lyutikova & Pereltsvaig 2015), the crucial fact about Hill Mari bare DOs is that they can be possessed, and it is the presence of the overt marker that makes the difference. The structure of the bare DO in (573) is in Figure 59.

Figure 59: *The structure of a Hill Mari bare DO in a nominalized clause*



Further evidence against the presence of the DP layer comes from restrictions on modification, which do not seem to correlate with the presence or absence of a feature like the definiteness feature [def*]. Bare DOs I am discussing here cannot combine with strong universal quantifiers that require definite descriptions (574). However, bare DOs can be interpreted as definite, which is supported by grammaticality of (575), where the DO has a definite interpretation due to the presence of a demonstrative.

(574)	**ävä-m-ən*	***cilä***	***əškal***	*šəpš-əl-mə-m*	*mən'*	*päl-em*
	mother-AGR.1SG-GEN	all	cow	pull-ITER-NMLZ-ACC	1SG[NOM]	know-NPST.1SG

Int.: ‘I know that my mother milked all cows.’

(575)	*kəm*	*ərvezäš-ən*	***ti***	***kn'igä***	*ləd-mə-štə-m*	*mən'*	*päl-em*
	three	boy-GEN	this	book	read-NMLZ-AGR.3PL-ACC	1SG[NOM]	know-NPST.1SG

‘I know that three boys read this book.’

Having determined the upper limit for the functional structure of bare DOs in nominalized and adjunct infinitival clauses, it is worth considering the question whether there is a lower limit: the largest non-DP structure that is allowed in the DO position in these constructions is NumP, but can an FP, nP or even an NP occupy this position? The answer appears to be negative. For those consultants that allow for plural marking (and therefore allow for NumP), it is obligatory for a plural interpretation of the bare DO. Similarly, the presence of a definite demonstrative is strongly preferred for the definite anaphoric reading of the bare DO in (576).

(576) *tə̂mdə̂-šə̂* *tet'ä-vlä-län* *knigä-m* *lə̂d-aš* *šüd-en*
teach-PTCP.ACT child-PL-DAT book-ACC read-INF order-PRET
'The teacher told his students to read a book.'

tədə *tidə-m* *ik* *ärn'ä-eš* *nənə-län* *pu-en*
3SG[NOM] this-ACC one week-LAT 3PL-DAT give-PRET
'He gave them one week for this.'

kə̂m *tet'ä-n* ??(***ti***) ***kn'igä*** *lə̂d-mə̂-m* *mən'* *päl-em*
three child-GEN this book read-NMLZ-ACC 1SG[NOM] know-NPST.1SG
'I know that three boys read the book.'

Thus, nominalized clauses and adjunct infinitival clauses allow for their DOs not to be full DPs, and the small DOs are as large as NumP (or smaller, depending on how much functional structure is needed).

6.4.1.2. Bare DOs in participial clauses

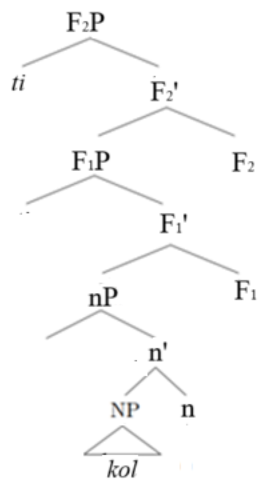
While bare DOs in participial clauses share all the restrictions observed in nominalized and goal/purpose infinitival clauses, they impose an additional morphological constraint — bare DOs in participial clauses cannot take a plural marker:

(577) ***toma*** / ****toma-vlä*** *və̂žalə̂-šə̂* *edem*
house house-PL sell-PTCP.ACT man
'the man selling houses'

Since number markers are hosted by the Num head, I argue that the inability of bare DOs in participial clauses to bear plural markers stems from the absence of NumP in their structure. However, the fact that cardinal numerals and demonstratives are allowed suggests that these bare

DOs are F_2Ps, because numerals are generated in F_1P (see discussion on the low structural position of numerals in Chapter 4, Sections 4.2.2 and 4.5), and demonstratives are generated in F_2P (see Chapter 2, Section 2.3.4, Chapter 3, Section 3.3.1, and Section 6.1.3 of the current Chapter). Figure 57 illustrates the structure of the bare DO *ti kol* [this fish] 'this fish'.

Figure 60: *The structure of a Hill Mari bare DO in a participial clause*



In contrast to nominalized clauses and adjunct infinitival clauses, participial clauses do not allow for NumP in their small DOs.

6.4.1.3. Bare DOs in infinitival arguments of evaluative predicates and converbial adjuncts

Bare DOs in infinitival arguments of evaluative predicates and converbial adjuncts exhibit all the restrictions of bare DOs observed in participial clauses. Additionally, they cannot be modified by possessors, numerals, or demonstratives, as shown in (578)–(580). In the noun phrase model outlined in Chapter 2, numerals and demonstratives are generated (and can stay) in F_1P and F_2P, respectively. Their absence in these structures suggests that FP layers are missing. Since possessors originate in nP, their ungrammaticality in these bare DOs indicates the absence of nP as well. Therefore, bare DOs in these contexts are as small as NP.

(578) POSSESSOR

**maša*	*posudə̂-m,*	*tən'-ən*	***mə̂rə̂***	*mə̂r-en,*	*mə̂šk-ən*
Masha[NOM]	dishes-ACC	2SG-GEN	song	sing-CVB	wash-PRET

Int.: 'Masha was washing the dishes singing songs.'

(579) NUMERAL

**so,*	***kə̂m***	***mə̂rə̂***	*mə̂r-en,*	*ävä-m*	*ätə̈der-əm*
always	three	song	sing-CVB	mother-AGR.1SG[NOM]	dishes-ACC
mə̂šk-eš					
wash-NPST.3SG					

Int.: 'My mother usually washes dishes singing three songs.'

(580) DEMONSTRATIVE

****ti***	***mə̂rə̂***	*mə̂r-en,*	*ät'ä-m*	*rokolma-m*	*itəräj-ä*
this	song	sing-CVB	father-AGR.1SG	potato-ACC	clean-NPST.3SG

Int.: 'My father peels potatoes singing this song.'

Unlike bare DOs in nominalized, adjunct infinitival, and participial clauses, those in adjunct converbial clauses can refer to a discourse antecedent (i.e., they can have a definite interpretation) even without the demonstrative *ti* 'this' (581). My analysis accounts for this distinction by considering the number of functional layers in these structures. I argue that the impossibility of inserting a demonstrative facilitates the definite interpretation of a *ti*-less DO. In contrast, in structures where *ti* could have been inserted, its absence implies an indefinite reading.

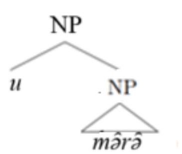
(581) *mä* *tol-ən-na,* *amasa-štə̑* *sə̑ravač* *keč-ä*
1PL[NOM] come-PRET-1PL door-IN lock hang-NPST.3SG
mä ***sə̑ravač-ə̑m*** / ***sə̑ravač*** *pə̑də̑r-t-en* *pə̑rə-š-na*
1PL[NOM] lock-ACC lock break-CAUS-CVB enter-AOR-1PL
'We came; there is a lock on the door. We entered having broken the lock.'

However, these bare DOs are not smaller than NP, as they still allow adjectival modification (582). The structure of the DO in (582) is represented in Figure 61.

(582) ***u*** ***mərə̑*** *mə̑r-en,* *ät'ä-m* *rokolma-m* *itəräj-ä*
new song sing-CVB father-AGR.1SG[NOM] potato-ACC clean-NPST.3SG
'My dad is peeling potatoes singing a new song.'

Thus, infinitival arguments of evaluative predicates and converbial adjuncts allow only for the smallest phrasal structures as their DOs unless a full DP is projected.

Figure 61: *The structure of a Hill Mari bare DO in a converbial adjunct*



6.4.1.4. Bare DOs in converbial arguments of phasal verbs and infinitival arguments of phasal and modal verbs

In addition to the restrictions observed for other bare DOs, bare DOs in converbial arguments of phasal verbs and in infinitival arguments of phasal and modal verbs, when accepted at all, do not permit adjectival modification (583). The only modifiers that are marginally acceptable are bare nouns (584).

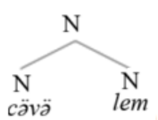
(583)	****kərän***	***škal***	*šəpš-əl-aš*	*kel-eš*	
	brown	cow	pull-ITER-INF	be_needed-NPST.3SG	
	Int.: 'It's necessary to milk the/a brown cow.'				

(584)	??*ävi-län*	***cəvə***	***lem***	*šol-t-aš*	*kel-eš*
	mother.KIN-DAT	chicken	soup	boil-CAUS-INF	be_needed-NPST.3SG
	'My mom must make chicken soup.'				

A comprehensive analysis of Hill Mari bare adnominal modifiers is beyond the scope of this study; however, I assume that their modification does not involve a full phrasal structure but rather restricted to interaction between heads.

The structure of a bare DO in converbial arguments of phasal verbs and infinitival arguments of phasal and modal verbs is illustrated in Figure 62.

Figure 62: *The structure of a Hill Mari bare DO in converbial arguments of phasal verbs and infinitival arguments of phasal and modal verbs*



In contrast to other constructions, where small DOs are allowed to project functional layers or at least a lexical NP, bare DOs in converbial arguments of phasal verbs and infinitival arguments of phasal and modal verbs must be heads.

6.4.2. Deriving the common properties of bare DOs

In this section, I account for the common properties of all bare direct objects (DOs) in Hill Mari non-finite clauses, arguing that they should be analyzed as SNs rather than cased full DPs. The empirical generalizations reflecting the common properties of Hill Mari bare DOs are presented in (585).

(585) a. **Overt marking generalization**:
If nominal agreement is expressed overtly, the DO must be accusative.
b. **Category restriction generalization**:
If the DO is a personal pronoun, it must be accusative.
c. **Featural restriction generalization** (effective only in some idiolects):
If the DO is a proper name or a kinship term, it must be accusative.
d. **Modification generalization**:
In the presence of an adnominal universal quantifier, the DO must be accusative.
e. **Adjacency generalization**:
If the DO is not adjacent to the predicate, it must be accusative.

The SN analysis effectively captures the shared characteristics of bare DOs. Below, I discuss each property in turn.

6.4.2.1. Overt marking and modification generalizations

The correlation between nominal agreement and case arises from the fact that both appear only in full DPs (see Chapter 1, Section 1.2.2). Case features are hosted by the D head, and, as argued in Section 6.4.1.1, the presence of D is also required for the Num head to function as a probe, which results in presence of nominal agreement. Additional support for this analysis comes from the observation that the only universally prohibited modifier in bare direct objects is the universal quantifier, which typically attaches in higher functional layers. If this restriction were tied to definiteness, bare direct objects prohibiting universal quantifiers would also prohibit demonstratives—but they do not. A broader examination of modification restrictions provides further evidence: bare direct objects that disallow demonstratives also disallow cardinal numerals.

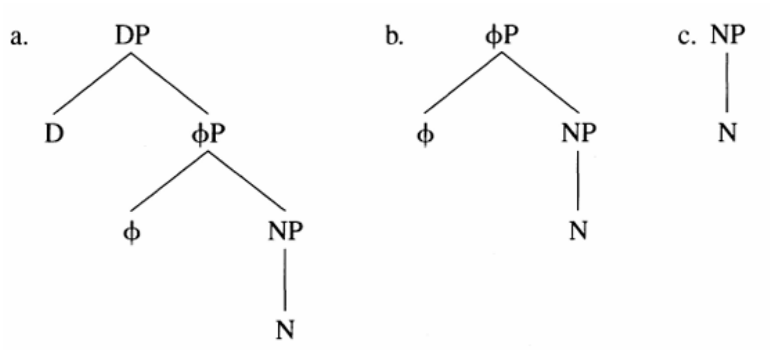
Since numerals are not inherently linked to specificity or definiteness (e.g., in bare direct objects in converbial clauses), the restriction must be structural. Specifically, the absence of the universal quantifier follows from the absence of D, while other adnominal modifiers remain possible because they are generated in lower structural positions.

6.4.2.2. Category and featural restriction generalizations

When it comes to the category restriction and featural restriction generalizations, the requirement of personal pronouns, proper names and kinship terms to bear overt case is less straightforward. For my analysis to hold, I must assume that Hill Mari personal pronouns are always DPs. The same assumption should extend to proper names and kinship terms. While this stipulation is neither counterintuitive nor unprecedented, independent evidence would be preferable, as pronouns, proper names, and kinship terms do not necessarily have to be DPs (Déchaine & Wiltschko 2002). For now, I leave the structural classification of Hill Mari personal pronouns for future research. Although the literature proposes diagnostics for identifying the structural size of pronouns, these tests yield conflicting results for Hill Mari. Furthermore, there is little consensus among researchers regarding the relation among different projections, so it is unclear how they translate to the nominal projections outlined in Chapter 2.

Déchaine and Wiltschko (2002) propose that pronouns have an internal structure comprising three layers: NP, φP, and DP (see Figure 63). Pronouns function as full DPs if they are excluded from predicative positions, exhibit the syntactic properties of R-expressions (i.e., subject to Condition C), and contain a D position. In contrast, φP-pronouns can appear in both predicative and argument positions and may be bound outside their local domain or even locally.

Figure 63: *Internal structure of pronouns* (Déchaine and Wiltschko 2002: 410)



Unlike Déchaine and Wiltschko, who position DP above φP, Cowper and Hall (2009) argue that DP and φP are in complementary distribution, attributing differences in their properties to the features present on each node. This raises the question of whether the DP–φP distinction reflects a difference in structural size or simply in feature composition.

Even if we adopt Déchaine and Wiltschko's model, their diagnostics produce contradictory results for Hill Mari personal pronouns. On the one hand, first- and second-person pronouns in Hill Mari are restricted to argument positions and require an agreeing copula when used as predicates (586), which suggests they are DP-pronouns. On the other hand, these pronouns can be bound within a clause and are not subject to Condition C (587), which suggests they are φP-pronouns. Additionally, while pronouns such as *mən'* 'I' and *tən'* 'you(SG)' are not easily decomposable, this does not rule out the possibility of an articulated internal structure. Therefore, I refrain from drawing conclusions based on this criterion.

(586) PERSONAL PRONOUNS IN PREDICATIVE POSITION

ti	*izi*	*ədəräš*	[***mən'***	**(əl-am)*]PRED
this	little	girl[NOM]	1SG[NOM]	be-NPST.1SG

'This little girl is me.'

(587) PERSONAL PRONOUN BOUND WITHIN A CLAUSE
mən'$_i$ *tengečə* *ške*$_i$ / ***mən'-ən***$_i$ *sumka-em* *jamd-en-äm*
1SG[NOM] yesterday REFL$_i$ 1SG-GEN$_i$ bag-AGR.1SG[ACC] lose-PRET-1SG
'Yesterday, I$_i$ lost my$_i$ bag.'

A potential argument for the obligatoriness of DP projection with kinship terms is the obligatory presence of a nominal agreement marker on a subset of them (Pleshak 2017b; Pleshak & Remizova 2018; Pleshak 2023b:170), see (588):

(588) a. *mən'-ən* *äkä-**m*** / ******äkä*
1SG-GEN older_sister-AGR.1SG older_sister
'my older sister'
b. *mən'-ən* *šə̂žar-**em*** / *šə̂žar*
1SG-GEN younger_sister-AGR.1SG older_sister
'my older sister'

The obligatory expression of overt nominal agreement requires a probe on Num, which in turn can only be present if the D head is present, as I argued in Section 6.4.1.1. I hypothesize that this requirement extends to all kinship terms, regardless of whether they obligatory bear overt agreement. As argued in Section 6.1.3, the absence of overt nominal agreement should not be taken as evidence for the absence of a DP projection in Hill Mari.

6.4.2.3. Adjacency generalization

The requirement that bare direct objects remain adjacent to the predicate can also be accounted for under my SN analysis. Agreement probes search for φ-features — most commonly person and number. In the nominal architecture I assume, person features are hosted by D. Without D, nominals are simply invisible to probes. Although some probes that trigger movement may target features other than φ-features, all features relevant for syntactic processes external to the nominal must still be located on D; otherwise, they would not be accessible to probes in contexts where the

DP layer is present, i.e., in the majority of syntactic environments. Since SNs lack D, they also lack these features, and therefore no structural conditions exist under which they would undergo movement.[46]

6.4.3. Interim summary

In this section, I argued that bare direct objects (DOs) in Hill Mari lack the DP layer and, depending on the construction they occur in, may also lack other higher functional projections below DP, such as NumP, F_2P, F_1P, and nP. The absence of DP accounts for the shared properties of DOs that lack case, including restrictions on modification with strong quantifiers, the requirement for adjacency to the predicate, and category restrictions.

The variation in which projections may be absent explains construction-based differences in the properties of bare DOs. Some bare DOs permit number marking because their structure includes NumP, while others do not, because NumP is missing. Similarly, bare DOs that allow modification with numerals and demonstratives include F_2P, whereas those that prohibit such modifiers lack this projection. The most restrictive bare DOs, those that only allow bare nominals as modifiers, are non-projecting N heads.

[46] Alternatively, one might conceptualize the adjacency requirement in terms of licensing. Adjacency has been proposed as a licensing mechanism alternative to case (e.g., Levin 2015). On this view, SNs, which lack case, must be licensed via adjacency. While this perspective is not incompatible with my analysis, I intentionally avoid invoking the notion of "licensing", as I separated case from agreement and from licensing. I find the term "licensing" confusing and choose not to use it.

More specifically, I proposed that in nominalized and goal / purpose infinitival clauses, bare DOs are as large as NumP: they can bear plural markers and allow a range of adnominal modifiers up to demonstratives. Bare DOs in participial clauses are smaller, F_2P-size: they cannot bear plural markers but still allow demonstratives. Bare DOs in infinitival arguments of evaluative predicates and converbial adjuncts are NPs, permitting only adjectives and bare nouns as modifiers. Finally, bare DOs in converbial arguments of phasal verbs and infinitival arguments of phasal and modal verbs do not project phrases, prohibiting all modification except nominal compounding.

Turning to the question of why particular syntactic configurations require particular sizes of bare arguments, I hypothesize that the structural size of the non-finite clause constrains the size of the bare DO. As I showed in Section 6.2, nominalized and goal / purpose infinitival clauses share the greatest number of properties with finite clauses, apart from lacking overt tense markers, which suggests that they may be as large as CP. Participial clauses also preserve most clausal properties, including modification with temporal adverbials and the possibility of overt subjects, and may be as large as TP. Considered infinitival arguments and converbial adjuncts are vPs: they disallow overt subjects, temporal adverbials, and subject agreement. Finally, converbial arguments of phasal verbs and infinitival arguments of phasal and modal verbs exhibit the fewest clausal properties and resemble VPs. Table 17 represents the mapping between clausal type and bare DO size through the structural size of the clause.

Table 17: *Clausal types and corresponding bare DOs*

Clausal type	Size of the clause	Bare DO size
nominalized and goal / purpose infinitival clauses	CP	NumP
participial clauses	TP	F_2P
infinitival arguments and converbial adjuncts	vP	NP
converbial arguments of phasal verbs and infinitival arguments of phasal and modal verbs	VP	N

Notably, the larger the clause, the larger its bare DO. If this preliminary analysis is on the right track, it may shed more light on the parallelism of the clausal and nominal structures. However, a more thorough examination of clausal properties is necessary to confirm the structural analysis of the clausal types under discussion.

6.5. Advantages of a small nominal analysis

In the previous section, I argued for a small nominal analysis of Hill Mari bare DOs. In this section, I argue that the small nominal approach is superior to alternatives.

An alternative analysis would assume that all or at least some bare DOs are full DPs with a null realization of the object case. For instance, under this approach, either object in (589) would be analyzed as a full DP bearing case. This case marking could be considered a different allomorph of the accusative case or an allomorph of another object case.

(589) *vas'a* ***ti*** ***kol'mə̂-m*** / ***kol'mə̂*** *näl-äš* *tol-ə̂n*
Vasja[NOM] this shovel-ACC shovel take-INF come-PRET
'Vasja came to buy this shovel.'

In the following discussion, I will examine two analyses that attribute the case asymmetry to different allomorphs of the same accusative case, as well as two analyses that propose distinct structural configurations and case markings for marked versus bare DOs.

6.5.1. The absence of the accusative as the null allomorph of the accusative

I will begin by considering the possibility that the null allomorph is simply another allomorph of the same case—namely, the accusative. Under this analysis, both DOs in (589) are fully cased DPs marked as accusative. The difference between them lies in the realization of the accusative case: in one instance, it appears as *-m*, while in the other, it is realized as *Ø*, with *-m* and *Ø* functioning as allomorphs of the same case.

Setting aside domain-specific reasons, as discussed in Chapter 4, Section 4.4, I identify two primary motivations for the use of the null allomorph instead of the overt one. First, the absence of an overt case marker may be conditioned by adjacency to the verb, allowing the case node to be filled with a null element in accordance with the Empty Category Principle (Lamontagne & Travis 1987). Second, the null allomorph may reflect the absence of a feature that is present in DOs bearing an overt allomorph, such as animacy or definiteness.

6.5.1.1. The absence of the accusative as the usage of a null allomorph of the accusative under adjacency: the ECP analysis

Lamontagne and Travis (1987) proposed an explanation for the alternation between overt and null case realization based on adjacency to a specific syntactic head. Situations in which an overt case marker or complementizer is absent from the surface structure are referred to as case-drop and complementizer-drop, respectively. In both cases, the absence of an overt marker is conditioned by adjacency to a governing head. Specifically, adjacency to the V-head results in a null case due to the recoverability of the case features on K. The variation observed in (589) can be analyzed as case-drop.

To apply a similar analysis to Hill Mari, as I did for Moksha, I must adopt a different set of theoretical assumptions in line with Lamontagne and Travis (1987). First, I assume that KP is the highest nominal projection above DP and that KP is present in narrow syntax rather than being introduced post-syntactically. Under this approach, K functions as a syntactic head that may either have an overt realization or host an empty category. Second, I adopt the Empty Category Principle (ECP), as shown in (590). While ECP plays a crucial role in Government and Binding Theory, its relevance within Minimalism remains debated. Third, I assume that case is assigned via Agree with a governing head. Because ECP requires features to be recoverable from the governing head, configurational case assignment is incompatible with ECP.

(590) **Empty Category Principle** (Lamontagne & Travis 1987: 177-180):

(i) All empty categories (Xs and XPs) must be identified.

(ii) a. The position of the gap is identified iff

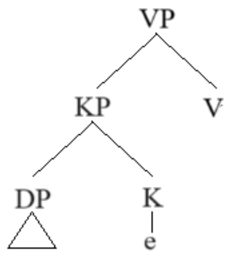
- it has a local antecedent, or
- it (or its maximal projection) must stand in the relation 'complement of'

b. The contents of the gap are identified through feature recoverability (subject to Head Feature Transmission Constraint)

(591) **Head Feature Transmission Constraint**: Head features may only be transmitted from a head to its sister.

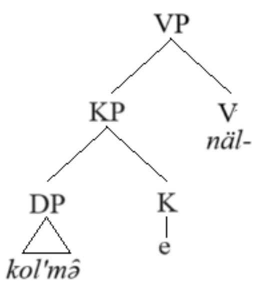
Figure 64 illustrates the structural configuration for case-drop.

Figure 64: *The structural configuration for case drop*



Applying an ECP analysis to the Hill Mari data, we can derive the DOM pattern. Here, V is the accusative-assigning head and carries the feature [ACC]. This feature can be recovered on the head of the DO, satisfying ECP. Consequently, the head K may be realized either with an overt marker or as an empty category.

Figure 65: *The bare DO as an EC*



However, this analysis encounters the same issue I previously identified for Moksha PPs. While it accounts for the existence of bare DOs, it fails to explain the ungrammaticality of certain bare DOs. For instance, it remains unclear why the presence of a nominal agreement marker in (592) prevents ECP satisfaction. Similarly, the analysis incorrectly predicts that personal pronouns (593), as well as proper names and kinship terms, should be able to surface as bare forms. More broadly, an ECP-based account is challenged by the Overt Marking, Category Restriction, and Feature Restriction generalizations presented in (594).

(592) DO WITH AN AGREEMENT MARKER

ədər	*mäm-nä-n*	***šarə̂k-na-m***	/	****šarə̂k-na***	*näl-äš*
girl[NOM]	1PL-AGR.1PL-GEN	sheep-AGR.1PL-ACC		sheep-AGR.1PL	take-INF

tol-ə̂n
come-PRET
'The girl came to buy our sheep.'

(593) PERSONAL PRONOUN DO

män'	***tən'-əm***	/	****tən'(ə)***	*kə̂mə̂l-ang-d-aš*	*tol-ə̂n-am*
1SG[NOM]	2SG-ACC		2SG	mood-INCH2-CAUS-INF	come-PRET-1SG

'I came to greet you.'

(594) a. **Overt marking generalization**:

If nominal agreement is expressed overtly, the DO must be accusative.

b. **Category restriction generalization**:

If the DO is a personal pronoun, it must be accusative.

c. **Featural restriction generalization** (effective only in some idiolects):

If the DO is a proper name or a kinship term, it must be accusative.

One possible solution to this issue is to assume that possessed nominals, pronouns, proper names, and kinship terms undergo object shift, moving out of their base positions. This movement disrupts the 'complement of' relation, resulting in an ECP violation. However, for shifted DOs to receive case, this movement must precede case assignment, which creates a theoretical conflict.

Thus, just as an ECP account failed to explain Moksha PPs, it also fails to account for Hill Mari due to this theoretical issue concerning the timing of case assignment and movement. This problem is avoided in my SN analysis, which does not rely on movement.

6.5.1.2. The “absence” of the accusative as the null allomorph of the accusative: different feature specification analysis

This section focuses on an analysis in which the null allomorph represents the absence of certain nominal features. I model this within the framework of Distributed Morphology (see Chapter 1, Section 1.2.1), where abstract features on syntactic nodes are lexicalized late in the derivation. The vocabulary consists of phonological realizations of morphemes paired with the abstract features they express. The Subset Principle determines morpheme selection when multiple forms correspond to the same abstract features: the morpheme specified for the largest subset of relevant features (without additional, extraneous features) is inserted.

In Hill Mari, semantically definite and indefinite, animate and inanimate nouns may receive overt accusative marking. However, the nature of the feature that distinguishes the two different accusative morphemes remains unclear. For the purposes of this analysis, I introduce a feature F and propose the following vocabulary items for the Hill Mari accusative case. In this system, the morpheme in (595) lexicalizes the accusative case when **F** is present, whereas the null morpheme in (595) functions as a default or elsewhere case. Combined with the number morphemes in (596), we derive the following forms in (598) for the direct objects in (597).

(595) **Vocabulary items for the accusative case**:

a. [F; accusative] < → /m/

b. [accusative] <-> /Ø/

(596) **Vocabulary items for number**:

a. [singular] ←> /Ø/

b. [plural] < → /vlä/

(597) *vas'a* **ti** ***kol'mə̂-m*** / ***kol'mə̂*** *näl-äš* *tol-ən*
Vasja[NOM] this shovel-ACC shovel take-INF come-PRET
'Vasja came to buy this shovel.'

(598) a. [root][singular][F; accusative] → *kol'mə̂-m*
b. [root][singular][accusative] → *kol'mə̂-Ø*

The morpheme in (595)a is more specified than (595)b because it contains the additional feature F. According to the Subset Principle, (595)a will always be chosen over (595)b when F is present in the nominal. Consequently, a bare DO is not a caseless nominal or one bearing a non-accusative case but rather a nominal that lacks F.

However, this analysis presents a challenge. It relies on an abstract feature (F) to account for the data while also overgeneralizing, predicting bare DOs in all contexts, including finite clauses where they are not always attested. If bare DOs are simply nominals lacking F, it remains unclear why such nominals cannot appear in finite structures.

In contrast, my SN analysis avoids introducing ungrounded features, deriving the absence of case from the absence of the corresponding syntactic structure. This approach is also more suitable for the situation with construction dependency: non-finite structures allowing for smaller DOs does not require the same behavior from finite structures.

6.5.2. The absence of accusative as the null allomorph of some other case

Now, let us consider the possibility that bare DOs bear the null allomorph of a case other than accusative. For this alternative analysis, the specific label of the case is not crucial, so I will refer to it as the object case. Under this approach, the two DOs in (599) receive different cases: one is marked with the accusative case, while the other bears the object case.

(599) *vas'a* **ti** ***kol'mə̂-m*** / ***kol'mə̂*** *näl-äš* *tol-ə̂n*
Vasja[NOM] this shovel-ACC shovel take-INF come-PRET
'Vasja came to buy this shovel.'

As I previously mentioned in the discussion of alternative analyses for Moksha PPs, I adopt a version of case theory in which there is a one-to-one correspondence between a syntactic configuration and the case assigned within it — whether assigned configurationally or by a syntactic head. Within this framework, variable case assignment necessitates the existence of two distinct case assignment positions.

6.5.2.1. Two case assignment DO positions

The idea of having two case assignment positions within the clause is not new and is commonly referred to as the object shift analysis. According to this approach, certain objects move upward due to the presence of specific features, which in turn affects the case they receive. If this analysis is correct for Hill Mari DOs, then bare DOs remain in situ and receive the object case, while accusative DOs move to be assigned the accusative case. This suggests that some DPs carry a feature that is attracted by a head in the clausal spine.

As mentioned in Section 6.5.1.2, it is difficult to identify a well-grounded feature that defines a natural class of obligatorily accusative DOs. Due to this uncertainty, I once again introduce a purely formal feature, F. I propose that personal pronouns, proper names, and kinship terms are inherently marked with F, whereas other DOs may receive F optionally. The structural difference between the two case markings in (599) is illustrated in Figure 66 and Figure 67. I further stipulate that the attracting head in this process is v, which attracts the DO to Spec,vP for accusative case assignment.[47]

Figure 66: *Accusative case assignment*

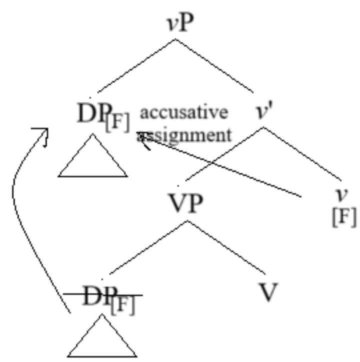
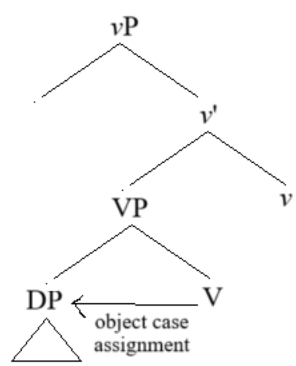


Figure 67: *Object case assignment*

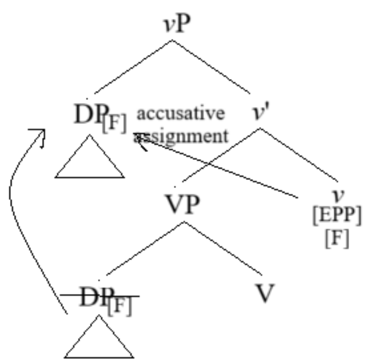


This analysis also explains why accusative marking is obligatory in finite clauses: in these environments, DOs must move, possibly due to an EPP feature on one of the heads in the clausal spine. The presence of this EPP feature may be conditioned by the finiteness. Accusative case

[47] Following Pylkkänen (2008) and Burukina and Polinsky (2025), I assume that the subject DP does not necessarily merge in Spec,vP. In the structures discussed here, Spec,vP is assumed to be empty to allow movement of the object DP into that position, while the subject DP merges higher, for example, in Spec,VoiceP in finite clauses.

assignment in finite clauses is shown in Figure 68, which can be compared to Figure 69. Given that finite and non-finite clauses share structural components such as VP, I rule out the possibility that the low object case is simply unavailable in finite structures. Instead, I propose that DO in finite clauses, due to obligatory movement, necessarily escape the structural configuration in which the object case would have been realized.

Figure 68: *Accusative case assignment in a finite clause*



However, this analysis has a significant limitation. While it accounts for the case-marking contrast between bare and marked DOs, it does not explain why bare DOs behave differently across various constructions. Specifically, this analysis has nothing to say regarding bare DOs' possibility of being modified with demonstratives in participial, nominalized and goal / purpose infinitival clauses and the impossibility of the same modification in the rest of non-finite environments.

In contrast, my SN analysis offers two advantages over an object-shift analysis. First, it does not require the stipulation of a purely formal feature. Second, it naturally accounts for the subtle differences among various bare DOs in different constructions.

6.5.2.2. Null case — default case

The final alternative I consider here is analyzing the null allomorph as an allomorph of the default case. This approach requires two distinct syntactic configurations. In one of them accusative gets assigned, while in the other one accusative assignment fails. Under this analysis, Hill Mari bare DOs reflect the absence of a source for accusative case.

Variation in the possibility of accusative assignment may arise due to differences in the v heads available for merge. One type of v facilitates case assignment (either via Agree or by forming a case-assigning domain, as outlined in Chapter 1, Section 1.2.2.1). The other v does not participate in accusative case assignment. Consider again our example, now repeated as (600). The DO surfaces as *ti kol'məm* if the v that assigns accusative is merged. The DO surfaces as *ti kol'mə̂* if the v that does not assign accusative is merged, in which case the DO receives the default case realized as null.

(600)	*vas'a*	***ti***	***kol'mə̂-m*** /	***kol'mə̂***	*näl-äš*	*tol-ə̂n*
	Vasja[NOM]	this	shovel-ACC	shovel	take-INF	come-PRET

‘Vasja came to buy this shovel.’

At this point, it is important to note that this formulation assumes that accusative case is assigned by v. However, if we adopt DCT — as I do elsewhere — the logic must be revised.

In DCT, accusative is a dependent case, assigned to a DP that is c-commanded by another DP within a case assignment domain. From this perspective, accusative assignment can fail only if there is no other DP in the same domain.

Returning to our example, the DO *ti kol'mə̂m* appears when both the subject and the DO exist in the same case assignment domain (not due to movement; see Section 6.5.2.1 for a movement-based analysis). The DO *ti kol'mə̂* appears when the DO is the only DP within its case domain.[48]

This interpretation would require two types of v: one v that does not form a case domain (Figure 69), and one v that does form a case domain (Figure 70).

[48] Technically, it would be more accurate to say that an unmarked case is assigned to this DO, as the unmarked case is assigned to the sole DP within a given domain. By contrast, default case is assigned to DPs located outside of any syntactic domain. I do not distinguish between unmarked and default case here, as the distinction is irrelevant for the analysis. What matters is that both are opposed to the dependent accusative.

Figure 69: *DCT: assignment of a dependent accusative to a c-commanded DP*

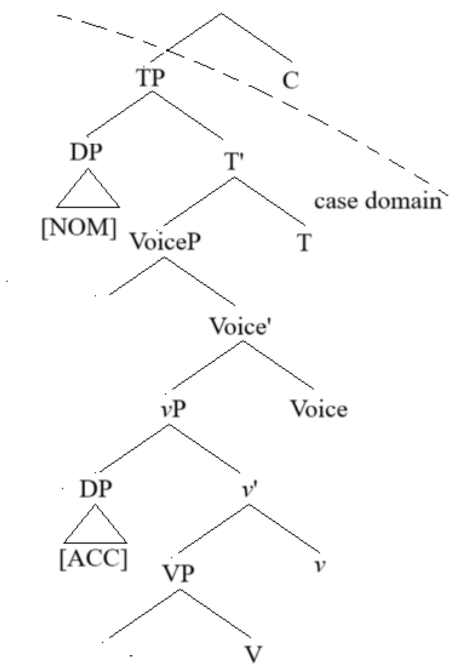
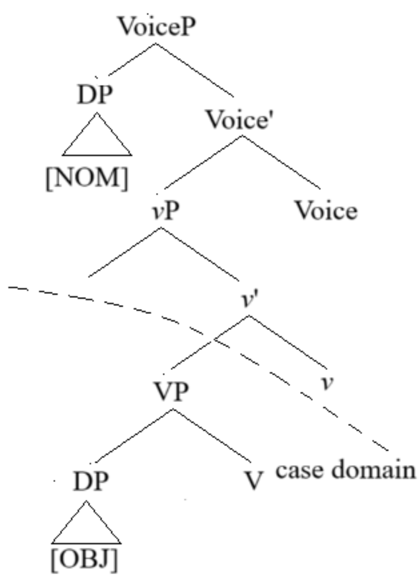


Figure 70: *DCT: assignment of a default case to the only DP*



There are two major issues with this approach. The first issue is inconsistency with standard assumptions about v. The second issue is the failure to explain variation among bare DOs.

I will start with the first issue. It is widely accepted that transitive v assigns accusative case (as well as unergative v), while intransitive unaccusative v does not (Burzio 1986; Yip, Maling & Jackendoff 1987; Woolford 2003, a.o.). If this analysis were correct, it would imply that an unaccusative v can be merged in otherwise transitive non-finite clauses. Under DCT, it would require transitive v to be either domain-forming (resulting in bare DOs) or non-domain-forming (resulting in accusative case). Given that case domains are typically defined as complements of phase heads (Baker 2015a), it is difficult to justify a system in which merging a phasal transitive v results in a bare DO, while merging a non-phasal transitive v results in an accusative DO.

Now let us address the second issue. Simply postulating a v head that fails to participate in accusative case assignment does not explain why, depending on the construction, bare DOs exhibit different syntactic and semantic properties. If the absence of the accusative case on the DO were solely the result of differences in case-assignment domains determined by the properties of v, all bare DOs would behave the same way with respect to their internal syntax. Yet this is not the case: bare DOs differ across clausal types in their ability to combine with certain modifiers and to bear number marking.

Thus, my SN analysis is not only more parsimonious than a default case analysis in that it does not require modifications to the widely accepted properties of v, it is also more explanatory in that it derives the variation in bare DOs across different constructions.

6.5.3. Interim summary

In this section, I outlined four alternative analyses of the Hill Mari DOM pattern. Two of these analyses derive the contrast between accusative and bare DOs through purely morphological processes — suggesting that the same case is realized differently depending on context or underlying features. The other two analyses propose distinct syntactic configurations, leading to the assignment of a case other than accusative.

Each approach faces specific challenges. Morphological analyses struggle to account for the restrictions on the nominal category of the DO and the constraint on the finiteness of the clause in which bare DOs can occur. Structural analyses, while addressing these constraints, fail to explain the variation among different types of bare DOs.

6.6. Placing demonstratives and universal quantifiers in the nominal spine

In previous sections, I demonstrated that bare DOs in Hill Mari are best analyzed as SNs. In this section, I use their properties to further refine the nominal structure between nP and NumP. Specifically, I provide a more explicit explanation for why SNs can combine with demonstratives but not with strong universal quantifiers in their definite readings.

This pattern suggests that demonstratives occupy a position below DP, challenging the common assumption that they are always associated with DP. I argue that demonstratives originate in lower projections within the nominal spine and do not require movement to higher positions, whereas strong universal quantifiers combine with full DPs.

6.6.1. Demonstratives

As I mentioned in Chapter 2, Section 2.3, demonstratives are often analyzed as adnominal modifiers in Spec,DP. Initially, adnominal demonstratives were considered to occupy the same structural position as articles, since they are in complementary distribution in English (Jackendoff 1977). However, later research has shown that demonstratives are phrasal, unlike determiners, which are heads (Giusti 1997). Based on this fact, demonstratives have been considered to occupy Spec,DP, while determiners are located in D, see (Alexiadou et al. 2007) for discussion. In Hill Mari, demonstratives do occur in SNs, contradicting the idea that they must move to DP to be licensed. This supports an alternative approach in which demonstratives originate lower in the nominal structure and do not necessarily need to move to DP.

Some researchers proposed that Spec,DP is not the base-generation position of demonstratives. Variations in the location of the demonstrative within the noun phrase in one and the same language serve as evidence that demonstratives originate lower in the structure and move to Spec,DP later (Giusti 1997, 2002; Brugè 2002; Brugè & Giusti 1996; Panagiotidis 2000; Grohmann & Panagiotidis 2005; Shlonsky 2004). For instance, a low base-generation of demonstratives analysis has been proposed for Rumanian. According to this analysis, to derive (601), the noun moves to the head of a functional projection between NP and DP, presumably, D itself. This explains that the demonstrative is left behind: it stays in the specifier of the projection it originated in, which is below D. On the other hand, when the demonstrative precedes the noun, the article is absent (601). Under this analysis, Dem + Noun word order is derived by the movement of Dem to Spec,DP to satisfy the [DEF] feature on D, otherwise satisfied by the overt definite article.

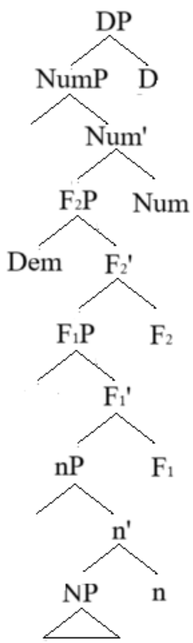
(601) RUMANIAN (Alexiadou et al. 2007: 110; from Giusti 2002: 71)

a. *acest* *băiat* *(frumos)* *al* *sau*
this boy nice of his

b. *băiat-ul* *acesta* *(frumos)* *al* *sau*
boy-the this nice of his
'this handsome boy of his'

Given that demonstratives occur in Hill Mari SNs, I reject the idea that they must move to DP for licensing. Instead, I propose that they are base-generated in a functional projection between nP and NumP, which I will call F_2P, see Figure 71.

Figure 71: *Base generation position of demonstratives*



This analysis explains the difference between Moksha and Hill Mari regarding the compatibility of demonstratives with SNs, as well as the variation among different bare DOs in Hill Mari. Specifically, I propose that in Moksha, small complements of postpositions are F_1Ps. This gives us the pattern observed in Moksha: bare postpositional complements allow for numerals but not for demonstratives, see Chapter 4 for more details.

In Hill Mari, those bare DOs that allow for modification with demonstratives are NumPs (in nominalized and adjunct infinitival clauses) or F_2Ps (in participial clauses). Bare DOs in the rest of the constructions are smaller: NPs or Ns, and we do not expect them to fit demonstratives.

If my SN analysis of bare complements in Moksha and Hill Mari is on the right track, the fact that one needs to adopt a low base generation position for demonstratives to account for the facts is another simple argument in support of the idea that demonstratives do not originate in Spec,DP.

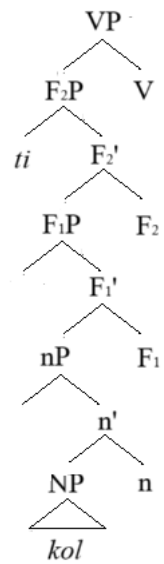
The reason for the movement of demonstratives has been attributed to both the needs of demonstratives themselves and the needs of D. According to Giusti (2002), demonstratives must check their referential features in D, while D has a referential feature that needs to be associated with an overt element.

Grammaticality of Hill Mari examples like (602) with the DO structure in Figure 72, where demonstratives could not have moved to D, is a strong argument in favor of analyses that explain the movement as being triggered by the needs of the probe rather than in order to satisfy the needs of the demonstrative itself. Indeed, if demonstratives moved to D due to some licensing requirements, the absence of the licensor (D) would result in ungrammaticality, which is not borne out. A question that arises here is whether demonstratives have to move in the presence of D. I leave this question for future research.

(602)	?***ti***	***kol***	*kâčə-šâ*	*ərvezäš*	*ängər-əš*	*ke-n*	*vaz-ən.*
	this	fish	catch-PTCP.ACT	boy[NOM]	creek-ILL	go-CVB	lay-PRET

‘The boy catching this fish fell in the water.’

Figure 72: *Bare DO with a demonstrative in the complement of a VP within a participial clause*



Thus, while the nominal functional structure between NP and NumP is not well established yet, Hill Mari provides strong support for having demonstratives originate in a functional projection between nP and NumP, which I call here F_2P, (F_1P is the projection of numerals, see the discussion below Figure 71 above), with a potential of further movement to DP.

6.6.2. Universal quantifiers

The situation observed with universal quantifiers is very different from what we saw with demonstratives. Even though both are only compatible with definite DPs in Moksha, demonstratives can occur in Hill Mari SNs without a D, whereas universal quantifiers cannot do so:

(603) a. *ävä-m-ən* **cilä** **ə̂škal-ə̂m** *šə̂pš-əl-mə̂-m* *mən'*
mother-AGR.1SG-GEN all cow-ACC pull-ITER-NMLZ-ACC 1SG[NOM]
päl-em
know-NPST.1SG

b. **ävä-m-ən* **cilä** **ə̂škal** *šə̂pš-əl-mə̂-m* *mən'*
mother-AGR.1SG-GEN all cow pull-ITER-NMLZ-ACC 1SG[NOM]
päl-em
know-NPST.1SG
'I know that my mother milked all the cows.'

This contrast suggests that universal quantifiers do combine with DPs, as usually assumed. According to the analyses proposed before, universal quantifiers should be treated as either adjuncts to full DPs (Sportiche 1988) or as heads that take full DPs as complements (Giusti 1990; Shlonsky 1991; Cirillo 2009, a.o). I adopt the adjunct analysis of universal quantifiers, which is preferrable for selection reasons given the model of the extended projection of the noun phrase outlined in Chapter 2. Having QP on top of DP would require all external heads to always select for either a DP or a QP. Still, the most important fact about the quantifiers for me is that they are not DP internal modifiers but require a full DP to combine with. The structure for the noun phrase in (603) is illustrated in Figure 73, and the impossible structure for the one in (603), is illustrated in Figure 74.

Figure 73: *Quantifier combining with a DP*

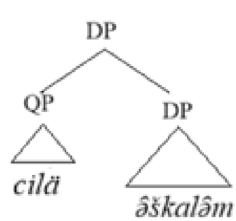
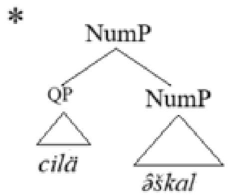


Figure 74: *Quantifier combining with a NumP (failure)*



Thus, the contrast between demonstratives and universal quantifiers in Hill Mari sheds light on the structure of the noun phrase. Demonstratives can be licensed in lower positions, while universal quantifiers require the full DP structure.

6.7. Summary

In this chapter, I have demonstrated that bare DOs in Hill Mari lack overt accusative marking due to the absence of corresponding projections in their syntactic structure. The variation in bare DO properties across different clause types — such as the possibility of number marking and the presence of adnominal modifiers (adjectives, numerals, and demonstratives) — reveals distinctions in the number of functional layers present. Specifically, the size of the SN in Hill Mari non-finite clauses ranges from very large (NumP) to non-projecting (N). Specifically, NumPs, F_2Ps, NPs and Ns are available, depending on the construction containing the DO. I also evaluated four alternative analyses of Hill Mari DOM, all of which fail to fully account for the observed pattern and its nuances.

This case-study of Hill Mari DOM reveals that the particular size of the SN that is allowed in a language does not only differ from language to language, but also depends on clause types within a single language.

One of the questions that still remain open is whether restrictions on a specific functional category within an extended projection are selectional restrictions. When a head can take either a DP or a NumP as a complement, it is tempting to frame this in terms of selection, similar to how a verb might select for either a DP or a CP. However, the restrictions of the selection that involves structures of completely different syntactic categories (nominal vs. clause) are usually attributed to the lexical head. For instance, certain verbs select for a DP or a CP. However, in cases where the restrictions apply to intermediate projections within the same extended projection, which are supposed to share categorical features, (i.e., different nominal layers), they should be attributed to the functional makeup of the selecting head, rather than to lexical selection. For instance, the ability of a non-finite clause to host an NP rather than a DP is not determined by the lexical verb but rather by the specific non-finite head involved.

Although the question of whether my analysis can be formulated in terms of categorial selection does not undermine its validity, I argue that such a formulation is indeed possible. I propose that the head of the extended projection enters the derivation specified for the number of functional heads that it will project. This specification, in turn, determines selectional restrictions of the projecting head. For instance, in Hill Mari, a head projecting only up to vP selects an NP, whereas a head projecting to CP selects a NumP. The precise mechanics of this analysis, however, remain a subject for future research.

Another open question concerns why bare DOs occur only in non-finite clauses in Hill Mari. My current analysis explains why finite clauses cannot contain bare DOs of very small sizes — such DOs are selected by heads that do not project a full structure, whereas finite clauses are CPs.

Still, since NumPs are selected by heads projecting CPs, it remains unclear why only non-finite CPs, and not finite ones, will select for NumPs.

Chapter 7: Diagnostics of small nominals

In this chapter, I explore potential universal diagnostics for identifying small nominals (SNs) cross-linguistically, addressing both their defining characteristics and the risk of misidentifications. Proper identification of small nominals is crucial for language-internal analysis (e.g., Pleshak 2022b on locative nominals) as well as for cross-linguistic comparison (see Pleshak & Polinsky 2025, on the importance of cross-linguistic comparison in formal syntax).

As I demonstrated in Chapter 4, the assumption that some arguments can be smaller than DPs is essential for analyzing bare complements of spatial postpositions in Moksha. However, in Chapter 5, I showed that an overgeneralization — assuming that all bare arguments in Moksha are SNs — can be misleading, particularly when explaining the absence of overt morphological case on Moksha indefinite direct objects (DOs).

In what follows, I propose several diagnostics based on existing arguments for nominal structures of different sizes across languages (Danon 2006; Pereltsvaig 2006; Lyutikova & Pereltsvaig 2015). These considerations, such as non-specific interpretation or inability to contain demonstratives were further taken as properties of small nominals (Pereltsvaig 2021). To transform these theoretical properties into empirical diagnostics, I critically assess how strongly they support an SN analysis and whether alternative explanations are possible.

A major challenge in identifying SNs is that many properties cited as evidence for smaller nominal structures allow for alternative analyses. This raises concerns that some reported properties of SNs might be based on nominal phrases that are not SNs at all. To address this issue, I emphasize the need for intralinguistic comparison — systematically examining nominals in different syntactic positions within a single language. Since such comparisons are often lacking in prior research, existing claims about SN properties should be approached with caution. My proposed diagnostics aim to reduce the possibility of alternative analyses by integrating intralinguistic comparison into their design.

Beyond the issue of misidentification, another challenge stems from different research traditions discussing smaller nominal structures. While some morphosyntactic accounts focus on deficient marking due to reduced structure, another body of literature examines the semantic effects of unmarked objects on predicates, often under the label of pseudo-incorporation (PNI). At the end of this chapter, I address the relationship between SNs and PNI.[49] While Lyutikova and Pereltsvaig (2015) oppose SNs to pseudo-incorporated objects, Serdobolskaya (2015) defines PNI through SNs. Here, I argue that pseudo-incorporated objects intersect with SN objects.

This chapter is organized as follows. Section 7.1 provides a working definition of SNs and examines the relationship between syntactic structure and meaning. Section 7.2 discusses how the SN analysis relates to Differential Object Marking (DOM) and PNI. Section 7.3 introduces diagnostics for detecting SNs cross-linguistically. Section 7.4 critically examines problems with existing SN properties and alternative analyses. Section 7.5 summarizes the key findings of the chapter.

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[49] Although pseudo-incorporated objects are not limited to DOs and may include instrumentals and locatives, I focus on pseudo-incorporated DOs, as they are the most frequently discussed in the literature.

7.1. Setting the stage

7.1.1. Small nominals and the hierarchical structure

Before turning to the diagnostics for identifying SNs, I begin with a general overview of what SNs are. In this thesis, SNs are defined as noun phrases that lack one or more of the higher layers in the extended nominal projection. The universal nominal structure I adopt is given in (604), with a brief summary of the functions of each projection in (605), see Chapter 2 for a more thorough discussion:

(604) NP < nP < F_1P < F_2P < NumP < DP

(605) **Functional projections and their roles**:

a. nP: introduces the possessor

b. F_1P: introduces cardinal numerals

c. F_2P: introduces demonstratives

d. NumP: responsible for number features and agreement probes; hosting number markers; source of morphological projection hosting nominal agreement markers

e. DP: responsible for case features and features relevant for agreement; hosting determiners; source of morphological projection hosting case markers

Under this definition, any structure smaller than DP qualifies as an SN. For instance, a structure as large as NumP (lacking only DP) is an SN, as are smaller structures like NP, nP or FPs. Crucially, intermediate projections cannot be skipped — in accordance with the principles of extended projection, no projection can be "cut out" from the middle of the hierarchy. Because of this, we expect to find SNs like (606)a-b but not (606)c, as in the latter FPs would have been skipped.

(606) a. NP < nP < F_1P < F_2P < NumP   b. NP < nP < F_1P   c. *NP < nP < NumP

Thus, by definition, SNs lack determiners, case and cannot trigger agreement — since all three are associated with the DP layer. However, the absence of these features is not sufficient to diagnose SNs, as these features may also result from surface-level processes rather than structural deficiency. For this reason, while SNs typically lack a determiner, case marking, and agreement, these properties are not, on their own, definitive indicators of SN status.

In the sections that follow, I examine properties that indicate reduced nominal size in nominals lacking a determiner, case morphology, and agreement features. Since SNs are defined in terms of the absence of overt morphology that is attested elsewhere in the language, they are more difficult to analyze in languages without articles or overt case and/or agreement morphology altogether. Acknowledging the limitations of an approach that relies on the overt expression of certain nominal categories, and overt articles in particular, I believe that the diagnostics developed in this thesis are useful not only for analyzing languages with articles but also for addressing the challenges posed by articleless languages. Although I do not adopt Bošković's (2008) DP vs. NP language classification based on the presence of articles (see Chapter 2, Section 2.4), I leave open the possibility that the behavior of different-sized nominals may vary systematically across these language types. Such variation may provide a first step toward developing a typology of SNs across the world's languages.

7.1.2. Structure and interpretation

The functions of the main projections I outlined in Section 7.1.1 are structural rather than semantic. While I do not deny that nominals of different sizes may exhibit interpretational differences, I do not consider such differences to be essential. In particular, I do not assume that DP is the projection of definiteness (contra Lyons 1999), or even of specificity. This structural perspective — eschewing a direct mapping between nominal height and interpretational specificity or definiteness — is central to my analysis of SNs. Under this view, SNs are not necessarily non-specific or indefinite, just as DPs are not necessarily specific. That is, I reject any deterministic link between the presence of DP and definiteness or specificity in interpretation. While this assumption is not the most traditional one, it is not unprecedented either. In her study of bare DOs in Meadow Mari, Serdobolskaya (2015) makes the same conclusion regarding the relation between semantics and DP (see Chapter 1, Section 1.1.2.5).

I consider familiar and/or unique noun phrases to be definite, including various anaphoric and situational uses (Hawkins 1978; Lyons 1999). Example (607) illustrates an anaphoric use of *this book* in (607)b given the antecedent *an interesting book* in (607)a. An immediate situation use is given in (608).

(607) ENGLISH: ANAPHORIC USE (Hawkins 1978: 87)

a. *Fred was discussing **an interesting book** in his class*.

b. *I went to discuss **this book** with him afterwards*.

(608) ENGLISH: IMMEDIATE SITUATION USE (Hawkins 1978: 103)

*Pass me **the/this/that bucket**, please*.

When I refer to formal definiteness marking, I mean the formal morphological marking systematically associated with definite interpretation in a given language. Following Hawkins (1978), I treat a formal marker that appears in the core definite contexts as a definite marker, even if its usage goes beyond these contexts. As discussed in Chapter 3, Moksha provides a case in point. In Moksha, topical objects that belong to a previously mentioned set — even if not strictly anaphoric or unique — are marked with the same definiteness marker used for more prototypical definite contexts.

The key analytical point here is that nominals obligatorily marked as definite in some syntactic contexts may resist definiteness marking in others. Consider an example from Moksha. As I discussed in Chapter 3, Moksha definite nominals must bear a definite determiner in the subject position (609), while non-genitive (bare) complements of postpositions lack a determiner even when interpreted as definite (510). I argue that this pattern reveals that subject nominals are full DPs in Moksha, whereas complements of postpositions are SNs.

(609) *mon* *mol'-ən'* *ul'c'a-va* *i* *n'ɛj-ən'* *pin'ə,*
1SG[NOM] walk-PST.1SG street-PROL and see-PST.1SG dog
i ***pin'ə-s'*** / ****pin'ə*** *uv-əman'*
and dog-DEF.SG[NOM] dog[NOM] bark-PST.1SG.O.3SG.S
'I walked down the street and saw a dog; and the dog barked at me.' (Kashkin 2018: 127)

(610) *komnata-sə* *ašč-ij̊-t'* ***fkɛ*** ***morkš*** *i* *fkɛ* *stul;*
room-IN be_situated-NPST.3-PL one table and one chair
morkš / ****morkš-s'*** ***lank-sə*** *ašč-i* *mazi* *vaza*
table table-DEF.SG top-IN be_situated-NPST.3SG nice vase
'There is a table and a chair in the room; there is a nice vase on the table.'

I refer to this pattern as the mismatch between definiteness marking and interpretation, hereafter MDMI: a familiar, unique, or otherwise definite object lacking formal definiteness marking. This mismatch provides crucial evidence for the SN analysis. Specifically, case marking and agreement appear to respond to formal definiteness marking, not to the interpretation of definiteness per se, because definite complements may be bare, and when bare, do not trigger agreement. This strongly supports the idea that morphosyntactic structure, rather than semantic content, determines whether a nominal can be marked for case or trigger agreement.

7.2. Small nominals in DOM and PNI

So far, I have not discussed the syntactic positions in which SNs can occur. Given that SNs lack the DP layer, and assuming that features in D drive movement, we expect SNs to appear only in positions that do not require feature-driven movement (see a more thorough discussion of why the presence of number features in NumPs does not make them proper goals for probes in Section 1.2.2.2). To my knowledge, no previous work has systematically addressed the distributional properties of SNs as a class. One syntactic position that is especially relevant to this discussion is the DO position, where language internal variation in marking and interpretation is widely attested.

Two prominent phenomena in this domain are Differential Object Marking (DOM) and Pseudo-Noun Incorporation (PNI). Both describe systematic variation in the morphosyntactic and semantic behavior of objects. In this section, I briefly review each phenomenon and argue that, while DOM and PNI are useful descriptive terms, they should not themselves to be treated as analyses. Instead, I suggest that variation in nominal size may underlie some of the patterns they capture.

7.2.1. Differential Object Marking

Across the world's languages, the DO position is often subject to DOM — a phenomenon whereby the marking of the object depends on semantic or pragmatic properties, most commonly animacy and definiteness (Bossong 1985; Aissen 2003, a.o.). Typically, some objects receive overt case-marking and/or trigger agreement, while others remain bare and fail to control agreement.

A classic example is Turkish, where definite and specific indefinites bear the accusative case (611)a, while non-specific indefinites do not (611)b:

(611) TURKISH (Enç 1991, ex. 14-15)
a. *Ali bir kitab-ɨ aldɨ.*
Ali one book-ACC bought
'A book is such that Ali bought it.'

b. *Ali bir kitap aldɨ.*
Ali one book bought
'Ali bought some book or other.'

While DOM approaches offer semantic or pragmatic explanations to the difference in the marking, I argue that a structural explanation (an SN analysis) is one possible approach to DOM. For instance, starting with (Givón 1978), DOM in Hebrew has been described as being dependent on the definiteness of the object: "in Hebrew, only objects high in prominence on the definiteness hierarchy are case-marked" (Aissen 2003: 3). However, Danon (2001) shows that semantically definite objects can lack overt marking (612), see Section 7.3 below for more details.

(612) HEBREW (adapted from Danon 2001: 1002)
ha-memsala daxata haca'a zo
the-government rejected proposal this
'The government rejected this proposal.'

Thus, what appears to be definiteness-based marking may, in fact, reflect the presence of a definite article, which is a distributional fact. The semantic/pragmatic aspects of DOM and its pure morphosyntactic realization are different. These differences may manifest themselves in the development of DOM; for example, Danon (2006) proposes that DOM starts out as sensitivity to semantic factors such as definiteness or animacy but then gets grammaticized as a strictly morphosyntactic feature of a language. Assuming this process of grammaticalization is unidirectional, languages may differ in the extent to which this grammaticalization of DOM has progressed. The variable and potentially heterogeneous nature of DOM types suggests that at least some DOM patterns may result from variation in nominal size. Specifically, overt case and agreement indicate a full nominal projection, while bare direct objects that lack agreement are structurally smaller.

7.2.2. Pseudo-Noun Incorporation

PNI captures cases where objects behave less autonomously than canonical nominal arguments, often surfacing bare, adjacent to the verb, and interpreted with narrow scope (Massam 2001). A canonical example comes from Niuean (613). In (613)a, the DO is accompanied by an absolutive and a plural particle, while in (613)b it is bare, not marked for case and number. The fact that the subject gets absolutive in (613)b indicated that the clause is intransitive, compare it to (613)a with the ergative subject. Notice also that while the absolutive object in (613)a follows the emphatic particle and the postverbal subject, the bare object in (613)b is adjacent to the verb, and all the other clausal elements follow it.

(613) NIUEAN (Massam 2001: 157)

a.	*takafaga*	*tūmau*	*nī*	*e*	*ia*	***e***	***tau ika***
	hunt	always	EMPH	ERG	3SG	ABS	PL fish
b.	*takafaga*	***ika***	*tūmau*	*nī*	*a*	*ia*	
	hunt	fish	always	EMPH	ABS	3SG	

‘He is always fishing.’

The lack of transitivity of the clause with the bare DO, as well as the adjacency of the DO to the verb makes (613)b resemble incorporation. However, the functional structure present in the DO suggests that we are not dealing with classic incorporation but with something else. Thus, constructions with objects that have some lower functional structure but otherwise resemble incorporation (whatever this means) are usually called PNI.[50]

The problem with PNI accounts is that its understanding is quite broad and varies from scholar to scholar. In her seminal work, Massam (2001) focused on morphosyntactic properties of the objects that she called pseudo-incorporated. She emphasized the fact that DOs like in (613)b are bare and adjacent to the verb, and then noticed that they carry non-specific interpretation. Later on, a lot of research was done on the semantics of such objects (Farkas & De Swart 2003; Dayal 2007, 2011), and the emphasis was made on the non-specificity of the object. Dayal (2015), who represents semantic accounts of PNI, discusses the morphosyntactic difference between classic incorporation and PNI and points out that, besides their ability to be modified (614)a, PNI objects in Hindi can bear plural morphology (614)b and be scrambled (614)c:

(614) HINDI (Dayal 2015: 49-50)[51]

a. *anu-ne bahut sundar laRkii cunii*
Anu very pretty girl chose-FEM
'Anu chose a very pretty girl.'

b. *anu bacce sambhaaltii hai*
Anu children manages
'Anu looks after children.'

---

[50] While locatives and instrumentals can also undergo PNI cross-linguistically, DOs serve as the clearest examples, given that expectations regarding canonical DOs are more straightforward than those for other syntactic positions. This makes it easier to identify the special properties associated with PNI. Accordingly, I will focus on DOs when discussing PNI objects.

[51] In these and the following examples from (Dayal 2015) I keep the original glossing and translation.

c. *baccaa anu bhii sambhaaltii hai*
child Anu also manages
'Anu also looks after children.'

Dayal (2015) also notes that some verb-object pairs yield non-compositional semantics, often tied to culturally conventionalized processes. In (615), the semantics of the predicate containing the verb and the object goes beyond the combination of the meanings of the two. In (615)a-b, the predicate denotes a specific cultural process, and the set of lexemes that can encode the possible bare objects is restricted. In (615)c-d, only the culturally conventionalized process like 'choosing a wife' (615)c allows for a bare object, not just any verb (615)d.

(615) HINDI (adapted from Dayal 2015: 55)
a. *laRkii / laRkaa-dekhnaa* b. **aurat / aadmii-dekhnaa*
girl boy-see woman man-see
'to see a future bride/groom (prospective bride/groom is seen by potential in-laws)'

c. *biwii-cunnaa* d. **biwii-maarnaa*
wife-choose wife-beat
'choose a wife' Int.: 'beat a wife'

For researchers working within semantic approaches to PNI, narrow scope and name-worthiness are crucial diagnostics. In contrast, properties like the absence of case-marking, adjacency to the verb, or structural reduction are not required for something to be classified as PNI (Frey 2015; Driemel 2020).

The co-existence of morphosyntactic and semantic approaches makes PNI notoriously hard to define consistently, as there seems to be no property besides the obligatoriness of narrow scope that would hold for PNI for all researchers. For practical purposes in this study, I define PNI as a construction characterized by the absence of case marking and the obligatoriness of narrow scope.

7.2.3. Relationship between DOM, PNI and the SN analysis

DOM and PNI cover observable patterns of variation in the morphosyntactic and semantic behavior of internal arguments[52] (here, I focus on DOs) within one language that belong to two different traditions. However, they are not analyses: they describe recurring configurations without actually explaining their structural source.

The SN analysis, by contrast, offers a structural explanation for certain instances of DOM and PNI. Reduced nominal size naturally predicts the absence of determiners, case, and agreement, as well as restricted movement — properties commonly associated with both DOM and PNI. Importantly, given the broad empirical scope of both phenomena, I do not claim that all instances of DOM or PNI reduce to SNs, nor do I define DOM or PNI in terms of nominal size (in contrast to Serdobolskaya 2015). Rather, I propose that some patterns traditionally described under these labels arise from the presence of SNs, and in what follows, I outline diagnostics for identifying when this analysis is appropriate.

7.3. Morphosyntactic diagnostics

In this section, I propose four diagnostics for SNs based on the morphosyntactic properties typically associated with full DPs versus structurally smaller nominal projections. I begin with what I consider the most reliable diagnostic — the mismatch between definiteness marking and interpretation (MDMI) — which features nominals that lack overt definiteness marking where it

[52] This is a very general statement, but the current state of the empirical landscape does not allow for a more specific formulation at this point.

would otherwise be expected in the language. I then present additional marking–interpretation mismatch diagnostics, outlining the conditions under which they apply and noting their limitations. While I reject Bošković's (2008) binary classification of languages into NP- and DP-languages based on the presence or absence of articles (see Chapter 2, Section 2.4), I do not exclude the possibility that SNs may exhibit different behavior in languages with articles compared to those without. Accordingly, when possible, I illustrate each diagnostic using examples from both a language with articles (mostly Moksha) and a language without articles (mostly Hill Mari). I conclude the section with a discussion of several properties that have been proposed as characteristic of small nominals but argue that these should not be used as diagnostics.

7.3.1. The main property of small nominals: the mismatch between marking and interpretation

7.3.1.1. MDMI diagnostic

The diagnostic I consider the most reliable is the MDMI diagnostic, formulated in (616).

(616) **MDMI diagnostic**:

If a language has formal definiteness marking that is obligatory (or at least possible) in some syntactic configurations and is ruled out in others, then definite nominals lacking such marking are smaller than DP.

This diagnostic is especially robust because the variation in formal marking across syntactic contexts cannot be easily explained semantically. When both the semantics and the syntactic positions of differently marked nominals remain constant, a morphological explanation also becomes implausible. The most convincing account, therefore, must appeal to structural differences.

To illustrate, consider Hebrew. Danon (2001) observes that the direct object marker *et* in Hebrew only occurs with nominals that include the definite article, as in (617)a, as well as with proper names and pronominal suffixes. Demonstratives, which require a definite interpretation, do not independently trigger *et*, as seen in (617)b.

(617) HEBREW (Danon 2001: 1002)

a.

ha-memsala	*daxata*	**(et)*	*ha-haca'a*	*ha-zo*
the-government	rejected	OM	the-proposal	the-this

b.

ha-memsala	*daxata*	*(*et)*	*haca'a*	*zo*
the-government	rejected	OM	proposal	this

'The government rejected this proposal.'

The status of *et* in Hebrew has received various interpretations: as a realization of accusative case itself (Shlonsky 1997), as an accusative case assigner (Falk 1991), or more generally as a structural case assigner (Danon 2001). However, for the purposes of the SN analysis, the exact nature of *et* is not crucial. What matters is that *et* is required with direct objects that show formal definiteness marking (i.e., the article *ha-*) and is incompatible with direct objects that lack such marking, regardless of their semantic definiteness.

There is no clear semantic feature that distinguishes the two classes of objects in (617). No matter how definiteness is defined, the object in (617)a and the object in (617)b should not be treated differently, yet they receive different marking. This discrepancy motivates Danon's (2001) structural analysis of Hebrew DOM. He introduces a formal feature [+def] associated with the presence of the definite article. The marker *et* is sensitive to this feature. However, not all semantically definite DPs carry [+def]. Danon suggests that only structurally encoded definiteness (e.g., via articles) resides in the D head, while other types of definite nominals (e.g., those with demonstratives) can be structurally smaller than DP and thus lack [+def].

This analysis aligns with my theoretical assumptions in Chapter 1 — namely, that only DPs require case and are visible to agreeing probes. It also reflects findings that demonstratives, while semantically definite, are generated below DP (Giusti 1994).

Adapting Danon's account to the framework adopted here, the structures of the two Hebrew direct objects in (617)a and (617)b can be schematically represented as Figure 75 and Figure 76 respectively.

Figure 75: *Syntactic structure for an object with et*

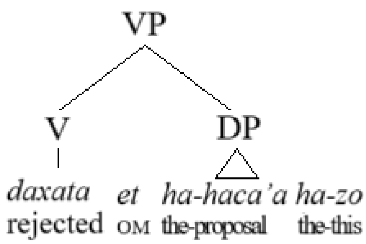
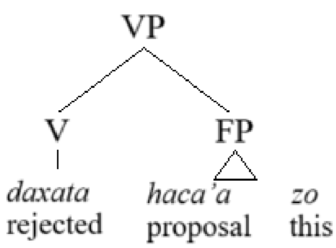


Figure 76: *Syntactic structure for an object without et*



These structures resemble those proposed for Moksha postpositional complements in Chapter 4. Indeed, my case study of Moksha bare complements serves as another successful application of this diagnostic, distinguishing bare postpositional complements from bare direct objects. Moksha has formal definiteness marking (see Chapter 3), which can be absent on definite postpositional complements (618), but is obligatorily present on definite direct objects (619). The former are small nominals, while the latter are full DPs (see Chapters 4 and 5 for details).

(618) MOKSHA

a.

komnata-sə	*ašč-ij̊-t'*	***fkɛ***	***morkš***	*i*	*fkɛ*	*stul*
room-IN	be_situated-NPST.3-PL	one	table	and	one	chair

'There is a table and a chair in the room.'

b.

morkš	***lank-sə***	*ašč-i*	*mazi*	*vaza*
table	top-IN	be_situated-NPST.3SG	nice	vase

'There is a nice vase on the table.'

(619) MOKSHA

a.

ves't'	*kir̥ks-s'*	*mu-**s'***	*makə-n'*	***vid'm'ə-n'ɛ.***
once	sparrow-DEF.SG	find-PST.**3SG**	poppy-GEN	seed-DIM

b.

... da	*i*	*səv-**əz'ə***	*optəm*	*vid'mə-n'ɛ-**t'***
and	and	eat-PST.**3SG.S.3SG.O**	whole	seed-DIM-**DEF.GEN**

c.

**... da*	*i*	*səv-**s'***	*optəm*	*vid'mə-n'ɛ*
and	and	eat-PST.**3SG**	whole	seed-DIM

'Once the sparrow found a poppy seed. And he ate the whole seed.'

The limitation of this diagnostic lies in its reliance on the presence of formal definiteness marking. In articleless languages, other strategies must be used to distinguish between SNs and full DPs, assuming one adopts the universal DP hypothesis (see Chapter 2 for the debate around the universality of the DP hypothesis).

An alternative diagnostic for languages without definite articles is in (620), relying on the examination of whether bare nominals can receive a definite interpretation without morphological case-marking — the mismatch between case-marking and definiteness interpretation (MCMDI).

(620) **MCMDI diagnostic**:

If nominals appear without case marking in a position where case is normally present, yet can be interpreted as definite, then these nominals are smaller than DP.

For instance, in Hill Mari, not only accusative but also bare direct objects in certain non-finite clauses can be definite, as indicated by (621), which includes a demonstrative. See Chapter 6 for a detailed analysis of bare direct objects in Hill Mari.

(621) HILL MARI
kə̂m ərvezäš-ən ***ti kn'igä*** *lə̂d-mə̂-štə̂-m mən' päl-em*
three boy-GEN this book read-NMLZ-AGR.3PL-ACC 1SG[NOM] know-NPST.1SG
'I know that three boys read this book.'

Thus, when a language provides two morphosyntactic options for encoding an argument — one marked and one unmarked — and the unmarked form is still capable of supporting a definite interpretation, it is likely that the unmarked nominal is a small nominal.

7.3.1.2. MNMI diagnostic

This MDMI diagnostic, based on the D head as the locus of formal definiteness marking, can be extended to number, since the Num head is the locus of formal plural marking (see Chapter 2, Section 2.3.3). Consider the mismatch between number-marking and interpretation — MNMI diagnostic:

(622) **MNMI diagnostic**:
If a language has overt number markers that are obligatory (or at least available) in some syntactic configurations but ruled out in others, then non-number-neutral nominals that lack overt number marking are smaller than NumP (and, by transitivity, smaller than DP).

To my knowledge, this specific diagnostic has not been proposed in the literature. I have applied it to argue that Moksha bare complements of postpositions are small nominals. Even though Moksha requires overt plural marking for plural direct objects (623), plural marking is prohibited in bare postpositional complements (624). This is exactly the pattern expected under the proposed diagnostic: Moksha subjects and direct objects require plural marking when plural, while postpositional complements show a split — some bear plural marking, and others do not.

(623) MOKSHA

a. *son* *mi-s'* *akšə* *alaša*
3SG[NOM] sell-PST.3[SG] white horse
'He sold a white horse.'
*'He sold white horses.'

b. *son* *mi-s'* *akšə* *alaša-t* / **alaša*
3SG[NOM] sell-PST.3[SG] white horse-PL horse
'He sold white horses.'

(624) MOKSHA
morkš-n'ə-n' / **morkš-t* *lank-sə*
table-DEF.PL-GEN table-PL on-IN
'on (the) tables'

I also need to justify the diagnostic's assumption that these nominals are not number neutral. While many accounts have observed that nominals lacking overt number morphology tend to be number neutral (e.g., Dayal 2007; Farkas & De Swart 2003), number neutrality is also attested in nominals with overt plural morphology (see Görgülü 2018 for an overview). This suggests that number neutrality does not necessarily correlate with the absence of a Num head. What a purely structural account requires are nominals that lack overt number marking but still have a non-neutral interpretation. In Chapter 4, I argued that Moksha bare postpositional complements can be semantically plural while lacking plural morphology. For example, the numeral in (625) triggers a plural interpretation, yet plural morphology on the noun is absent:

(625) MOKSHA
[*kolmə* *morkš*] *lank-sə*
three table on-IN
'on three tables'

Importantly, the absence of plural morphology in (625) is not due to the presence of a numeral, as is the case in some other languages (see, e.g., Bale, Gagnon & Khanjian 2011 on Turkish, Nelson & Toivonen 2000 on Finnish, and Farkas & de Swart 2010 on Hungarian). In fact, Moksha requires plural marking with numerals below 10 in subject and direct object positions (626), see Chapter 3. Thus, I argued in Chapter 4 that the absence of number morphology in (625) must reflect the absence of a Num head (see my analysis in Section 4.2.2 of that chapter).

(626) MOKSHA (Sidorova 2018b: 329)

vel'ə-t'	*esə*	*vet'ə*	*katə-t*	/	**katə*
village-DEF.SG.GEN	in.IN	five	cat-PL[NOM]		cat[NOM]

‘There are five cats in the village.’

Having demonstrated how this diagnostic functions in Moksha — a language with articles — I now turn to Hill Mari to show that it also applies to articleless languages. As discussed in Chapter 6, non-generic Hill Mari nominals must bear plural marking when they are semantically plural (627). However, in certain non-finite clauses, such as converbial adjuncts, plural marking is prohibited (628). I take this restriction to indicate that the nominal is structurally smaller than NumP.

(627) HILL MARI (Bochkova 2018: 64)

stöl	*və̈l-nə̈*	*ta-maxan'*	***tärelkä***	*ki-ä*
table	top-IN2	INDEF-what	plate	lay-NPST.3SG

‘There is a plate on the table (lit.: on the table, some plate lays).’
*‘There are some plates on the table.’

(628) HILL MARI

maša	*posudə̂-m,*	***mə̂rə̂***	/	****mə̂rə̂-vlä***	*mə̂r-en,*	*mə̂šk-ə̂n*
Masha[NOM]	dishes-ACC	song		song-PL	sing-CVB	wash-PRET

Int.: ‘Masha washed dishes singing songs.’

Thus, this diagnostic seems to work the same way in languages with and without articles. However, a limitation of the MNMI diagnostic is that it only works in one direction. The absence of plural morphology can be used to diagnose an SN (i.e., smaller than NumP), but the presence of plural morphology does not necessarily mean that the nominal is *not* small. This is because NumPs are themselves small nominals. For example, Hill Mari nominalizations (discussed in Chapter 6) allow unmarked direct objects that still receive plural interpretations, and in most idiolects, these objects bear overt plural marking:

(629) HILL MARI

?*maša-n*	***tə̂gə̂r-vlä***	*ə̂rgə̂-mə̂-žə̂-m*	*už-Ø-ə̂m*
Masha-GEN	shirt-PL	sew-NMLZ-AGR.3SG-ACC	see-AOR-1SG

'I saw Masha sew shirts.'

In sum, number marking can serve as a diagnostic for identifying small nominals — but only in cases where it is absent. The presence of number morphology alone does not rule out a small nominal analysis.

7.3.1.3. MPMI diagnostic

The MDMI diagnostic should be transferable to possession/nominal agreement marking[53] as well. This yields a diagnostic based on the mismatch between possession marking and interpretation (MPMI), as formulated in (630). Recall that, despite being located on Num, the agreement probe is only available when D is present, due to feature inheritance (see Chapter 6, Section 6.4.1).

(630) **MPMI diagnostic**:

If a language has nominal agreement markers that are obligatory (or at least available) in some syntactic positions but ruled out in others, then possessed nominals lacking agreement are smaller than NumP (and, by transitivity, smaller than DP).

[53] I assume that the presence of an agreement probe is the crucial element for the diagnostic. The precise nature of the agreement marker — whether it is a bona fide agreement marker or a cliticized pronoun — does not affect the outcome of the diagnostic.

Let us return to the Moksha data. In Moksha, nominal agreement with genitive possessors is obligatory in subjects, objects, and complements of postpositions marked with the genitive (631). However, agreement with the possessor is impossible in bare complements of postpositions (632). As argued in Chapter 4, such bare complements are SNs.

(631) MOKSHA

mon'	*morkš-əz'ə-n'*	/	**morkš-t'*	*lank-sə*
1SG.GEN	table-1SG.AGR.SG-GEN		table-DEF.SG.GEN	on-IN

‘on my table’

(632) MOKSHA

a.

mon'	*morkš*	*lank-sə-n*
1SG.GEN	table	on-IN-AGR.1SG

b.

**mon'*	*morkš-əz'ə*	*lank-sə(-n)*
1SG.GEN	table-1SG.AGR.SG	on-IN-AGR.1SG

‘on my table’

In Hill Mari, nominal agreement is optional but possible in all syntactic positions — except in bare direct objects (633). This contrast again supports the use of agreement as a diagnostic for nominal size.

(633) HILL MARI

a.

ədər	*mäm-nä-n*	***šarə̂k-na-m***	/	****šarə̂k-na***	*näl-äš*
girl[NOM]	1PL-AGR.1PL-GEN	sheep-AGR.1PL-ACC		sheep-AGR.1PL	take-INF

tol-ə̂n.
come-PRET
‘The girl came to buy our sheep.’

b.

?ədər	*mäm-nä-n*	***šarə̂k***	*näl-äš*	*tol-ə̂n.*
girl[NOM]	1PL-AGR.1PL-GEN	sheep	take-INF	come-PRET

‘The girl came to buy our sheep.’

Before moving on to the next diagnostic, it is important to note that this particular test cannot be applied in languages where possessed nominals cannot appear bare at all — where examples like

(633) would be ungrammatical by default. For instance, in Tatar, possessed nominals always require both nominal agreement and overt case marking. Therefore, while one can argue that possessed nominals in Tatar are DPs, no conclusions can be drawn about the size of other types of nominals based on this alone.

Lyutikova and Pereltsvaig (2015) compellingly argue that in Tatar, the presence of nominal agreement marking indicates the presence of DP. This follows from the observation that Tatar nominals bearing possessive affixes must also be case-marked, even when they are interpreted as non-specific and take narrow scope (634). According to the theoretical assumptions outlined in Chapter 1, Section 1.2.2.1, the requirement for case marking itself is an indicator of DP. Furthermore, the fact that nominal agreement triggers case marking supports the conclusion that agreement implies the presence of DP. Given the assumptions about feature inheritance and the necessity of D for Num to host an agreement probe, it is expected that agreement markers will only occur within DPs.

(634) TATAR

marat	*alsu-nɤŋ*	*ber*	*fɤtɤgrafijä-se-*(n)*	*dä*	*it-m-ä-de*
Marat	Alsu-GEN	one	photo-3-ACC	EMPH	do-NEG-ST-PST

‘Marat has not taken any picture of Alsu.’
1. Neg > ∃ {i.e. there is no picture of Alsu made by Marat}
2. *∃ > Neg {i.e. there are pictures of Alsu not made by Marat} (Lyutikova 2019: 393)

However, this reasoning tells us nothing about structures that lack nominal agreement, and thus does not allow us to draw conclusions about their size. In particular, comparing non-possessed bare nominals to possessed accusative nominals is methodologically unsound. While the impossibility of bare direct objects being possessive phrases might stem from a constraint on nominal size, alternative explanations must be ruled out. A related question is why Tatar disallows bare phrases with overt possessors altogether, despite their availability in languages like Moksha and Hill Mari.

Although Lyutikova and Pereltsvaig focus primarily on identifying DPs in an articleless language, they also seem to suggest that one can detect SNs by contrasting cased and caseless noun phrases. They propose that nominals lacking overt case are structurally smaller than DPs. Based on their observations and conclusions, it seems that they used the logic that could be formulated as the following diagnostic:

(635) **Case–agreement correlation diagnostic (putative)**:

If overt morphological agreement triggers obligatory case marking, then nominals without overt case marking are SNs.

I argue that while obligatory case marking in the presence of agreement does support the DP status of the nominal, it tells us little about SNs. In other words, (635) cannot serve as a general diagnostic. As stated at the beginning of this chapter, the absence of case marking may suggest a nominal is smaller than DP — but independent evidence is required to confirm that. I do not intend to dispute Lyutikova and Pereltsvaig's claim that bare direct objects in Tatar are smaller than DPs; rather, I argue that the evidence they provide is insufficient. Like Moksha, Tatar requires nominal agreement on all direct objects with dependent genitive possessors. It remains unclear whether the impossibility of agreement on bare direct objects simply reflects the absence of a DP needed to host the agreement probe.

To sum up this section, I proposed that SNs can be reliably diagnosed through mismatches between formal marking and interpretation. The most robust of these is the MDMI diagnostic: when a language has formal definiteness marking that is obligatory (or available) in some syntactic configurations but ruled out in others, definite nominals that lack such marking are structurally smaller than DP. In languages without definite articles, the same logic can be applied by examining whether nominals that lack expected case-marking can nevertheless receive a definite interpretation. If they can, this too points to structural reduction. The same marking–interpretation mismatch reasoning extends beyond definiteness. First, it applies to number: if a language requires overt plural marking on plural nominals in some contexts but prohibits it in others, yet the interpretation remains plural, the latter nominals are smaller than NumP (and thus smaller than DP). Second, it also applies to nominal agreement: if nominal agreement is possible or obligatory in some positions but ruled out in others, possessed nominals without agreement are structurally smaller as well.

These diagnostics converge on a single principle: when two nominal types share the same semantics and syntactic distribution but differ in their ability to host certain formal markers, the most plausible explanation is a structural one. In languages with and without articles alike — as shown with Hebrew, Moksha, and Hill Mari — such mismatches reveal that some nominals lack higher functional projections (D, Num) that others possess. The diagnostics thus provide a cross-linguistically applicable toolkit for distinguishing full DPs from structurally reduced nominals.

7.3.2. Restrictions on Modification

Just as number, nominal agreement, and definiteness markers are typically located in the Num and D heads — serving as indicators of the presence of those projections — adnominal modifiers are also associated with specific positions in the nominal spine and can shed light on the structure of the noun phrase. In this section, I discuss how the distribution of modifiers can help diagnose the presence or absence of DP and other functional projections within the extended nominal projection.

Because demonstratives and possessors occupy the left periphery of the nominal in English, it is commonly assumed that they are located in the DP layer, and that their absence indicates the absence of DP. However, recent research into nominal structure shows that neither demonstratives nor possessors must be located in DP (see Chapter 2 for the hierarchical model of the noun phrase). In this dissertation, I further argued against their strict association with DP (see Chapters 4 and 6). This means we must be cautious when drawing structural conclusions based solely on the distribution of these elements.

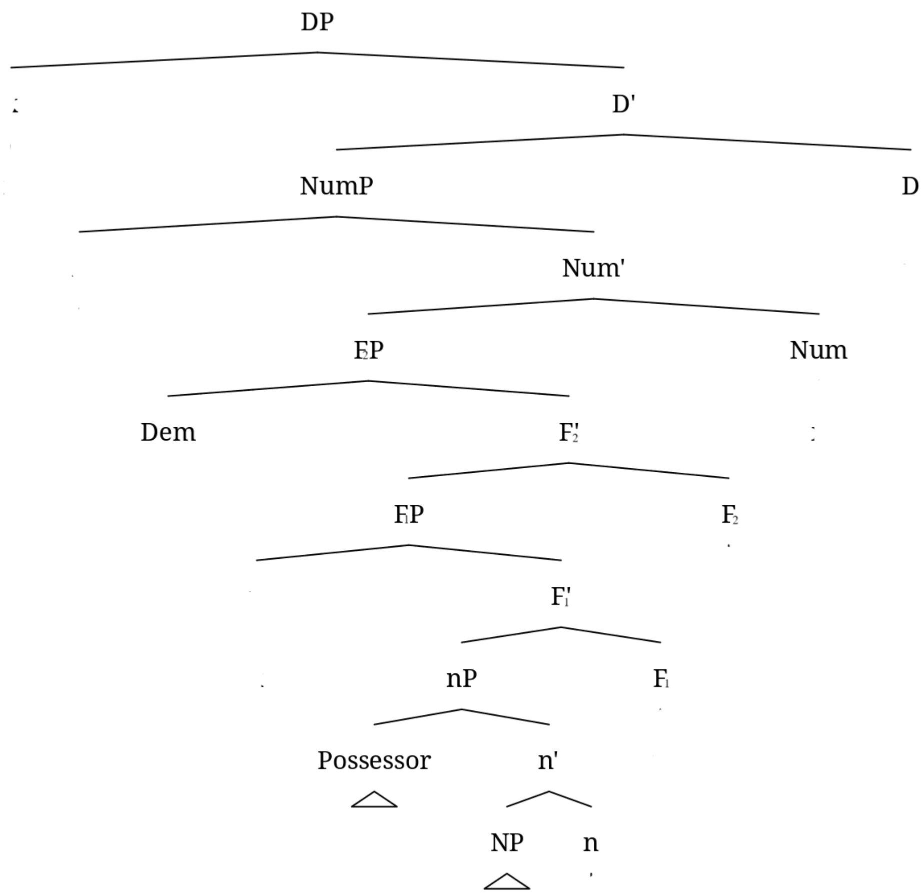
Let us begin with previous claims in the literature and then adjust them accordingly. In her overview of small nominal properties, Pereltsvaig (2021) argues that small nominals cannot combine with demonstratives, strong quantifiers, or certain possessors. I refer to this as the Restrictions on Modification diagnostic, see my formulation in (636).

(636) **Restrictions on Modification diagnostic (putative)**:

If a nominal cannot co-occur with demonstratives, strong quantifiers, or possessors, then it is an SN.

Pereltsvaig's reasoning, as I mentioned above, is that demonstratives, strong quantifiers, and possessors are located in DP, and SNs — lacking DP — cannot accommodate them. However, as I showed in Chapter 4 and Chapter 6, neither possessors nor demonstratives need to occupy Spec,DP; rather, they may originate in lower structural positions and remain there. Figure 77 illustrates this revised structure, with both possessors and demonstratives generated in low positions.

Figure 77: *Basic nominal structure with possessors and demonstratives generated low*



This means that in many languages, SNs may indeed contain demonstratives and possessors. Consider the Hebrew example in (637), repeated from Section 7.3.1.1, where the object lacks overt definiteness marking but still contains a demonstrative. A similar argument applies to possessors, as in the Hill Mari example in (638). These cases show that the presence of a demonstrative or possessor does not necessarily signal the presence of DP.

(637) HEBREW (adapted from Danon 2001: 1002)
ha-memsala *daxata* *haca'a* ***zo***
the-government rejected proposal this
'The government rejected this proposal.'

(638) HILL MARI
ˀədəräš ***ti*** ***ləpä*** *kə̂č-aš* *kərgə̂ž-ə̂*
girl[NOM] this butterfly catch-INF run-AOR.3SG
'The girl ran to catch this butterfly.'

Furthermore, the absence of demonstratives or possessors does not guarantee the absence of DP. These modifiers carry significant interpretational weight and may be excluded for semantic, rather than structural, reasons. For instance, bare direct objects in Moksha must be indefinite and are incompatible with both possessors (639) and demonstratives (640). As argued in Chapter 5, this incompatibility stems from semantic constraints — definiteness and possession are prohibited — not from the absence of DP. In fact, argument marking in causative clauses supports the analysis of Moksha bare DOs as full DPs. Thus, we cannot use the absence of demonstratives or possessors alone as evidence for a reduced nominal structure.

(639) MOKSHA
**mon'* *mird'ə-z'ə* *ašəz'* *t'ijə* *min'*
1SG.GEN husband-1SG.AGR.SG[NOM] NEG.PST.3SG do[CN] we.GEN
s't'ər'-n'əkə-n' *fkɛ-vək* *fotografijɛ*
girl-AGR.1PL-GEN one-ADD picture
Int.: 'My husband hasn't taken a single picture of our daughter.'

(640) MOKSHA (Kashkin 2018: 126)
son* *dasad'ɛ-s'* ***t'ɛ* ***s'ora-n'ɛ***
3SG[NOM] offend-PST.3SG this boy-DIM
Int.: 'He offended this boy.'

To summarize so far: possessors and demonstratives can appear in structurally low positions, and their presence or absence is not a reliable diagnostic for DP.

However, strong quantifiers do behave differently. Because they are structurally high — typically adjoining to DP (see Chapters 4 and 6 for my assumptions regarding the adjunction of universal quantifiers) — their absence in a nominal structure is more telling. If a position allows demonstratives and possessors but disallows strong quantifiers, it should be analyzed as an SN (641). Another type of modifiers that are more telling are numerals. Unlike possessors and demonstratives, numerals do not interfere with definiteness and specificity requirements, and, assuming they are generated in F_1P (see Figure 78) their absence strongly suggests the lack of that syntactic layer, see Chapter 2, Section 2.3 and Chapter 4, Section 4.5 for the nominal structure and the placement of adnominal modifiers.[54]

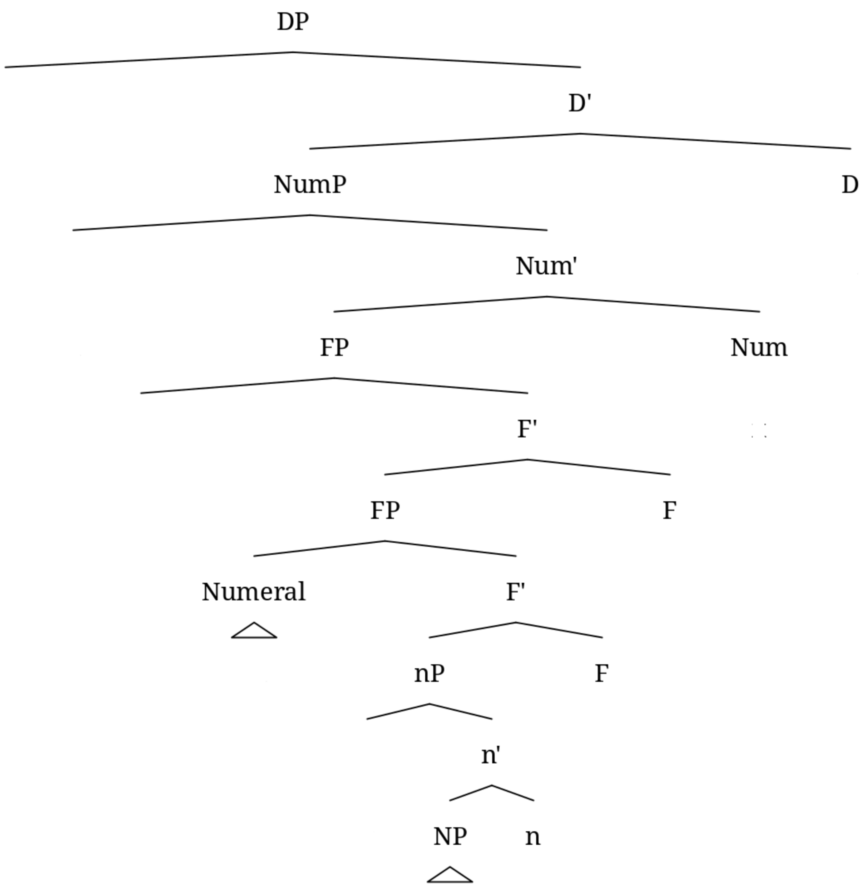
(641) **Restrictions on modification diagnostic revisited**:

a. If a nominal allows modification with all modifiers except strong quantifiers, it is an SN.

b. If a nominal does not allow modification by numerals (and all higher modifiers), it is an SN.

[54] I am aware of the considerable variation in the syntactic analysis of numerals. The diagnostic is assumed to be valid insofar as the language data are compatible with the numeral structure proposed in this dissertation.

Figure 78: *Basic DP structure with numerals originating in F_1P*



These revisions are supported by both theoretical considerations and empirical data from Moksha and Hill Mari. In what follows, I discuss each part of the diagnostic in turn.

In Moksha, bare complements of postpositions — which I analyze as SNs in Chapter 4 — permit possessors but disallow demonstratives and strong quantifiers:

(642) MOKSHA
a. *[*t'ɛ* *morkš*] *lank-sə*
this table on-IN
Int.: 'on this table'

b. *[*s'embə* *morkš*] *lank-sə*
all table on-IN
Int.: 'on all tables'

Similarly, in Hill Mari, bare DOs of nominalizations — also analyzed as SNs in Chapter 6 — can be definite and modified by a possessor (643) or demonstrative (644), but not by a universal quantifier (645).

(643) HILL MARI
ti *ədär* ***mämnän*** ***koti*** *pukš-ə̂mə̂-m* *mən'* *už-ə̂n-am*
this girl[NOM] 1PL.GEN cat feed-NMLZ-ACC 1SG[NOM] see-PRET-1SG
'I saw that this girl fed our cat.'

(644) HILL MARI
kə̂m *ərvezäš-ən* ***ti*** ***kn'igä*** *lə̂d-mə-štə̂-m* *mən'*
three boy-GEN this book read-NMLZ-AGR.3PL-ACC 1SG[NOM]
päl-em
know-NPST.1SG
'I know that three boys read this book.'

(645) HILL MARI
ävä-m-ən* ***cilä* ***ə̂škal*** *šə̂pš-əl-mə̂-m* *mən'* *päl-em*
mother-AGR.1SG-GEN all cow pull-ITER-NMLZ-ACC 1SG[NOM] know-NPST.1SG
Int.: 'I know that my mother milked all (the) cows.'

Thus, the absence of all high modifiers (possessors, demonstratives, quantifiers) does not necessarily indicate an SN, as it may be semantically motivated (as in Moksha bare DOs, see Chapter 5). However, the presence of demonstratives and/or possessors with the exclusion of quantifiers is a more reliable structural diagnostic.[55]

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[55] I acknowledge that some quantifiers may be adverbial in their nature rather than adnominal. The diagnostic discussed here applies only to quantifiers that can be shown to be adnominal.

Unfortunately, I cannot currently identify a language with articles that illustrates the second part of the revised diagnostic. Moksha, with which I exemplified my other diagnostics, is not informative in this regard, as numerals are allowed in all bare complements (see Chapters 4 and 5). I leave the search for an appropriate language for future research. Hill Mari, on the other hand, provides an example as an articleless language. Bare DOs in Hill Mari non-finite clauses, specifically converbial adjuncts (discussed in detail in Chapter 6, Section 6.3.2.4) cannot be modified by demonstratives (646) or cardinal numerals (647), even though they clearly project at least NP, as they can be modified by adjectives (648).

(646) HILL MARI

***ti**	***mə̂rə̂***	*mə̂r-en,*	*ät'ä-m*	*rokolma-m*	*itəräj-ä*
this	song	sing-CVB	father-AGR.1SG	potato-ACC	peel-NPST.3SG

Int.: 'My dad peels potatoes humming this song.'

(647) HILL MARI

**so,*	***kə̂m***	***mə̂rə̂***	*mə̂r-en,*	*ävä-m*	*ätəder-əm*
always	three	song	sing-CVB	mother-AGR.1SG	dishes-ACC

mə̂šk-eš
wash-NPST.3SG

Int.: 'My mother usually washes dishes humming three songs.'

(648) HILL MARI

u	***mə̂rə̂***	*mə̂r-en,*	*ät'ä-m*	*rokolma-m*	*itəräj-ä*
new	song	sing-CVB	father-AGR.1SG	potato-ACC	peel-NPST.3SG

'My father peels potatoes humming a new song.'

Thus, the impossibility of numerals in a nominal suggests that its structure is very small, smaller than F_1P, where numerals get generated. The absence of numerals cannot be explained by the restriction on the animacy of the object, as numerals can combine with both animate and inanimate objects.

7.4. Less reliable diagnostics

Pereltsvaig (2021) outlines several properties that are characteristic of SNs, based on her research on articleless languages. While these properties may indeed characterize certain instances of SNs, they are neither necessary nor sufficient to define a structure as an SN and should be considered only alongside more reliable diagnostics. First, each of them functions as a one-way implication. Second, they are open to alternative analyses. In this section, I argue that these properties cannot serve as reliable diagnostics for identifying SNs.

7.4.1. Category restriction

One property that has been proposed as indicative of small nominals is the inability of pronouns to occur in certain syntactic configurations (Pereltsvaig 2021). This observation motivates the formulation of the following diagnostic in (649), based on the assumption that pronouns are uniformly DPs (which I do not adopt in this thesis) and therefore cannot occupy positions that require smaller structures.

(649) **Category restriction diagnostic (putative)**:
If pronouns cannot be bare, then bare nominals are SNs.

This pattern is exemplified in Tatar, where pronouns do not participate in DOM and must always appear with accusative case marking:

(650) TATAR
marat a-lar-(nɤ) kür-de*
Marat 3SG-PL-ACC see-PST
'Marat saw them.' (Lyutikova 2019: 390)

This property appears to hold for the small nominals discussed in this dissertation as well. Specifically, personal pronouns cannot appear as bare complements of postpositions in Moksha (651), nor can they function as bare direct objects in Hill Mari (652).

(651) MOKSHA

a.

mon'	/	**mon*	*lank-sə-n*
1SG.GEN		1SG	on-IN-AGR.1SG

b.

**mon*	*lank-sə*
1SG.GEN	on-IN

'on me'

(652) HILL MARI

män'	***tän'-äm***	/	****tän'(ä)***	*kâmâl-ang-d-aš*	*tol-ân-am*
1SG[NOM]	2SG-ACC		2SG	mood-INCH2-CAUS-INF	come-PRET-1SG

'I came to greet you.'

However, the fact that pronouns cannot be bare only tells us that pronouns in Tatar (and potentially other languages) are DPs. It does not provide evidence about the structural status of other nominals. Consider again Moksha bare direct objects. Personal pronouns cannot appear bare in this position and must instead appear in their genitive form (653). At first glance, this appears to replicate the pattern identified by Pereltsvaig. However, as I demonstrated in Chapter 5, Moksha bare DOs must be indefinite. Given that personal pronouns are inherently definite — referring to entities already introduced in the discourse — their incompatibility with indefinite positions can be explained semantically, not structurally.

(653) MOKSHA

a.

mon	*ton'*	*muj-t'ɛ*
1SG[NOM]	2SG.GEN	find-NPST.2SG.O.1SG.S

b.

**mon*	*ton*	*muj-an*
1SG[NOM]	2SG	find-NPST.1SG

'I will find you.'

Furthermore, not all pronouns are necessarily full DPs. Pronouns may vary in structural size both cross-linguistically and within a single language, as argued by Déchaine and Wiltschko (2002, 2010), see also (Cardinaletti & Starke 1994) on structural deficiency in pronouns. Therefore, while the incompatibility of pronouns with certain syntactic positions may suggest that those positions are reserved for SNs, one should make sure that the restriction is not tied to the definite status of personal pronouns.

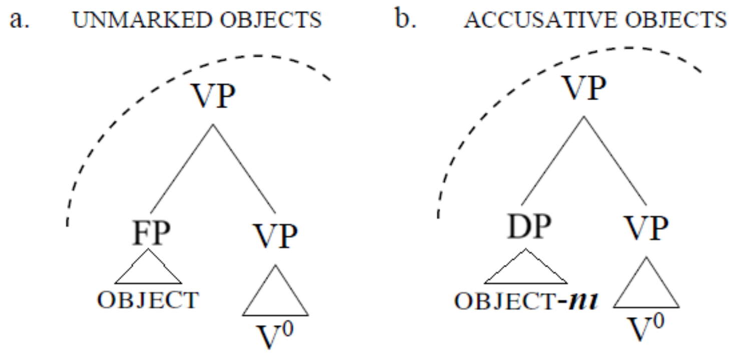
7.4.2. Movement

Another putative diagnostic for the SN analysis is based on the impossibility of tying morphological case marking to a specific syntactic position. This idea is formulated as a diagnostic in (654). The reasoning goes as follows: if case-marked nominals appear in different structural positions — e.g., both inside and outside the VP — then the absence of overt case marking cannot be attributed to the unavailability of case in the base-generated position. Instead, it must result from the nominal's structural inability to bear case, which implies reduced syntactic size.

(654) **The movement for case diagnostic (putative)**:
If a bare nominal occurs in a case position, it is an SN.

This logic is illustrated in Figure 79, which compares two structures with VP-internal objects. The tree on the right shows a full DP in object position, which receives case. The tree on the left shows a small nominal in the same position, which fails to receive case, resulting in a bare form.

Figure 79: *VP-internal object* (adapted from Lyutikova & Pereltsvaig 2015: 304)



Let us examine a concrete application of this logic. Lyutikova and Pereltsvaig (2015, 2023) argue that in Tatar, object markedness cannot be straightforwardly tied to a VP-internal versus VP-external distinction, as has been proposed for languages such as Sakha and Hindi (Baker & Vinokurova 2010). While it is true that VP-external objects must bear accusative case, VP-internal objects in Tatar may appear either marked or unmarked:

(655) TATAR (Lyutikova & Pereltsvaig 2015: 303)

a. *marat mašina-nı sat-ıp al-dı*
Marat car-ACC sell-CONV take-PAST
'Marat bought a (specific)/the car.'

b. *marat mašina sat-ıp al-dı*
Marat car sell-CONV take-PAST
'Marat bought a car/cars.'

To demonstrate that case marking is not dependent on syntactic position, Lyutikova and Pereltsvaig use the adverb *tiz* 'quickly' to diagnose VP boundaries in (656). In their analysis, (656)a illustrates a case-marked object that has moved out of the VP. In (656)b, movement of an unmarked object results in ungrammaticality. Example (656)c shows that a case-marked object may remain within the VP. These derivations are shown in Figures 80–82. Lyutikova and

Pereltsvaig do not give an example with a bare object following the adverb, but their prose suggests that it is grammatical.

(656) TATAR (Lyutikova & Pereltsvaig 2015: 303-305)

a. *Marat botka-nı tiz aša-dı*
Marat porridge-ACC quickly eat-PAST
'Marat ate (the) porridge quickly.'

b. **Marat botka tiz aša-dı*
Marat porridge quickly eat-PAST
'Marat ate porridge quickly.'

c. *Marat tiz botka-nı aša-dı*
Marat quickly porridge-ACC eat-PAST
'Marat ate porridge quickly.'

Figure 80: *Accusative DO moving out of VP* (Lyutikova & Pereltsvaig 2015: 304)

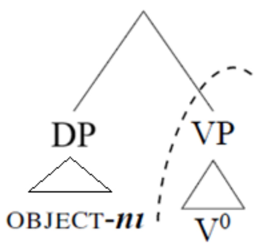


Figure 81: *Bare DO not being able to move out of VP* (constructed based on Lyutikova & Pereltsvaig 2015: 304)

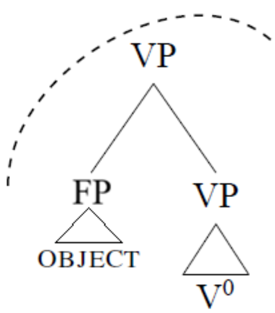
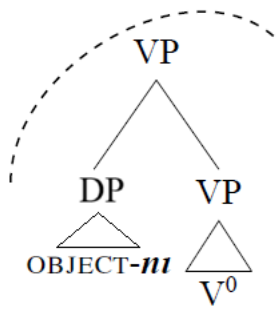
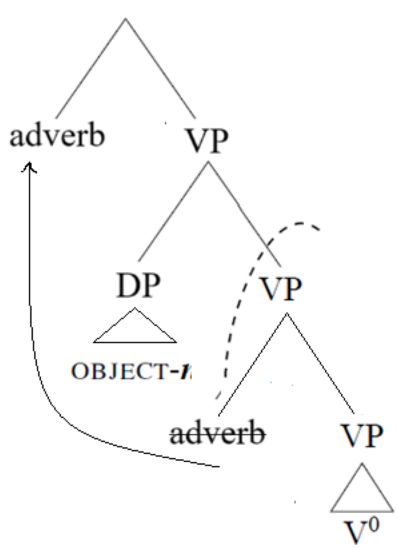


Figure 82: *Accusative DO staying in-situ* (constructed based on Lyutikova & Pereltsvaig 2015: 304)



On this account, the pattern in (656) shows that accusative case can be assigned within the VP, and hence, the differential object marking in Tatar cannot be reduced to structural position alone. However, this argument is problematic due to the presence of scrambling in Tatar (Tatevosov et al. 2019). If adverbs can scramble out of the VP, as shown in Figure 83, then one cannot reliably use their surface position to diagnose VP boundaries. Consequently, both the object and the adverb in (656)c may have moved out of the VP.

Figure 83: *Scrambling of an adverb* (constructed based on Lyutikova & Pereltsvaig 2015)



Moreover, the presence of scrambling, combined with the fact that accusative direct objects in Tatar can take either wide or narrow scope (unlike in languages with obligatory object shift), further undermines the claim. In languages with object shift, object movement is driven by syntactic features and results in obligatory scope effects (Baker 2013). In contrast, in Tatar, scope is flexible, as shown in (657). Because of this, one cannot guarantee that the object in (656)c remains in-situ.[56]

(657) TATAR (Lyutikova & Pereltsvaig 2015: 305)

a. *Här ukučı [Tukaj-**nıŋ** ike šigır-e-**n**] ukı-dı.*
Every student Tukay-GEN two poem-3-ACC read-PAST
'Every student read two poems by Tukay.'
2 > ∀: 'There are (certain) two poems by Tukay that every student read.'
∀ > 2: 'Every student read (some) two poems by Tukay.'

b. *Marat [Alsu-**nıŋ** fotografijä-se-**n**] kür-me-de.*
Marat Alsu-GEN photo-3-ACC see-neg-PAST
'Marat didn't see a photo of Alsu.'
∃ > Neg: 'There is a photo of Alsu that Marat didn't see.'
Neg > ∃: 'It is not the case that Marat saw a photo of Alsu.'

In order to properly test Lyutikova and Pereltsvaig's diagnostic, one needs a language in which object movement out of VP is unrelated to case-marking or wide-scope interpretation, yet also not attributable to scrambling. It is unclear what kind of syntactic mechanism this would entail, or whether such a language exists. Alternatively, a language with variable object marking, but lacking both object shift and scrambling, might provide more reliable data. At present, I am not aware of such a language.

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[56] The observation that bare DOs cannot undergo scrambling may also be relevant. I discuss this issue further in Section 7.4.3.1.

Thus, it is impossible to predict whether bare direct objects in a language are SNs based on the general data on the possibility of object movement. Whether bare DOs are morphologically unmarked DPs or genuinely reduced nominals likely depends on whether the language has object shift. In Moksha, both scrambling and object shift are attested, and bare DOs are not SNs. Instead, they are DPs that lack the features necessary to undergo object shift and receive overt case marking. This is supported by scope asymmetries between genitive and bare objects, as shown in (658), and see more arguments in favor of Moksha bare DOs being full DPs in Chapter 5.

(658) MOKSHA (Toldova 2018: 604)

a. *ɛr'* *s't'ər'-n'ɛ-s'* *sɛv-əz'ə* *mar'-t'*
each girl-DIM-DEF.SG[NOM] take-PST.3SG.O.3SG.S apple-DEF.SG.GEN
i. 'Every girl took an apple (one and the same apple; the girls took turns taking it).'
ii. *'Every girl took an apple (each girl took her own apple).'

b. *ɛr'* *s't'ər'-n'ɛ-s'* *sɛv-**s'*** ***mar'***
each girl-DIM-DEF.SG[NOM] take-PST.3[SG] apple
i. 'Every girl took an apple (each girl took her own apple).'
ii. *'Every girl took an apple (one and the same apple; the girls took turns taking it).'

Once again, the classification of bare DOs as full DPs or SNs must ultimately be determined through thorough empirical investigation, not based on general theoretical considerations alone.

7.4.3. Diagnostics referring to PNI

Among the properties of SNs, Pereltsvaig (2021) includes obligatory narrow scope, as well as the inability to control anaphora or PRO. Broadly speaking, this list brings together various phenomena that result from the non-specificity requirement. If we recall the key diagnostics for PNI outlined in Section 7.2 — namely, the lack of case marking and narrow scope — we observe a significant overlap between small nominals and PNI objects. For this reason, Serdobolskaya (2015) defines PNI objects through SNs. Specifically, she assumes that PNI is a phenomenon involving SNs in argument positions. In this section, I review the relationship between PNI and SNs arguing that some PNI objects can be analyzed as SNs with an additional constraint of obligatory narrow scope on those SNs (the other PNI objects may be full DPs with scope constraints). I also examine the role of adjacency to the verb, which, as I will show, is too controversial to serve as a reliable diagnostic for PNI. Furthermore, I argue that control of anaphora, control of PRO, and number neutrality are not necessary properties of either PNI or small nominals and should not be used as core definitional criteria.

Regarding SNs in Tatar, Lyutikova and Pereltsvaig (2015) explicitly argue that the phenomenon under discussion is not PNI. Their reasoning rests on the observation that the objects in question are referential, specified for number, can bear plural morphology, and do not need to be adjacent to the predicate. However, I would like to point out that all of these properties have, at various points, been attributed to PNI by different researchers (see Section 7.2 for a review). Moreover, while Lyutikova and Pereltsvaig do not elaborate on the possible relationship between SNs and PNI, I propose that PNI objects, as defined in my working hypothesis, can be analyzed as SNs. In other words, while not all SNs are PNI objects, some PNI objects may indeed receive an SN account (compare this view with Lyutikova and Pereltsvaig's, who treat SN and PNI as mutually exclusive, and with Serdobolskaya 2015, who defines PNI in terms of SN). At the same time, an SN account is also suitable for phenomena that fall outside the empirical domain of PNI.

In what follows, I examine PNI properties that may intersect with the empirical coverage of the SN analysis. This approach allows us to identify the subset of PNI objects that can be explained via an SN analysis. It is important to recall that not all researchers consider adjacency to the verb to be a necessary condition for PNI (see Dayal 2015). I argue that the only requirement imposed on SNs by PNI is that of obligatory narrow scope. In contrast, the inability to control anaphora or PRO, as well as number neutrality, appear to be optional properties of PNI and SNs, and thus should not be treated as definitional. The distribution of the properties is shown in Table 18.

Table 18: *The empirical coverage of PNI and SN*

Property	PNI (non-SN) objects	SN (non-PNI) objects	Non-PNI & non-SN objects	PNI&SN objects
Verb Adjacency	no	no	yes	yes
Number neutrality	yes	no	yes	yes
Narrow Scope	yes	no	no	yes
Control of PRO and anaphora	yes	yes	yes	yes

In summary, the properties of adjacency to the verb, control of anaphora, control of PRO, and number neutrality do not appear to be reliable diagnostics for distinguishing PNI or SNs. Because PNI can be defined by the narrow scope, PNI-type SNs will be characterized by obligatory narrow scope. However, non-PNI-type SN can still take wide scope.

7.4.3.1. Adjacency

Ever since Massam (2001) established that PNI objects in Niuean must be adjacent to the verb, some researchers — including Lyutikova and Pereltsvaig (2015) — have taken verb adjacency to be an inherent property of PNI. However, this view has been challenged by subsequent work. Dayal (2011) demonstrates that Hindi PNI objects can be scrambled, and Driemel (2020) provides further evidence for the movement of PNI objects in Turkish and German.

Despite this variation in verb adjacency requirements of PNI, Lyutikova and Pereltsvaig use adjacency as part of their argument against a PNI analysis for Tatar bare DOs, emphasizing that those DOs can be separated from the verb. This reasoning implicitly sets up an opposition between SNs and PNI, where the former differ from the latter by lacking the adjacency restriction.

In this section, I re-evaluate Lyutikova and Pereltsvaig's argument and propose two points: (i) the adjacency requirement is irrelevant for defining PNI objects, and (ii) since PNI is a descriptive label rather than an analytical framework, it cannot serve as a reference point against which to motivate an SN analysis.

The argument in (Lyutikova & Pereltsvaig 2015) goes as follows. In Tatar, bare DOs exhibit some properties associated with PNI (although, as discussed above, the only reliable PNI property is narrow focus), but they should not be treated as PNI because they can be separated from the verb. Consequently, they propose an SN analysis as an alternative — an "alternative" to something that is not in fact an analysis.

Indeed, Tatar bare DOs can appear apart from the verb, either due to intervening nominal elements of the predicate (659) or focus particles (660) (Lyutikova 2019):

(659) TATAR (Lyutikova 2019: 309)

äti-s	*Marat-ka*	*mašina*	*büläk*	*it-te*
father-3	Marat-DAT	car	gift	make-PAST

'His father gave Marat a car as a gift.'

(660) TATAR (Lyutikova 2019: 394)

marat	*bala-ga*	*jaŋa*	*kitap*	*kɤna*	*uk-ɤ-dɤ*
Marat	child-DAT	new	book	FOC	read-ST-PST

'Marat read the child only a new book.'

However, this pattern should be viewed as a descriptive generalization about bare DOs in Tatar rather than as an argument in favor of an SN analysis (and especially an argument against their PNI status). It is true that SNs do not require verb adjacency: for example, Hebrew bare objects, which are argued to be SNs based on the markedness–interpretation mismatch diagnostic (Section

7.3.1.1), can be separated from the verb. An adverbial phrase such as *berov gadol* 'by a large majority' can intervene between the verb *daxata* 'rejected' and the bare object *haca'a nosefet* 'another proposal' (661). Notably, objects in Tamil — which have been analyzed as PNI by Baker (2014), Levin (2015), and Driemel (2020) — also exhibit this behavior: they may be separated from the verb by focus particles, indicating that adjacency does not consistently hold for PNI either (662).

(661) HEBREW (G. Danon, p.c.)
ha-memshala *daxata* ***be-rov*** ***gadol*** [*haca'a* *nosefet*]$_{DO}$.
the-government rejected in-majority big proposal additional
'The government rejected another proposal by a large majority.'

(662) TAMIL (Driemel 2020: 7)
Maala *pustagam-kuuṭa* *paḍi-cc-aa*
Mala.NOM book-MIR read-PST-3SG.F
'Mala even read a book/books.'

In sum, while SNs indeed do not require adjacency to the verb, adjacency is not a reliable diagnostic for either PNI or SNs. Although often assumed to be central to PNI, it does not consistently apply across languages and thus cannot be relied upon as a meaningful distinction in this domain.

7.4.3.2. Non-referentiality and number neutrality

The aim of this section is to show that neither referentiality nor number neutrality, which have been often associated with PNI and reduced structure, are reliable diagnostics for identifying SNs.

Referentiality in nominals is typically manifested through their ability to control anaphora and PRO. While there is considerable variation in referentiality among nominals analyzed as PNI (see Massam 2009), Pereltsvaig (2021) nonetheless treats non-referentiality as a core property of PNI.

She identifies the inability to control anaphora and PRO — reflecting a lack of referential force in the bare nominal — as a defining characteristic of SNs as well.

However, this claim appears inconsistent with her earlier work co-authored with Lyutikova (2015), where they explicitly show that what they classify as SNs in Tatar can, in fact, support anaphora:[57]

(663) TATAR (Lyutikova & Pereltsvaig 2015: 308)

sin	*anarga*	*kitap*	*ala*	*ala-sıŋ*
you	that.DAT	book	take.IPFV	can.PRES-2SG

häm	*a-nı*	*matur*	*it-ep*	*ter-ep*	*büläk*	*it-ergä*	*bula*
And	that-ACC	beautifully	make-CONV	wrap-CONV	gift	make-INF	be.PRES

‘You can buy him a book. You can wrap it beautifully and give it to him as a gift.’

From a theoretical perspective as well, it is not necessary for non-referential nominals to lack a DP projection. As discussed in Section 7.1.2, there is no strict one-to-one mapping between syntactic structure and semantic interpretation. This undermines Pereltsvaig’s assumption that non-referentiality must correlate with a reduced syntactic structure. Consequently, I argue that non-referentiality is not an essential property of SNs, and that the ability to support anaphora or control PRO should not be used as a diagnostic criterion.

Turning to number neutrality, similar issues arise. While some researchers have proposed number neutrality as a hallmark of PNI, it has been shown that PNI objects can, in fact, be marked for number and projected as NumPs. See Farkas and de Swart (2003) and Dayal (2015) for detailed discussion. Consider the following example from Hindi, where the bare DO is clearly not number

[57] Unfortunately, the authors do not provide data regarding the control of PRO.

neutral. Dayal argues that when the verb is perfective and telic, the PNI object does not exhibit number neutrality.

(664) HINDI (Dayal 2015: 66)
*anu-ne tiin ghanTe meN / *tiin ghanTe tak* ***kitab*** *paRh Daalii*
Anu-ERG 3 hours n 3 hours for book read COMPL
'Anu read a book in three hours.' = exactly one book

Given that the absence of number neutrality does not reliably correlate with PNI status, and does not directly inform us about the presence or absence of a DP layer, I consider discussions focusing on number neutrality in bare objects not to be revealing. However, see Section 7.3.1.2 for a discussion of the MNMI diagnostic, which does offer a more nuanced approach to number interpretation in these contexts.

7.4.3.3. Movement restrictions and narrow scope

Pereltsvaig (2021) claims that one of the defining properties of SNs is their obligatory narrow scope. For instance, Tatar small nominals, according to Lyutikova and Pereltsvaig (2015), exhibit this pattern:

(665) TATAR (Lyutikova & Pereltsvaig 2015: 307)
marat ike kitap ukı-ma-dı
Marat two book read-NEG-PAST
Neg > 2: 'It is not the case that Marat read two books.'
*2 > Neg: 'There are (certain) two books that Marat didn't read.'

While obligatory narrow scope may indeed characterize SNs in languages such as Tatar or Russian — the primary empirical basis for Pereltsvaig's generalization — it cannot be assumed to hold universally. As discussed in Section 7.3, SNs can be definite and can clearly take wide scope, as seen in Hebrew:

(666) HEBREW (adapted from Danon 2001: 1002)
ha-memsala *daxata* *haca'a* ***zo***
the-government rejected proposal this
'The government rejected this proposal.'

One might hypothesize that this difference arises due to the presence or absence of articles, with only articleless languages requiring small nominals to be non-specific. However, data from Hill Mari — a language without articles — contradict this assumption: SNs in Hill Mari can also be definite, as shown below:

(667) HILL MARI
kə̑m *ərvezäš-ən* ***ti*** ***kn'igä*** *lə̑d-mə̑-štə̑-m* *mən'* *päl-em*
three boy-GEN this book read-NMLZ-AGR.3PL-ACC 1SG[NOM] know-NPST.1SG
'I know that three boys read this book.'

This example shows that narrow scope is not a necessary property of SNs, even in articleless languages. Consequently, I argue that scope behavior does not reliably indicate SN status. Moreover, since I do not adopt a theoretical stance that opposes PNI to SNs, the scope data should not be treated as a decisive diagnostic.

To summarize, I have shown that properties commonly associated with PNI do not consistently hold across all instances of PNI, reflecting the heterogeneity encompassed by the label "pseudo noun incorporation." I also argued that, even though an SN analysis can account for patterns outside of the domain of PNI, it is also is well-suited for accounting for some PNI objects, and should not be categorically opposed to the concept of PNI. In light of this, I reject several properties identified by Pereltsvaig as necessary for identifying SNs. Specifically, verb adjacency, anaphora control, PRO control, number neutrality, and obligatory narrow scope are not reliable diagnostics for SN status.

7.5. Summary

In this chapter, I shaped my understanding of SNs as nominals that lack at least the DP layer, or more of the higher functional projections. I emphasized that the absence of such structure does not necessarily preclude definite interpretation, and conversely, that a non-specific interpretation does not entail a lack of DP structure.

Building on this foundation, I argued that DOM and PNI are best treated as descriptive phenomena that do not have single analyses. Some cases can be analyzed with SNs but some should not be.

To facilitate the identification of SNs cross-linguistically, I introduced a set of diagnostics, grounded in both the existing literature and my own analysis of Moksha and Hill Mari. These diagnostics were selected specifically because they reflect structural reductions and resist alternative explanations. Importantly, they should be applied to nominals that lack overt case marking or fail to trigger agreement in positions where such features are otherwise expected. Here are the diagnostics once again:

(668) Diagnostics for detecting SNs:

A. **MDMI diagnostic**:

If a language has formal definiteness marking that is obligatory (or at least possible) in some syntactic configurations and is ruled out in others, then definite nominals lacking such marking are smaller than DP.

A'. **MCMDI diagnostic**:

If nominals appear without case marking in a position where case is normally present, yet can be interpreted as definite, then these nominals are smaller than DP.

B. **MNMI diagnostic**:

If a language has overt number markers that are obligatory (or at least available) in some syntactic configurations but ruled out in others, then non-number-neutral nominals that lack overt number marking are smaller than NumP (and, by transitivity, smaller than DP).

C. **MPMI diagnostic**:

If a language has nominal agreement markers that are obligatory (or at least available) in some syntactic positions but ruled out in others, then possessed nominals lacking agreement are smaller than NumP (and, by transitivity, smaller than DP).

D. **Restrictions on Modification diagnostic revisited**:

a. If a nominal allows modification with all modifiers except strong quantifiers, it is an SN.

b. If a nominal does not allow modification by numerals (and all higher modifiers), it is an SN.

In addition to establishing these diagnostics, I reviewed other properties that have been proposed as indicative of SNs. I argued that many of these are analytically ambiguous and may mislead if used as diagnostics, since they are compatible with multiple syntactic analyses. The properties turned into uninformative/misleading diagnostics are the following:

(669) Diagnostics that are uninformative or misleading:

A. **Category restriction diagnostic**:

If pronouns cannot be bare, then bare nominals are SNs.

B. **Movement for case diagnostic**:

If a bare nominal occurs in a case position, it is an SN.

Finally, I critically examined the claim that certain properties oppose SNs to PNI. I showed that such an opposition is empirically unmotivated, given the wide cross-linguistic variation found under the label "PNI". Thus, properties often referred to as PNI properties do not provide reliable diagnostics for identifying SNs. I further demonstrated that most properties often associated with PNI — such as adjacency to the verb, anaphora control, PRO control, number neutrality, and scope — do not hold consistently across languages or constructions in first place.

Chapter 8: Conclusions

In this dissertation, I investigated the empirical domain of reduced nominal structures — small nominals (SNs) — which appear in argument positions yet lack properties typically associated with full DP arguments. I focused on two primary case studies: bare complements in Moksha and bare direct objects in non-finite clauses in Hill Mari. Through a detailed comparison of the morphosyntactic behavior of bare nominals versus cased arguments, I identified core structural properties of SNs and proposed a set of cross-linguistic diagnostics for their identification.

8.1. Key findings of the dissertation

The presence of SNs alongside full DPs in a language with a definite article, such as Moksha, provides further evidence against the binary approach that classifies languages as either DP or NP languages (see also Oda 2022 on a scalar approach to NP/DP-language parametrization). The Moksha case study also highlights that not all nominals that appear caseless on the surface should be assumed to be structurally reduced. The co-existence of small bare complements of postpositions and bare direct object DPs in Moksha demonstrates that even in languages that allow SNs in argument positions, some bare nominals can still be full DPs (even though someone could assume that only in the languages that do not allow for SNs in argument positions can bare nominals be DPs).

The Hill Mari case study further illustrates that not only can SNs co-exist with full DPs, but also that SNs of varying structural sizes may coexist within a single language. This finding shows that the size of nominal arguments can vary depending on the construction. This, in turn, suggests that our understanding of categorial selection should be reconsidered: the structural size of nominal arguments is influenced by the larger syntactic context in which the selecting head occurs, rather than by the lexical properties of the head itself.

The main theoretical contribution of this dissertation is the recognition that not all bare nominals are small nominals, and that careful consideration must be given to distinguishing between properties that require a structurally reduced analysis and those that are merely compatible with such an analysis. I propose cross-linguistic diagnostics based on properties that entail a reduced nominal structure — specifically, markedness-interpretation mismatches and restrictions on modification. I also critically examine other properties often cited in support of an SN analysis — such as category restrictions, movement restrictions, number neutrality, support of anaphora, and obligatory narrow scope — and argue that these do not in themselves necessitate a reduced structure.

8.2. Implications

8.2.1. Implications for the general nominal architecture

One of the key theoretical contributions of this dissertation is the argument in favor of a low base-generation position for possessors, numerals, and demonstratives, and the claim that these elements do not necessarily undergo movement to higher positions.

The investigation of different sizes of SNs offers new empirical support for a strictly hierarchical architecture of the extended projection, in which intermediate projections cannot be skipped. That is, if a nominal lacks a particular projection, all projections structurally higher than it must also be absent. For example, a nominal cannot have a DP layer without also having a NumP layer beneath it. On the other hand, a verbal head (V) may select a NumP directly as its complement, skipping the DP layer. This asymmetry indicates a crucial distinction between complementation within a single extended projection (which must be continuous) and complementation across different extended projections (which may permit the absence of certain functional layers). This finding highlights important constraints on nominal structure and informs our understanding of syntactic selection and projection.

8.2.2. Implications for the theory of case

The availability of SNs as complements of adpositions has significant implications for the theory of case. One such implication concerns inherent cases and their analysis as syntactically Ps. The other one addresses the issue of the source of the case for adnominal DPs (such as possessors).

8.2.2.1. Inherent cases as Ps

Given that most inherent cases are argued to be better analyzed as syntactic Ps rather than as morphological case assigners, it is important to keep in mind that not only DPs, but also smaller structures may be complements of Ps (McFadden 2004; Asbury 2008, a.o.). This consideration changes completely the way one might think about nominals in oblique cases when they exhibit properties restricted compared to ones exhibited by other DPs. If one is convinced that PPs are always built on top of DPs, which would follow from a widespread assumption that P is a head in the extended projection of P (Grimshaw 1993), then the restricted modification and referential properties sometimes observed in oblique phrases pose a problem. Such cases would require additional stipulations to explain why the full range of DP-level features are absent. However, if we instead adopt a model in which Ps can select SNs — i.e., projections smaller than DP — these restrictions fall out naturally from the reduced structure. Crucially, SN selection by P is possible only if P forms its own projection, rather than serving as a functional head within the extended projection of the nominal.

This theoretical perspective is supported by data from Moksha, which has several inherent cases with spatial semantics, such as inessive, elative and illative (Table 19), see Chapter 3 for an overview of Moksha nominal morphosyntax.

Table 19: *A fragment of the Moksha case paradigm: inessive, elative and illative* (adopted from Kholodilova 2018a: 66)

Case	Marker
Inessive	*-sə*
Elative	*-stə*
Illative	*-s*

What is particularly noteworthy about these spatial cases is that, unlike the genitive and dative, they do not allow overt marking for number or definiteness. When definiteness or plurality needs to be expressed, the language resorts to analytic constructions with postpositions instead, as demonstrated in (670)–(672). Example (670) shows that inessive case is incompatible with number and definiteness within one nominal. Examples (671) and (672) demonstrate the solution; complements of postpositions are genitival; genitival phrases can bear definiteness and number affixes.

(670) MOKSHA: INESSIVE (LOCATIVE, LEXICAL) → NO DEFINITENESS, NO PLURALITY
*vel'ə-t-n'ə-sə [village-PL-DEF-IN] 'in the villages'

(671) MOKSHA: PLURALITY IN LOCATIVE CONTEXTS → POSTPOSITIONAL PHRASE WITH DEFINITE PHRASE
vel'ə-t'-n'ə-n' esə [village-PL-DEF-GEN in.IN] 'in the villages'

(672) MOKSHA: SINGULARITY IN LOCATIVE CONTEXTS → POSTPOSITIONAL PHRASE WITH DEFINITE PHRASE
vel'ə-t' esə [village-DEF.SG.GEN in.IN] 'in the village'

These examples suggest that in Moksha, the nominals bearing spatial markers must be small, i.e., lacking the higher projections that host definiteness and number features. These features are only available when the phrase is structured as a PP with a genitive DP complement.

Additional support comes from the behavior of possessed nominals. Again, in the so-called possessive declension, number is only expressed in nominative, genitive and dative, while the other cases exhibit number neutrality. As shown in (673), genitive forms encode number distinctions, while in inessive forms (674), number is neutralized. Furthermore, the order of affixes differs: in genitive and dative forms, nominal agreement follows case, but in spatial cases, it precedes the spatial suffix.

(673) MOKSHA: NOMINATIVE POSSESSIVE → NUMBER DISTINCTION
a. *vel'ə-z'ə-n'* [village-1SG.AGR.SG-GEN] 'of my village'
b. *vel'ə-n'ə-n'* [village-1SG.AGR.PL-GEN] 'of my villages'

(674) MOKSHA: INESSIVE (LOCATIVE, LEXICAL) POSSESSIVE → NO NUMBER DISTINCTION
vel'ə-sə-n [village-IN-AGR.1SG] 'in my village / in my villages'

These morphological asymmetries are unexpected if all complements of Ps are full DPs, as shown schematically in Figure 84. In contrast, if we allow for SN complements of Ps, as in Figure 85, these patterns follow straightforwardly from the reduced structure: number and definiteness features are simply not projected and thus cannot be morphologically realized.

Figure 84: *Moksha inessive phrases as DPs*[58]

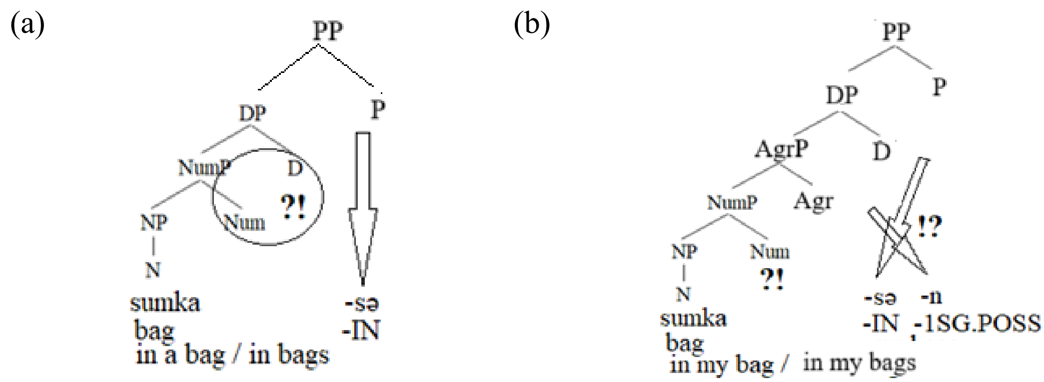

Figure 85: *Moksha inessive phrases as SNs*

In sum, recognizing that SNs can serve as P-complements allows us to capture the restricted morphological behavior of oblique nominals in Moksha without ad hoc stipulations. This supports a broader theoretical move: not all oblique phrases are built on DPs, and the structure of nominal complements in PPs should be evaluated in light of the availability of reduced nominal projections.

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[58] For illustrative purposes, I omit functional projections that lack overt morphological realization.

8.2.2.2. Case assignment to adnominal DPs

Now I can proceed to the second implication. Case assignment domains are defined through phases, and DP is a phase in which genitive is assigned as the unmarked case (Baker 2015b). Since small nominals may be nPs, and possessors are generated low in the structure and may stay low, the question arises how these possessors get case. Because nPs do not constitute case-assigning domains, the prediction is that possessors in small nominals should receive case from the next enclosing phase, typically the one headed by a P (or V, depending on the structure). This prediction is borne out in the behavior of possessor case in Izhma Komi, a Uralic language (Komi Zyryan branch, Finno-Ugric family).

In Izhma Komi[59], possessors of subject nominals are obligatorily marked with the genitive case, with no option of the lack of case morphology (675). In contrast, possessors of oblique nominals, such as complements of PPs, cannot be genitive-marked; instead, they appear bare, with genitive morphology resulting in ungrammaticality (676). This asymmetry suggests that possessor case marking in Izhma Komi is not strictly determined by the internal structure of the nominal, but rather by the syntactic position of the possessed nominal itself. This is problematic under traditional case theories, which predict uniform genitive case for possessors across contexts. However, under the SN analysis, this pattern is expected.

[59] The Izhma Komi case study presented here is based on my own fieldwork data collected in the village of Samburg (Temnikovsky District, Yamalo-Nenets Autonomous Okrug, Russia) during the winter of 2015.

(675) IZHMA KOMI
[*vas'a-**len*** / ****vas'a*** *aj-ys*]SUBJ *ol-e* *čomj-yn*[60]
Vasja-**GEN** Vasja[**NOM**] father-AGR.3SG[NOM] live-PRS.3SG tent-ESS
'Vasja's father lives in a tent.'

(676) IZHMA KOMI
sya *mun-is* *ylla-a* [***aj-ys*** /
that[NOM] go-PST.3SG street-ILL father-AGR.3SG[**NOM**]
aj-ys-len*** *šuba-an*]OBLIQUE
father-AGR.3SG-**GEN** fur-ESS
'He went out in his father's fur coat.'

I argue that this possessor case marking pattern reflects the difference in the nominal size between subjects and complements of Ps (oblique phrases with free-standing postpositions and oblique phrases with spatial "cases"). Specifically, subjects are DPs, while Izhma Komi complements of Ps are smaller. Because of this, the possessor DP in a subject gets assigned genitive, which is the unmarked case of the nominal domain (the complement of D), see Figure 86. Non-pronominal complements of Ps, on the other hand, are SNs in Izhma Komi, so there is no nominal domain to assign genitive. As a result, the possessor must receive case from the next higher phase — the P domain — which in Izhma Komi assigns morphologically unmarked case, leading to bare possessors, see Figure 87.

[60] In the examples, I use a transliteration of the Komi alphabet that reflects the phonological system of Izhma Komi. Most symbols correspond to the IPA, though some diverge: Komi y = /ɨ/, ɨ = /i/, after velarized consonants, ö = /ə/, š = /ʃ /, č = /t͡ʃ/. The symbol ' indicates palatalization.

Figure 86: *Possessor case assignment in subject DPs*

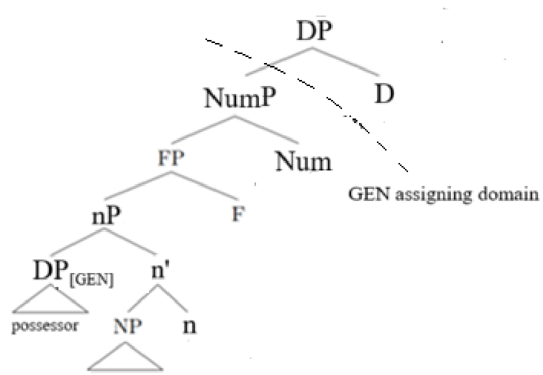
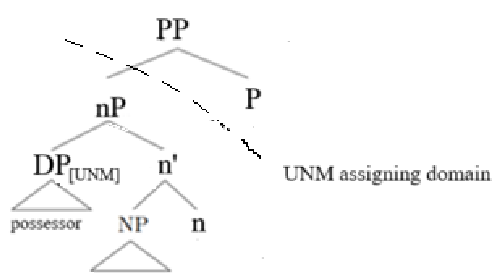


Figure 87: *Possessor case assignment in small complements of Ps*[61]



The evidence that the case of the P-domain is indeed morphologically unmarked comes from the fact that all complements of Ps including pronominal ones surface as bare or nominative (677). I stipulate that only nominals headed by regular nouns must be small nominals in the complement of a P; pronouns are always DPs, and they get case.

[61] In the tree diagram, UNM denotes the morphologically unmarked case assigned within PPs. I use this label descriptively, without committing to a specific theoretical interpretation of what the name of this case should be.

(677) IZHMA KOMI

te	/	****tejad***	***meste-ad***
2SG[NOM]		2SG.GEN	place-AGR.2SG

‘instead of you’

If we look at Moksha data discussed in Chapter 4, we will see that they do not contradict this analysis of adnominal DP case assignment. Unlike in Izhma Komi, in Moksha, all adnominal DPs are always genitive, and there is no asymmetry in the adnominal DP case assignment in subjects and direct objects (DOs) (678) vs. bare complements of postpositions (679). However, as I discussed in that chapter, DP complements of Moksha postpositions get genitive (680), which means that the unmarked case of the P domain in Moksha is genitive, the same as in the nominal domain. This results in the uniform possessor case assignment in Moksha, unlike in Izhma Komi, where the two domains differ in the unmarked case they assign.

(678) MOKSHA

[***mon'***	*morkš-əz'ə*]	*ašč-i*	*s'a-də*	*mala-sə*	*s'embə*
1SG.GEN	table-1SG.AGR.SG[NOM]	be_situated-NPST.3SG	that-ABL	close-IN	all
morkš-n'ə-n'	*kor'as*				
table-PL-GEN	among				

‘My table is the closest out of all the tables.’

(679) MOKSHA

[***mon'***	*morkš*]	*šir'ə-sə-n*
I.GEN	table	side-IN-1SG.AGR

‘near my table.’

(680) MOKSHA

[*kolmə*	*morkš-n'ə-n'*]	*šir'ə-sə*
three	table-DEF.PL-GEN	side-IN

‘near the three tables’

Thus, the small nominal analysis makes very specific predictions regarding the case assignment within a nominal, since adnominal case is associated with the D head. In particular, the absence of the D head should result in an alternative case assignment. In Dependent Case Theory (DCT), the case should be assigned within the enclosing case domain. As I showed, based on Izhma Komi data, this prediction is borne out. Furthermore, different unmarked cases assigned in particular domains (DP or P) in different languages lead to cross-linguistic variation in possessor morphology.

8.3. Further questions

While this dissertation has provided a detailed analysis of SNs and their syntactic behavior, several key questions remain open and deserve further exploration.

8.3.1 The Need for Larger Structures

One important question that remains is: Why do larger nominal structures need to be built if smaller ones are allowed? As I have demonstrated, the largest, full structures (i.e., full DPs) are not always necessary for syntactic completeness. Smaller structures, such as nPs, can often suffice, provided there are no additional requirements that necessitate the presence of a larger structure. However, it remains unclear what these additional requirements might be.

In particular, it seems that certain syntactic positions — subject position, for example — require a full DP, even if smaller nominal structures are syntactically permissible. Moreover, other contexts in which a full projection might be required include the overt realization of nominal agreement and definiteness marking. These phenomena suggest that while smaller structures are viable in many syntactic positions, there are still specific configurations where larger structures are obligatory.

8.3.2 The Acquisition of Small Nominals

The question of why languages have different nominal structures and what these structures reveal about language acquisition is another intriguing aspect of this study. If languages permit varying sizes of nominals in different constructions, and if they impose different syntactic requirements regarding when full structures must be built, this raises the issue of how children acquire such knowledge. It seems plausible that the allowance of SNs could be treated as a parameter that children must acquire. In this framework, a child must learn whether the language allows smaller structures, which positions accommodate SNs, and which nominal size is appropriate for each syntactic configuration, based on linguistic cues provided by the language input.

For example, if a language permits SNs in certain positions (e.g., complements of adpositions) but not in others (e.g., DOs), a child must learn this pattern. The child's task is to determine the distributional restrictions on SN structures and adjust their grammar accordingly. This type of acquisition would require certain cues from the input, as well as an understanding of the syntactic contexts in which each structure is permissible.

8.3.3 Predicting the Positions of SNs

Another intriguing question concerns the predictability of the syntactic positions in which SNs can occur. From a theoretical standpoint, we do not expect SNs to appear in derived positions, since their lack of features limits their ability to move. However, there is no clear reason to assume that SNs cannot be base-generated in specifier positions, provided they remain in situ. This empirical issue warrants further investigation.

A related question concerns the cross-linguistic distribution of SN-allowing positions. One possible hypothesis is that there exists an implicative hierarchy of syntactic positions in which SNs can appear. Specifically, one might imagine a hierarchy of positions, such as A > B > C > D, where the following implicature holds:

- If SNs are allowed in position B, they are also allowed in position A, but not in positions C and D.
- Conversely, if SNs are allowed in position A, they may or may not be allowed in position B, but they cannot appear in C or D.

One might propose the following hierarchy in (681)a, suggesting that if small nominals are allowed in DOs, they are also allowed in complements of postpositions. It is worth noting that the reverse order in (681)b is definitely wrong. Based on Moksha data discussed in Chapters 4 and 5, SNs are allowed in complements of adpositions, but they are not allowed in DOs.

(681) a. Complements of adpositions > Direct Objects > …

b. Direct Objects > Complements of adpositions > …

This hypothesis remains speculative and requires further empirical investigation. More cross-linguistic data are needed to determine whether such a hierarchy exists and, if so, how it might be structured across languages. Additionally, it would be valuable to examine how languages combine SNs with other grammatical properties, such as movement, case assignment, and agreement, to better understand their distribution and the constraints on their occurrence.

8.3.4. Interaction between Num and D

As I showed in Chapter 3, in Moksha, nominal agreement markers (associated with Num) and formal definiteness markers (associated with D) are in complementary distribution, which I attributed to post-syntactic processes. At the same time, number (associated with Num) is always expressed together with definiteness (associated with D). It might not be coincidental that in such a language, we do not find SNs of the NumP size. In other words, in Moksha, not only are Num and D always expressed together, but the presence of Num also entails the presence of D. In contrast, Hill Mari allows NumPs to function as SNs, and this correlates with the availability of an independent number marker that does not encode any additional features.

The contrast between Moksha and Hill Mari thus suggests that the range of possible nominal sizes, such as NumP, depends on language-specific relationships between adjacent syntactic heads. Specifically, in a language where Num and D are obligatorily fused, bare NumPs are ruled out because their morphological realization is impossible. Conversely, in a language where Num is lexicalized independently, bare NumPs are permitted. The idea that syntactic nodes require independent phonological realization is promising, as it finds parallels in other domains. For instance, whether a language allows independent clauses as small as a VP (lacking all functional projections above VP) appears to correlate with the Bare Stem Parameter (Hyams 1986) — that is, whether the language permits bare verb stems as grammatical forms (Tamás Halm, p.c.).

Unfortunately, Hill Mari lacks overt expression of formal definiteness, leaving open the question of whether the fusion of Num and D is limited to languages that morphologically encode definiteness (as in a reformulated NP/DP-languages hypothesis; Bošković 2008), or whether it represents an independent parameter.

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